CIVL6415

TRAFFIC ANALYSIS AND SIMULATION

MODULE 1

Traffic flow fundamentals

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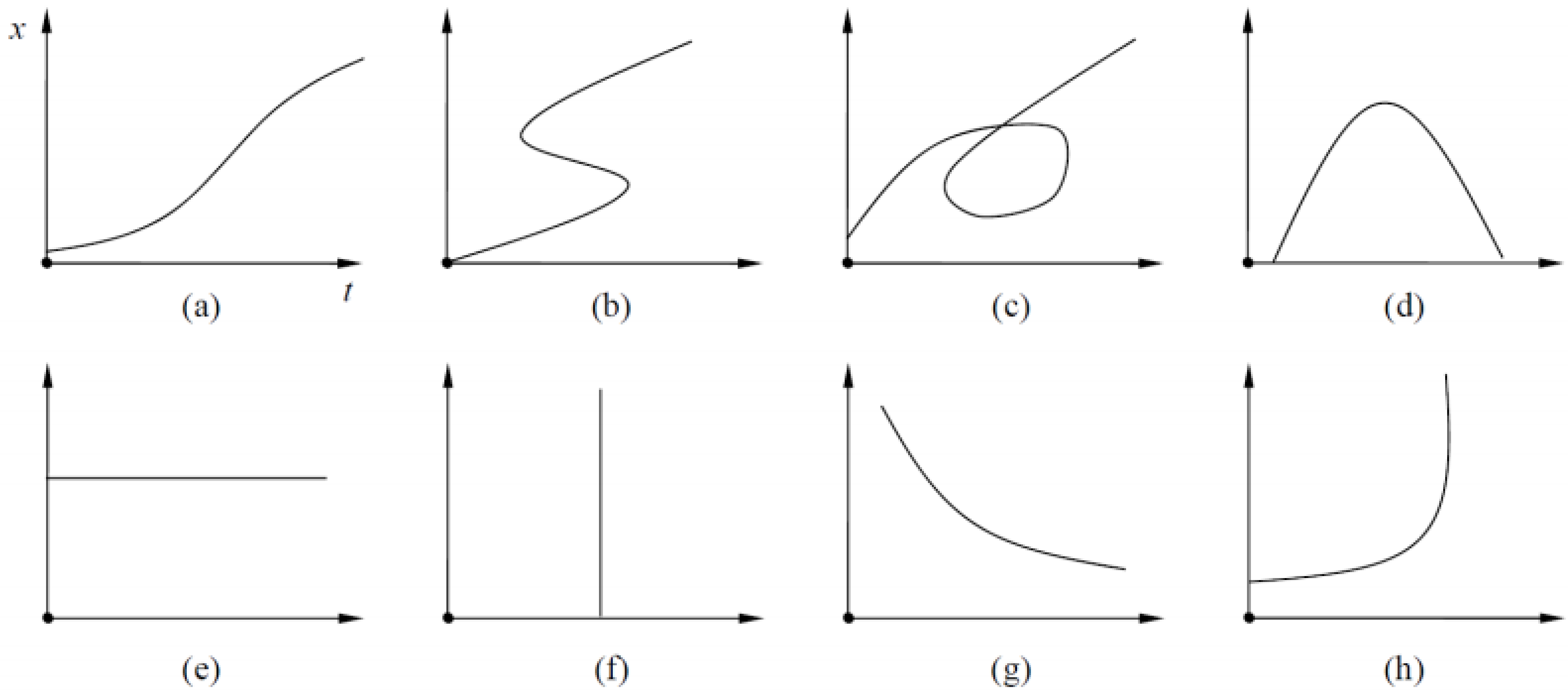
Outline

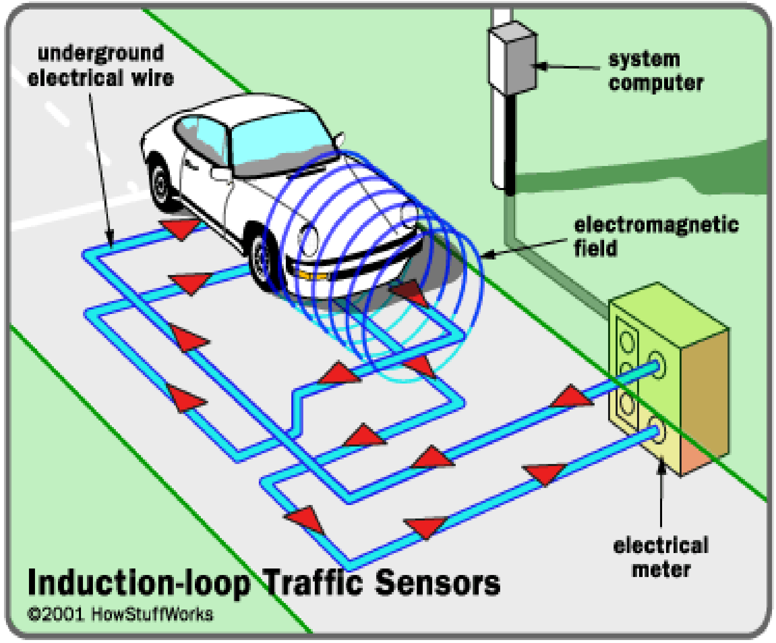
- **Basic Descriptors of Traffic Flow**
  - Flow, Density (Occupancy), Speed
  - Headway, Spacing
- **Relationships between Traffic Flow Parameters**
  - Flow, Density, and Speed
  - Flow vs. Headway
  - Density vs. Spacing
  - Space mean speed vs. Time mean speed
- **Graphical Relationships for Uninterrupted Flow**
  - Speed vs. Density
  - Flow vs. Density
  - Speed vs. Flow
- **Generalized Definitions of Flow, Density and Speed**

Space-time diagram



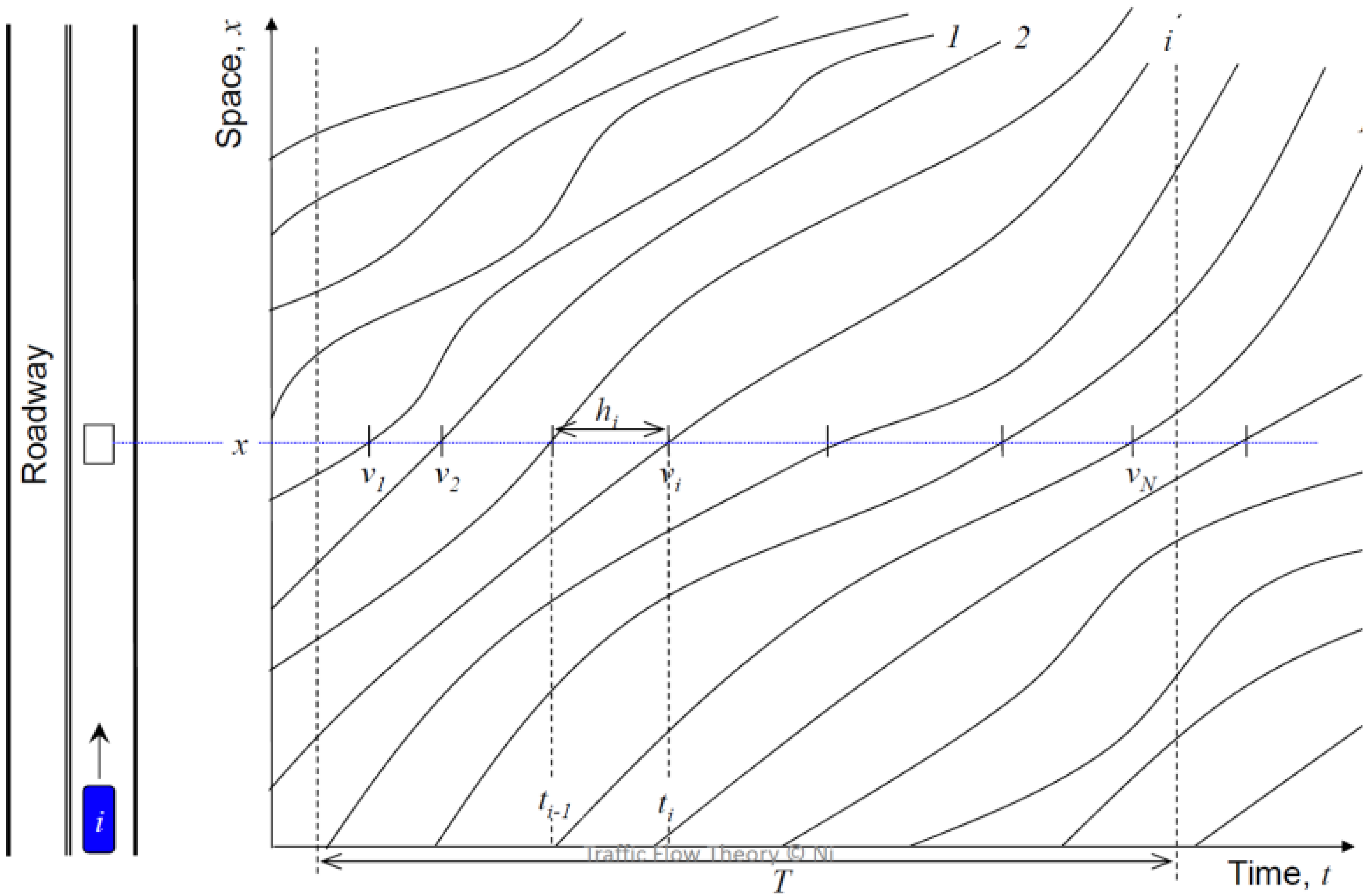
Point sensors

- Inductive loops
  - Coil of wire in pavement
  - Energised by AC
  - Detector senses the ‘presence’ of vehicle
  - Single loop: volume & occupancy
  - Dual loops: vol, occ & speed
  - typically spaced at 500 m on motorways
  - actuated signals
  - dominant technology (low cost)

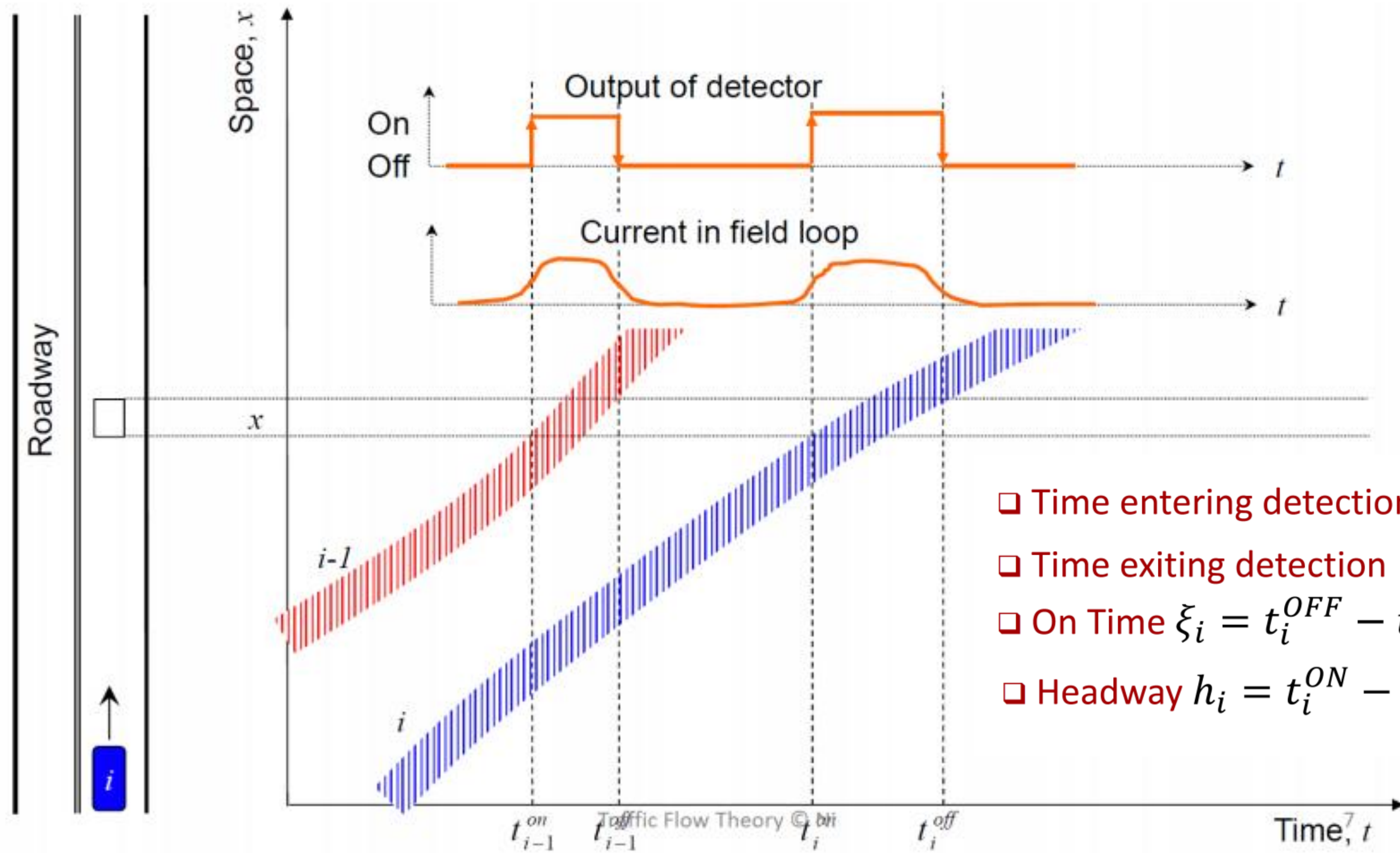


- CCTV cameras

Point sensor data



Point sensor data



- Time entering detection t_i^{ON}
- Time exiting detection t_i^{OFF}
- On Time $\xi_i = t_i^{OFF} - t_i^{ON}$
- Headway $h_i = t_i^{ON} - t_{i-1}^{ON}$

Basic Descriptors of Traffic Flow

Basic Traffic Flow Parameters

- **Three Principal Traffic Flow Descriptors**
  - Volume (q)
  - Density (k) + Occupancy (Occ)
  - Speed (v)
  - Headway (h)
  - Spacing (s)

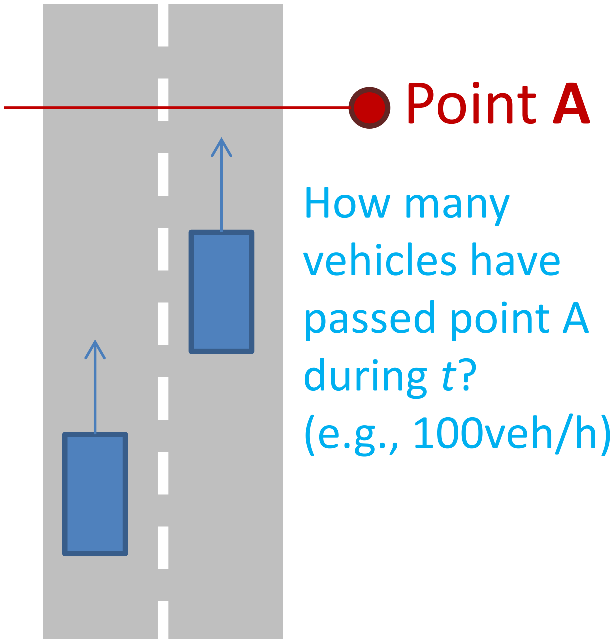
Volume or Flow

- **Flow (q)** (also called 'Volume' or 'Flow rate')
  - Number of vehicles per unit time passing a given point on a road
  - *vehicles per second* (veh/s), *vehicles per hour* (veh/h), or *vehicles per day* (veh/d).

Flow $$q = \frac{N}{T}$$

where:

- *N = number of vehicles passing some designated roadway point during time t, and*
- *T = duration of time interval.*



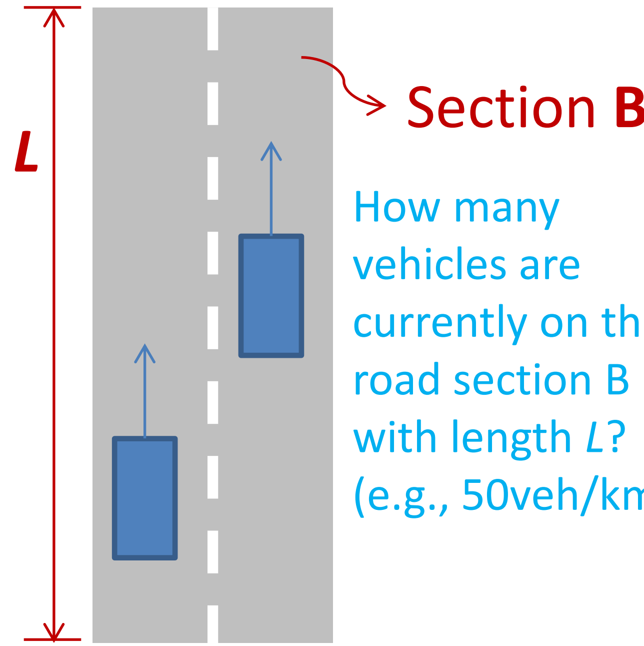
Density

- **Density (k)** (also known as 'Concentration')
  - Number of vehicles present within a unit length of lane or road at a given instant of time
  - *vehicles per kilometre* (veh/km) or *vehicles per metre* (veh/m)

Density $$k = \frac{N}{L}$$

where:

- *N = number of vehicles occupying some length of roadway at some specified time, and*
- *L = length of roadway.*



How many vehicles are currently on the road section B with length *L*? (e.g., 50veh/km)

Question 1

Flow and Density

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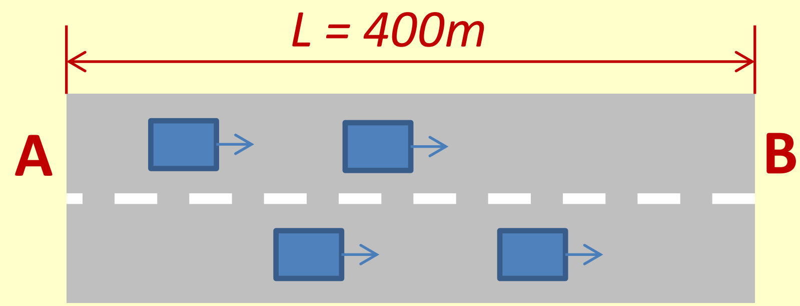
- An observer located at point A observes four vehicles passing point A during 30 sec. The figure shows their positions at an instant of time by photography. Calculate flow (veh/h) and density (veh/km).



A. q = 0.13 veh/h, k = 0.01 veh/km

B. q = 8 veh/h, k = 1 veh/km

C. q = 240 veh/h, k = 10 veh/km

D. q = 480 veh/h, k = 10 veh/km

E. None of the above

Headway

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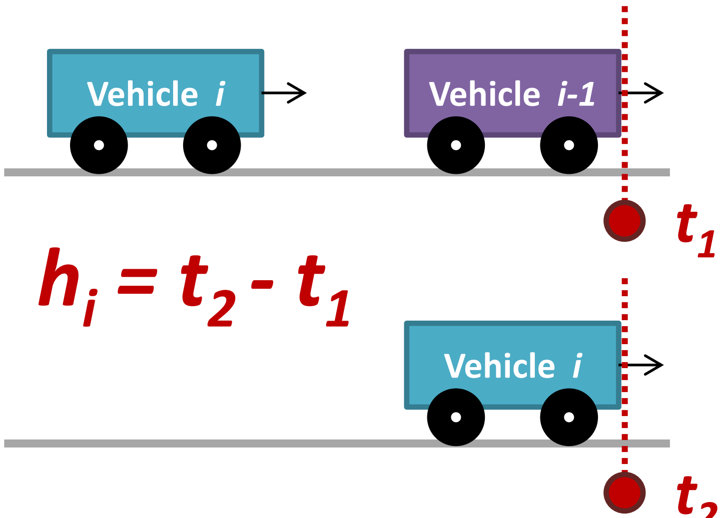
- **Headway (h)**
  - Time interval separating the passing of a fixed point by two consecutive vehicles in a traffic stream, expressed as *seconds per vehicle* (s/veh)
  - The average headway (h) of the stream over a given time interval is the arithmetic mean of the series of headways occurring over that interval

Average headway

$$h = \frac{1}{N}\sum_{i=1}^{N} h_i$$



$$h_i = t_2 - t_1$$

where:

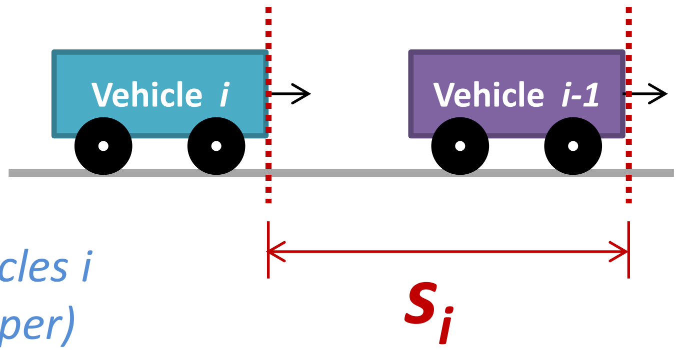
- *h_i = time headway of vehicle i (the elapsed time between the arrivals of vehicles i and i-1)*
- *N = number of measured vehicle time headways*

Spacing

- **Spacing (s)**
  - Distance between the fronts of two consecutive vehicles in a traffic stream at a given instant of time, expressed as *metres per vehicle* (m/veh)
  - The average spacing (s) of the stream over a given length of lane or carriageway is the arithmetic mean of the individual spacings occurring over that length at that instant of time

Average spacing

$$s = \frac{1}{N}\sum_{i=1}^{N} s_i$$



where:

- *s_i = spacing of vehicle i (the distance between vehicles i and i-1, measured from front bumper to front bumper)*
- *N = number of measured vehicle spacings*

Speed

- **Speed (v)**
  - Distance travelled by a vehicle during a unit of time
  - Travel distance / travel time
  - *metres per second* (m/s) or *kilometres per hour* (km/h or kph)
- **Two Ways in which average speeds can be obtained:**
  - Time Mean Speed (v_t):
    - The average speed for vehicles that travel an equal amount of time
    - Average speed over time = Average travel distance / Fixed time
  - Space Mean Speed (v_s):
    - The average speed for vehicles that travel an equal amount of distance
    - Average speed over space = Fixed distance / Average travel time

Speed

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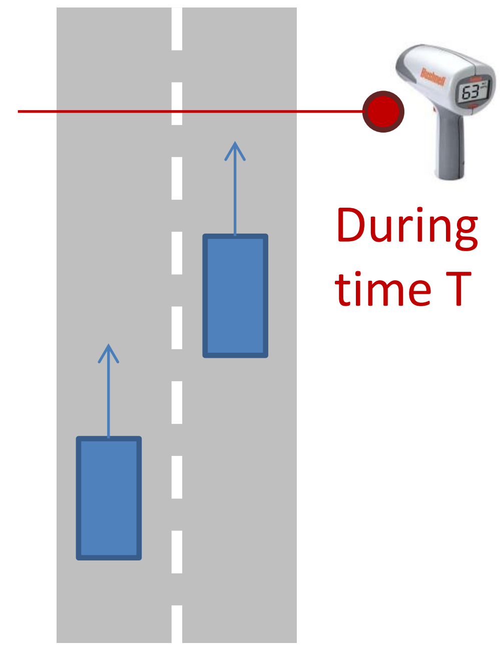
- **Time Mean Speed (v_t)**
  - Arithmetic mean of the measured speeds of all vehicles passing a given point during a given time interval
  - Measured by taking a reference point on the roadway over a fixed period of time T
  - Average travel distance / Fixed time T:

Time mean speed $$v_t = \frac{\frac{1}{N}\sum_{i=1}^{N} d_i}{T} = \frac{\frac{1}{N}\sum_{i=1}^{N} T v_i}{T} = \frac{1}{N}\sum_{i=1}^{N} v_i$$



During time T

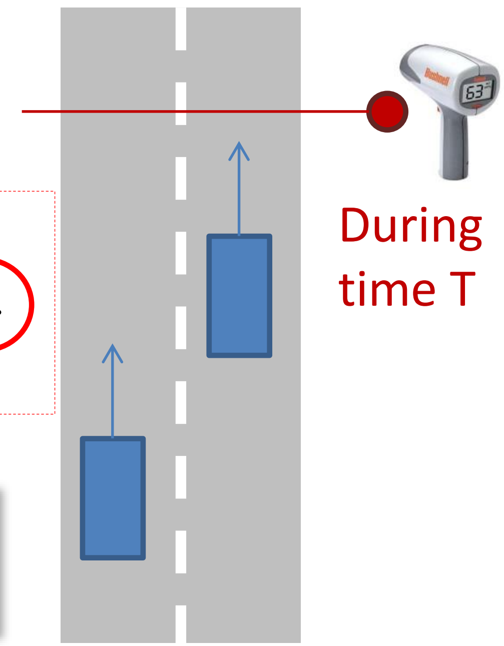
where

- d_i = *travel distance of vehicle i during T*
- v_i = *individual speed (or 'spot speed') of vehicle i*
- N = *number of vehicles during T*

Speed

- **Time Mean Speed (v_t)**
  - Arithmetic mean of the measured speeds of all vehicles passing a given point during a given time interval
  - Measured by taking a reference point on the roadway over a fixed period of time T
  - Average travel distance / Fixed time T:

Time mean speed $$v_t = \frac{\frac{1}{N}\sum_{i=1}^{N} d_i}{T} = \frac{\frac{1}{N}\sum_{i=1}^{N} Tv_i}{T} = \frac{1}{N}\sum_{i=1}^{N} v_i$$



where

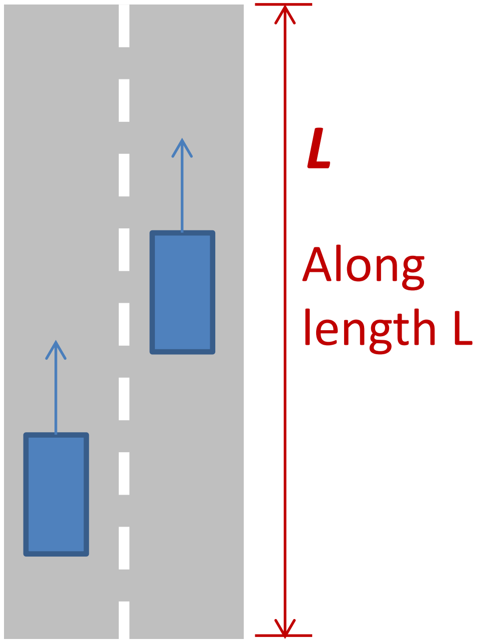
- d_i = travel distance
- v_i = individual spee
- N = number of vehicles during T

Arithmetic mean of spot speeds

Speed

- **Space Mean Speed (v_s)**
  - Arithmetic mean of the measured speeds of all vehicles within a given length of lane or carriageway, at a given instant of time
  - Measured by taking the whole roadway segment into account
  - Fixed length L / Average travel time:

Space mean speed $$v_s = \frac{L}{\frac{1}{N}\sum_{i=1}^{N} t_i} = \frac{L}{\frac{1}{N}\sum_{i=1}^{N} \frac{L}{v_i}} = \frac{1}{\frac{1}{N}\sum_{i=1}^{N} \frac{1}{v_i}}$$



where:

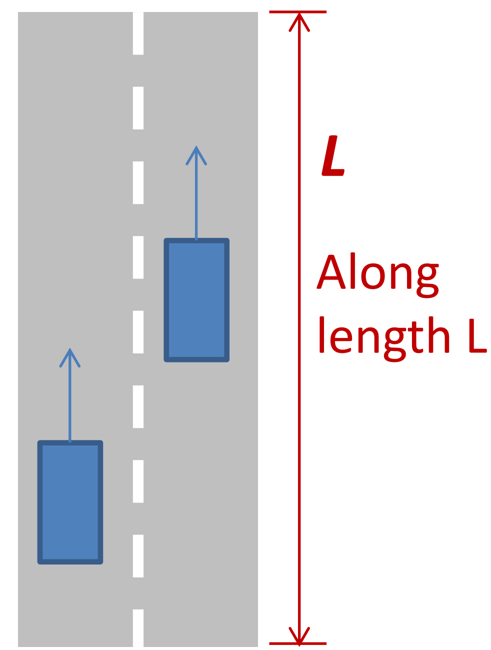
- *t_i = travel time of vehicle i to traverse length L*
- *v_i = individual speed (or 'spot speed') of vehicle i*
- *N = number of vehicles along L at a given time*

Speed

- **Space Mean Speed (v_s)**
  - Arithmetic mean of the measured speeds of all vehicles within a given length of lane or carriageway, at a given instant of time
  - Measured by taking the whole roadway segment into account
  - Fixed length L / Average travel time:

Space mean speed

$$v_s = \frac{L}{\frac{1}{N}\sum_{i=1}^{N} t_i} = \frac{L}{\frac{1}{N}\sum_{i=1}^{N} \frac{L}{v_i}} = \frac{1}{\frac{1}{N}\sum_{i=1}^{N} \frac{1}{v_i}}$$



where:

- *t_i = travel time of vehicle i to traverse length L*
- *v_i = individual spee*
- *N = number of veh*

Harmonic mean of spot speeds

Question 2

Speed

- Three vehicles traverse a 1 km segment of freeway in 1.2, 1.5, and 1.0 minutes respectively. What are the time-mean and space-mean speeds (km/h) ? (Assume that the speeds are constant along the segment).

A. v_t = 41.8 km/h, v_s = 40.9 km/h
B. v_t = 40.9 km/h, v_s = 41.8 km/h
C. v_t = 50.0 km/h, v_s = 48.7 km/h
D. v_t = 48.7 km/h, v_s = 50.0 km/h
E. None of the above

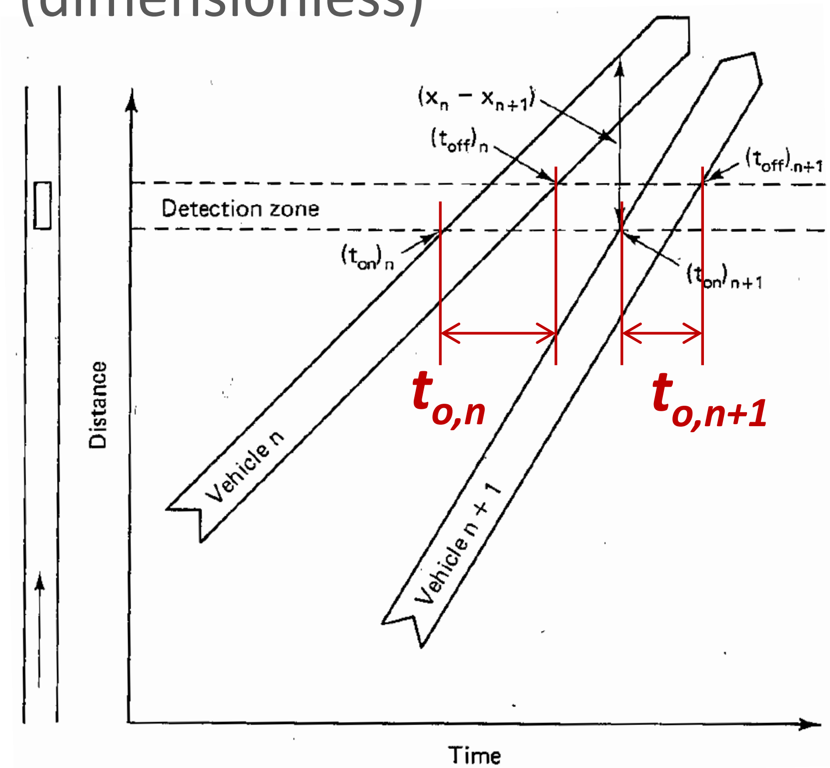
Lane Occupancy

- **Occupancy (Occ)**
  - The proportion of time for which a detector is "occupied" or covered by a vehicle at a given location over a defined period of time T, expressed as a proportion (dimensionless)

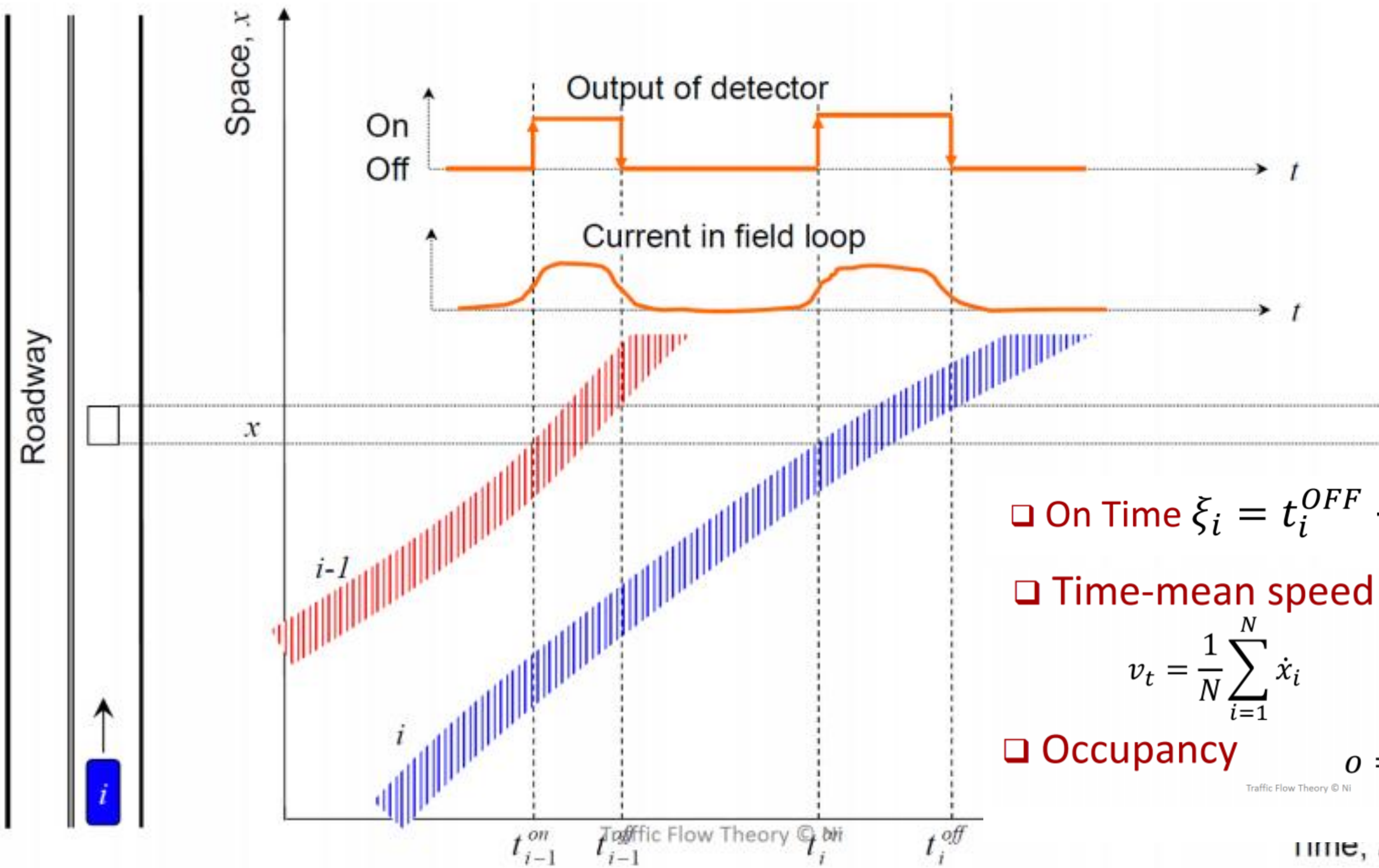
Occupancy $$Occ = \frac{\sum_{i=1}^{N} t_{o,i}}{T}$$

where:

- *$t_{o,i}$ = duration of the period when the detection zone is occupied by vehicle i*
- *N = number of vehicles detected during time period T*
- *T = observation time period*



Point sensor data



- On Time $\xi_i = t_i^{OFF} - t_i^{ON}$
- Time-mean speed

$$v_t = \frac{1}{N}\sum_{i=1}^{N} \dot{x}_i$$

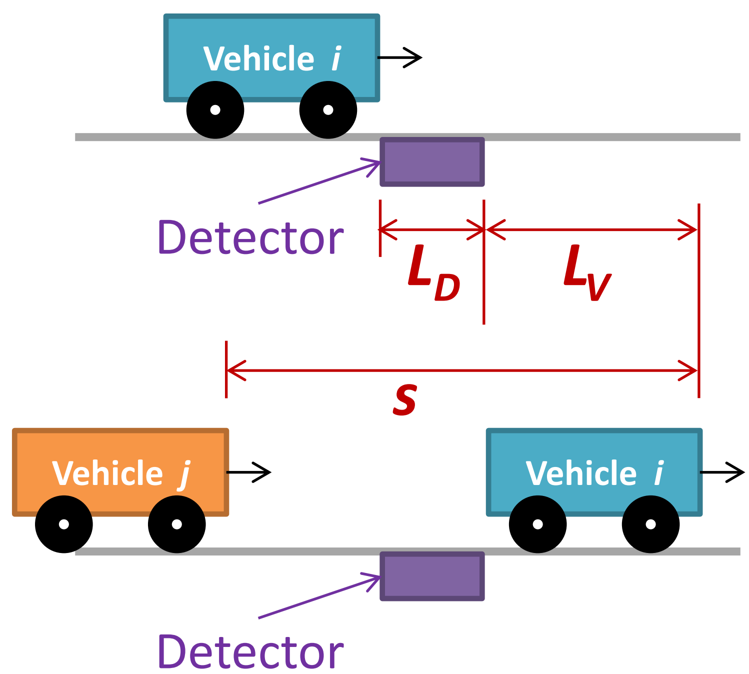
- Occupancy

$$o = \frac{\sum_{i=1}^{N} \xi_i}{T}$$

Lane Occupancy

- **Occupancy (Occ)**
  - Related to the average spacing (s) as follows:

Occupancy $$Occ = \frac{\sum_{i=1}^{N} t_{o,i}}{T} = \frac{L_V + L_D}{s}$$



where:

- $t_{o,i}$ = *duration of the period when the detection zone is occupied by vehicle i*
- L_V = *average length of a vehicle (m)*
- L_D = *effective length of detector (m)*
- *s = average spacing of vehicles (m/veh)*

Lane Occupancy

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▪ Density and Occupancy

– Occupancy is often used as a 'surrogate' for density (k) because density is difficult to measure directly (e.g., need aerial photography)

– Obtain Density from Occupancy:

$$Occ = \frac{L_V + L_D}{s} \qquad \leftarrow s[\mathrm{m/veh}] = \frac{1000}{k[\mathrm{veh/km}]}$$

$$k = \frac{1000}{s} = \frac{1000 \cdot Occ}{L_V + L_D} = \frac{10 \cdot (\%Occ)}{L_V + L_D}$$

where:

- *%Occ = occupancy expressed as a percentage*
- *L_V, L_D = vehicle length and detector length, respectively (m)*

Question 3

Density and Occupancy

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- A detector records a %occupancy of 20% for a 15-minute analysis period. If the average vehicle length is 9m and the detector is 1m long, what is the density?

A. 0.2 veh/km
B. 4 veh/km
C. 10 veh/km
D. 20 veh/km
E. None of the above

$$Occ = \frac{\sum_{i=1}^{N} t_{o,i}}{T}$$

$$k = \frac{1000 \cdot Occ}{L_V + L_D} = \frac{10 \cdot (\%Occ)}{L_V + L_D}$$

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Relationships between Traffic Flow Parameters

Macroscopic vs. Microscopic

- **Macroscopic Parameters**
  - Describing the entire traffic stream

- **Microscopic Parameters**
  - Describing individual vehicles

Macroscopic	Microscopic
Flow (q)	**Headway (h_i)**
Density (k)	**Spacing (s_i)**
Speed (v_s, v_t)	**Speed (v_i)**

Fundamental Relationships

- **Flow is the product of Density and (Space Mean) Speed**

$$\boxed{q = k \times v_s} \qquad \left[\frac{\text{veh}}{\text{h}}\right] = \left[\frac{\text{veh}}{\text{km}}\right] \times \left[\frac{\text{km}}{\text{h}}\right]$$

- **Flow is the inverse of Average Headway**

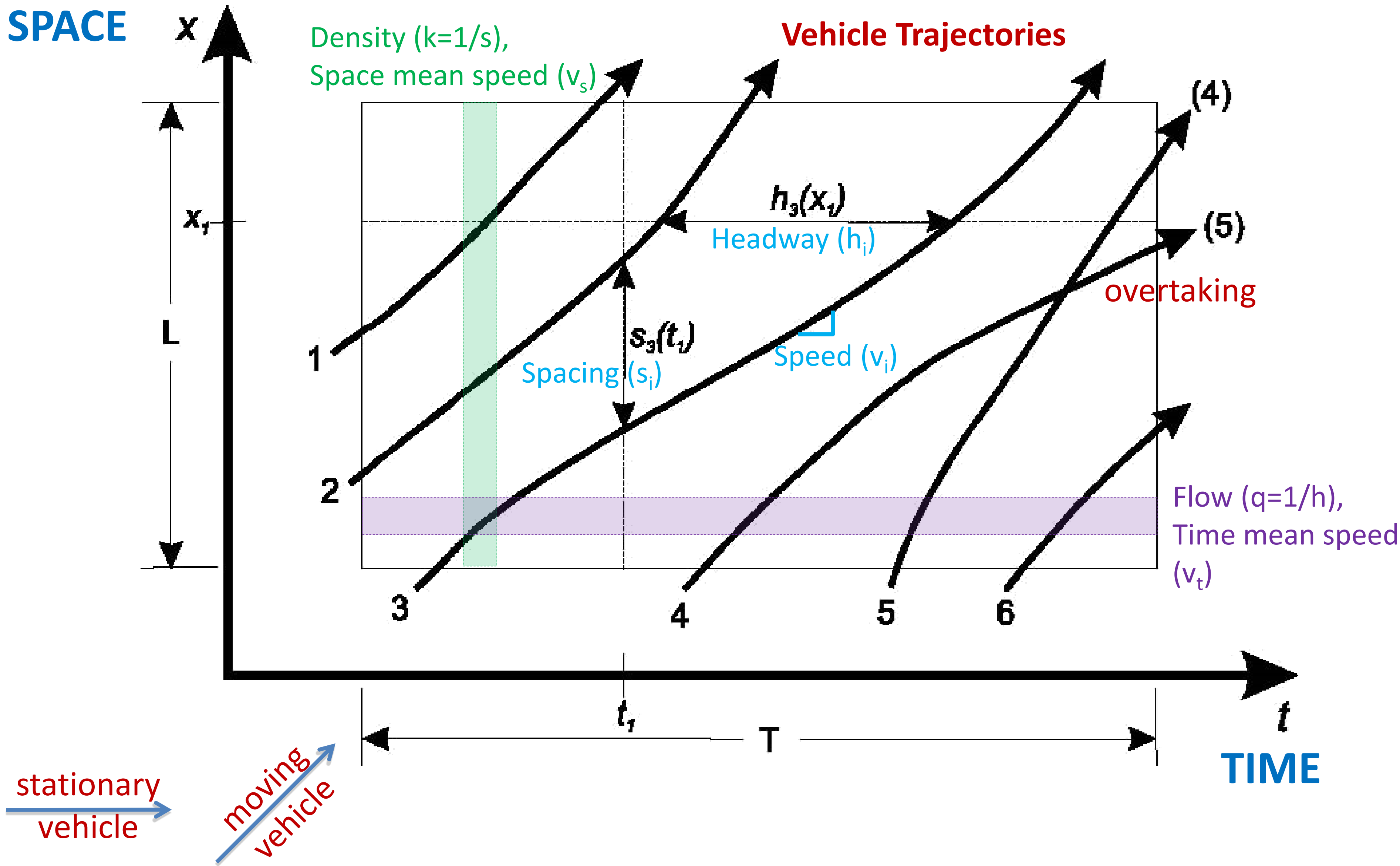
$$q = 1/h \qquad q = \frac{3600}{h} \quad \left[\frac{\text{veh}}{\text{h}}\right] = 1 \Big/ \left[\frac{\text{s}}{\text{veh}} \times \frac{1\text{h}}{3600\text{s}}\right]$$

- **Density is the inverse of Average Spacing**

$$k = 1/s \qquad k = \frac{1000}{s} \quad \left[\frac{\text{veh}}{\text{km}}\right] = 1 \Big/ \left[\frac{\text{m}}{\text{veh}} \times \frac{1\text{km}}{1000\text{m}}\right]$$

- **Other derived relations:** $h = s/v \qquad s = h \cdot v$

Space-Time Diagram



Question 4

Headway and Flow

- Traffic flow on a one lane ramp is 360 veh/h, what is the average headway (seconds)?

A. 0.1 s

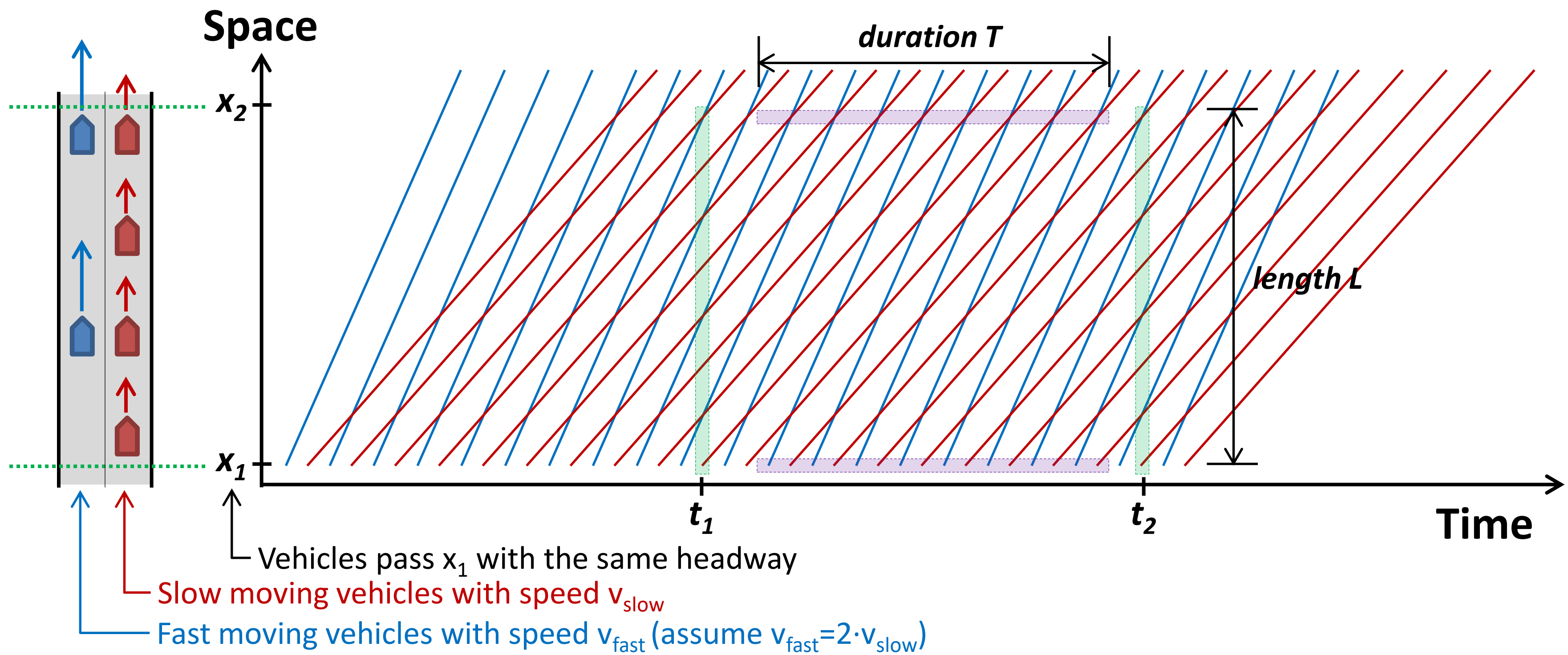
B. 10 s

C. 36 s

D. 60 s

E. None of the above

Why $V_s \leq V_t$??



Slower vehicles occupy any given segment of road for a longer period of time than faster vehicles, and therefore receive a greater weighting in the calculation of space mean speed than they do in the calculation of time mean speed.

[Time Mean Speed] During ***T*** at any given location ***x***, the sample contains an equal number of slow and fast vehicles

$\rightarrow v_t = \frac{1}{2} v_{slow} + \frac{1}{2} v_{fast}$

[Space Mean Speed] Over ***L*** at any given time ***t***, the sample contains twice as many slow vehicles as fast vehicles

$\rightarrow v_s = \frac{2}{3} v_{slow} + \frac{1}{3} v_{fast}$

Relationships between V_t and V_s

- **The Approximate Relationship between Time Mean Speed and Space Mean Speed**

$$v_t = v_s + \frac{\sigma_s^2}{v_s}$$

σ_s^2 = the variance of space mean speed,

i.e., $\sigma_s^2 = \frac{\sum_{i=1}^{N}(v_i - v_s)^2}{N-1}$

- **Space mean speed is always less than or equal to time mean speed ($V_s \leq V_t$)**

  When all vehicles have exactly the same speed, the two mean speeds are equal ($V_s = V_t$).

Graphical Relationships for Uninterrupted Flow

Types of Traffic Flow

- **<u>Uninterrupted Flow</u>** Vs. **Interrupted Flow**
  - Uninterrupted flow occurs in a traffic stream that is not delayed or interfered with by factors external to the traffic stream itself
  - Uninterrupted flow facilities
    - No intersections at grade, traffic signals, STOP or GIVE WAY signs, direct property access (e.g., controlled-access roads such as motorways)
    - The characteristics of the traffic stream are based solely on the interactions among vehicles and with the roadway and the general environment.

Types of Traffic Flow

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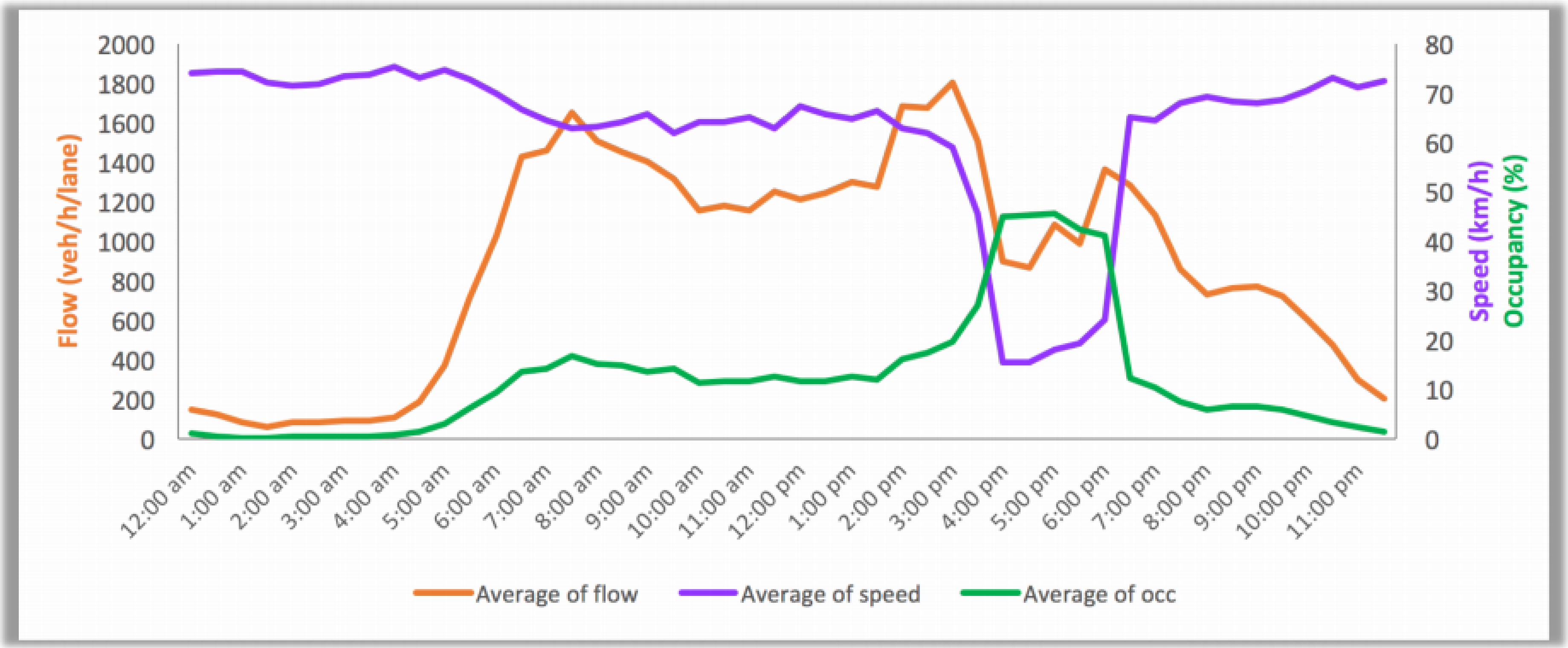
- **Uninterrupted Flow** Vs. **Interrupted Flow**
  - Interrupted flow occurs when external factors have significant effects on the traffic flow.
  - Interrupted flow facilities
    - Incorporate fixed external interruptions into their design and operation: traffic signal, STOP or GIVE WAY signs, unsignalized at-grade intersections, land access operations (e.g., arterials and other urban streets)
    - The characteristics of the traffic stream are primarily based on external interruptions (e.g., "red" signals stop traffic periodically and create a platoons of vehicles); vehicle-vehicle interactions and vehicle-roadway interactions play a secondary role in defining the traffic flow.

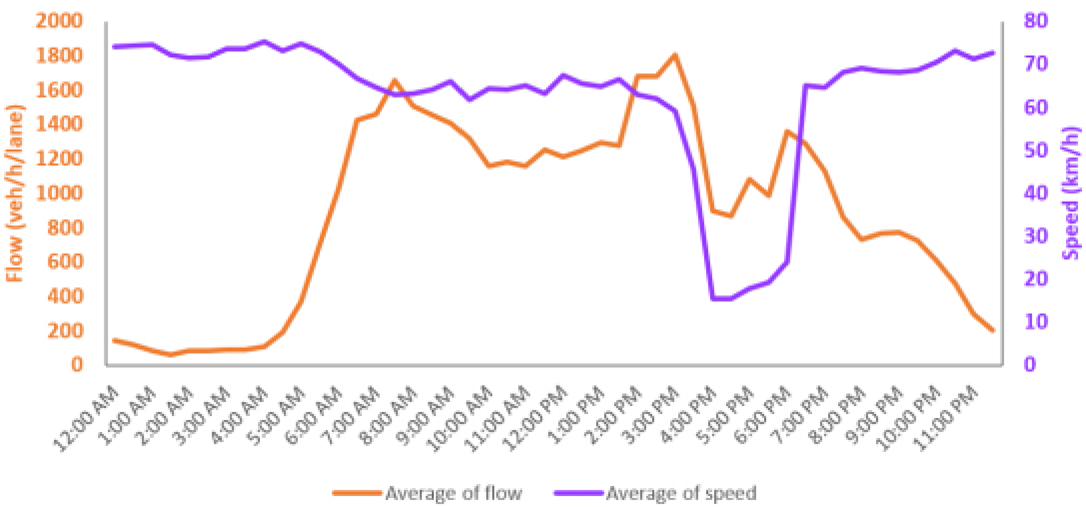
Traffic Characteristics

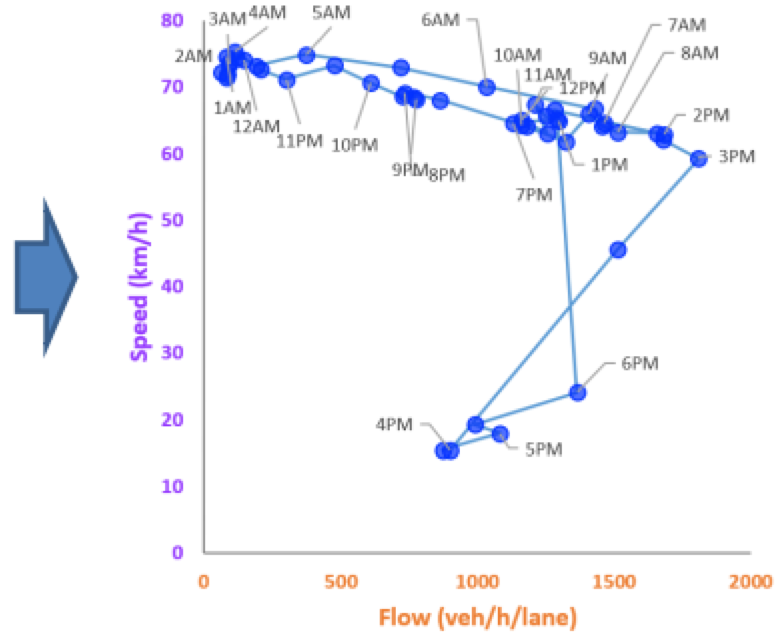
- **Flow, Density, and Speed Relationship**



Traffic Characteristics

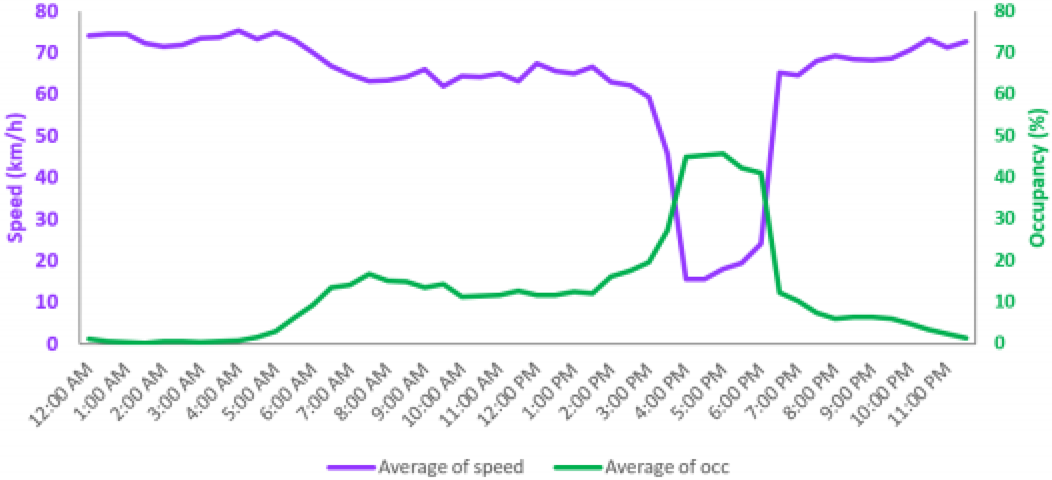
- **Speed (v) – Flow (q) Plot**

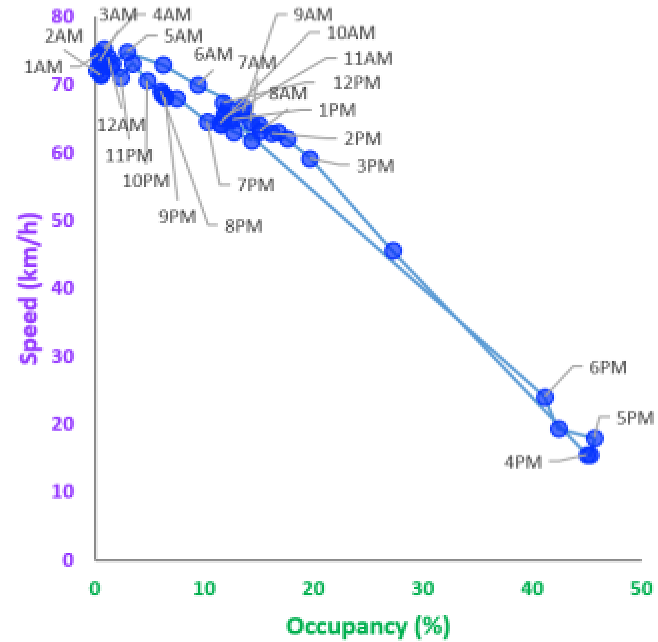




Traffic Characteristics

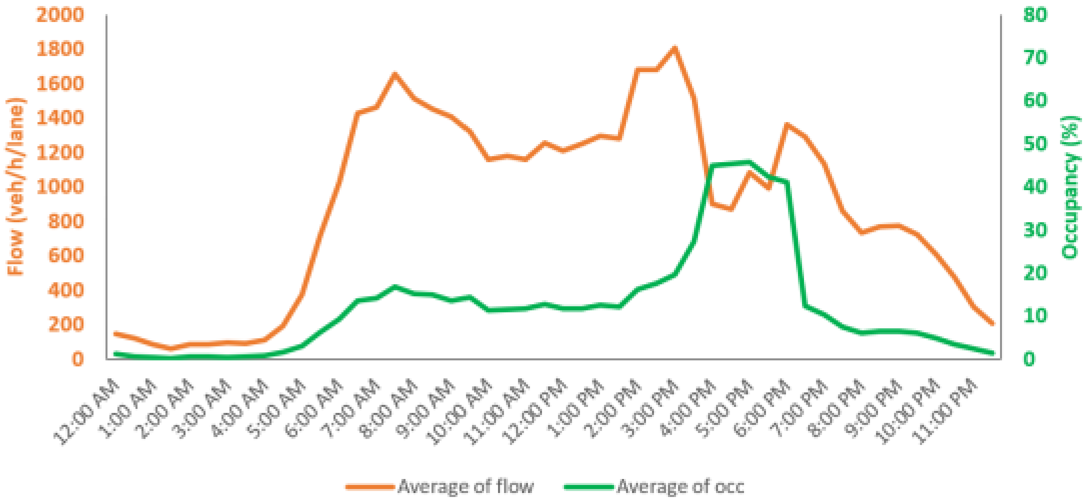
- **Speed (v) – Density (k) Plot**

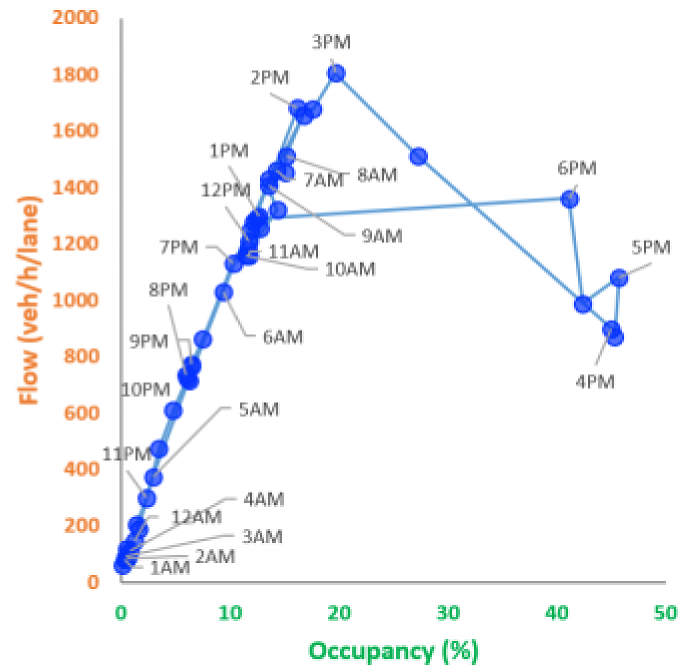




Traffic Characteristics

- **Flow (q) – Density (k) Plot**





Traffic Characteristics

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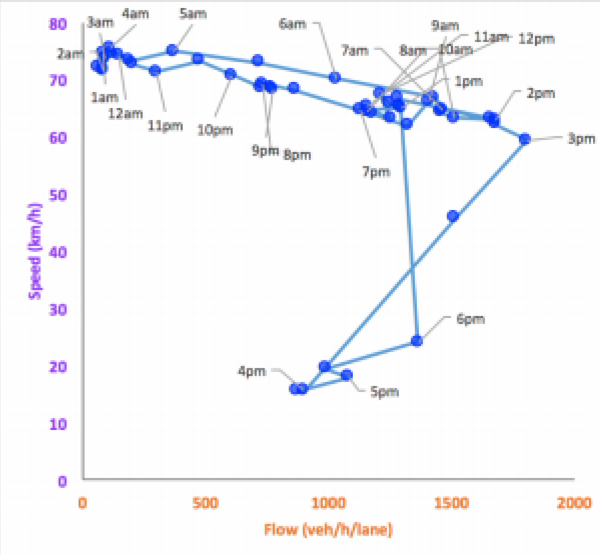
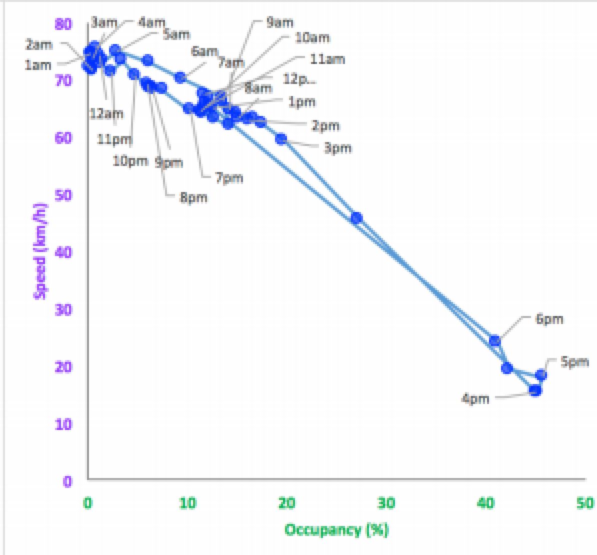
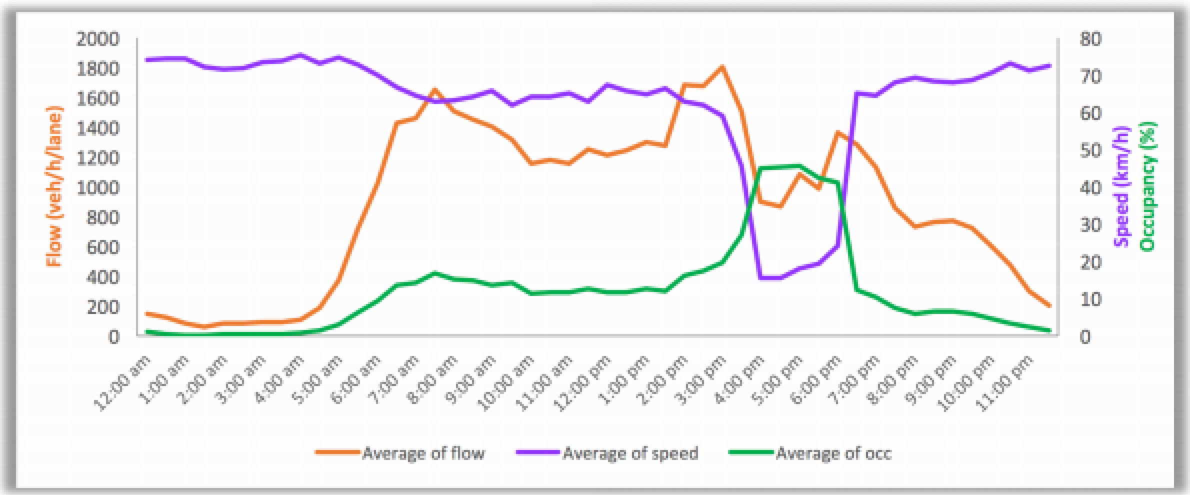
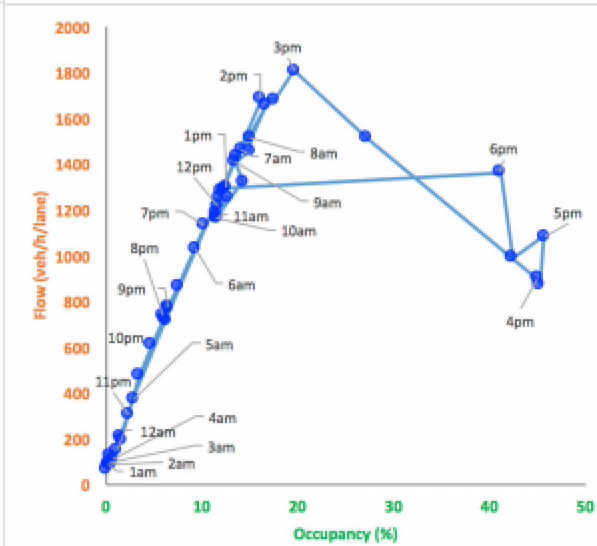
- **Fundamental relationships of 'uninterrupted' traffic flow**

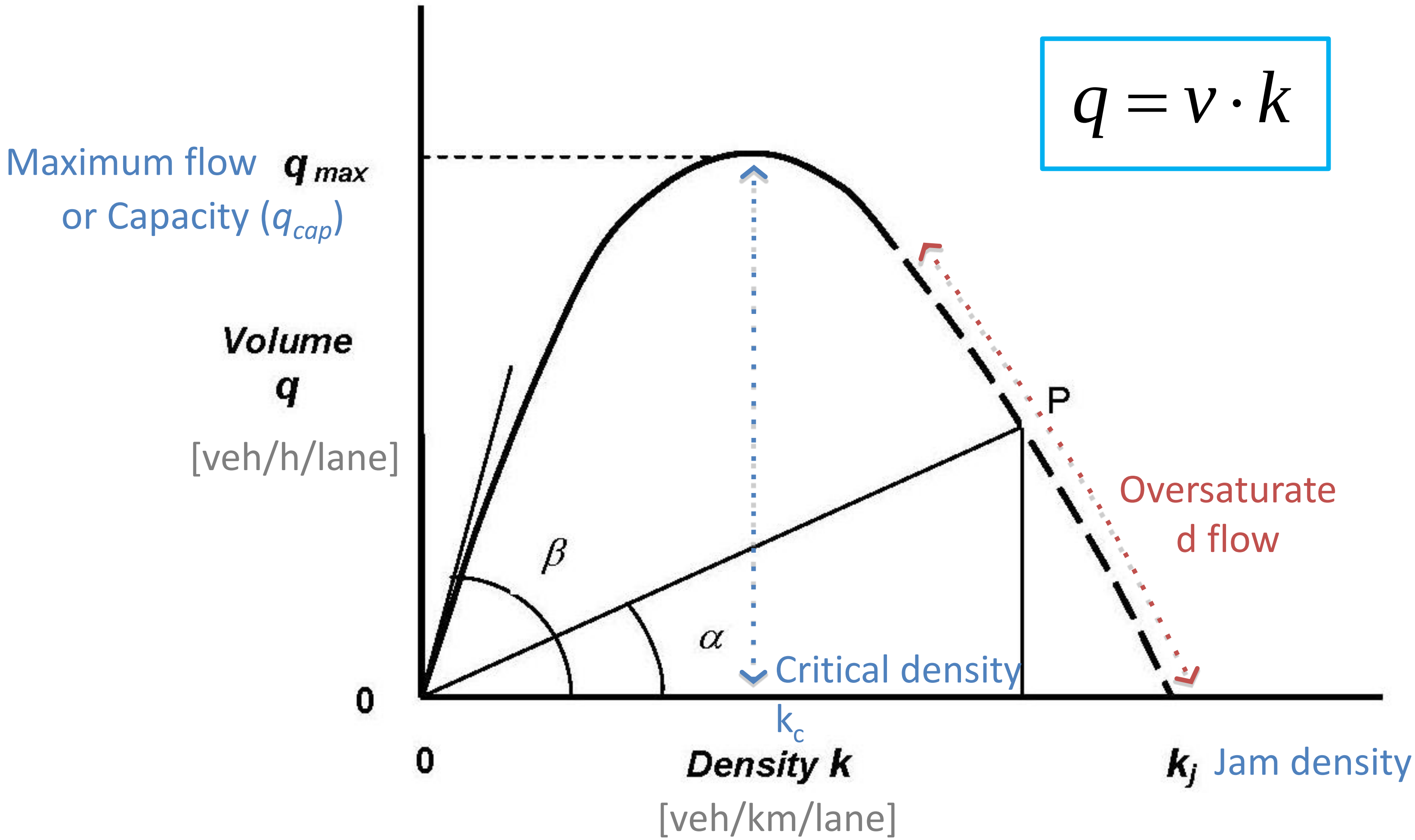




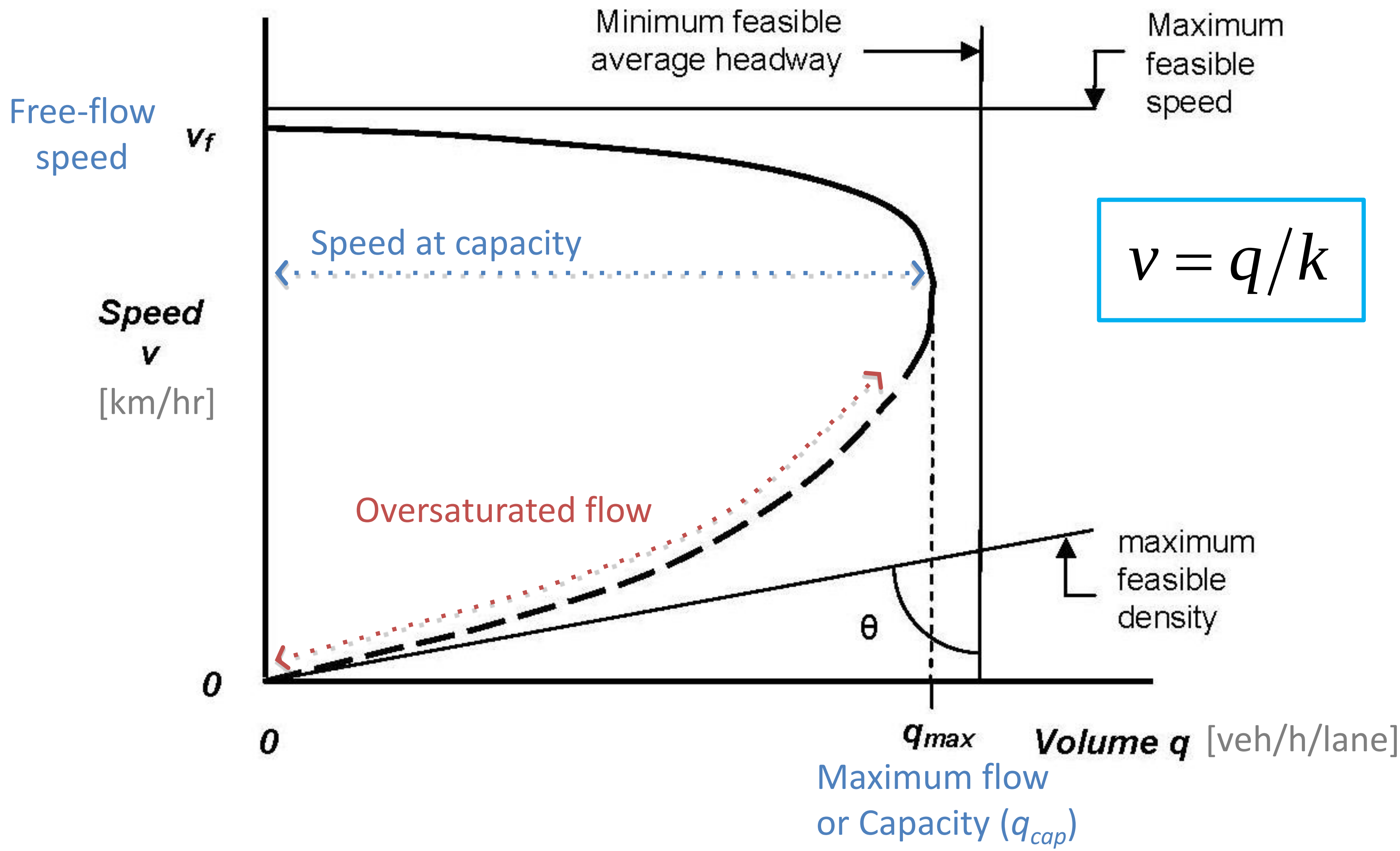




Flow-Density Relationship



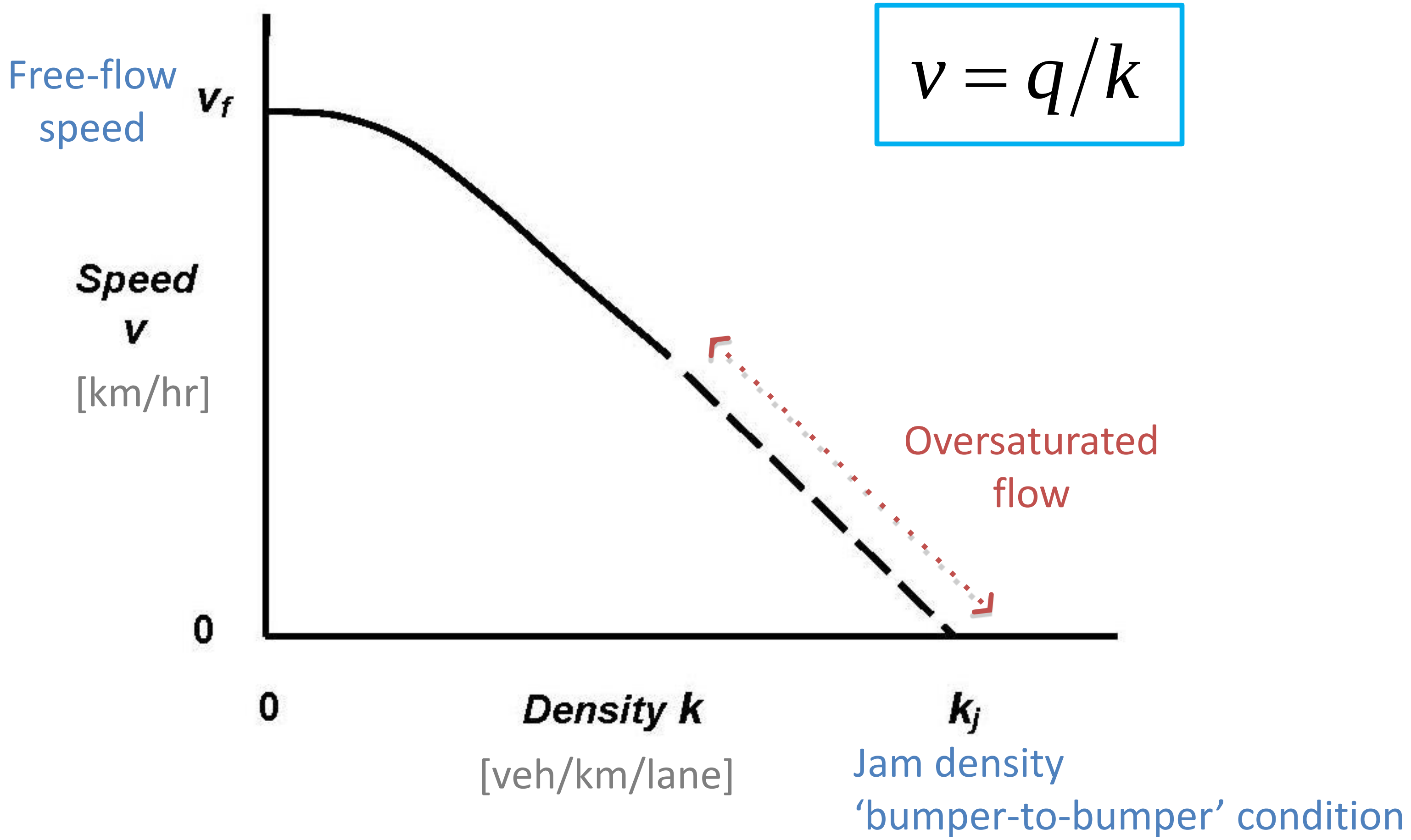
Speed-Flow Relationship



$$\boxed{v = q/k}$$

Speed-Density Relationship

$$v = q/k$$

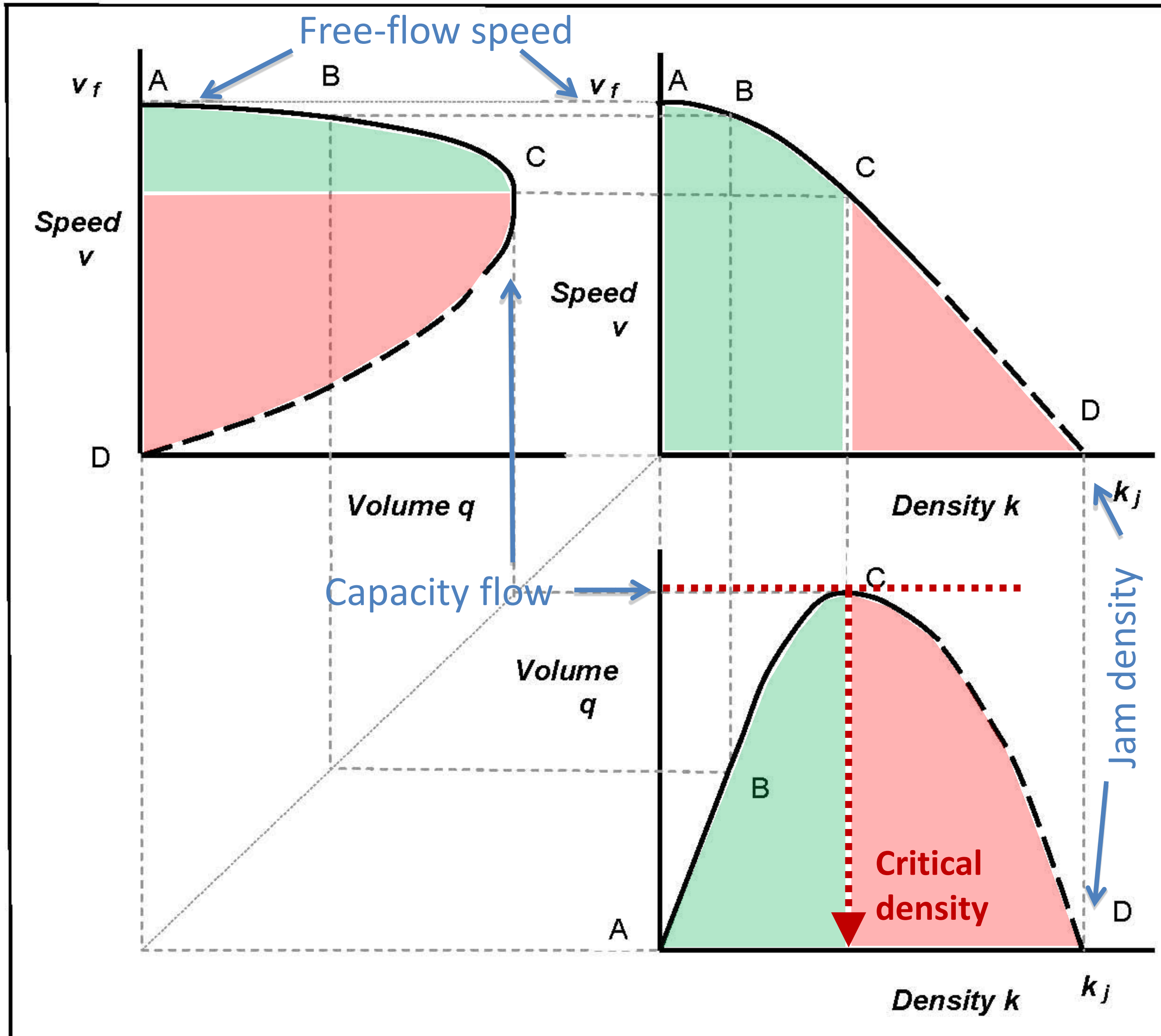


Flow-Density-Speed Relationships

Fundamental Diagrams (FD)

'Fundamental relationships' of 'uninterrupted' traffic flow

At **A**, no interaction
A-**B**, free flow
B-**C**, steady flow
At **C**, traffic unstable
C-**D**, forced flow
At **D**, traffic stationary

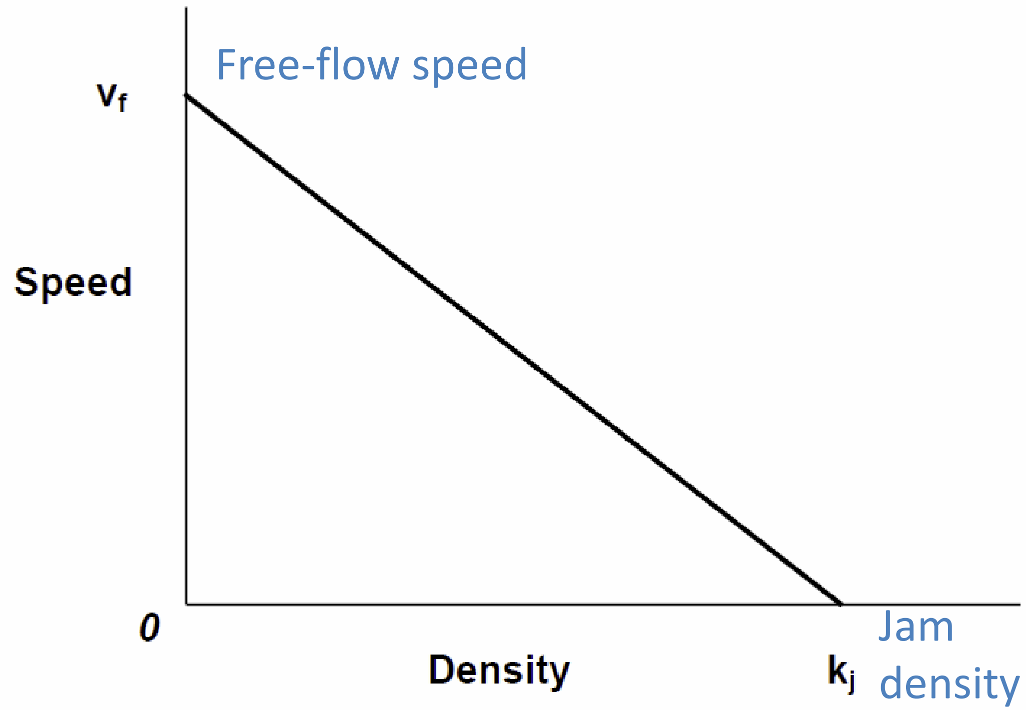


Traffic Stream Model

- *Linear* **Speed-Density Relationship**
  - The first traffic stream model, developed by Greenshields (known as 'Greenshields model')
  - Assumes that the space mean speed of the traffic would decrease linearly from the mean free speed, v_f, at zero density, to zero speed at the jam density, k_j.

$$v = v_f\left(1 - \frac{k}{k_j}\right) \quad \cdots\cdots (1)$$

where:
*v = **space-mean** speed (km/h)*
v_f = free-flow speed (km/h)
k = density (veh/km)
k_j = jam density (veh/km)



Traffic Stream Model

- *Parabolic* **Flow-Density Relationship**
  - $q=k \cdot v$ ← Eq. (1)

$$q = v_f\left(k - \frac{k^2}{k_j}\right) \quad \cdots\cdots (2)$$

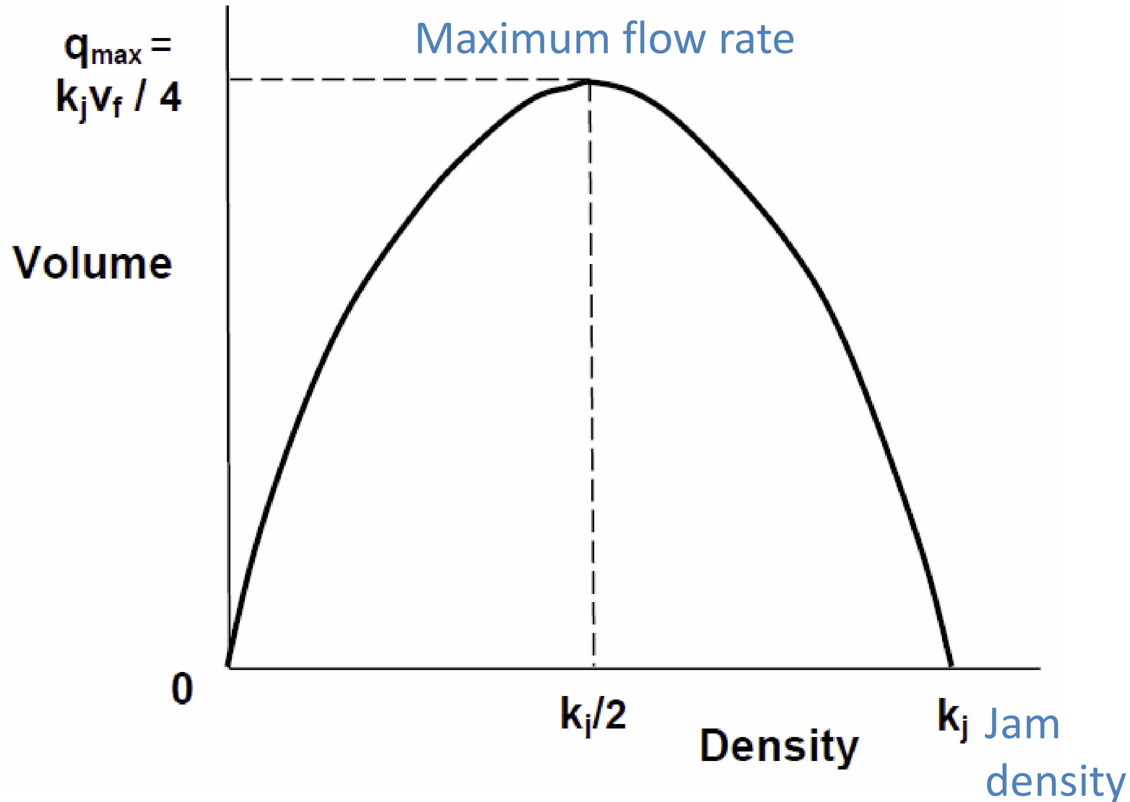
where:

q = flow rate or volume (veh/h)

v_f = free-flow speed (km/h)

k = density (veh/km)

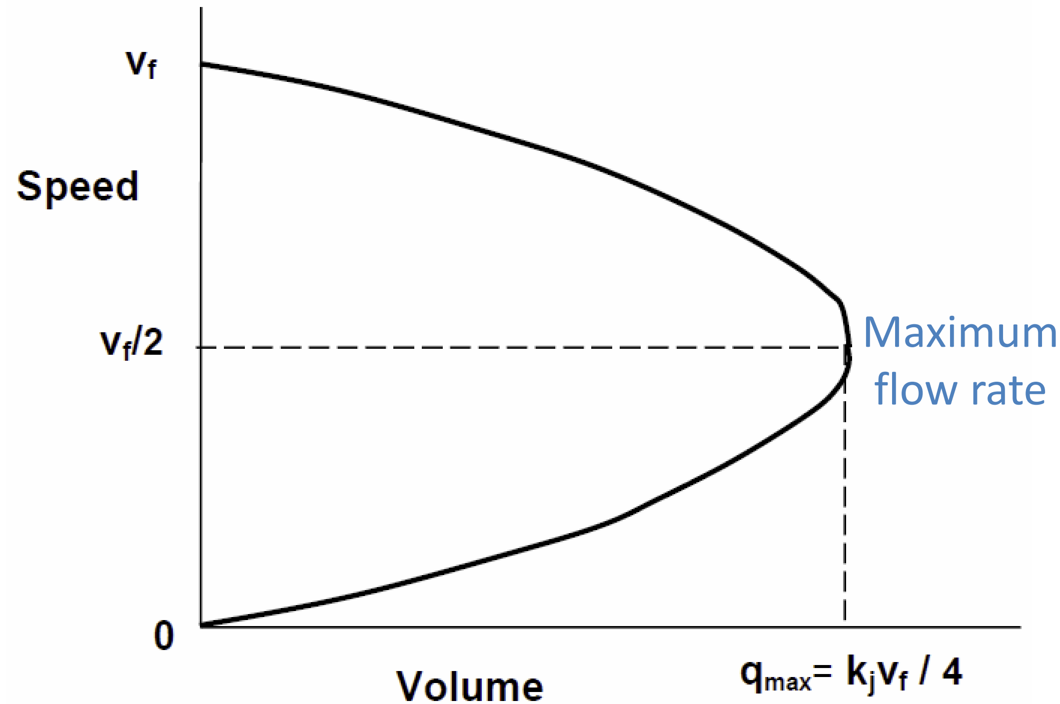
k_j = jam density (veh/km)



Traffic Stream Model

- *Parabolic* **Speed-Flow Relationship**
  - Rearranging Eq.(1) → $k = k_j \left(1 - v/v_f\right)$
  - *q=k·v* ← *k*

$$q = k_j \left(v - \frac{v^2}{v_f} \right) \quad \text{..... (3)}$$



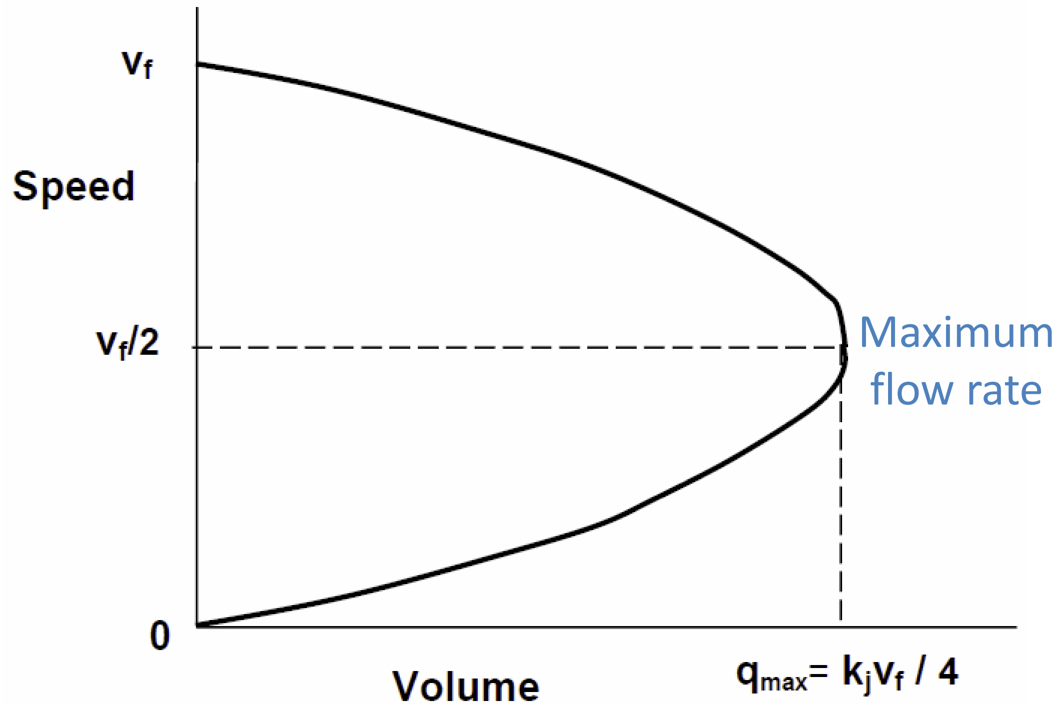
where:
q = flow rate or volume (veh/h)
*v = **space-mean** speed (km/h)*
v_f = free-flow speed (km/h)

Traffic Stream Model

- *Parabolic* **Speed-Flow Relationship**
  – Rearranging Eq.(1) → $k = k_j\left(1 - v/v_f\right)$
  – *q=k·v* ← *k*

$$q = k_j\left(v - \frac{v^2}{v_f}\right) \quad \cdots\cdots (3)$$

$$\frac{dq}{dv} = 0 \Rightarrow \boxed{v_{cap} = \frac{v_f}{2}}$$

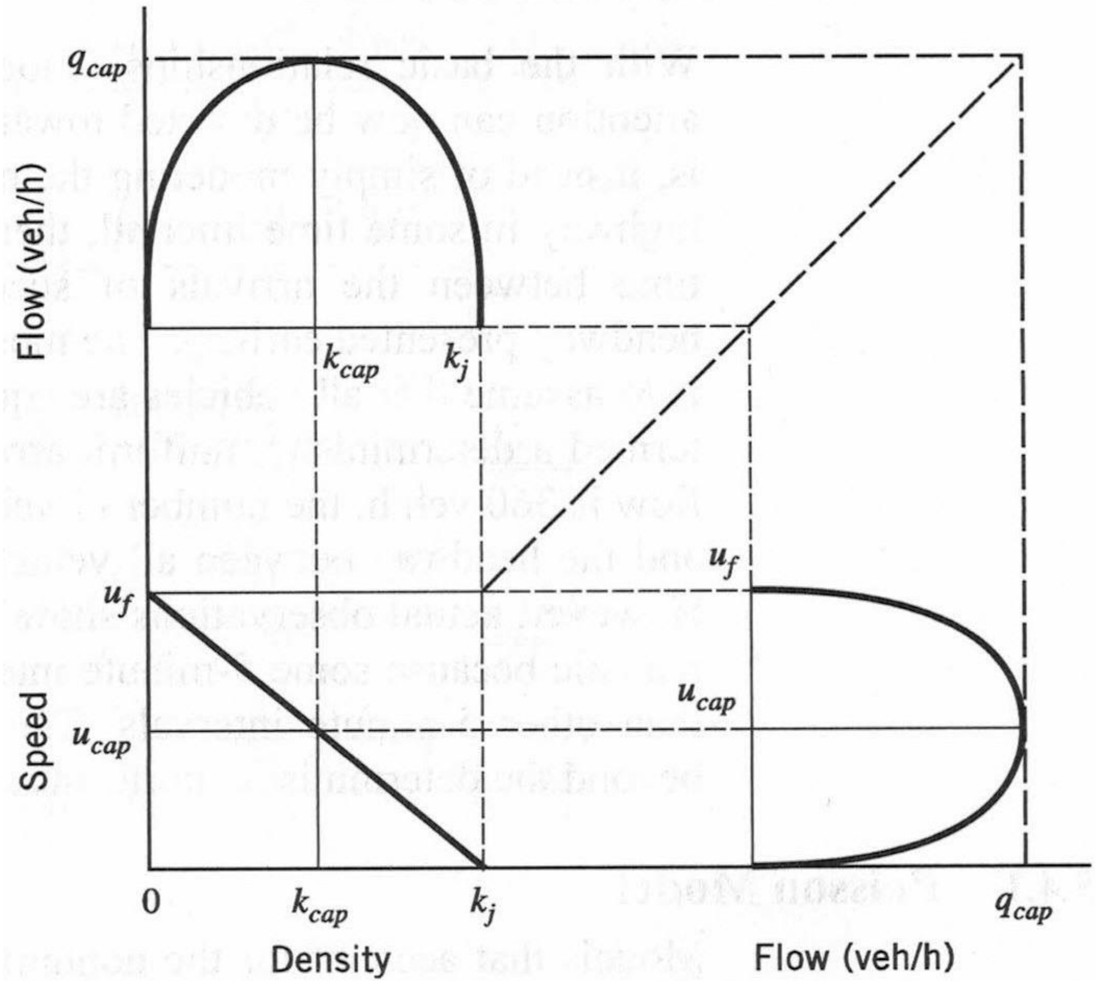


where:
q = flow rate or volume (veh/h)
*v = **space-mean** speed (km/h)*
v_f = free-flow speed (km/h)
v_{cap} = speed at capacity or maximum flow

Traffic Stream Model

- **Speed-Flow-Density Relationships**

$$q = v_f\left(k - \frac{k^2}{k_j}\right) \quad \cdots\cdots (2)$$



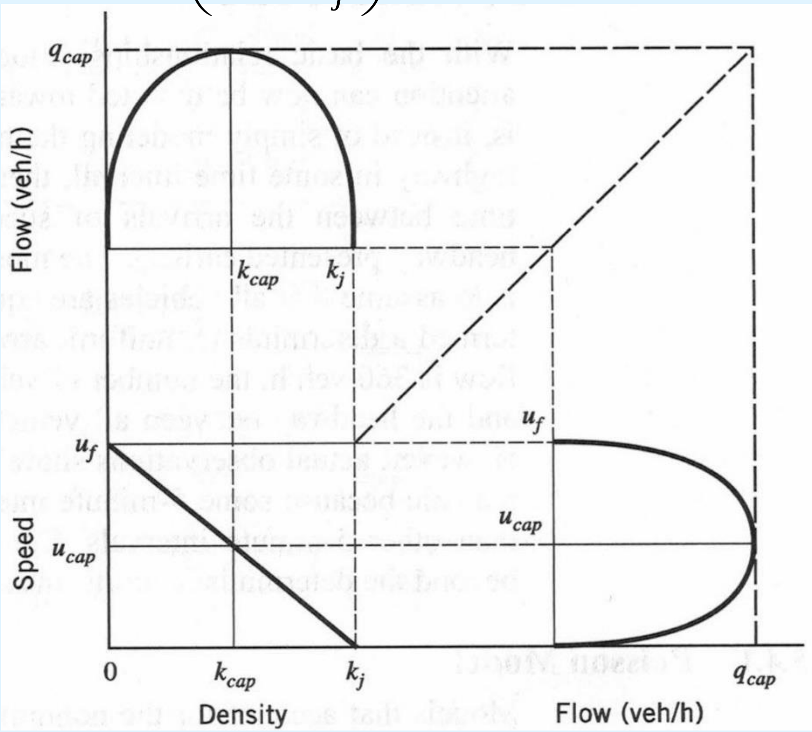
$$v = v_f\left(1 - \frac{k}{k_j}\right) \quad \cdots\cdots (1)$$

$$q = k_j\left(v - \frac{v^2}{v_f}\right) \quad \cdots\cdots (3)$$

Exercise: Traffic Stream Model

- A section of highway is known to have a free-flow speed of 88 km/h and a capacity of 3300 veh/h. In a given hour, 2100 vehicles were counted at a specified point along this highway section. If the linear speed-density relationship in Eq. (1) applies, what would you estimate the space-mean speed of these 2100 vehicles to be?

$$q = v_f\left(k - \frac{k^2}{k_j}\right) \quad \text{..... (2)}$$



$$v = v_f\left(1 - \frac{k}{k_j}\right) \quad \text{..... (1)}$$

$$q = k_j\left(v - \frac{v^2}{v_f}\right) \quad \text{..... (3)}$$

Exercise: Traffic Stream Model

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$$q_{cap} = \frac{v_f k_j}{4} \Rightarrow k_j = \frac{4q_{cap}}{v_f} = \frac{4 \times 3300}{88}$$

$$\Rightarrow k_j = 150 \text{ veh/km}$$

$$\begin{cases} q = 2100 \text{ veh/h} \\ k_j = 150 \text{ veh/km} \end{cases} \Rightarrow v = ?$$

Rearranging Eq. (3):

$$q = k_j \left(v - \frac{v^2}{v_f} \right) \Rightarrow \frac{k_j}{v_f} v^2 - k_j v + q = 0$$

$$\Rightarrow \frac{150}{88} v^2 - 150v + 2100 = 0$$

$$\Rightarrow v = 70.53 \text{ km/h or } 17.47 \text{ km/h}$$

Generalized Definitions of Flow, Density, and Speed

Generalized Definition for *q* and *k*

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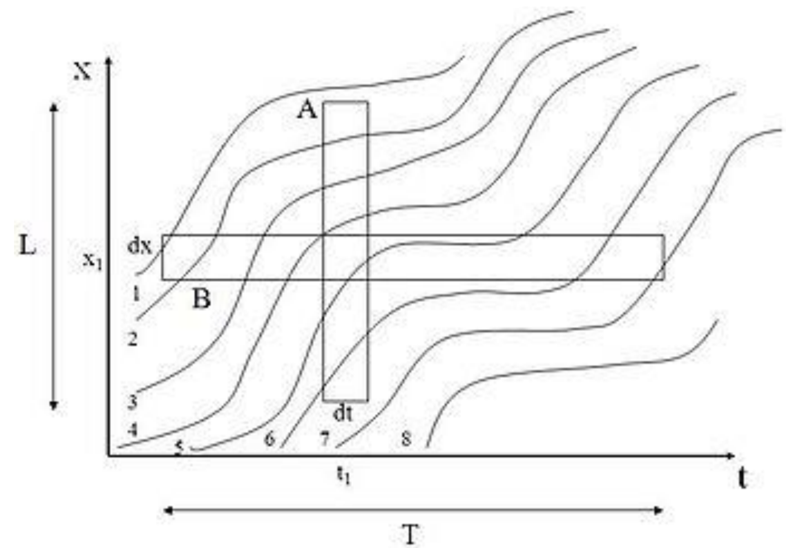
- **A More Generalized Definition of Flow and Density:**

$$q = \frac{N}{T} = \frac{N \times dx}{T \times dx} = \frac{d(B)}{|B|}$$

dx : infinitesimal distance
d(B) : sum of distances travelled by all vehicles during T
|B| : area of time-space rectangle B bounded by T and dx
Flow (q) : total distance travelled by all vehicles within B divided by the area of B



$$k = \frac{N}{L} = \frac{N \times dt}{L \times dt} = \frac{t(A)}{|A|}$$

dt : infinitesimal duration
t(A) : sum of times spent by all vehicles within A
|A| : area of time-space rectangle A bounded by L and dt
Density (k) : total travel time spent by all vehicles within A divided by the area of A

Generalized Definition for *q* and *k*

- **Unifying the Generalized Definition of q, k, and v:**
  – "Edie's generalized definition" (Edie, 1965)

$$q(C) = \frac{d(C)}{|C|}$$

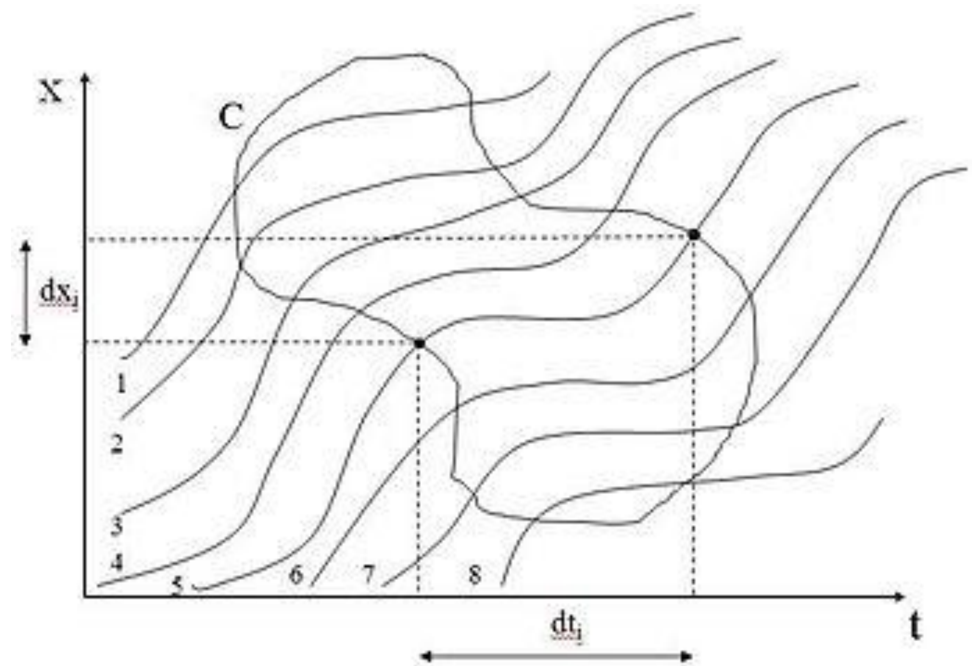
Total travel distance (veh·km) / Total space-time region (km·hrs)

$$k(C) = \frac{t(C)}{|C|}$$

Total travel time (veh·hrs) / Total space-time region (km·hrs)

$$v(C) = \frac{d(C)}{t(C)}$$

Total travel distance (veh·km) / Total travel time (veh·hrs)

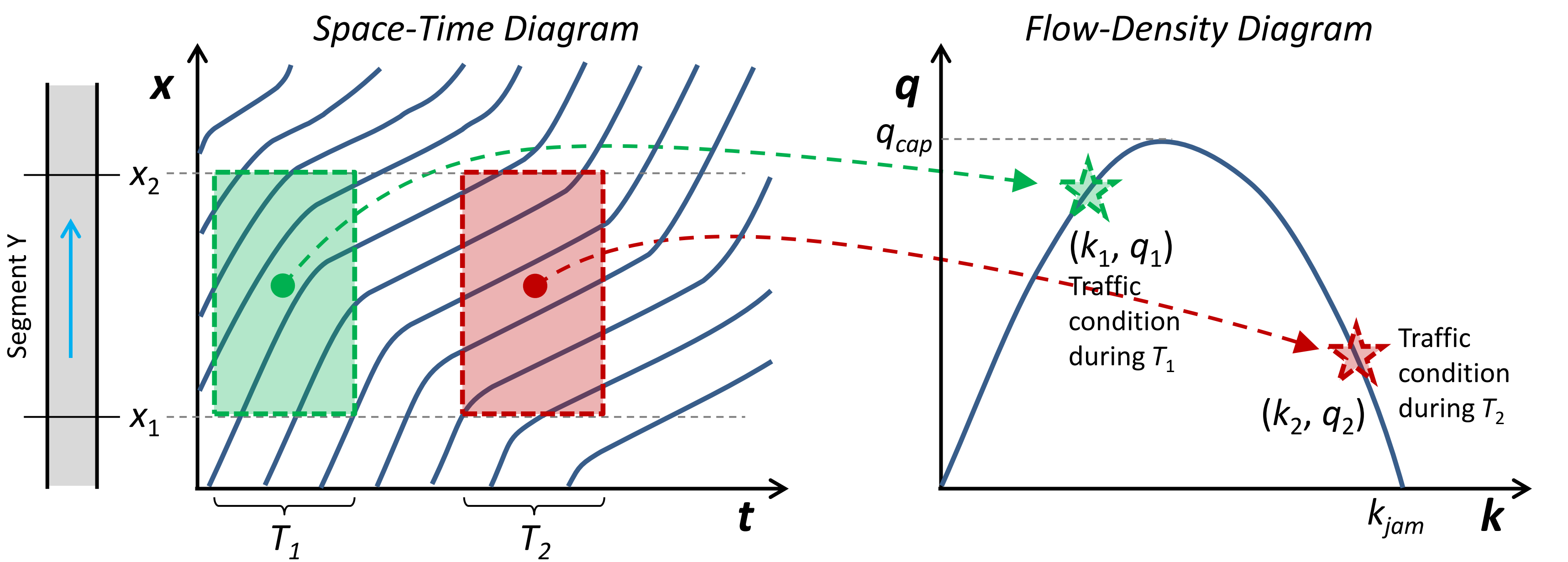


where

$$d(C) = \sum_{i=1}^{N} dx_i \qquad t(C) = \sum_{i=1}^{N} dt_i$$

Space-Time Diagram (x-*t* plot) vs. Fundamental Diagram (*q*-*k* plot)

- **Relation between Space-Time Diagram and Flow-Density Diagram**



Points on the fundamental diagram (FD) describe all possible traffic conditions on the road segment Y.

→ ***FD is a (steady-state) property of the road.***

Clicker Question 5

Edie's Generalized Definition

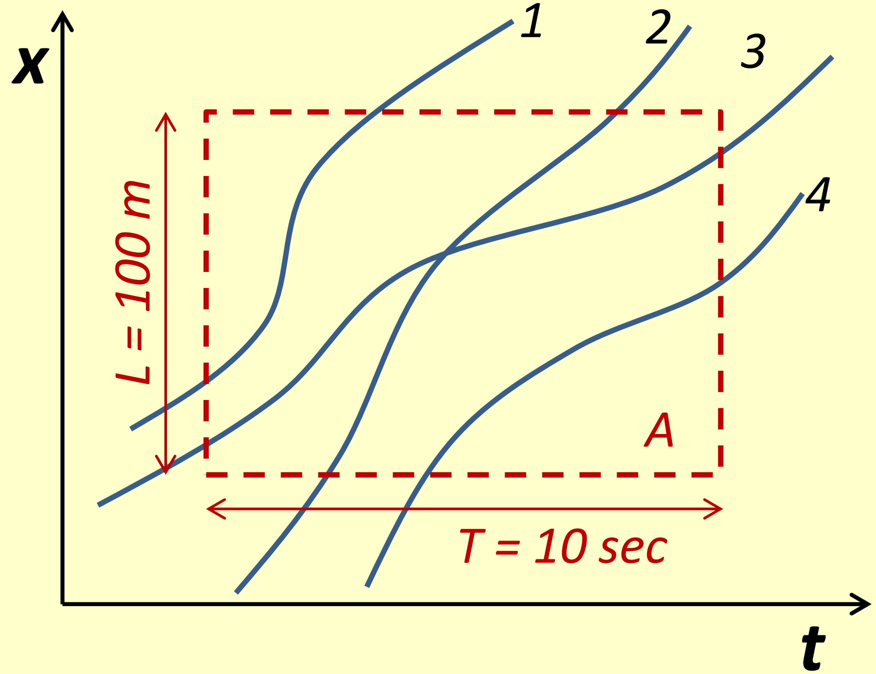
- Calculate flow q in region A using Edie's generalized definition.

Veh i	dt_i	dx_i
1	3 sec	70 m
2	6 sec	100 m
3	10 sec	80 m
4	5 sec	50 m

$$q(A) = \frac{d(A)}{|A|}$$

$$k(A) = \frac{t(A)}{|A|}$$

$$v(A) = \frac{d(A)}{t(A)}$$



A. 1080 veh/h
B. 1440 veh/h
C. 1880 veh/h
D. 2400 veh/h
E. None of the above

Next

- Tutorial on Wednesday
- Please don’t forget to request the Aimsun License!

CIVL6415

TRAFFIC ANALYSIS AND SIMULATION

MODULE 2

Introduction to traffic simulation

Lecture Week 4

Dr. Mehmet Yildirimoglu
School of Civil Engineering
The University of Queensland
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Outline

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- **Transportation systems**
  - **Supply systems**
  - **Demand modelling**
  - **Decisions on transportation systems**
    - **short-term, medium-term, long-term**

- **Role and structure of transport models**
  - **Time-Space representation**
  - **Static-Dynamic models**
  - **Micro-Meso-Macro models**

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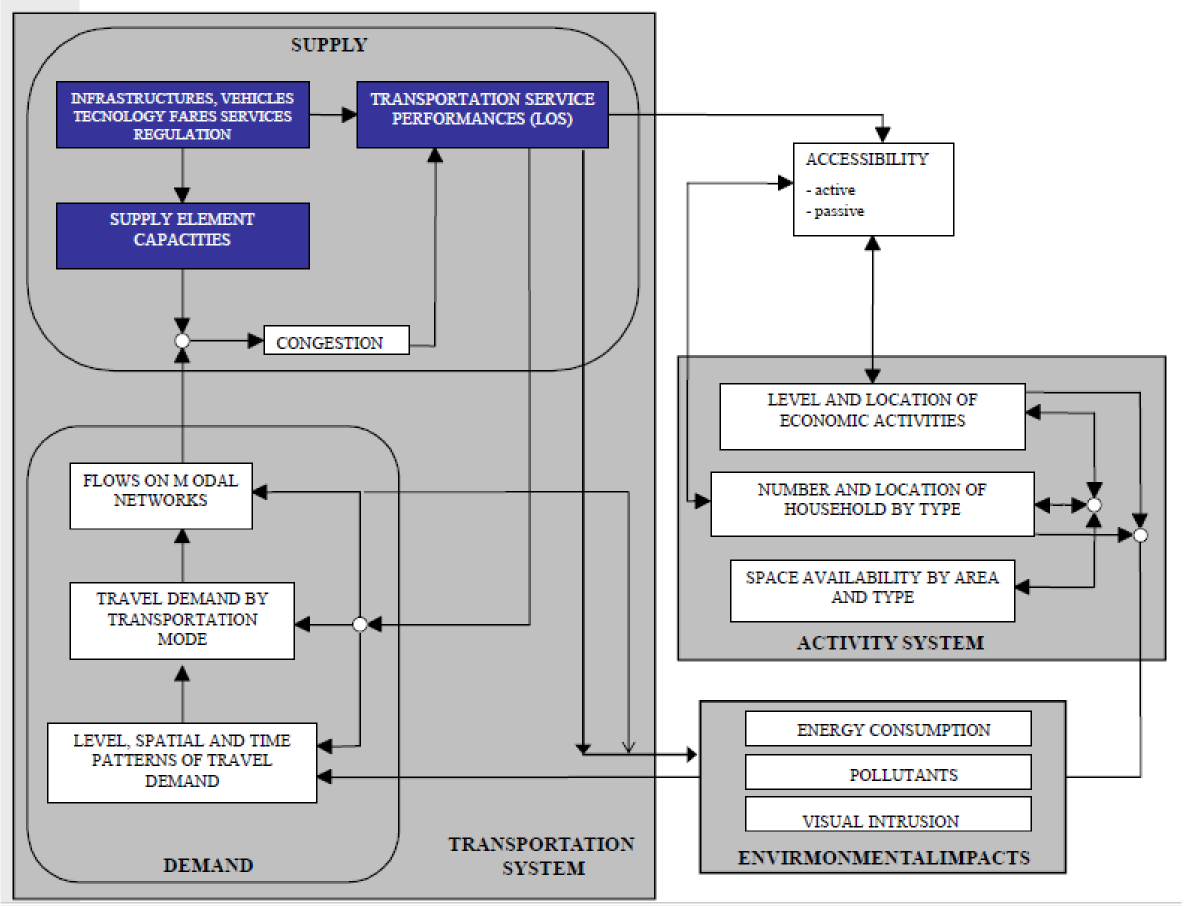
  - Micro-Meso-Macro models

Transportation systems

- Complex relationships
- Supply, Demand, Activity (Land use)



Supply systems

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- Infrastructure
  - – Roadways, Public transport facilities,
  - – Parking, etc.

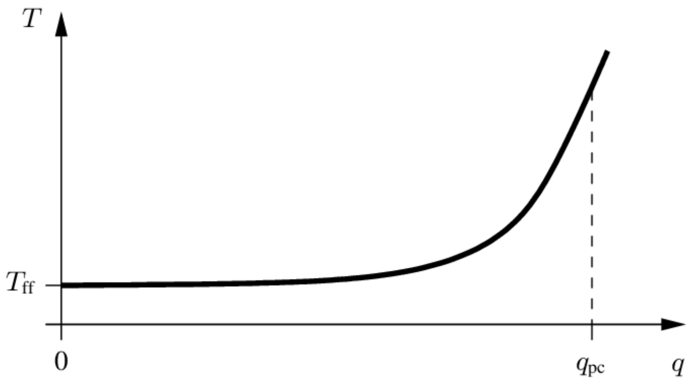
- Vehicles
  - – Private cars, Public transport vehicles
  - – Bicycles, Active transport

- Technologies
  - – Traffic signals, Smart management
  - – Traveller information systems

Supply systems

- Capacity
- Congestion and Travel times

- Congestion influences
  – Travel times
  – Reliability of the system
  – Emissions
  – Wellbeing



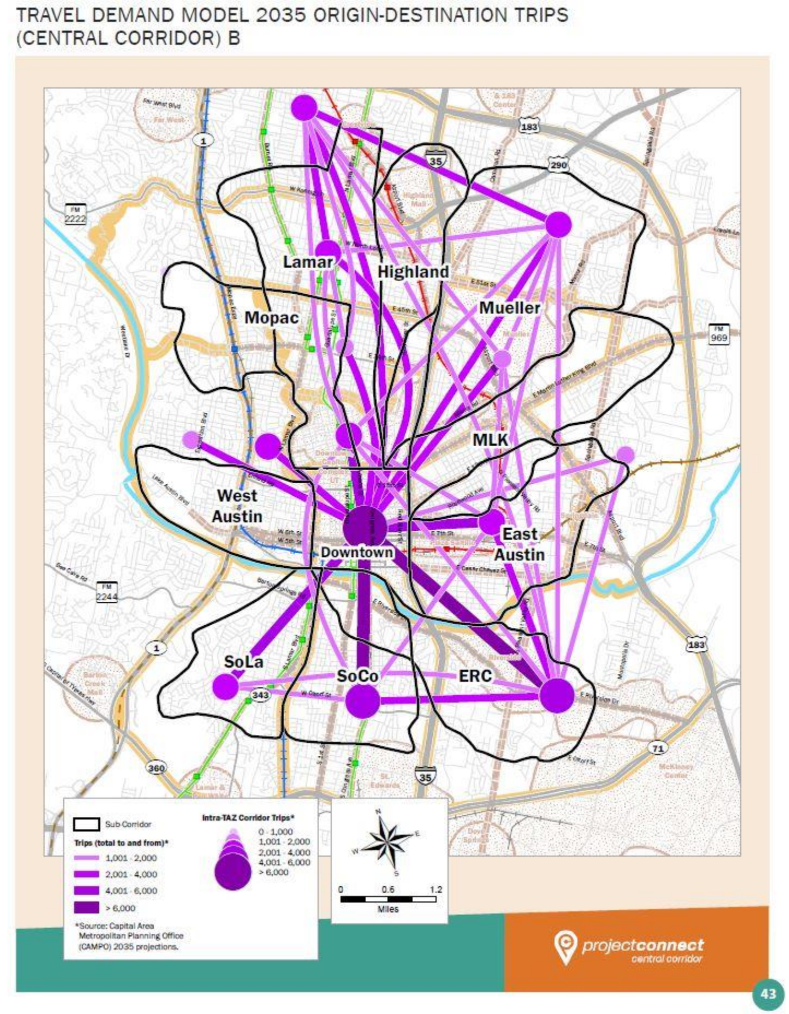
Bureau of Public Roads (BPR) function – Travel time vs. Volume

Travel demand

- Travel demand is a “derived function” from the participation of people in activities

- Land use (residential and work areas)

- Average number of daily trips
- Mode distribution
- Probabilistic choices
  – Route
  – Departure time
  – Mode



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  - Static-Dynamic models

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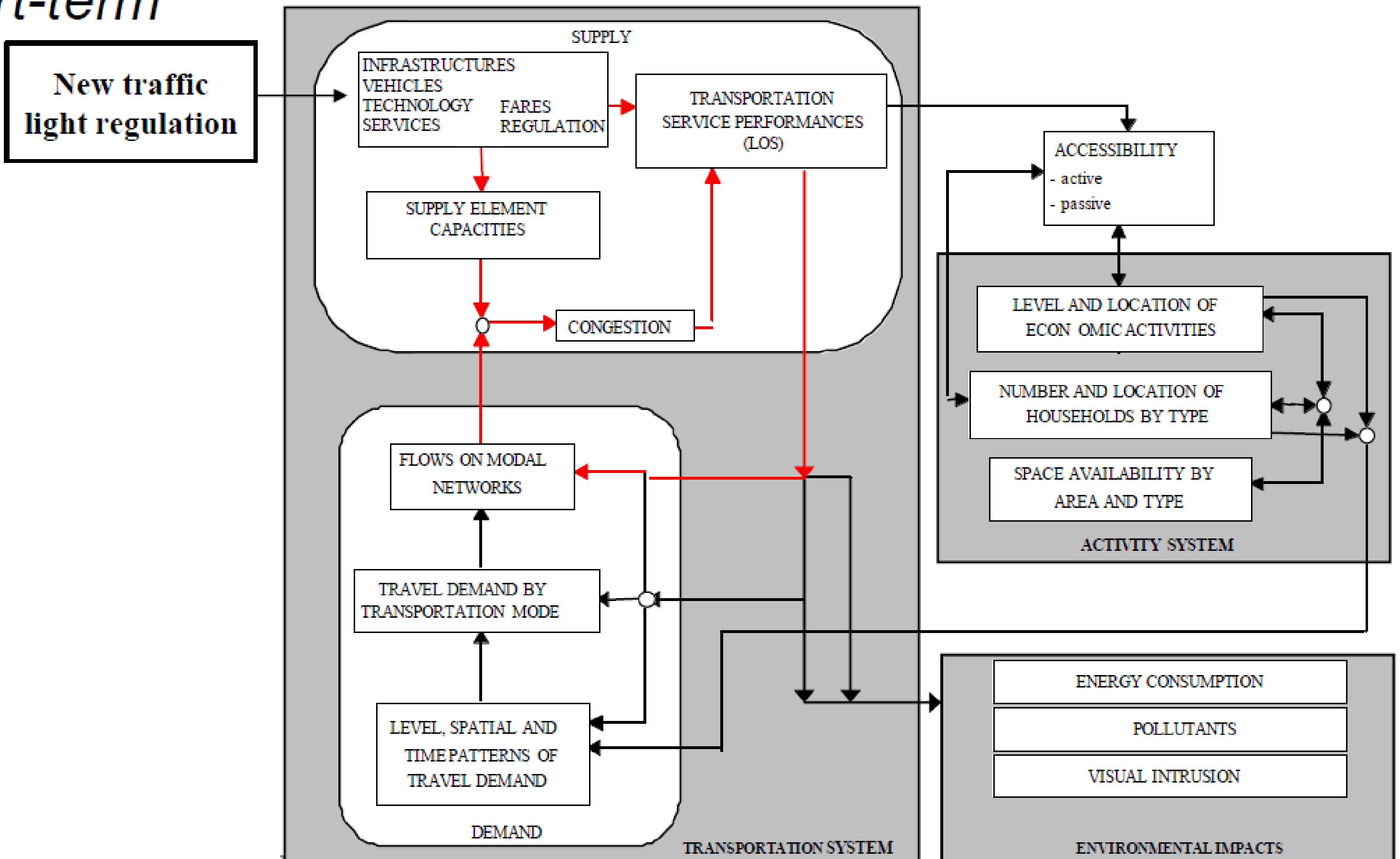
Feedback loops

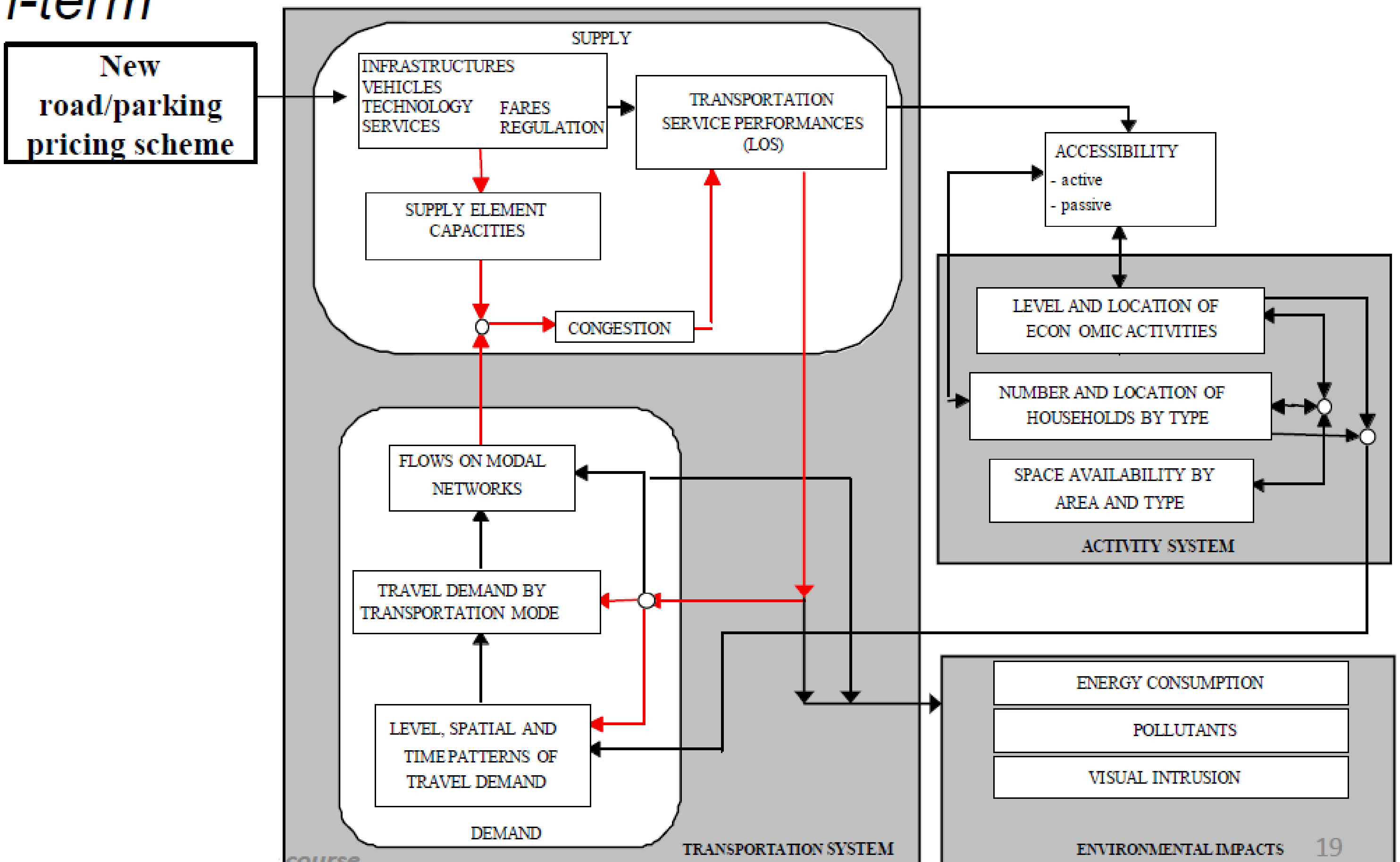
Short-term



From Cascetta
Introduction to traffic modelling

Feedback loops

Medium-term

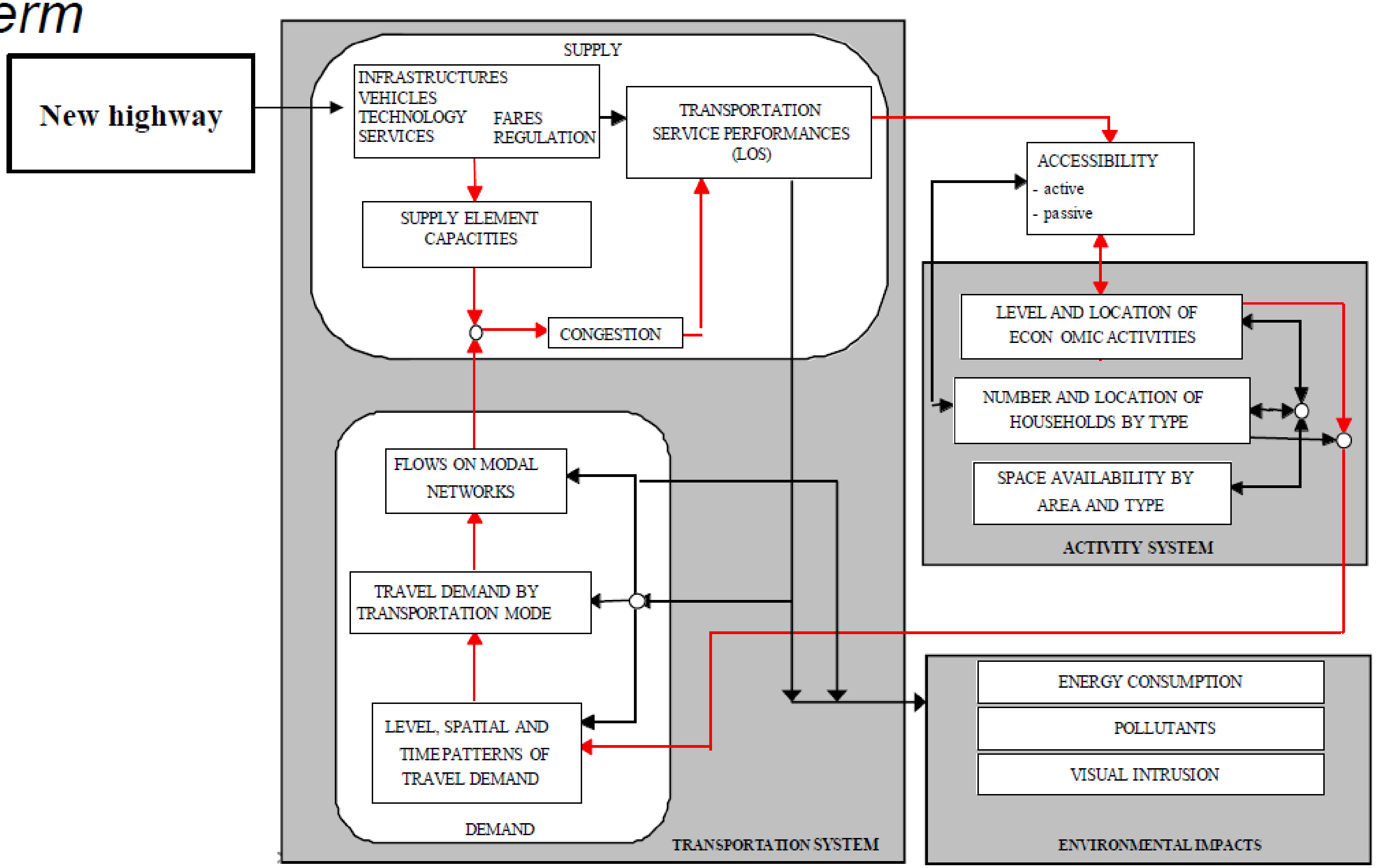


From Cascetta
Introduction to traffic modelling

Decisions

Feedback loops

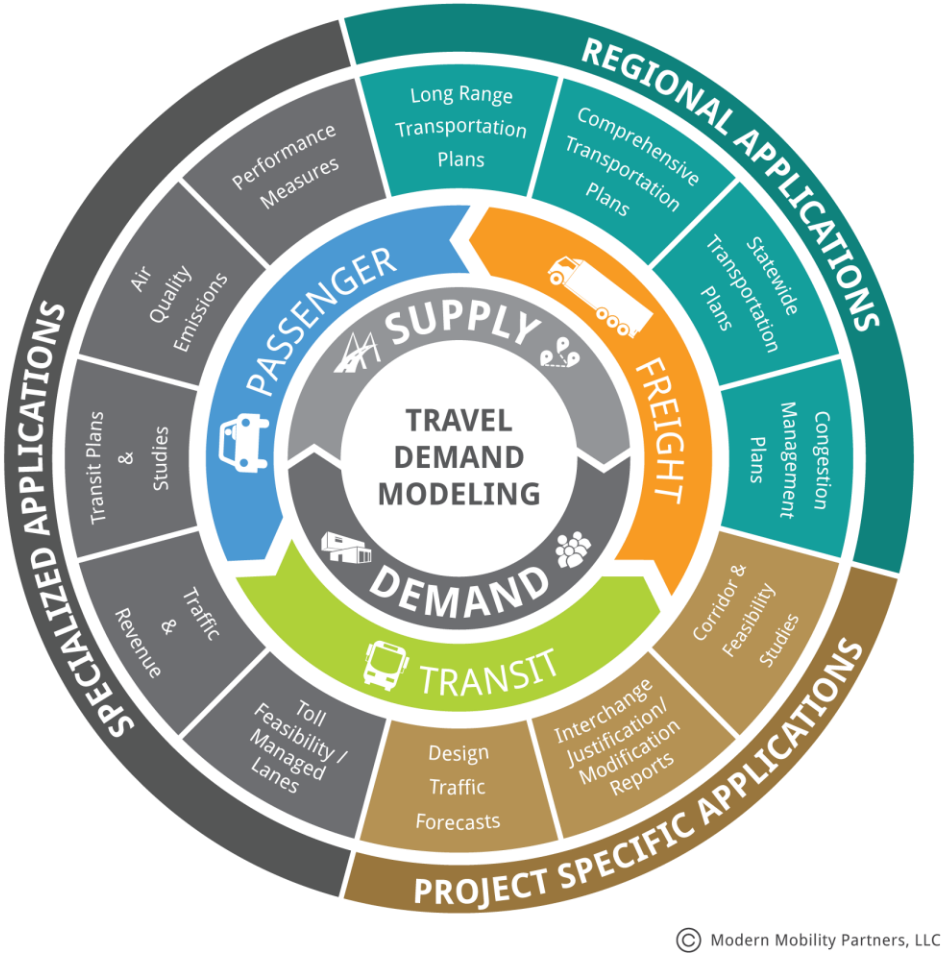
Long-term



From Cascetta
Introduction to traffic modelling

Decisions

- Transport Infrastructure
  – Building new roadways
  – Upgrading existing facilities
- Public Transport
  – Fares, timetables
- Vehicles & Technologies
  – Fleet composition, car-sharing
  – ITS technologies
- Policy / Regulations
  – Parking policy
  – Land use regulations, economic activities



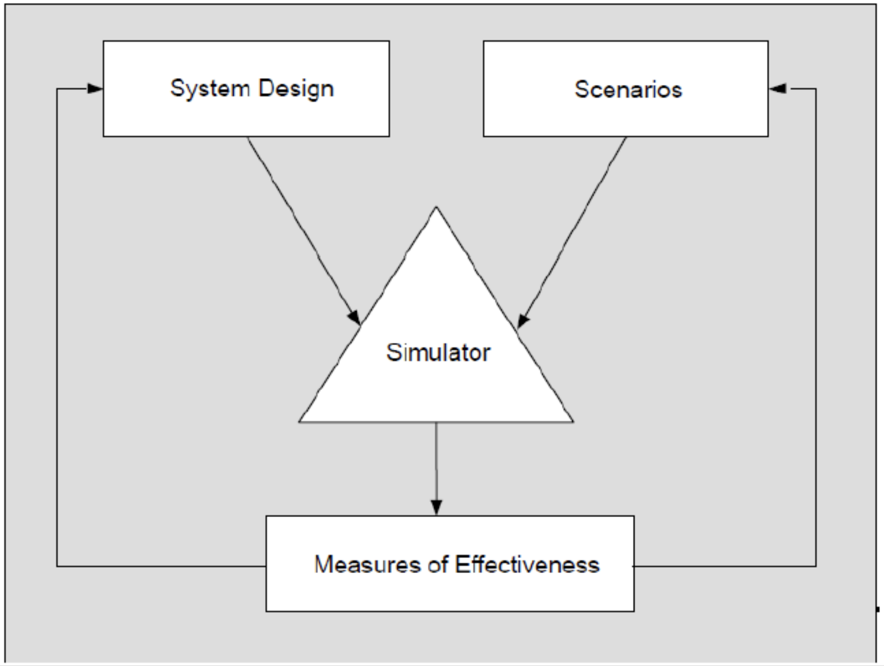
Applications

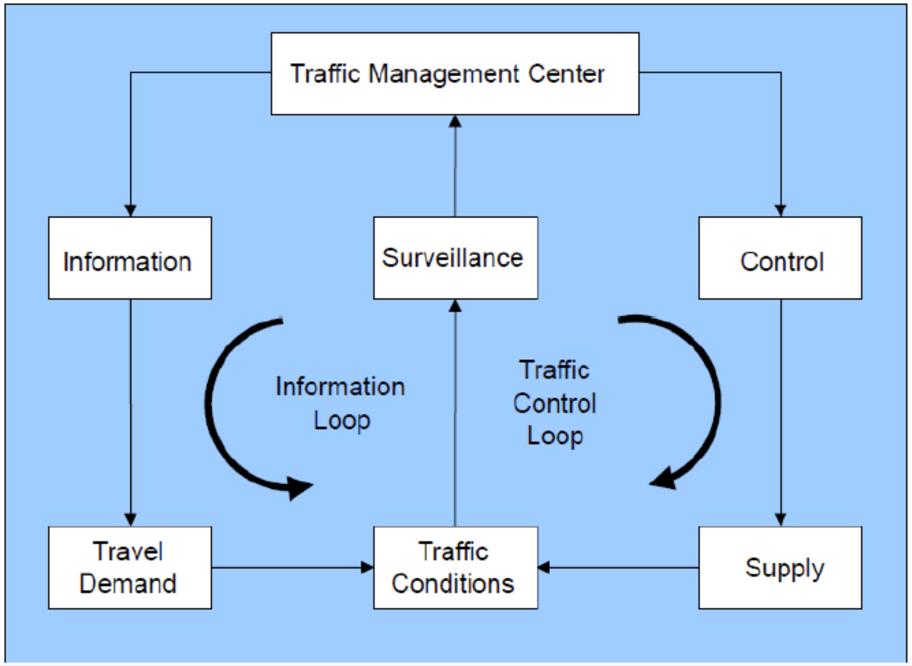
- Off-line evaluation
  - Future system and network modifications
  - Pricing policy changes
  - Before-After comparison

- On-line or Real-time decision support systems
  - Route guidance
  - Adaptive traffic control
  - Traffic management

Offline vs Online

- Offline evaluation
- Online decisions





Outline

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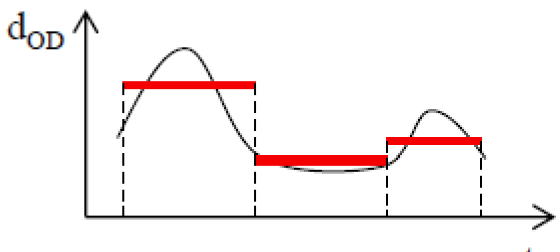
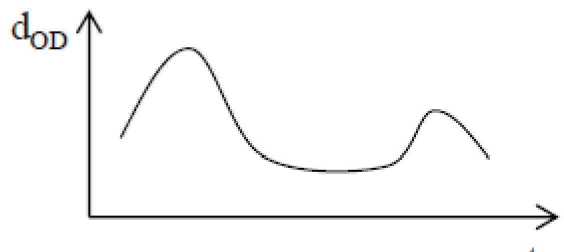
- **Role and structure of transport models**
  - **Time-Space representation**
  - **Static-Dynamic models**

Levels of representation

- Time representation



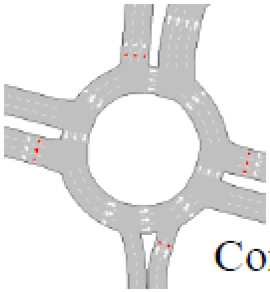
Within-day static



Within-day dynamic

- Space representation

Discrete



Continuous

Regional

- User representation
  - Discrete particles
  - Continuous fluid

STATIC TRAFFIC MODELS

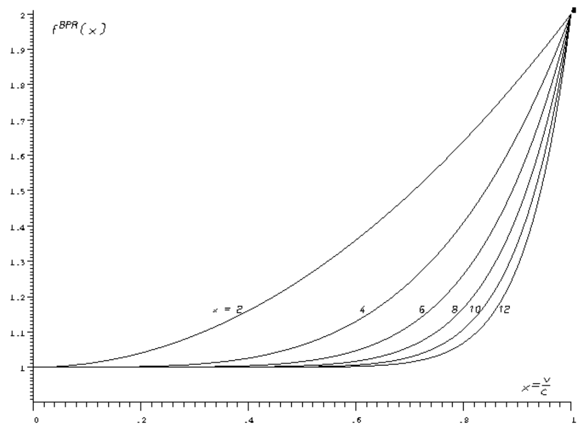
Static traffic models

- Off-line evaluation
- Planning purposes

- ***4-Step Modelling***
  - Trip generation (Estimate trips produced and attracted in each zone)
  - Trip distribution (Find O-D flows)
  - Mode split (Consider alternative transport modes)
  - **Traffic assignment (Route choice for O-D car flows)**
    - **Static traffic models with travel time functions**
    - **Equilibrium**

Roadway performance function

- Travel time is assumed to increase with traffic flow
  - Linearly (not realistic)
  - Nonlinearly (e.g., BPR)



Bureau of Public Roads (BPR) - travel time function

$$T_f = T_o * \left(1 + \alpha * \left[\frac{V}{C}\right]^{\beta}\right)$$

where:

T_f = final link travel time
T_o = original (free-flow) link travel time
alpha = coefficient (often set at 0.15)
V = assigned traffic volume
C = the link capacity
beta = exponent (often set at 4.0)

Graph, Links, Paths

- **GRAPH**

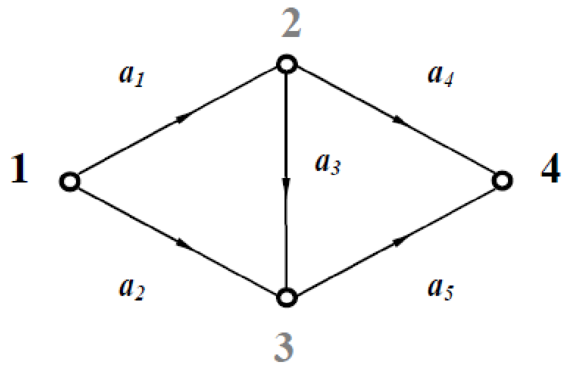
Ordered pair of sets G(N,L) representing network connections

N = {1, 2, 3, 4}

L = {a1, a2, a3, a4, a5}

Origin centroid: 1

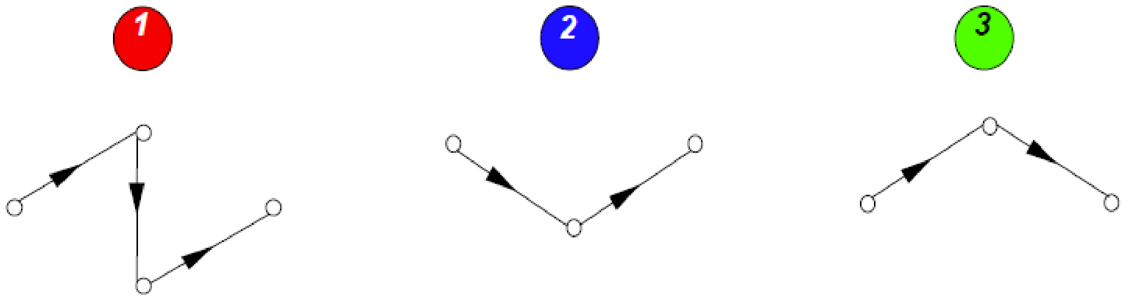
Destination centroid: 4



- **Path k:**

Sequence of links representing "typical" trips allowed by the supply system modelled

Path set K = {1, 2, 3}

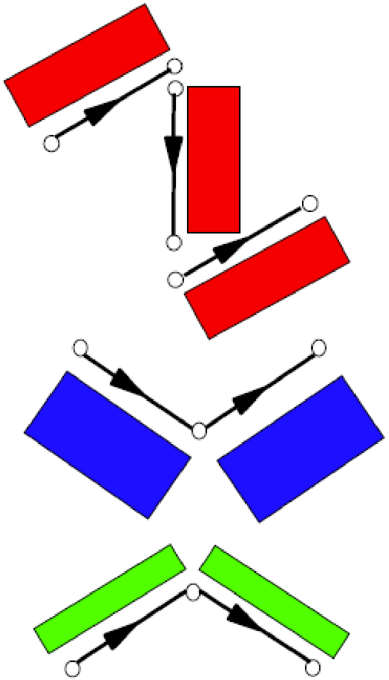


Path flows

Simulation period = simulation life= 1 hour

- **PATH FLOWS**

  $\overline{h}_k$: (average) flow along path *k* during the simulation period T [veh/h]



$\Rightarrow \quad \overline{h}_1$ = **800 veic/h**

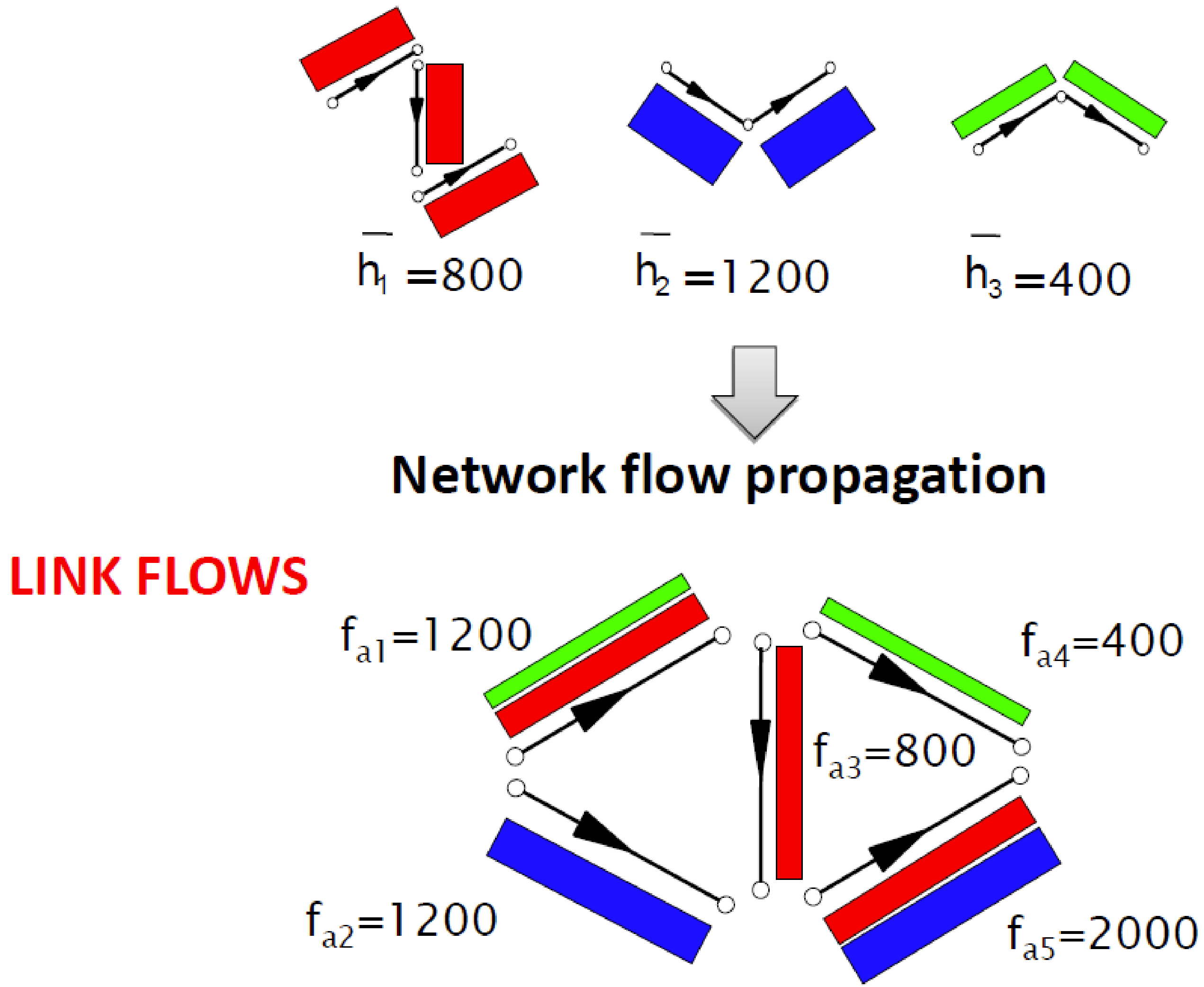
$\Rightarrow \quad \overline{h}_2$ = **1200 veic/h**

$\Rightarrow \quad \overline{h}_3$ = **400 veic/h**

2.400 veic/h

Path flows are propagated simultaneously on each link belonging to the path

Link flows



$\bar{h}_1 = 800$
$\bar{h}_2 = 1200$
$\bar{h}_3 = 400$
Network flow propagation
LINK FLOWS
$f_{a1}=1200$
$f_{a4}=400$
$f_{a3}=800$
$f_{a2}=1200$
$f_{a5}=2000$

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DYNAMIC TRAFFIC MODELS

Time-dependent flows

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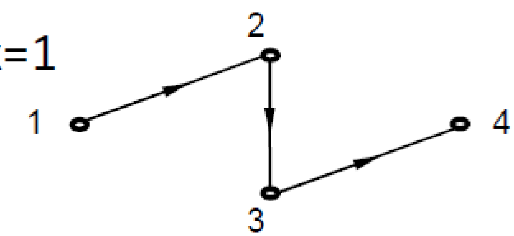
Path flows (time dependent)

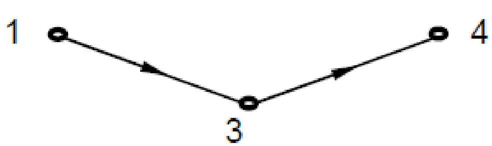
(2/2)

Path k=1



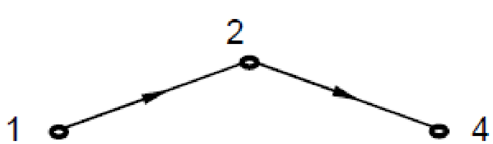
$$h_1 = \begin{array}{c} \begin{bmatrix} 200 \\ 10 \\ 400 \\ 100 \end{bmatrix} \end{array} \begin{array}{l} 1_1 \\ 1_2 \\ 1_3 \\ 1_4 \end{array} \qquad \bar{h}_1 = 800 \text{ veic/h}$$

Path k=2



$$h_2 = \begin{array}{c} \begin{bmatrix} 300 \\ 300 \\ 200 \\ 400 \end{bmatrix} \end{array} \begin{array}{l} 2_1 \\ 2_2 \\ 2_3 \\ 2_4 \end{array} \qquad \bar{h}_2 = 1200 \text{ veic/h}$$

Path k=3

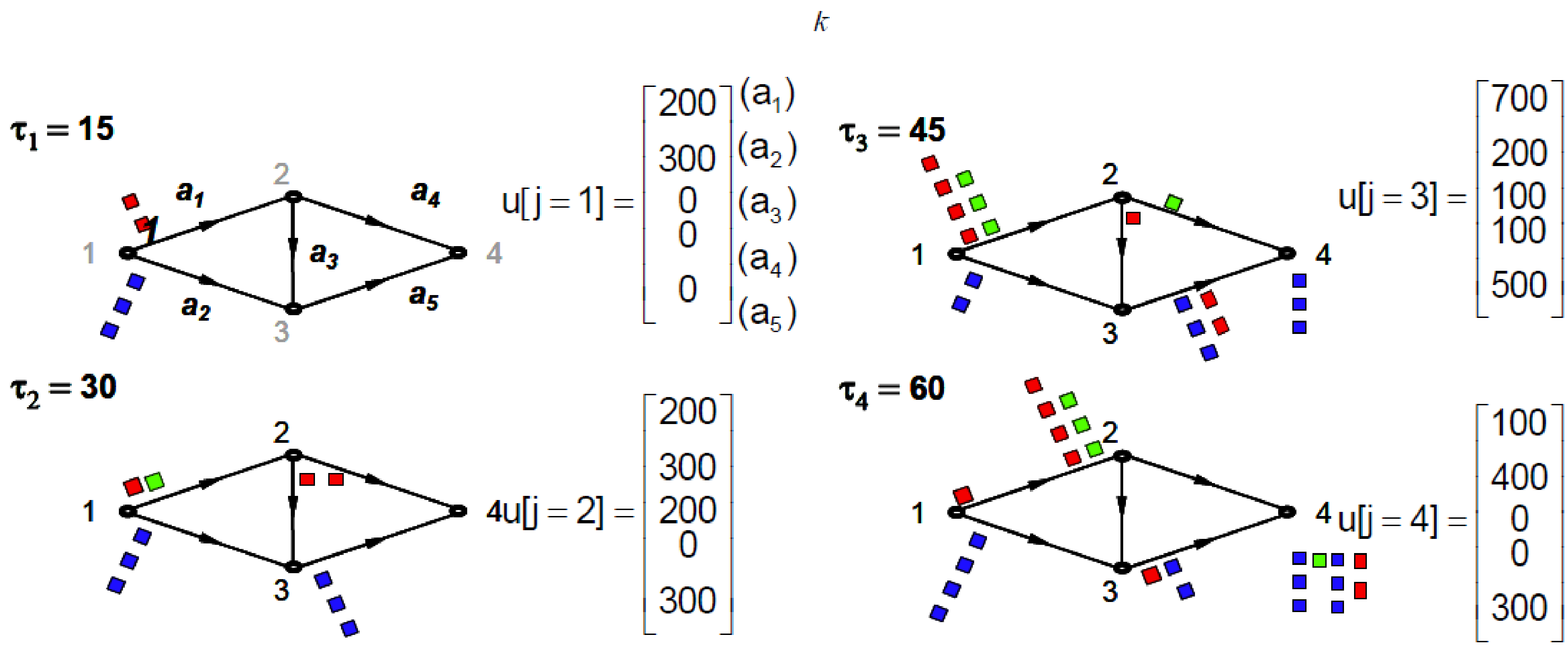


$$h_3 = \begin{array}{c} \begin{bmatrix} 0 \\ 100 \\ 300 \\ 0 \end{bmatrix} \end{array} \begin{array}{l} 3_1 \\ 3_2 \\ 3_3 \\ 3_4 \end{array} \qquad \bar{h}_3 = 400 \text{ veic/h}$$

Each square represents 100 vehicles

Time-dependent flows

Total inflows at links



note: in the pictures, each unit represents 100 vehicles

Assume vehicles move one link at each time step

Representation

- **User variables** *how vehicles are simulated*
  - AGGREGATE
    - i.e. traffic stream as a continuous fluid
  - DISAGGREGATE
    - i.e. traffic stream composed by single vehicles

- **Speed variables** *how network performance are simulated*
  - AGGREGATE
    - i.e. each link with an average speed for all vehicles on it
  - DISAGGREGATE
    - i.e. each single vehicles with its own speed

Modelling approaches

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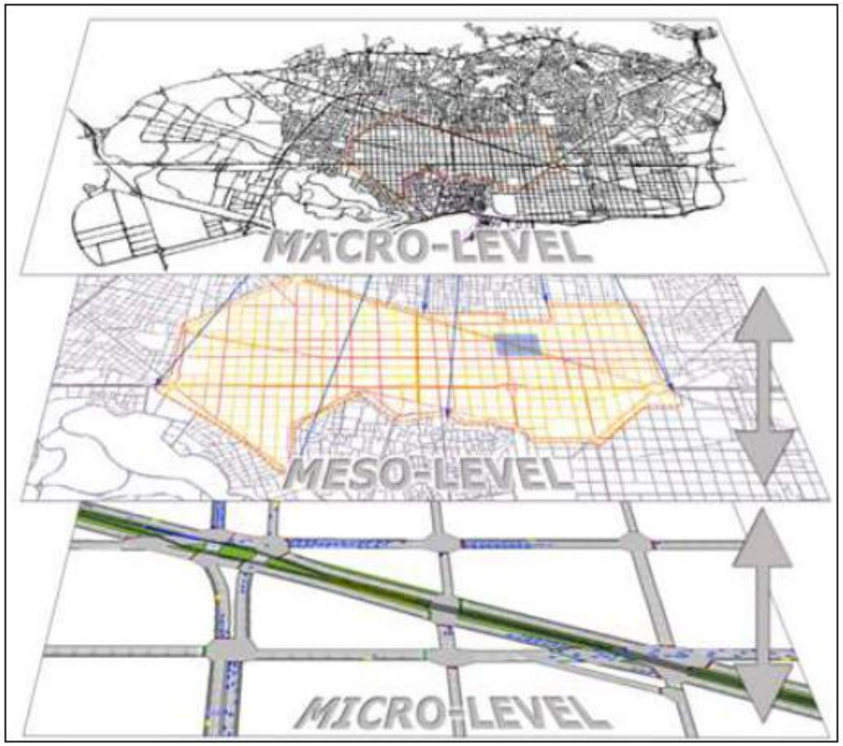
Speed variables / ***user variables***	Aggregate $u(x,t)$	Disaggregate $u_i(t)$
Aggregate $q(x,t)$, $k(x,t)$	MACRO scopic	
Disaggregate $x_i(t)$	MESO scopic	MICRO scopic

Modelling approaches



Source: Barcelo, J. Casas J, Garcia D & Perarnau J (2005)

❑Static:

- Large scale, strategic, no detailed representation of congestion.

❑Meso:

- Medium scale, models intersections in detail, capable of dynamic assignment.

❑Micro:

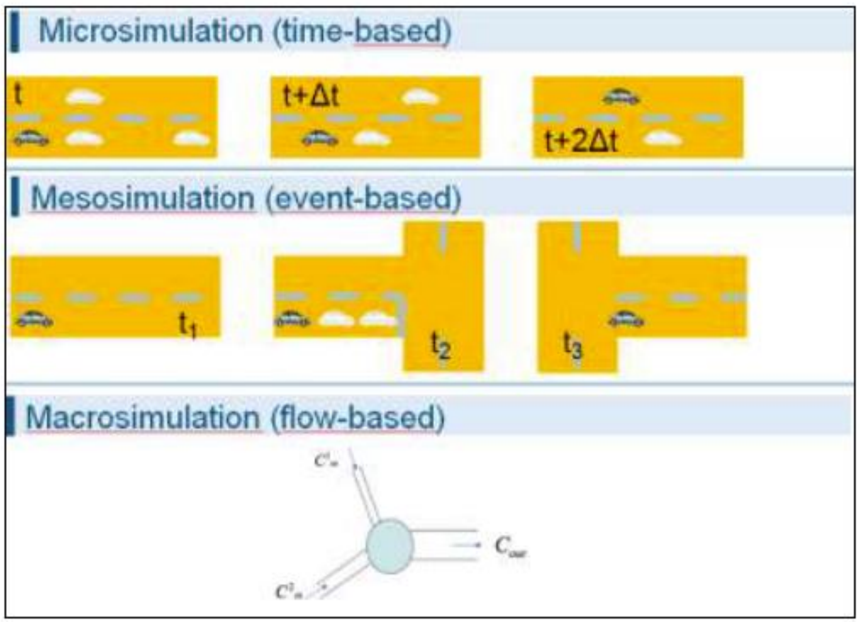
- Finest level of detail, complex signal operation, models individual vehicle movements.

Traffic models

- True/False questions:

  - All traffic models capture the time dependent nature of traffic.
  - Mesoscopic models are used in the four-step planning framework.
  - Microscopic models build on car following and lane changing models.
  - Static traffic models model individual vehicles in the network.

Modelling approaches

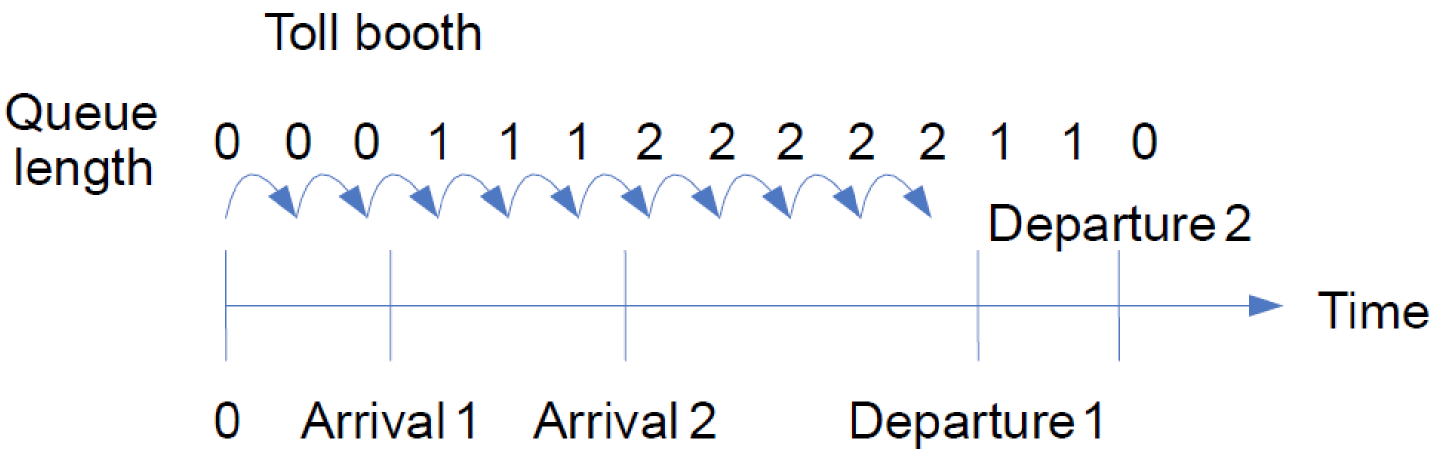


Source: TSS Aimsun 8.1.4 User's Manual 2016

- Discrete event simulation: Simulation time changes when an event occurs.
  - Vehicle generation, vehicle node movement, traffic signal change.
  - Simplified car-following and gap acceptance model
- Vehicle considered only as it enters and exits a node section.
- Does not consider acceleration/ deceleration or details lane changing behaviour.

Time-based simulation

- Pre-defined equal time intervals
  - at each time step update system state
  - process all events occurring at *(t-Δt, t]*



Event-based simulation

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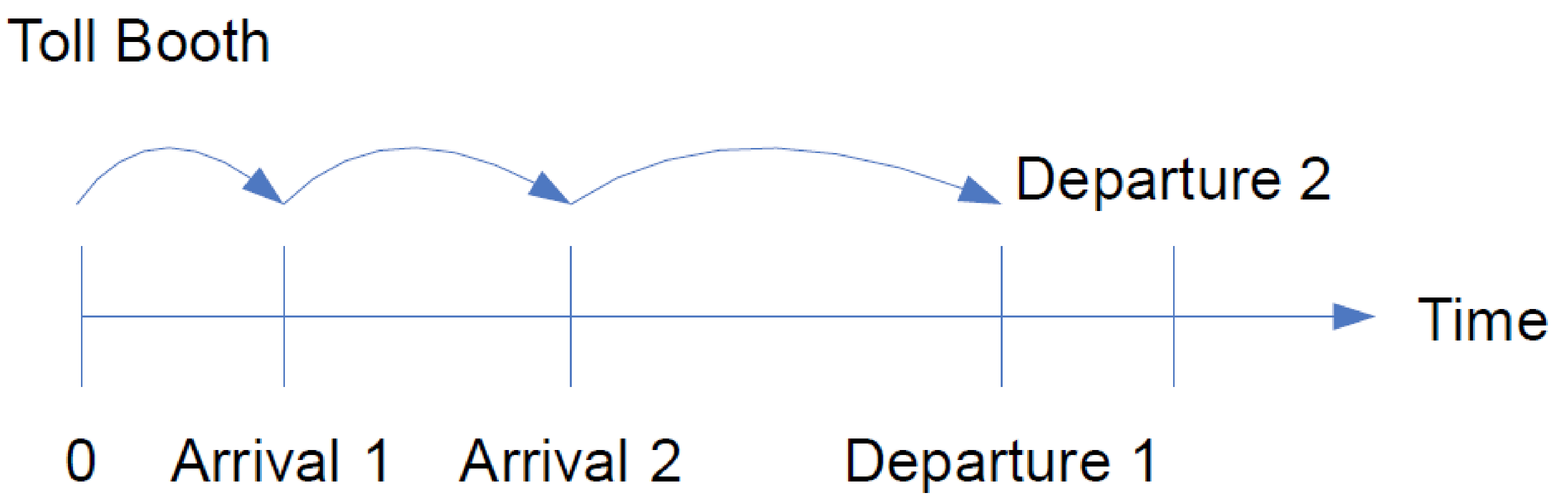
- Maintain list of upcoming events
- Process events of current time
- Advance clock to time of the next event



Stochastic simulation

- Inputs, models are probabilistic

Monte Carlo simulation

- Calculate probability of event
- Draw random numbers to realize event

Implications

- Input distributions
- Outputs are random variables

Traffic models

- True/False questions:

  - Event-based simulation requires a list of upcoming events.
  - In time-based simulation, time interval can vary.
  - In time-based simulation, system state has to be updated only when a new event occurs.
  - Static traffic models build on an event-based simulation technique.

Macroscopic models

- Common for evaluation of freeway corridor operations

- Traffic dynamics
  - Macroscopic traffic flow
- Basic approach
  - Numerical solution
  - Discretization of space and time

Macroscopic models

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- **Objective**: Provide fundamental relationships among macroscopic traffic stream characteristics for uninterrupted flow conditions
  - *Speed-density*
  - *Flow-density*
  - *Speed-flow*

➢ *Does not consider individual vehicles*

➢ *Fluid approximation*

Macroscopic models

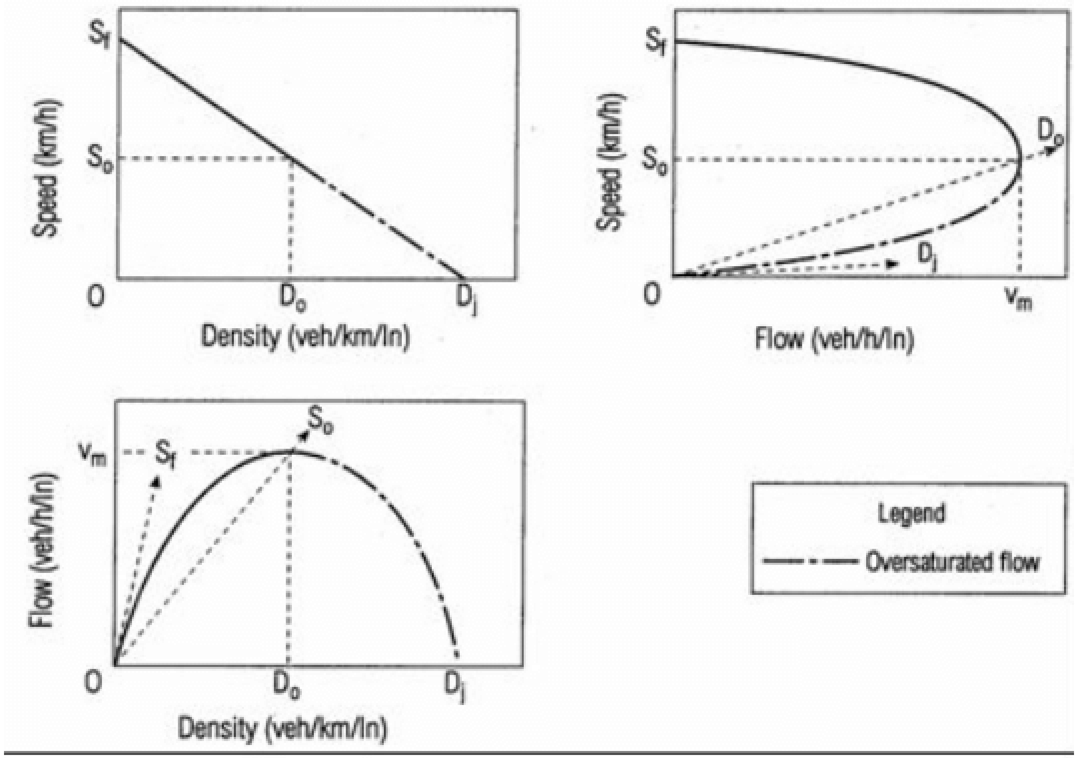
- **Fundamental relationship**

$$q = v\,k$$

q = flow [veh/h]

v = speed [km/h]

k = density [veh/km]



Macroscopic models

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- **Models to represent the fundamental diagram**

Greenshield's model:

$$u = u_f\left(1 - \frac{k}{k_j}\right) \qquad q = uk = ku_f\left(1 - \frac{k}{k_j}\right)$$

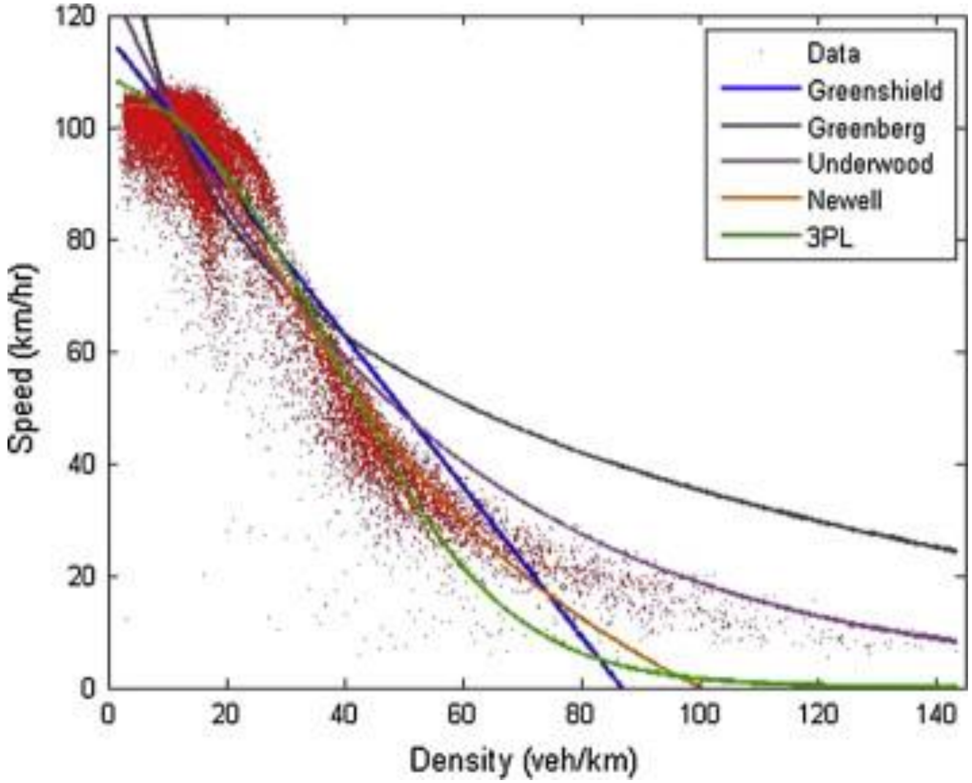
Greenberg's model:

$$u = c\ln\left(\frac{k_j}{k}\right) \qquad q = uk = k_j u\, e^{-u/c}$$

u_f, k_j are the parameters to be calibrated

Macroscopic models

- **Models to represent the fundamental diagram**



from Qu *et al*. (2015)

All models are wrong, but some are useful!!

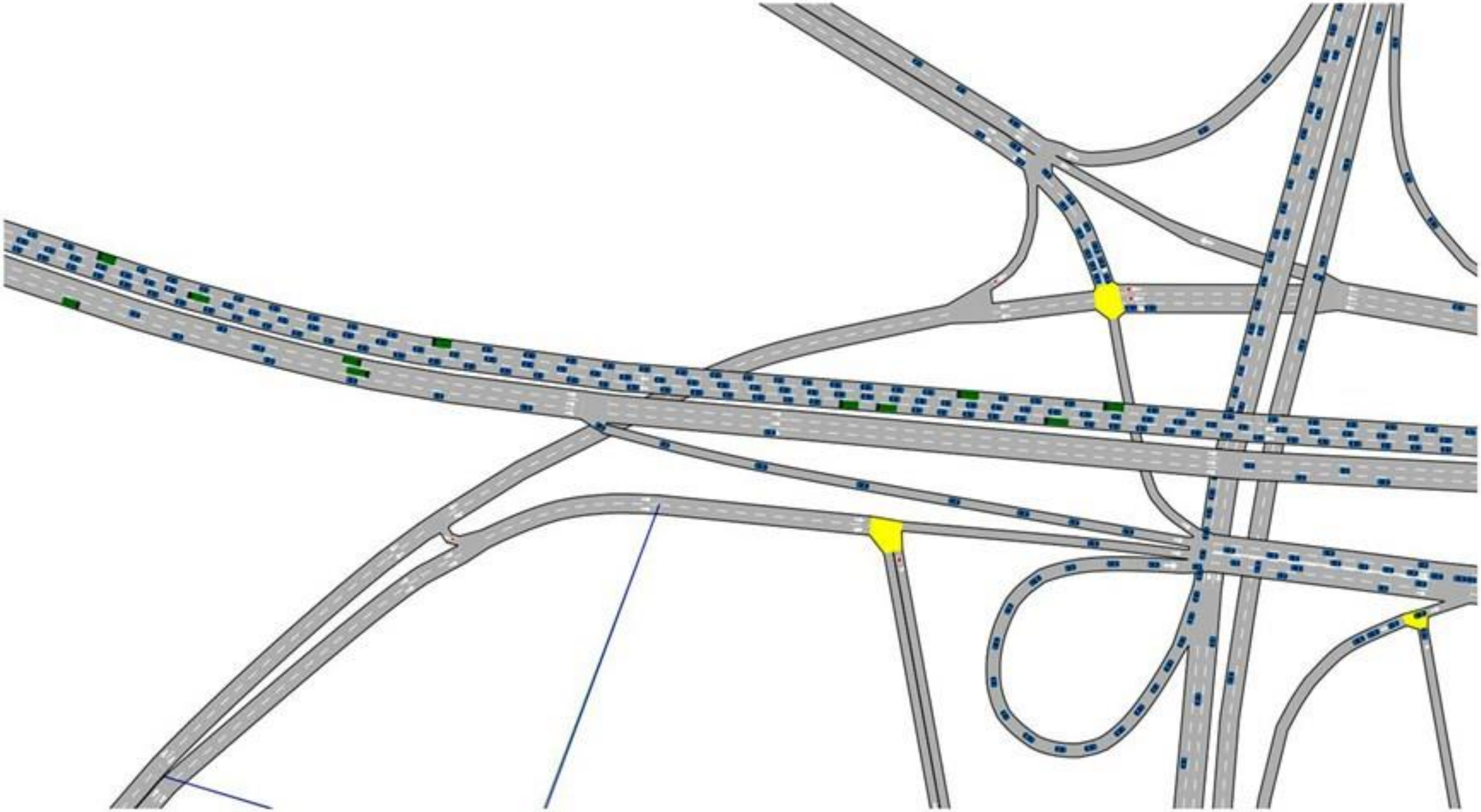
Macroscopic models

- **Traffic stream examples**

- FREFLOW (Payne et al, 1973)
- NETCELL (Daganzo et al., 1994)
- METANET (Papageorgiou et al., 2006)

- Freeway networks
- Ramp metering strategies
- Traffic control
- **1st order vs 2nd order models**

Microscopic models



Microscopic models

- Detailed models
- Synthesis of
  - Driving behaviour (e.g, car following)
  - Control strategies (e.g., traffic signals)

- Time-based simulation (e.g., 0.1-1s)
- Both freeways and urban networks can be modelled
- Multiple driver classes (with different characteristics)
- Stochastic

Microscopic models

- **Commercial models**
  - Aimsun
  - Vissim
  - Paramics
  - Etc.
- **Research-oriented**
  - MITSIMLab
  - SUMO (Open-Source)
  - etc.

aimsun. A Siemens Company

PTV VISSIM the mind of movement

Paramics microsimulation

Microscopic models

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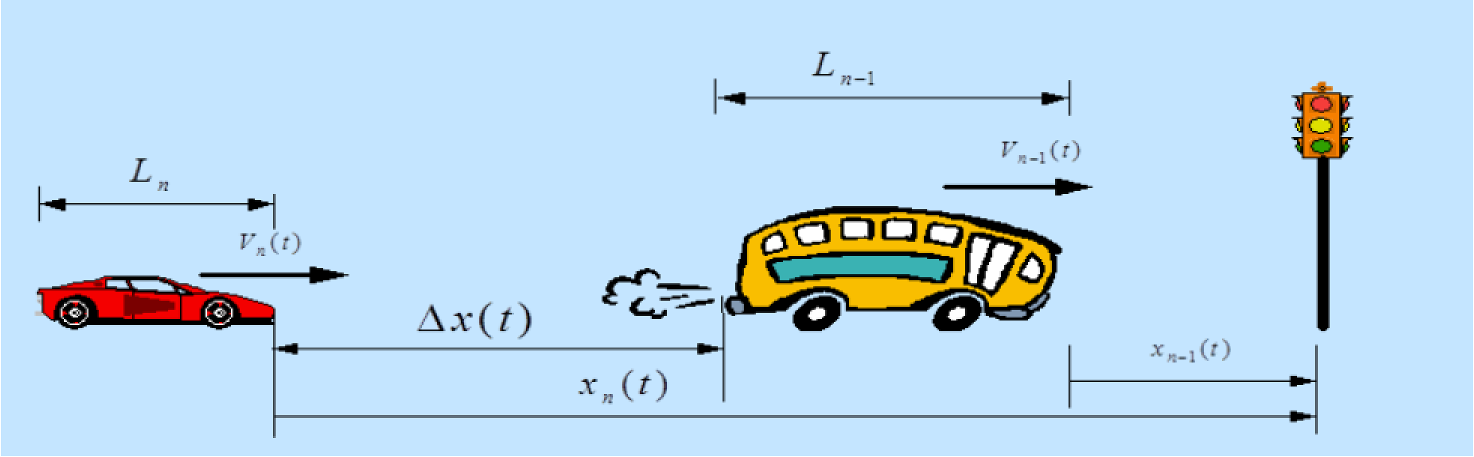
- Driving behaviour

- Car following (Acceleration)
- Lane changing
- Gap acceptance

Microscopic models

- **Car following**



Common Model: *Response(t) = sensitivity • stimulus(t-τ)*

τ : reaction time
Stimulus: Δv (relative speed), Δx (relative distance)
Sensitivity: function of Δx, speed, traffic conditions
Response: acceleration, speed

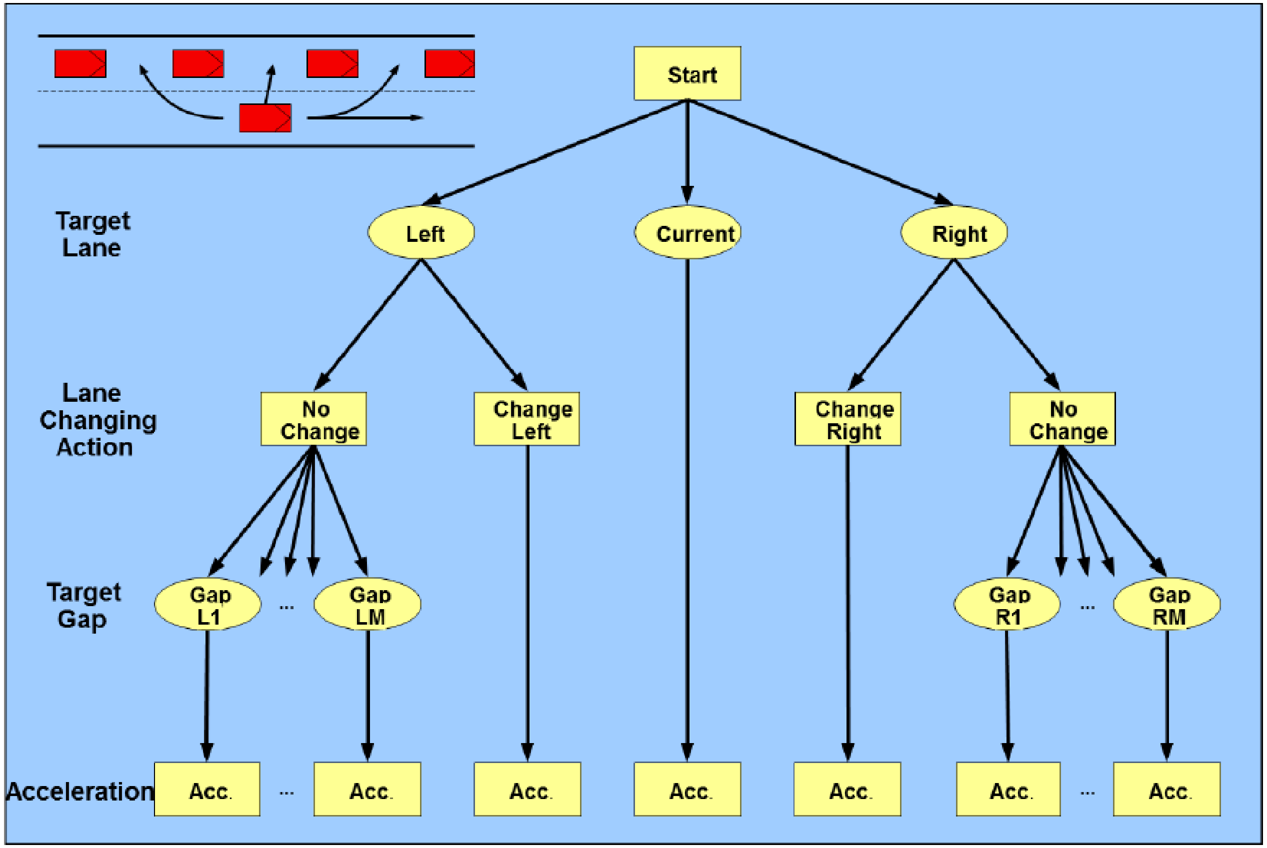
Microscopic models

- **Lane changing**

- Mandatory and discretionary lane-changing
- ***Mandatory***: getting off the current lane in order to continue on the desired path (e.g. exiting), or to avoid lane closure
- ***Discretionary:*** attempting to achieve desired speed, avoid following trucks, avoid merging traffic, etc.

Microscopic models

- **Integrated model**
- Toledo, 2002



Mesoscopic models

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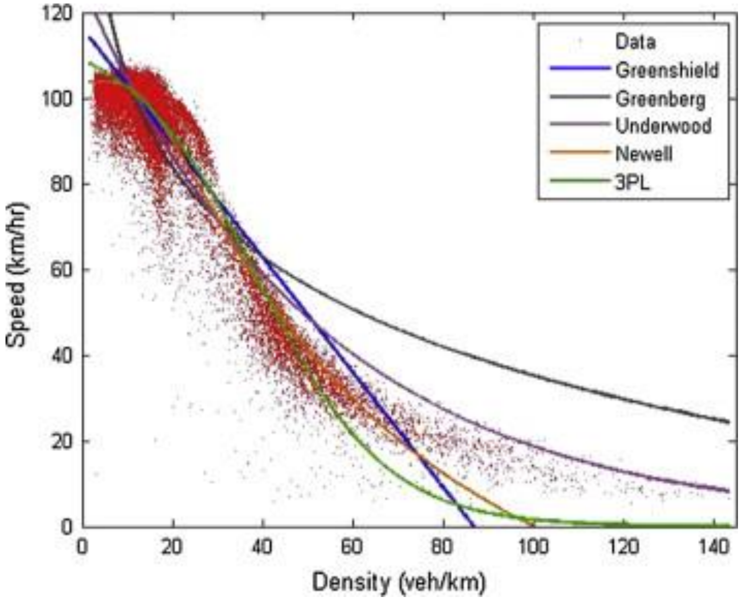
- **User representation**
  - Disaggregate
  - Individual vehicles or packets of vehicles
- **Speed representation**
  - Aggregate
  - Fundamental relationship
  - Queuing models
- **Traffic control**
  - Adjusting the capacities (e.g., traffic signals)
- ***Event-based simulation***



Mesoscopic models

- **Application**

- Dynamic traffic assignment (DTA)
  – Network performance
  – Prediction
  – State estimation
- Planning and policy analysis
- Network design

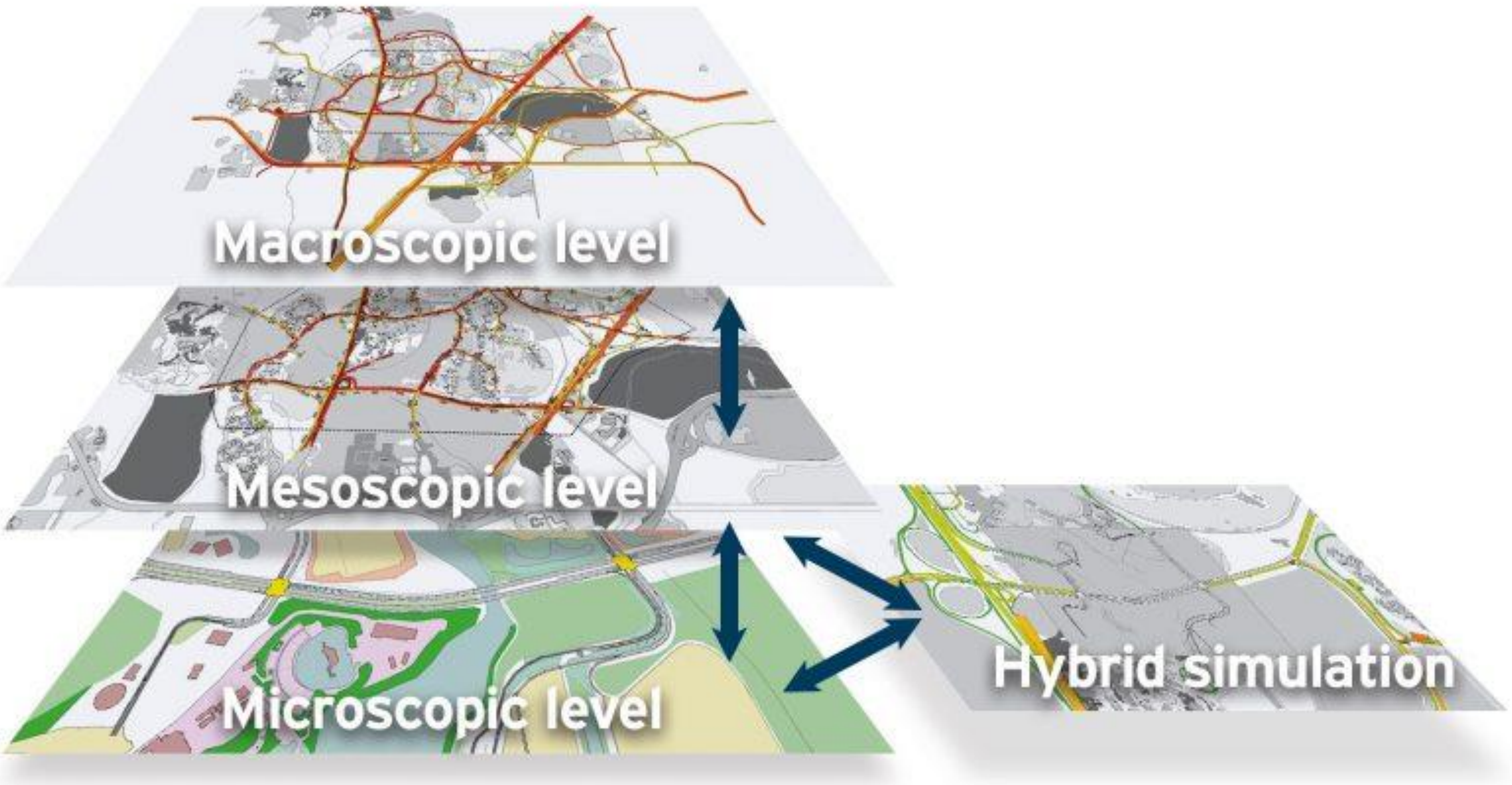
- **Fast computation!!**

Mesoscopic models

- **Examples**

CONTRAM	TRRL	Operations evaluation, planning, DTA
DYNAMEQ	Mahut, Florian	ITS, DTA, planning
DynaMIT	Ben-Akiva, Koutsopoulos, …	ITS, DTA, evaluation, prediction, short-term planning
DYNASMART	Mahmassani	
Dynemo	Schwerdtfeger PTV	ITS, prediction
INTEGRATION*	Van Arde	ITS, evaluation
METROPOLIS	De Palma	Planning
Mezzo and BusMezzo	Burghout, Koutsopoulos, Cats	DTA, short-term planning

Hybrid Models



Hybrid Models

- Combine different levels of resolution at different parts of the network
  - Microscopic in areas of interest, high congestion
  - Mesoscopic in remaining network
- Many advantages
  - Computational
  - Data preparation
  - Calibration
  - Scope of applications
- Example

Next

This week
- Tutorial on Thursday
- Computer Exercise 1 will be posted on Thursday

Next week
- A new module on microscopic models
- A/Prof Zuduo Zheng

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CIVL6415: TRAFFIC ANALYSIS AND SIMULATION

Microscopic Traffic Flow Modelling (Part I: Introduction & Basics)

Professor Zuduo Zheng
School of Civil Engineering
University of Queensland
Semester 2 2025

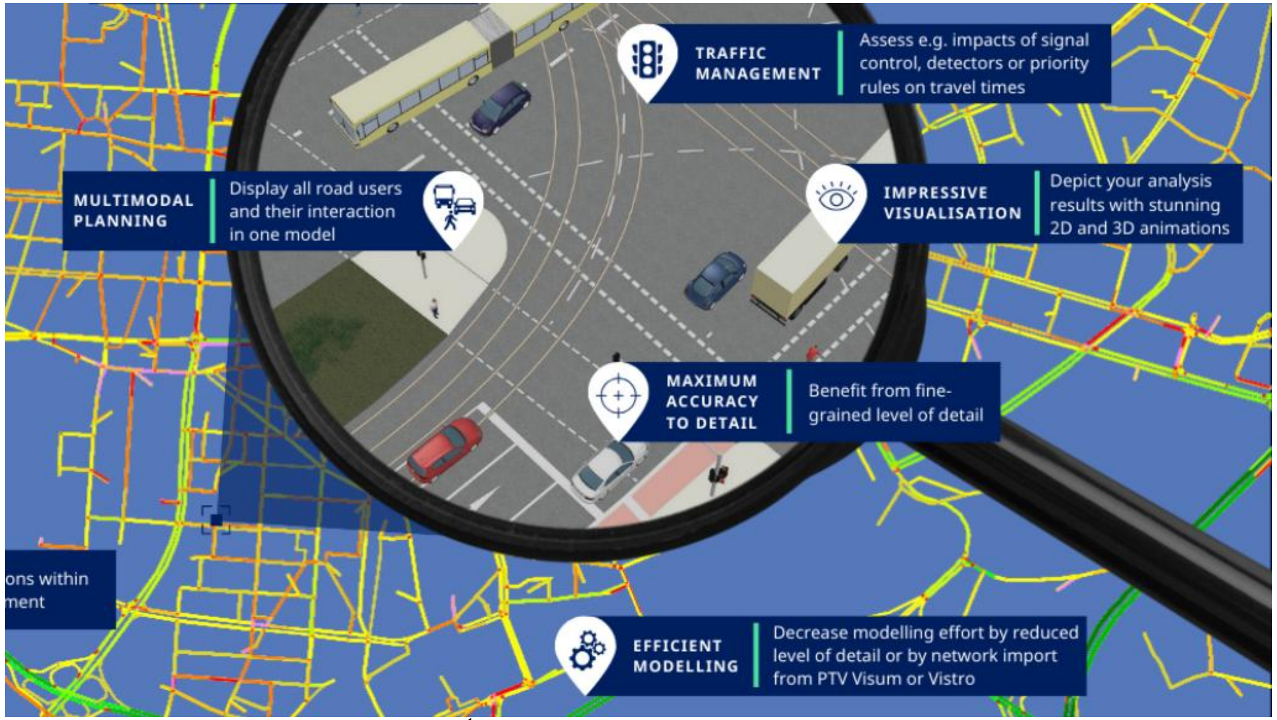
Agenda

- Overview of Microscopic Traffic Flow Modelling
- Car Following Modelling Basics

Overview of Microscopic Traffic Flow Modelling

Levels of Traffic Modelling

- Microscopic (Vehicles & Drivers)
  - ✓ Models each vehicle/driver behavior (car-following, lane-changing).
  - ✓ Detailed simulation, intelligent transport systems, local junction analysis.
- Macroscopic (Traffic Flows)
  - ✓ Treats traffic like fluid flow (density, speed, flow).
  - ✓ Network modelling, strategic planning, regional/national models.
- Mesoscopic (Vehicle Groups)
  - ✓ Models flows of vehicles in groups with some individual characteristics.
  - ✓ Corridor-level studies, intermediate detail.

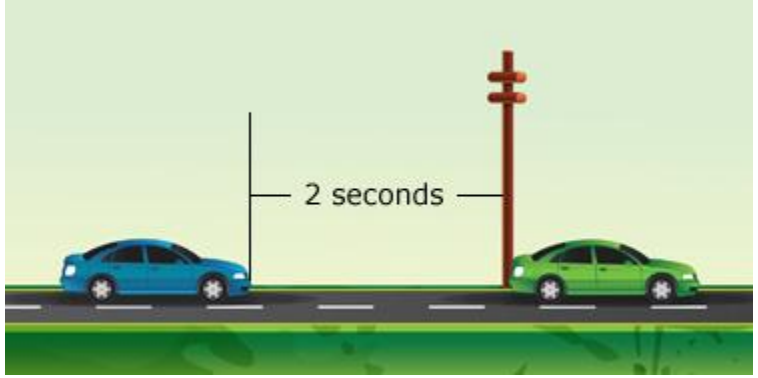


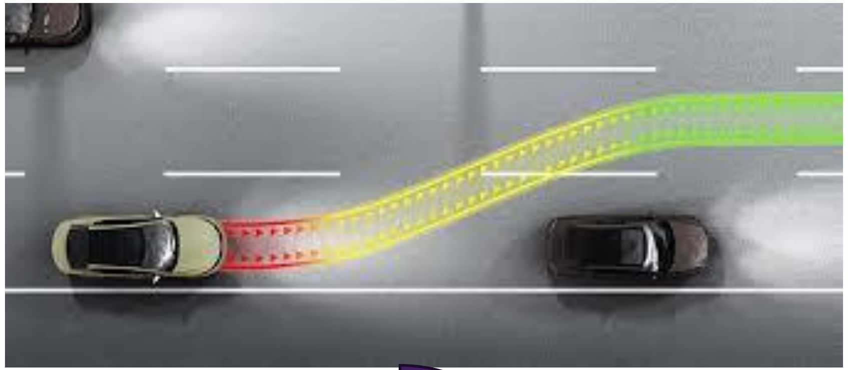
ptvgroup.com

Characterisation of Microscopic Models

- Microscopic models considers the smallest objects that play an important role in a system, e.g., atoms in physics, individual decision makers in economics.
- In traffic flow, this smallest object usually is **the driver-vehicle unit** but it can also be a cyclist, a pedestrian, etc.
- Microscopic models are more detailed than the macroscopic ones which locally aggregate the microscopic quantities.
- Microscopic models are less detailed than models for the vehicle dynamics which include details related to brake, engine control path, stability control, etc.

Two primary tasks of a driver-vehicle unit





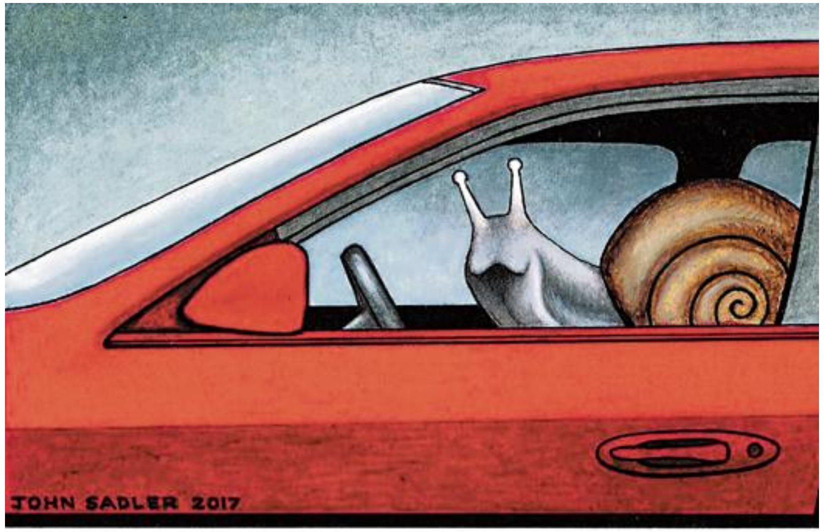
Car following

Lane changing

Every rational driver can be a good car-following modeller!

A good starting point is to think how you normally drive.

Safety: Not too close



Efficiency: Not too far away

What makes car following modelling complicated is due to the simple fact

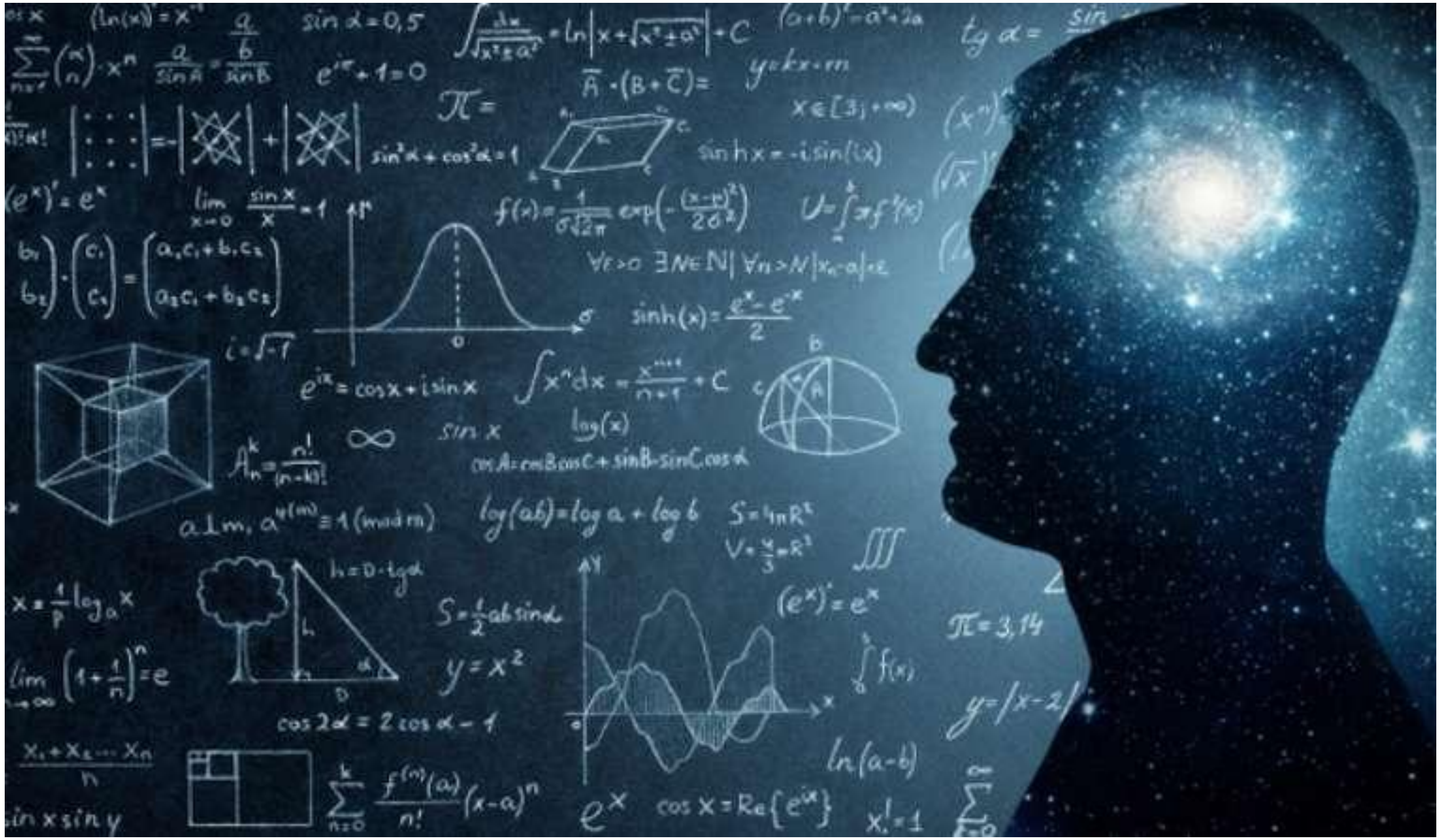
Each driver is different!

Car following in real life

Source: online

A traffic congestion that seemingly comes from nowhere

So, we need properly developed car-following models



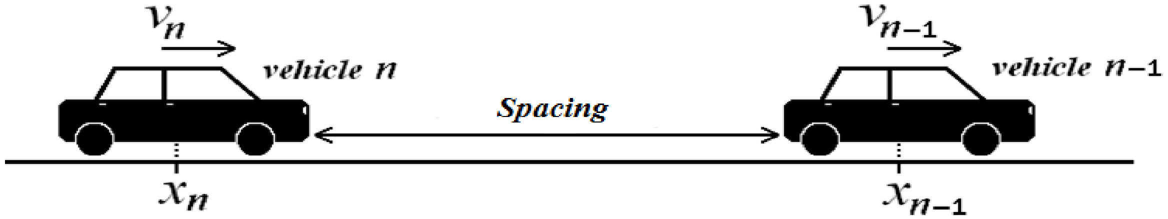
Car Following Modelling Basics

Introduction

- A large number of CF models have been developed in an attempt to describe CF behavior under a wide range of traffic conditions, ranging from free-flow to extreme situations.
- In a strict sense, car-following models describe the driver's behavior only in the presence of interactions with other vehicles while free traffic flow is described by a separate model. In a more general sense, car-following models include all traffic situations such as car-following situations, free traffic, and also stationary traffic.
- A car-following model is ***complete*** if it is able to describe all situations including acceleration and cruising in free traffic, following other vehicles in stationary and non-stationary situations, and approaching slow or standing vehicles, and red traffic lights.

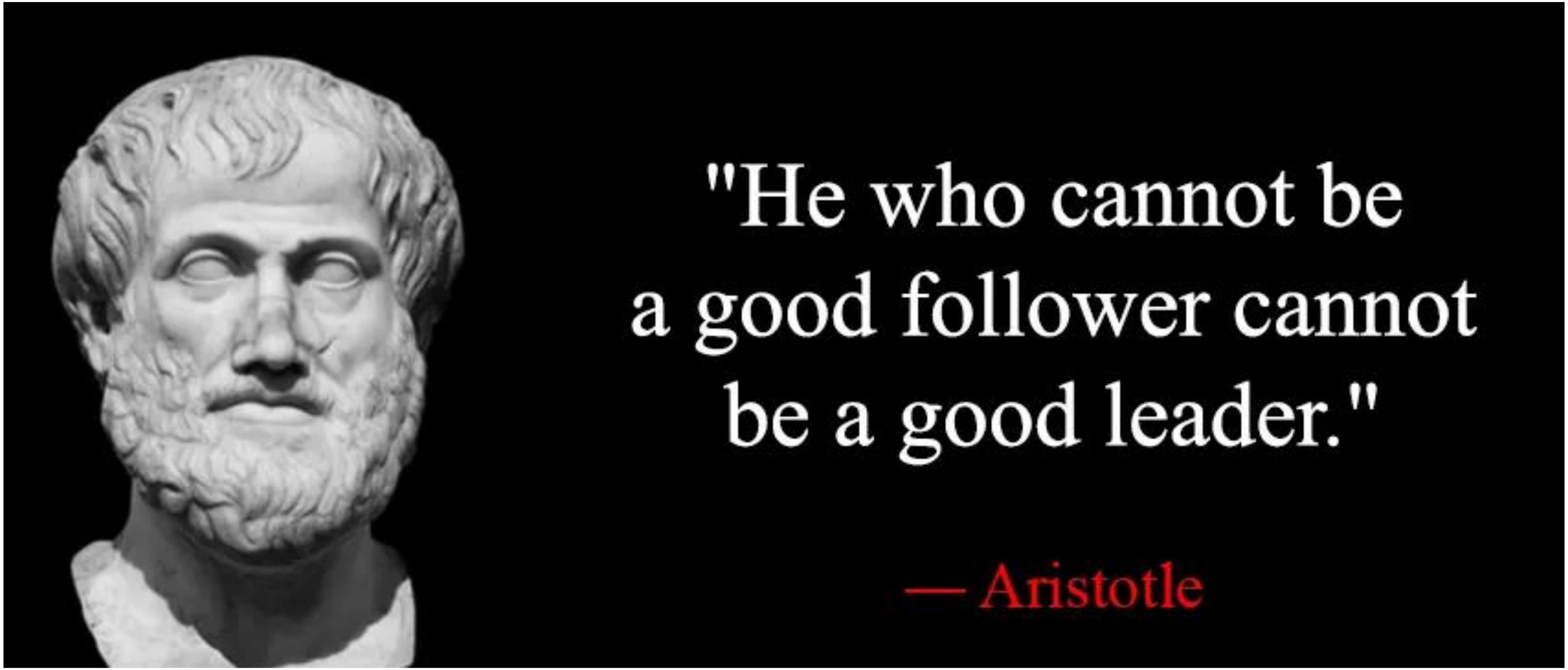
Car-following model

- **Car-following (CF) model** describes the decision of the driver to follow the preceding vehicle **efficiently** and **safely**.

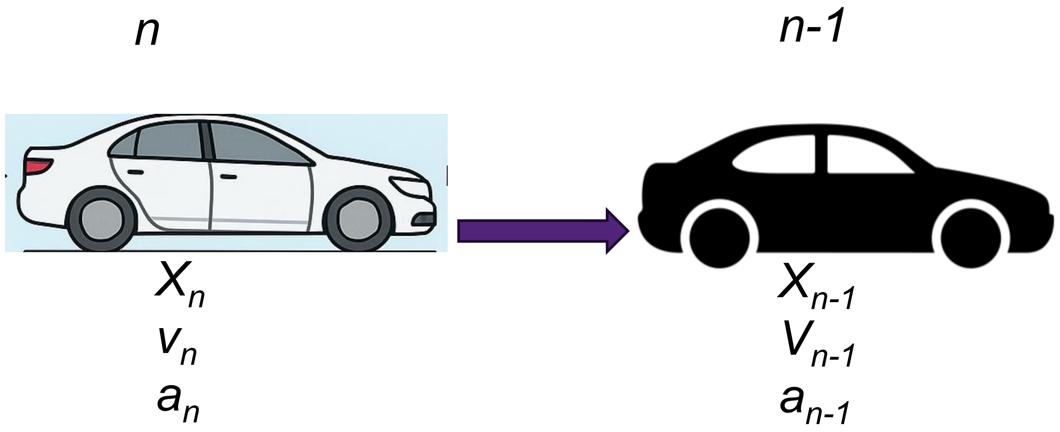


- Rules governing CF are typically simplified as a set of mathematical models that aim for reproducing the longitudinal interactions in the driver-vehicle system.
- As one of the oldest topics in traffic engineering, numerous CF models have been proposed in the literature.

Think about how you follow a car in front of you



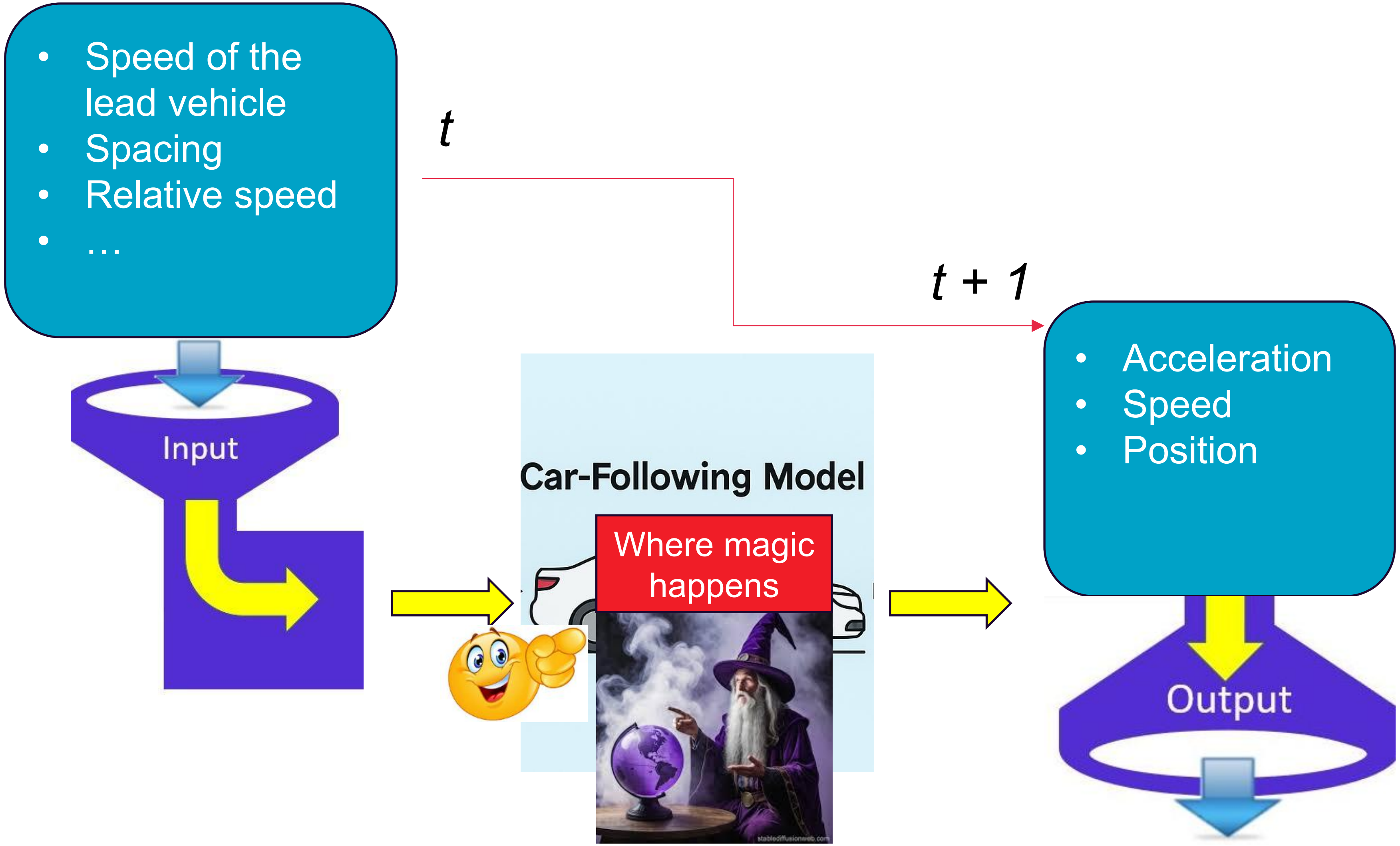
The state variables of car-following models



Each driver-vehicle unit n is described by the state variables:

- location $x_n(t)$, and speed $v_n(t)$ as a function of the time t, and their derived variables such as spacing and relative speed.
- by the attribute "vehicle length" l_n .

We define the vehicle index n such that vehicles pass a stationary observer (or detector) in ascending order, i.e., the first vehicle has the lowest index.



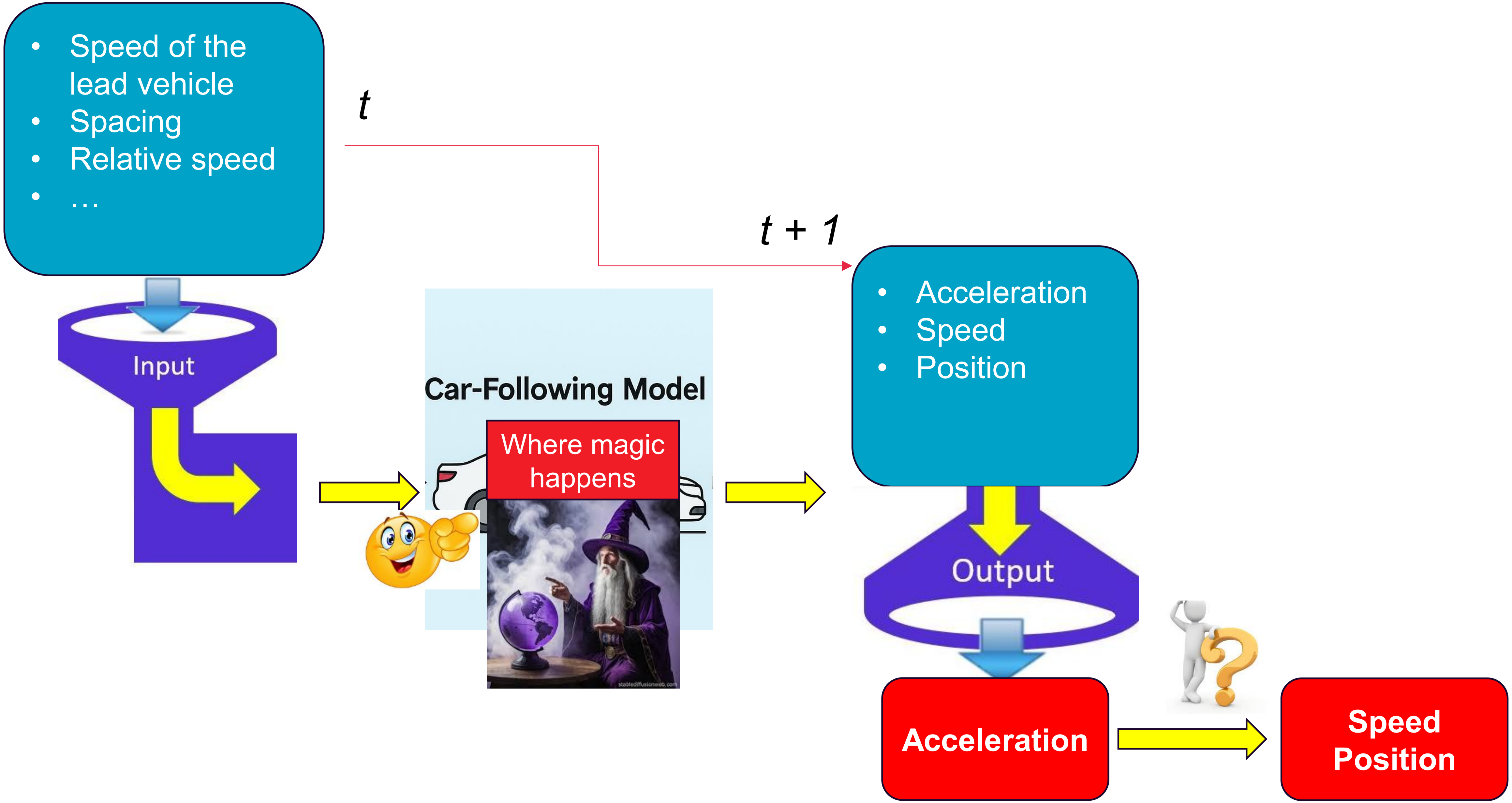
Speed of the lead vehicle
Spacing
Relative speed
…
Input
t
t + 1
Car-Following Model
Where magic happens
Acceleration
Speed
Position
Output

Continuous-time car-following models

In continuous-time models, the driver's response is directly given in terms of an acceleration function $a_{mic}(s_n, v_n, v_{n-1})$, leading to a set of coupled ordinary differential equations:

$$\dot{x}_n(t) = \frac{dx_n(t)}{dt} = v_n(t)$$

$$\dot{v}_n(t) = \frac{dv_n(t)}{dt} = a_{mic}(s_n, v_n, v_{n-1}) = a_{mic}(s_n, v_n, \Delta v_n)$$



Speed of the lead vehicle
Spacing
Relative speed
…
t
t + 1
Input
Car-Following Model
Where magic happens
Acceleration
Speed
Position
Output
Acceleration
Speed
Position

Numerical integration

- In general, time-continuous models cannot be solved analytically and an integration scheme is necessary for an approximate numerical solution of the system of equations.
- For traffic flow applications, only **explicit update schemes** with a fixed time step are practical.
- Assuming a constant update time step Δt, a simple but efficient explicit update method is given by the "ballistic" assumption of **constant accelerations during each time step**.

$$v_n(t+\Delta t) = v_n(t) + a_{mic}(s_n(t), v_n(t), \Delta v_n(t))\Delta t$$

$$x_n(t+\Delta t) = x_n(t) + \frac{v_n(t) + v_n(t+\Delta t)}{2}\Delta t$$

Discrete-time CF models

- Time is not modelled as a continuous variable but discretized into finite and generally constant time steps.
- The driver's response is no longer modelled by an acceleration function but by **a speed function**, indicating the speed that will be reached at the end of the next time step.
- A generic formulation of discrete-time CF models is shown:

$$v_n(t + \Delta t) = v_{mic}(s_n(t), v_n(t), v_{n-1}(t))$$

Discrete-time CF models

- Compared to continuous models, discrete-time car-following models are generally less realistic and less flexible but require less computing power for their numerical integration.
- Most discrete-time car-following models have been proposed at times where computing was more expensive. Nowadays, hundreds of thousand vehicles can be simulated with time-continuous models on a PC in real time, so this numerical advantage become less relevant.
- Most traffic simulation software uses time-continuous models.

Different CF models

- A specific time-continuous model is uniquely characterized by **its acceleration function a_{mic}**.
- Similarly, a specific discrete-time model is completely characterized by **its speed function v_{mic}**.
- When simulating heterogeneous traffic consisting of a variety of driving styles and vehicle classes (such as cars and trucks), each driver-vehicle combination is described by different acceleration functions $a_{mic}^{n}(s, v, v_l)$ or speed function $v_{mic}^{n}(s, v, v_l)$, respectively.

Prove the equivalence between continuous-time models and discrete-time models if the ballistic update scheme is used in updating continuous-time models.

Mathematical equivalence between continuous-time and discrete-time CF

If we assume $a_{mic}(s, v, v_l) = {}^{(v_{mic}(s,v,v_l)-v)}/_{\Delta t}$,

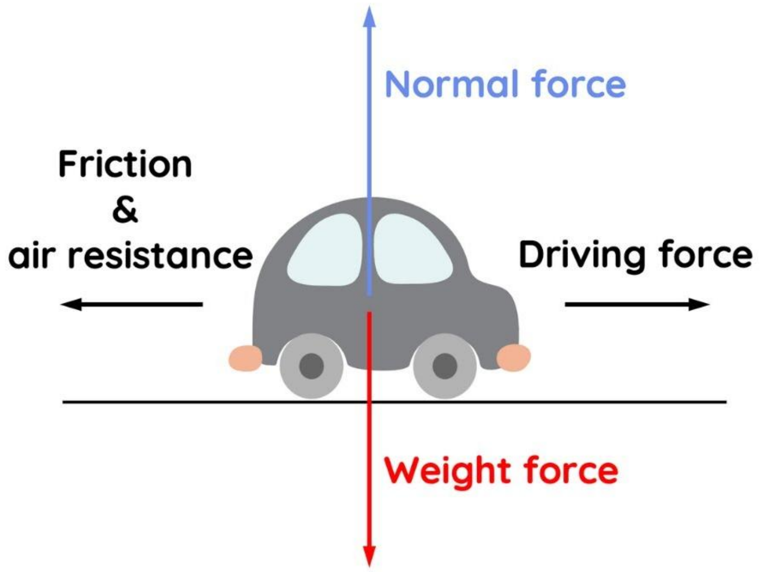
The combination of a continuous-time model with the ballistic update scheme is **mathematically equivalent** to discrete-time models.

Conceptual difference between continuous-time and discrete-time CF

- For discrete-time models, the time step Δt plays the role of a model parameter typically describing the reaction time, the time headway, or the speed adaptation time.
- For time-continuous models, the update time Δt is an auxiliary variable of the approximate numerical solution which preferably should be small as the true solution is obtained in the limit $\Delta t \to 0$ (at least if the numerical method is consistent and stable).

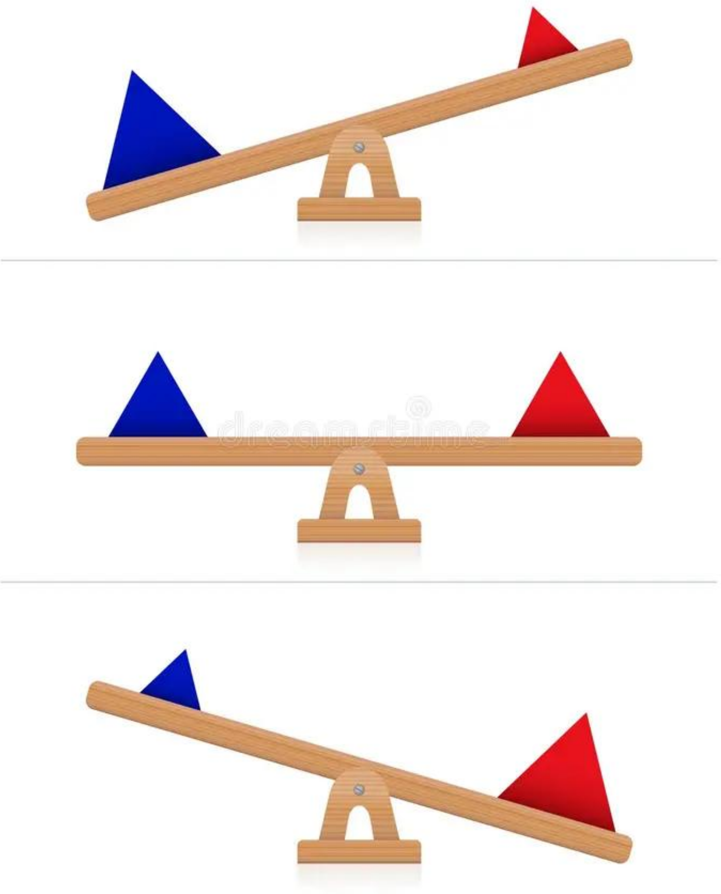
Steady State Equilibrium

- Since the driver-vehicle units of microscopic models are equivalent to *driven particles* of physical systems, there is no equilibrium in the strict sense. Instead, there is a stationary state where the forces and the entering and exiting energy fluxes are balanced.
- Strictly physically, this can be interpreted in terms of a balance of the forces (the sum of friction, wind drag, and engine driving force equates to zero) or energy fluxes (engine power equals change in potential and kinetic energy plus energy dissipation rate by friction and wind drag).



Steady State Equilibrium: balancing the social forces

- More relevant for traffic flow, however, is the concept of balancing the *social forces*:
  - ✓ The desire to go ahead generates a positive (accelerating) social force while the interactions with other vehicles generally lead to negative social forces in order to avoid critical situations and crashes.
- In any case, such a balanced state is denoted as ***steady-state equilibrium***. For microscopic models, a consistent description of the steady-state equilibrium requires **identical** driver-vehicle units on a **homogeneous road**.



Conditions of Steady-State Equilibrium

- Technically, this implies that the model parameters are the same for all drivers and vehicles, i.e., the acceleration or speed functions characterizing the respective model do not depend on the vehicle index, $a^i_{mic}(s, v, v_l) = a_{mic}(s, v, v_l)$, and $v^i_{mic}(s, v, v_l) = v_{mic}(s, v, v_l)$, respectively.
- From the modelling point of view, the steady-state equilibrium is characterised by the following two conditions:
  - ✓ Homogeneous traffic: All vehicles drive at the same speed ($v_i = v$) and keep the same gap behind their respective leaders ($s_i = s$).
  - ✓ No accelerations: $\dot{v}_i = 0$ or $v_i(t + \Delta t) = v_i(t)$ for all vehicles i.

Derive the Steady-state Relations of a CF Model

- For time-continuous models, this implies $a_{mic}(s, v, v_l) = 0$,
- For discrete-time models, this implies $v_{mic}(s, v, v_l) = v$
- Depending on the model, the microscopic steady-state relations above can be solved for
  - ✓ The equilibrium speed $v_e(s)$ as a function of the gap;
  - ✓ The equilibrium gap $s_e(v)$ for a given speed.

Bridging Microscopic and Macroscopic Relations

- The key is to use the relation between the distance gap *s* and the density *k*: $s_i = s = \frac{1}{k} - l$
- Meanwhile, the steady-state equilibrium implies that the speed of all vehicles is the same and equal to the macroscopic speed $V(x,t) = v_i(x,t) = v_e(s)$.
- So, we can derive the steady-state speed-density relation and the fundamental diagram:

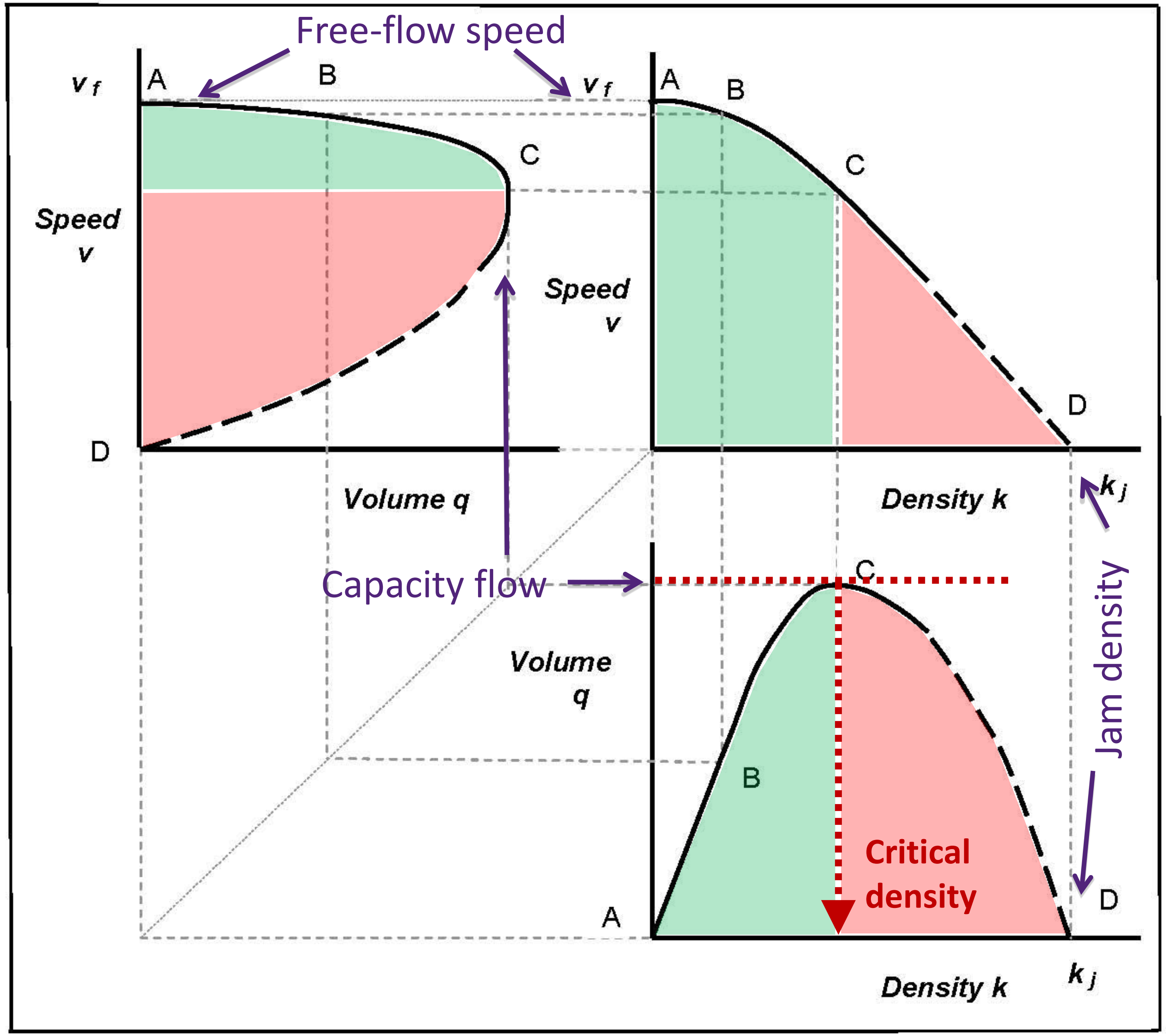
$$V_e(k) = v_e\left(\frac{1}{k} - l\right),\ Q_e(k) = k v_e\left(\frac{1}{k} - l\right).$$

Flow-density-speed relationships

Fundamental Diagrams (FD)

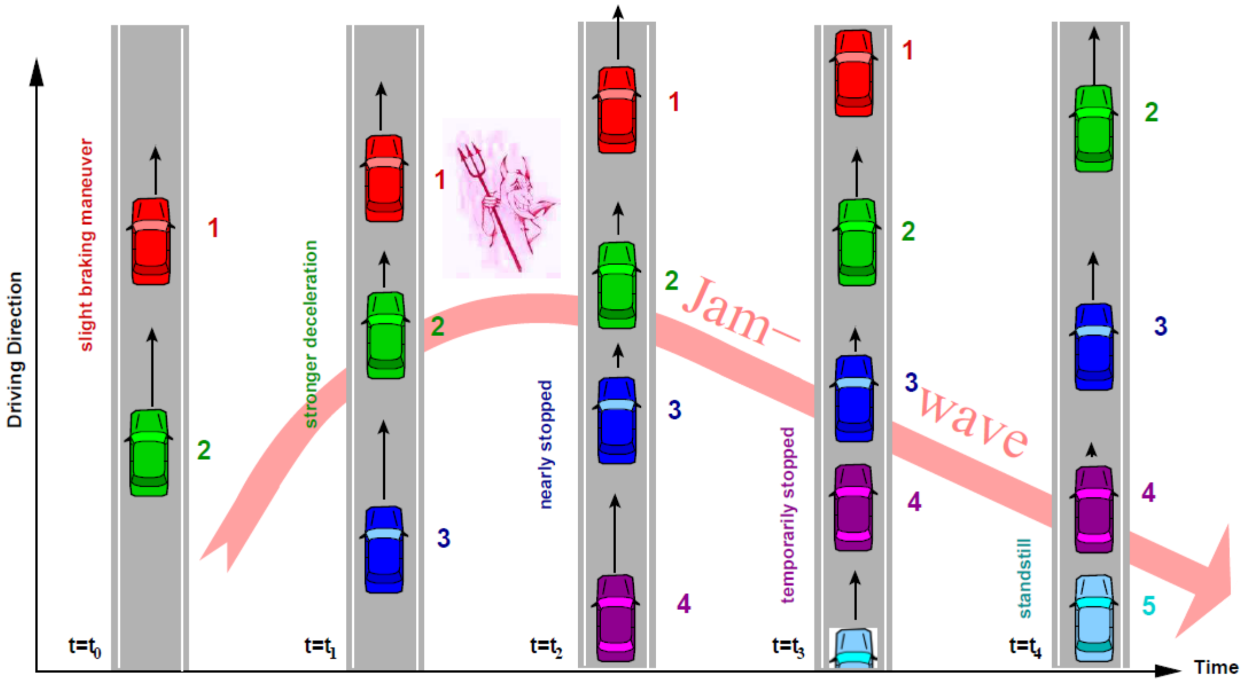
'Fundamental relationships' of 'uninterrupted' traffic flow

At **A**, no interaction
A-**B**, free flow
B-**C**, steady flow
At **C**, traffic unstable
C-**D**, congested flow
At **D**, traffic stationary



Discussion: how to simulate heterogeneous traffic using car following model?

Stability Analysis



The vicious circle: In order to regain the safety gap, the driver of every following vehicle needs to brake harder than his or her predecessor. The numbers beside the vehicles denote the vehicle index.

A traffic congestion that seemingly comes from nowhere

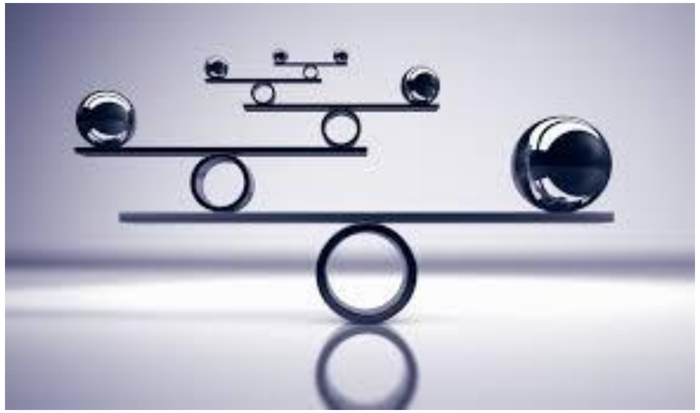
Phantom Traffic Jam Explained.

The Japanese Experiment.

Phantom Intersection.

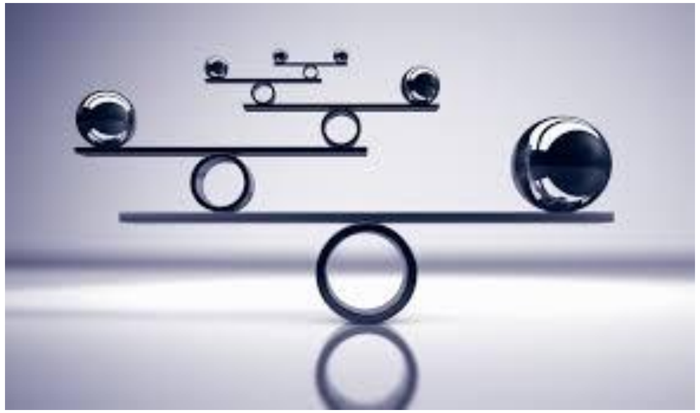
Local stability

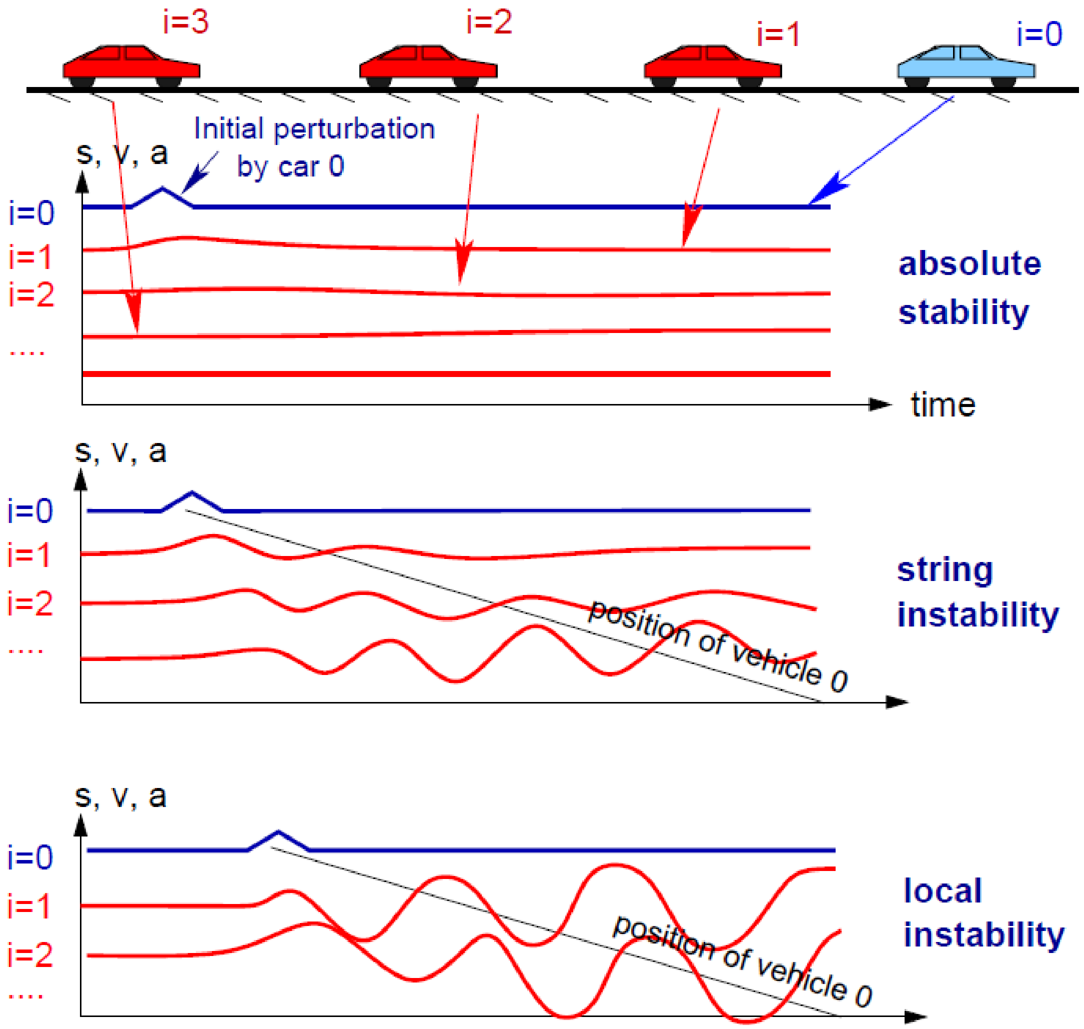
- Local stability is to investigate the stability of a single vehicle's movement over time under the influence of a small perturbation that is often originated from the leading vehicle's movement.
- It is locally stable if the perturbation diminishes over time, which corresponds to asymptotic stability in dynamic systems.



String stability

- String stability focuses on the stability of a platoon of vehicles over space under the influence of a small perturbation that is originated from the first vehicle of the platoon.
- It is asymptotically stable if the perturbation strictly diminishes over space/vehicles.




i=3
i=2
i=1
i=0
Initial perturbation
by car 0
s, v, a
i=0
i=1
i=2
....
absolute
stability
time
s, v, a
i=0
i=1
i=2
....
position of vehicle 0
string
instability
s, v, a
i=0
i=1
i=2
....
position of vehicle 0
local
instability

For a good CF model,
1) shall it be locally stable?
2) shall it be string unstable?

Implications on CF Modelling

- Being locally stable is an important feature that a CF model should have because typically a driver is able to easily recover from small disturbances in real traffic and return to a steady CF state over time.
- The string stability analysis is particularly relevant to traffic flow modelling as the stop-and-go oscillation in real traffic is a manifestation of the string instability of traffic flow.
- From the CF model development perspective, for a realistic CF model, it should be able to depict the string instability in order to replicate characteristics related to traffic oscillations;
- However, from traffic operation perspective, string instability should be minimized or avoided. Thus, the string stability analysis has been frequently implemented since the early-stage of CF model development.

Further readings

Chapter 10, Traffic flow dynamics: Data, models and simulation by Martin Treiber and Arne Kesting.

van Wageningen-Kessels, F., van Lint, H., Vuik, K. et al., 2015. *EURO J Transp Logist* 4: 445. https://doi.org/10.1007/s13676-014-0045-5

Saifuzzaman, M., Zheng, Z., 2014. Incorporating human-factors in car-following models: A review of recent developments and research needs. *Transportation Research Part C* 48, 379 - 403.

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Thank you

Questions?

Microscopic Traffic Flow Modelling (Part II)

Professor Zuduo Zheng
School of Civil Engineering
University of Queensland
Semester 2 2025

Agenda

- Main Car Following Models
- Car Following Models Used in Some Popular Traffic Simulation Packages

Main Car Following Models

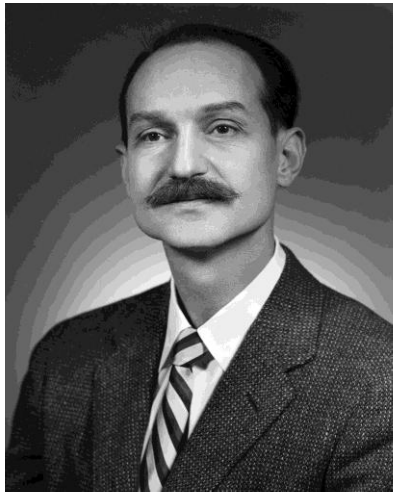
“In the beginning there was GM.”

Car Following Models - GM

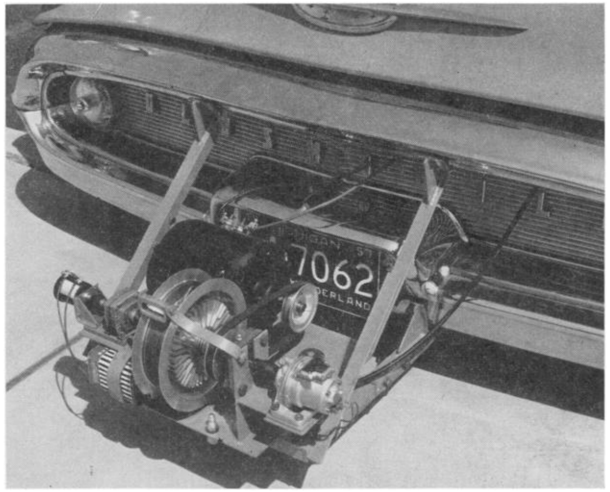
General Motors (GM) models

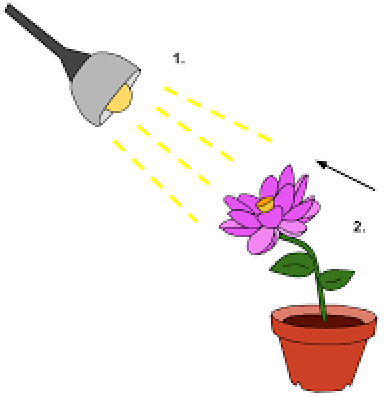
Back in the 1950's, General Motors sponsored a team of scientists in its Research Laboratories, from which pioneering work was done that broke the ground for traffic flow theory. At the foremost of such efforts was the family of General Motors models (GM).

$$response = sensitivity \times stimulus$$



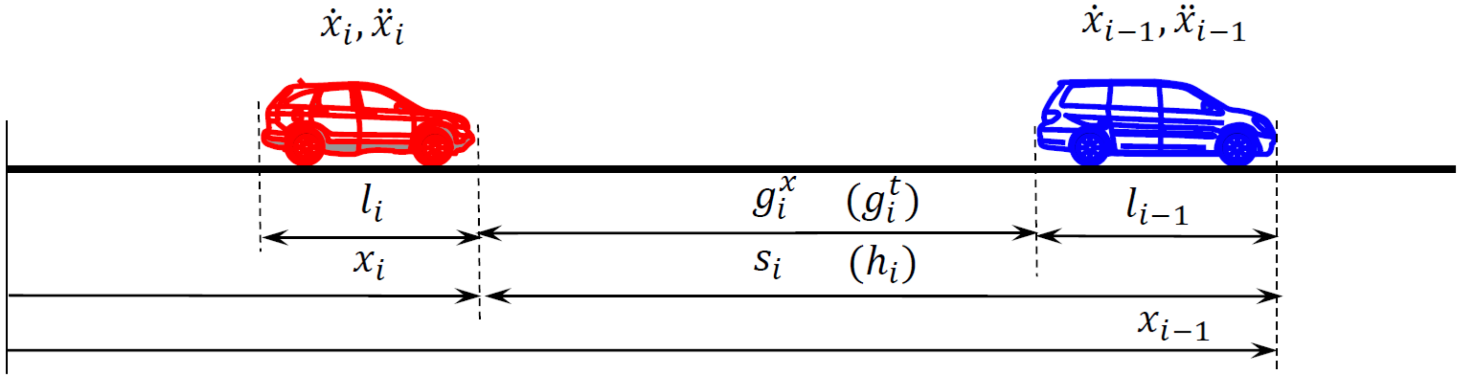
Robert Herman at GM
(Niels Bohr Library & Archives)





https://en.wikipedia.org

GM1



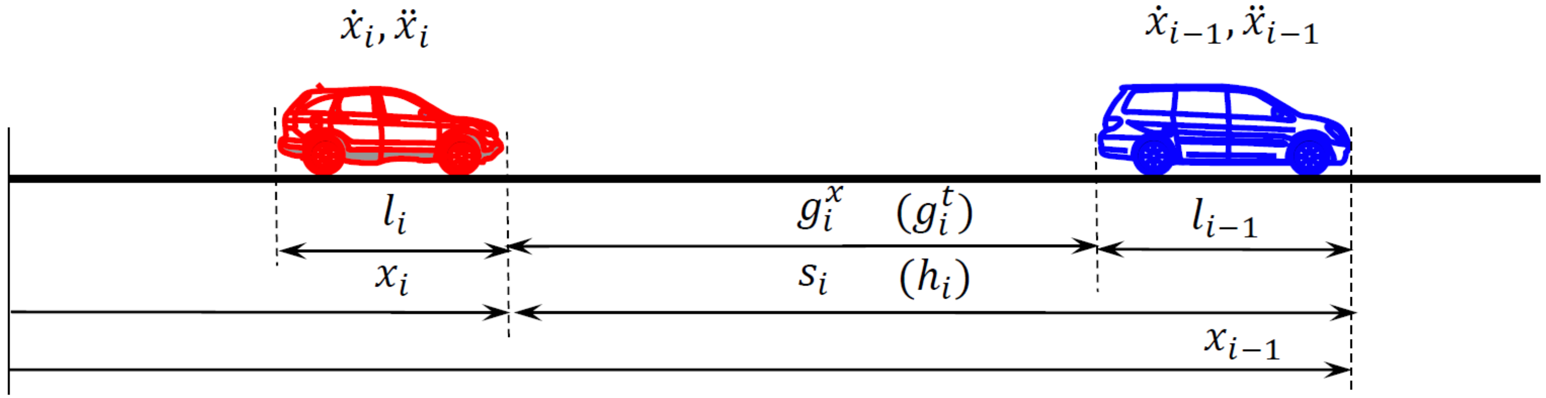
response = f (sensitivity, stimuli)

$$\ddot{x}_i(t+\tau_i) = \alpha[\dot{x}_{i-1}(t) - \dot{x}_i(t)]$$

- Scenario 1: $\dot{x}_i(t) = 120\,\text{km/h}$, $\dot{x}_{i-1}(t) = 100\,\text{km/h}$, and $s_i(t) = 50\,\text{m}$
- Scenario 2: $\dot{x}_i(t) = 120\,\text{km/h}$, $\dot{x}_{i-1}(t) = 100\,\text{km/h}$, and $s_i(t) = 5000\,\text{m}$

What will be the predicted accelerations in these two scenarios? Is it reasonable?

GM2



$$\ddot{x}_i(t+\tau_i) = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} [\dot{x}_{i-1}(t) - \dot{x}_i(t)]$$

Field experiments revealed that sensitivity coefficient α ranges between (0.17, 0.74). In GM2, a high sensitivity value α_1 is chosen when the two vehicles are close, while a low sensitivity value α_2 is employed when the two vehicles are far apart.

GM3

$$\ddot{x}_i(t+\tau_i) = \alpha \frac{[\dot{x}_{i-1}(t) - \dot{x}_i(t)]}{[x_{i-1}(t) - x_i(t)]}$$

- Scenario 1: In Brisbane CBD, one vehicle is following another at a spacing of 100 m with speeds $\dot{x}_i(t) = 30$ km/hr, $\dot{x}_{i-1}(t) = 10$ km/hr.
- Scenario 2: In M1 motorway, one vehicle is following another at a spacing of 100 m with speeds $\dot{x}_i(t) = 130$ km/hr, $\dot{x}_{i-1}(t) = 110$ km/hr.

What will be the predicted accelerations in these two scenarios? Is it reasonable?

GM4

$$\ddot{x}_i(t+\tau_i) = \alpha \frac{\dot{x}_i(t+\tau_i)[\dot{x}_{i-1}(t) - \dot{x}_i(t)]}{[x_{i-1}(t) - x_i(t)]}$$

$$F = G\frac{m_1 m_2}{r^2}$$

Similarity between GM4 and gravity model

GM5

$$\ddot{x}_i(t+\tau_i) = \alpha \frac{[\dot{x}_i(t+\tau_i)]^m}{[x_{i-1}(t)-x_i(t)]^l}[\dot{x}_{i-1}(t)-\dot{x}_i(t)]$$

- Exponents *m* and *l* generalises GM model.
- By taking different value combinations of *m* and *l*, GM5 can be reduced to earlier models.
- GM5 serves as the “bridge” that connects microscopic and macroscopic models.

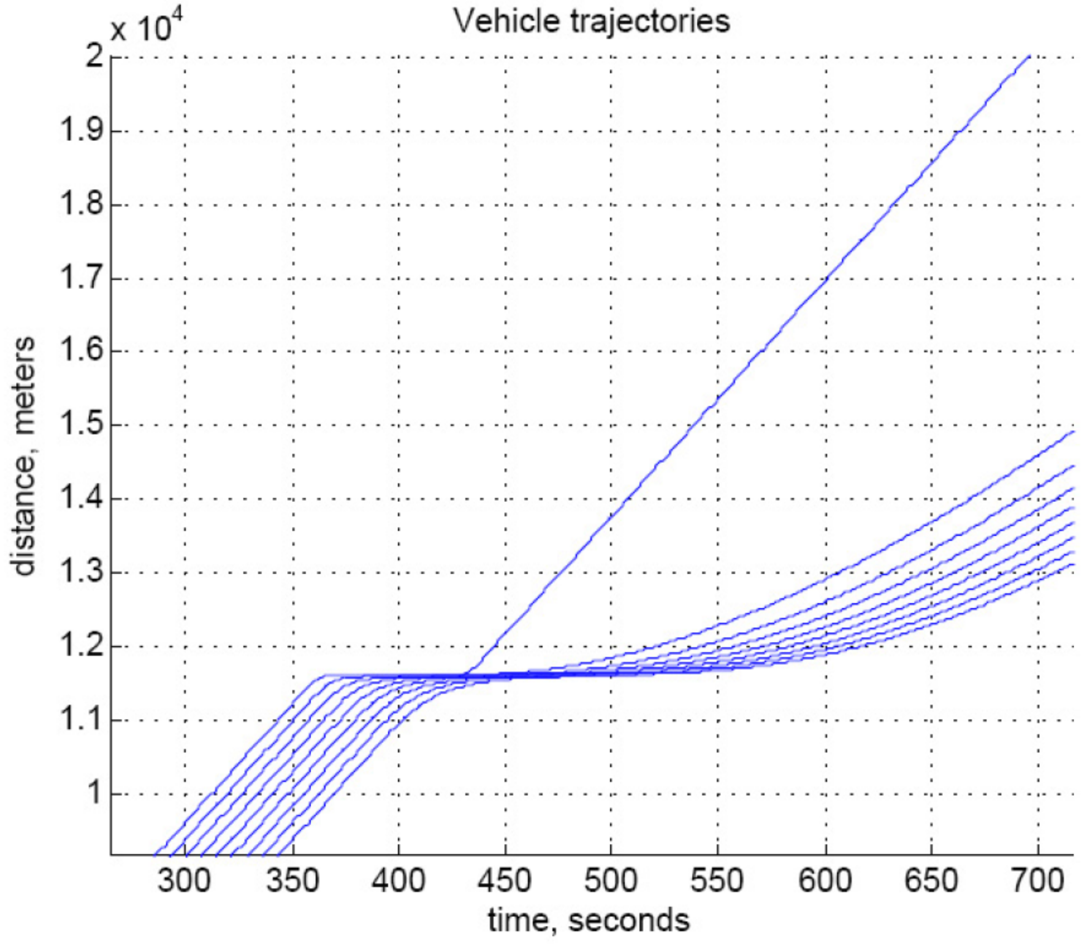
Discussion

- If you use GM4 to simulate a vehicle who stops at a red traffic light and then starts moving when the traffic light turns green. What would happen?
- According to GM4, if two vehicles are travelling at the same speed, what is the implication on their spacing?

Limitations of GM Models: Slow Start

According to GM4, a vehicle at a stopped position is unable to get started. This is because the vehicle's speed ($\dot{x}_i(0)$ = 0) determines its acceleration in the next step.

Therefore, the vehicle has to maintain a non-zero speed at any time in order to avoid being trapped. As such, the vehicle on red light has to slow down to an infinitesimal speed rather than a complete stop in order to avoid being trapped. When the light turns green or the leading vehicle resumes motion, the subject vehicle will take a long time to get up to speed. Figure illustrates such a scenario where a noticeable gap is resulted between the first and second vehicle.



Limitations of GM Models: An Intimate Pair

According to GM4, two vehicles can get arbitrarily close as long as they are traveling at the same speed, which is certainly not true - no one would dare to follow another with one inch apart at 120 km/h! The reason why GM4 allows such a ridiculous car-following distance is because, regardless of how close the two vehicles are, the model predicts no response as long as the two vehicles are moving at the same speed.

Extensions of GM Models

- **Memory functions**: Assuming that a driver reacts to the relative speed of the preceding vehicle over a period of time, rather than in an instant, a memory function was introduced into the linear GM model to store the information of relative speed during CF. Although the model removes unrealistic peaks in acceleration profile, the implementation of the model in traffic simulation is considerably more complex due to the need of maintaining an array of past conditions for each vehicle.
- **Acceleration and deceleration asymmetry**: 1) most passenger cars have a greater deceleration than acceleration capacity. And 2) in congested traffic, drivers are more sensitive to deceleration than to acceleration.
- **Multiple-vehicle interaction**: The models discussed above are based on the assumption that each driver reacts in some specific manner to some stimuli from the preceding vehicle. In the real world, however, drivers most likely adjust their behaviors according to their observations of more than one vehicle ahead.

CF Models Used in Some Popular Traffic Simulation Packages

Car Following Models – Gipps' model

ELSEVIER

Transportation Research Part B: Methodological

Volume 15, Issue 2, April 1981, Pages 105-111

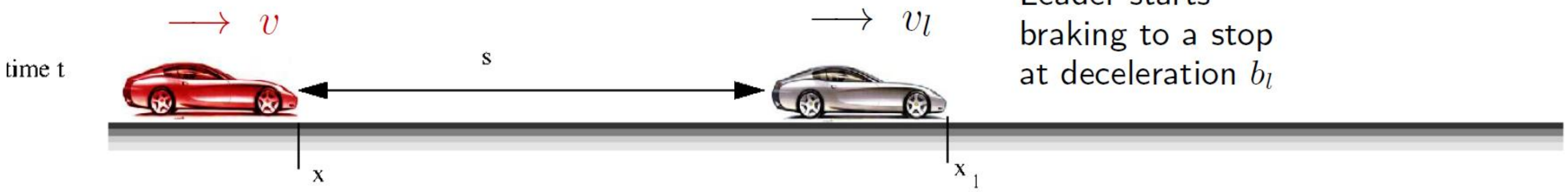
A behavioural car-following model for computer simulation

P.G. Gipps

Gipps' model

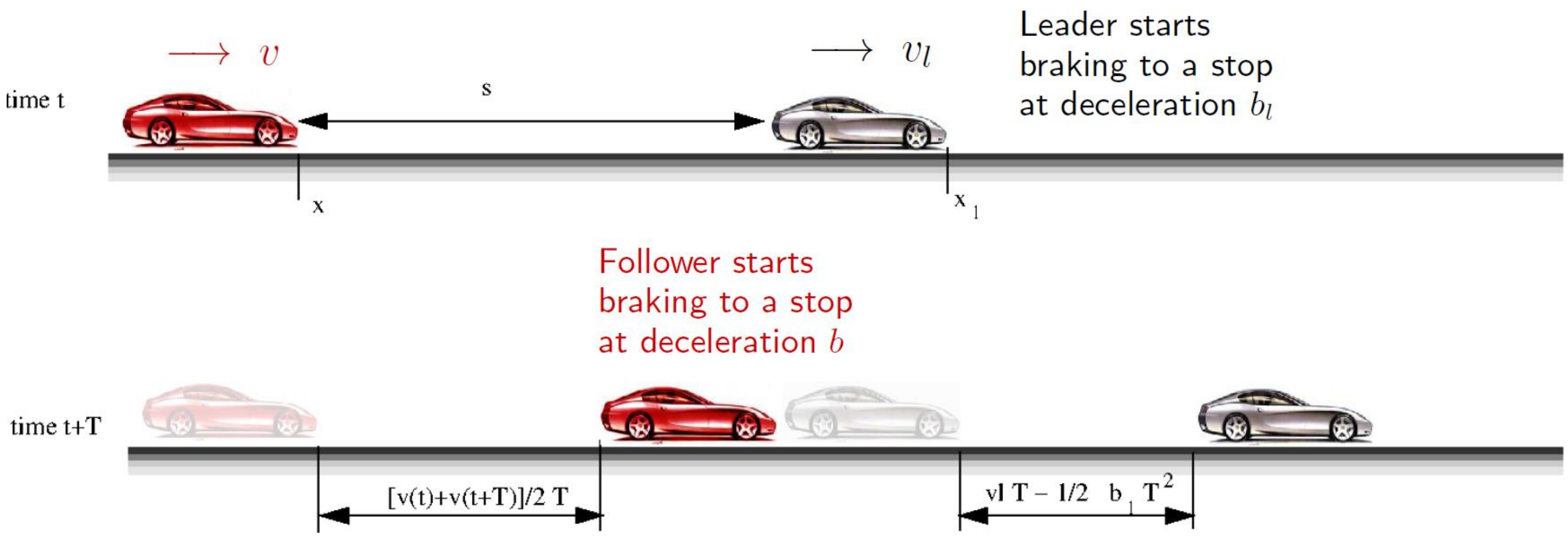
- Gipps' model is the most popular safety distance model and developed by Gipps in 1981.
- Safety distance models differ from GM models by hypothesizing that the driver reacts to spacing relative to the preceding vehicle, rather than to the relative speed. This idea was first proposed by Kometani and Sasaki (1959).
- Gipps' model assumes that the speed is selected by the driver in a way to ensure that the vehicle can be safely stopped in case the preceding vehicle should suddenly brake.
- Gipps' model includes two modes of driving: free-flow and CF. The driver chooses the smaller one from the speeds obtained from the free-flow and CF modes.

Derivation of Gipps' CF Model: Overview

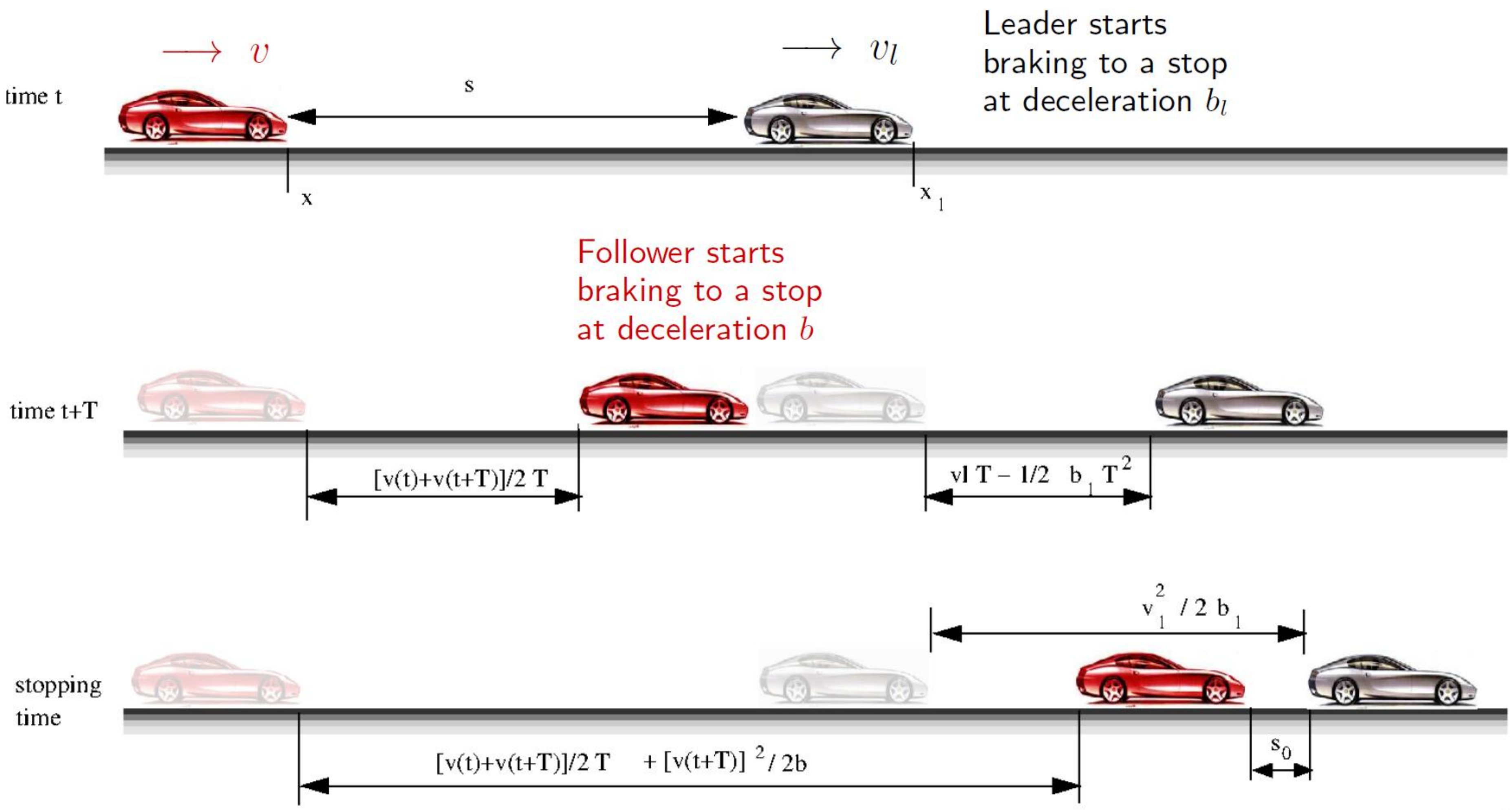


Leader starts braking to a stop at deceleration b_l

Derivation of Gipps' CF Model: Overview



Derivation of Gipps' CF Model: Overview



Gipps' CF model

- Modeling approach: considered limitations/constraints of vehicles.
  - In **free-flow case**:
    - It will not exceed its driver's desired speed.
    - Its free acceleration should first increase with speed as engine torque increases then decrease to zero as the desired speed is reached.

$$v_n(t+\tau) \leq v_n(t) + 2.5a_n\tau(1 - v_n(t)/V_n)(0.025 + v_n(t)/V_n)^{1/2}$$

Variables

- a_n- max acceleration that vehicle n wishes to take
- V_n - desired speed of vehicle n
- τ is the reaction time of vehicle n.

Gipps' CF model

- In **braking case**: To assure that if the leader brakes, the vehicle will not crash. This means that we should guarantee the worst, but still safe, condition; i.e., the subject vehicle is about to crash when it comes to stop.
  - This requires the following inequality:

$$x_{n-1}^* = x_{n-1}(t) - v_{n-1}(t)^2/2b_{n-1}$$

$$x_n^* = x_n(t) + [v_n(t) + v_n(t+\tau)] * \frac{\tau}{2} + v_n(t+\tau)\theta + \frac{v_n(t+\tau)^2}{-2b_n}$$

$$x_{n-1}^* - s_{n-1} \geq x_n^*$$

- τ: subject vehicle reaction time. θ is the delay in addition to the regular reaction time τ . This represents the safety margin.
- b_n: subject vehicle deceleration;
- b_{n-1} : leader deceleration; the subject vehicle has to estimate b_{n-1}. We use $\hat{b}$ to replace that.
- Both b_n and b_{n-1} are negative if vehicles are decelerating.
- s_{n-1}: the jam spacing, usually s_{n-1} equals to vehicle length and some extra spacing for buffer.
- If $\theta = \tau/2$, we can solve the inequality above to obtain speed $v_n(t+\tau)$:

Gipps' CF model

- In **braking case**:

$$x_{n-1}^{*} = x_{n-1}(t) - v_{n-1}(t)^2/2b_{n-1}$$

$$x_n^{*} = x_n(t) + \underbrace{\boxed{[v_n(t) + v_n(t+\tau)] * \frac{\tau}{2}}}_{\text{term1}} + \underbrace{\boxed{v_n(t+\tau)\theta}}_{\text{term2}} + \underbrace{\boxed{\frac{v_n(t+\tau)^2}{-2b_n}}}_{\text{term3}}$$

$$x_{n-1}^{*} - s_{n-1} \geq x_n^{*}$$

- **term1**: during the reaction time, the subject vehicle will continue its previous kinematic dynamics (e.g., acceleration or deceleration). Therefore, term 1 captures the traveled distance during from $[t, t+\tau]$.
- **term2**: at $(t+\tau)$, the subject vehicle has decided its next action (deceleration/acceleration), but it may have some delay in executing that, denoted as θ. During $[t+\tau, t+\tau+\theta]$, the subject vehicle cruises at its speed at $v_n(t+\tau)$. Term 2 is the traveled distance during this process.
- **term 3**: from time $(t+\tau+\theta)$, subject vehicle starts to decelerate at b_n until it stops. Note that there is a negative sign because it's decelerating. b_n is negative. Term 3 is the travel distance during this process.

Gipps' CF model

- In **braking case**:

$$x_{n-1}^{*} = x_{n-1}(t) - v_{n-1}(t)^2/2b_{n-1}$$

$$x_n^{*} = x_n(t) + \underbrace{\boxed{[v_n(t) + v_n(t+\tau)] * \frac{\tau}{2}}}_{\text{term1}} + \underbrace{\boxed{v_n(t+\tau)\theta}}_{\text{term2}} + \underbrace{\boxed{\frac{v_n(t+\tau)^2}{-2b_n}}}_{\text{term3}}$$

$$\boxed{x_{n-1}^{*} - s_{n-1} \geq x_n^{*}}$$

To assure that if the leader brakes, the vehicle will not crash.

- **term1**: during the reaction time, the subject vehicle will continue its previous kinematic dynamics (e.g., acceleration or deceleration). Therefore, term 1 captures the traveled distance during from $[t, t+\tau]$.
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- **term 3**: from time $(t+\tau+\theta)$, subject vehicle starts to decelerate at b_n until it stops. Note that there is a negative sign because it's decelerating. b_n is negative. Term 3 is the travel distance during this process.

Gipps' CF model

- Solving the inequality gives us:

$$v_n(t+\tau) \leq b_n\left(\frac{\tau}{2}+\theta\right)+\sqrt{{b_n}^2\left(\frac{\tau}{2}+\theta\right)^2-b_n\left[2s-v_n(t)\tau-\frac{v_{n-1}(t)^2}{\hat{b}}\right]}$$

where $s = x_{n-1}(t) - x_n(t) - s_{n-1}$, is the net-spacing (i.e., excluding jam spacing)

- If $\theta = \tau/2$, speed $v_n(t+\tau)$ becomes:

$$\boxed{v_n(t+\tau) \leq b_n\tau+\sqrt{{b_n}^2\tau^2-b_n\left[2s-v_n(t)\tau-\frac{v_{n-1}(t)^2}{\hat{b}}\right]}}$$

The University of Queensland Australia

Gipps' CF model is the base model in Aimsun.

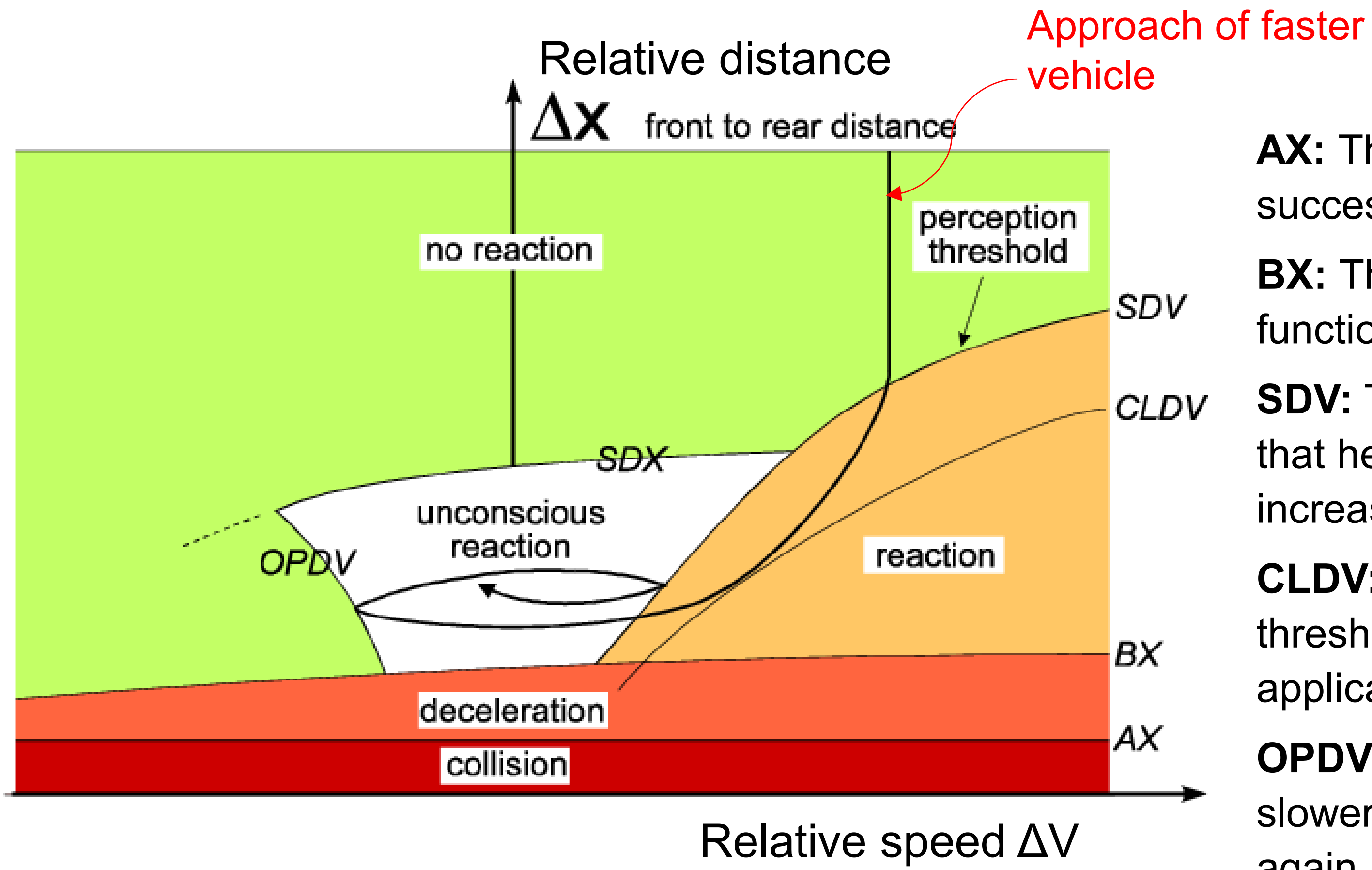
aimsun.

Discussion

Which type does Gipps CF model belong to?
Continuous time or discrete time?

Wiedemann's CF model/'psycho-physical' models

- Engineering CF models unrealistically assume that drivers can perceive and react even to small changes in the driving environment (for example, to slight change in speed difference or spacing).
- To overcome this problem, Wiedemann (1974) introduces the term 'perceptual threshold' to define the minimum value of the stimulus a driver can perceive and will react to.
- The models based on perceptual threshold are also known as 'psycho-physical' models. The threshold is expressed as a function of speed difference and spacing between the preceding and subject vehicles, and is different for acceleration and deceleration decisions. It increases driver alertness when spacing is small, and provides more freedom when it is large.



Wiedemann's CF model (Source: Wiedemann, 1974)

AX: The desired spacing between the front sides of two successive vehicles in a standing queue

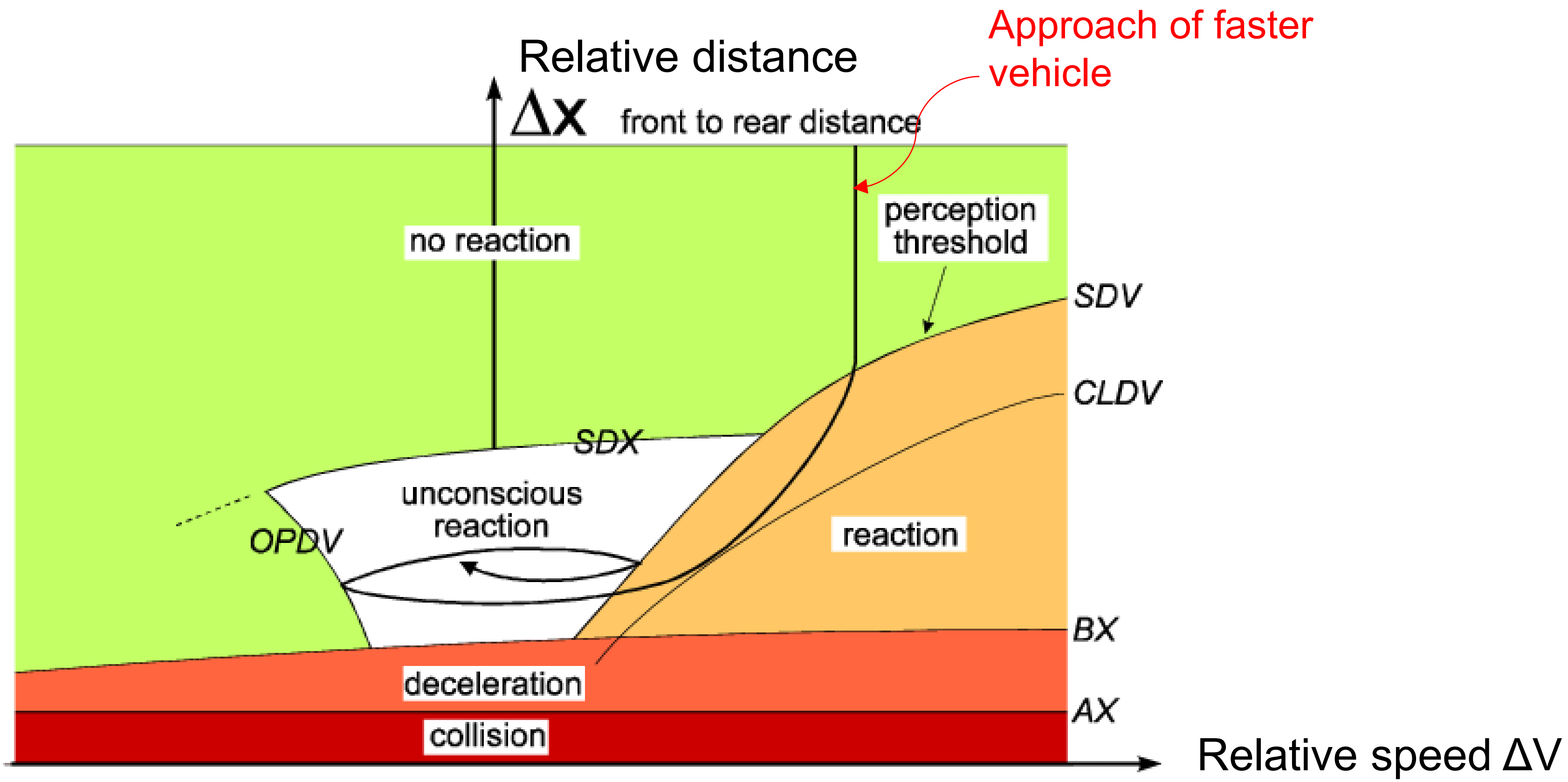
BX: The desired minimum following distance, which is a function of AX, the safety distance, and speed

SDV: The action point where a driver consciously observes that he/she is approaching a slower leading vehicle; SDV increases with increasing speed difference

CLDV: Closing delta velocity (CLDV) is an additional threshold that accounts for additional deceleration by the application of brakes

OPDV: The action point where a driver notices that he/she is slower than the leading vehicle and starts to accelerate again

SDX: A perception threshold to model the maximum following distance, which is approximately 1.5–2.5 times BX



The dark line in the figure shows the decision path of an approaching vehicle. A vehicle travelling faster than the leader will get close to it until the deceleration perceptual threshold (SDV) is crossed. The driver will then decelerate to match the leader's speed. However, as a human being, the driver is unable to accurately replicate the leader's speed, and spacing will increase until the acceleration perceptual threshold (OPDV) is reached. The driver will again accelerate to match the leader's speed and the process continues, as shown in the unconscious reaction zone.

A modified version of the original Wiedemann model has been used in the commercial microsimulation software **VISSIM**.

Intelligent driver model (IDM)

One of the most popular models using desired measures is the intelligent driver model (IDM) proposed by Treiber et al. (2000). This model considers both the desired speed and the desired space headway, as defined in

$$\boxed{\dot{v} = a\left[1 - \left(\frac{v}{v_0}\right)^{\delta} - \left(\frac{s^*(v, \Delta v)}{s}\right)^2\right] \quad \text{IDM.}}$$

where $a_{\max}^{(n)}$ is the maximum acceleration of the subject vehicle n, v_0 is the desired speed, S is spacing between two vehicles measured from the front edge of the subject vehicle to the rear end of the preceding vehicle, s^* is the desired spacing, and δ is a parameter.

Intelligent driver model (IDM)

- When preceding vehicle is far away, the third term in this equation becomes negligible and the model performs as a free flow model where the desired speed of the driver governs the acceleration.
- Use of one equation ensures a smooth transition between free-flow and car-following situations.
- The desired space headway (or following distance) in IDM is dependent on several factors and can be calculated using

$$s^*(v, \Delta v) = s_0 + \max\left(0, vT + \frac{v\Delta v}{2\sqrt{ab}}\right)$$

Discussion

a) Compared with GM models and Gipps' CF model, which parameter commonly used in many other CF models is not used in IDM?

b) If $\delta \rightarrow \infty$, which model (GM 1~4, Gipps' CF model, or Wiedemann) will IDM become similar to? Why?

Interpreting IDM

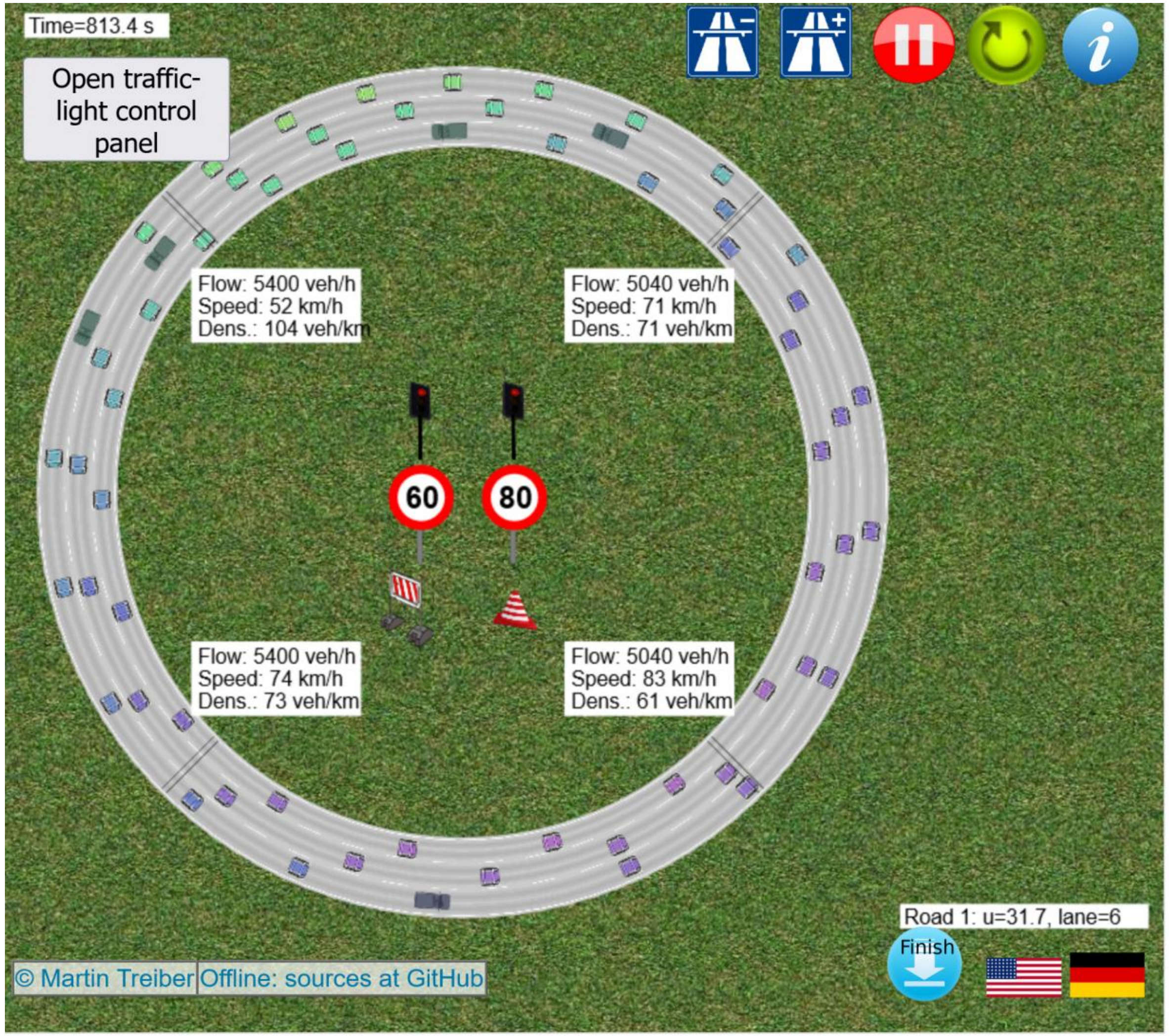
We can easily interpret the model parameters by considering the following three standard situations:

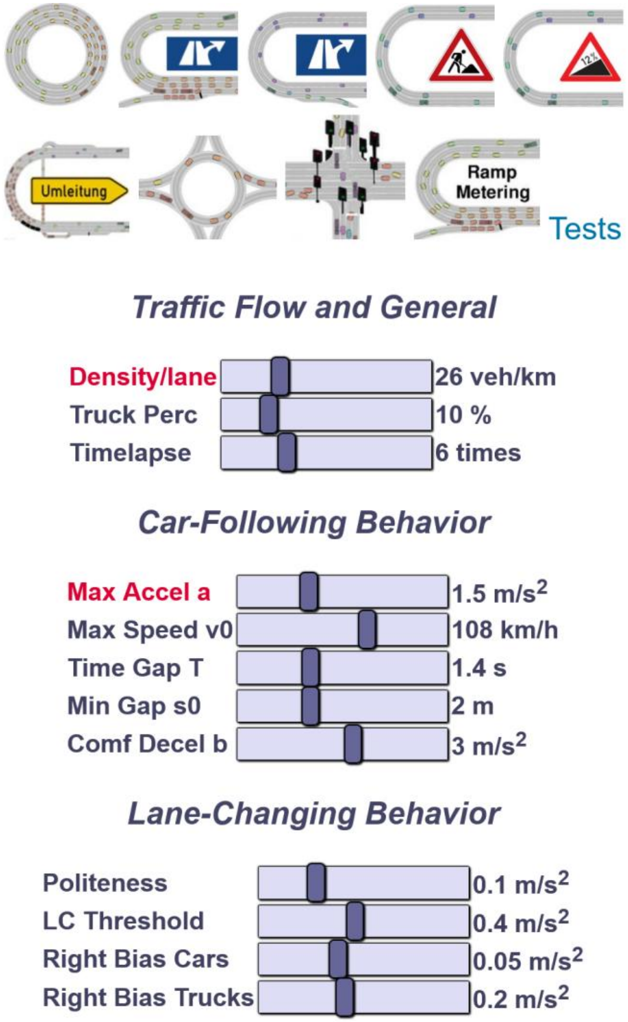
- When ***accelerating on a free road from a standstill***, the vehicle starts with the maximum acceleration a. The acceleration decreases with increasing speed and goes to zero as the speed approaches the desired speed v_0. The exponent δ controls this reduction: The greater its value, the later the reduction of the acceleration when approaching the desired speed.
- When ***following a leading vehicle***, the distance gap is approximatively given by the *safety distance* $s_0 + vT$.
- When ***approaching slower or stopped vehicles***, the deceleration usually does not exceed the comfortable deceleration b.
- The acceleration function is smooth during transitions between these situations.

IDM is implemented in SUMO

SUMO (Simulation of Urban MObility), an open source, highly portable, microscopic and continuous multi-modal traffic simulation package designed to handle large networks.

Online Traffic Simulating Using IDM





Further readings:

- Chapter 10, Traffic flow dynamics: Data, models and simulation by Martin Treiber and Arne Kesting.
- van Wageningen-Kessels, F., Van Lint, H., Vuik, K., & Hoogendoorn, S. (2015). Genealogy of traffic flow models. EURO Journal on Transportation and Logistics, 4(4), 445-473.
- Saifuzzaman, M., Zheng, Z., 2014. Incorporating human-factors in car-following models: A review of recent developments and research needs. *Transportation Research Part C* 48, 379 - 403.

Thank you

Questions?

CIVL6415, Semester 2, 2025

TRAFFIC ANALYSIS AND SIMULATION

MODULE 4

Macroscopic traffic models

Lecture Week 7

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Outline

- **Arrival / Departure or Input / Output curves**
- **Shockwaves**
- **Shockwave velocity equation**
- **Types of bottlenecks**
  - Temporary bottlenecks
  - Stationary bottlenecks
  - Moving bottlenecks

Modelling approaches

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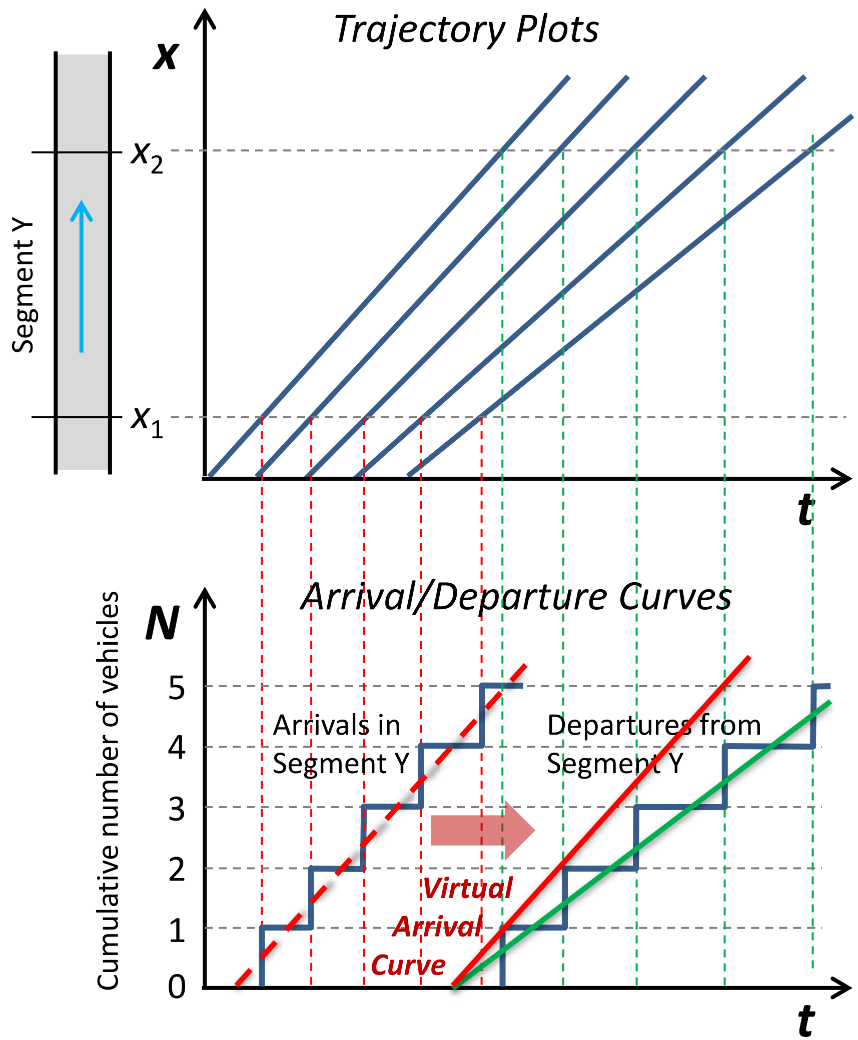
Speed variables / ***user variables***	Aggregate $u(x,t)$	Disaggregate $u_i(t)$
Aggregate $q(x,t)$, $k(x,t)$	MACRO scopic	
Disaggregate $x_i(t)$	MESO scopic	MICRO scopic

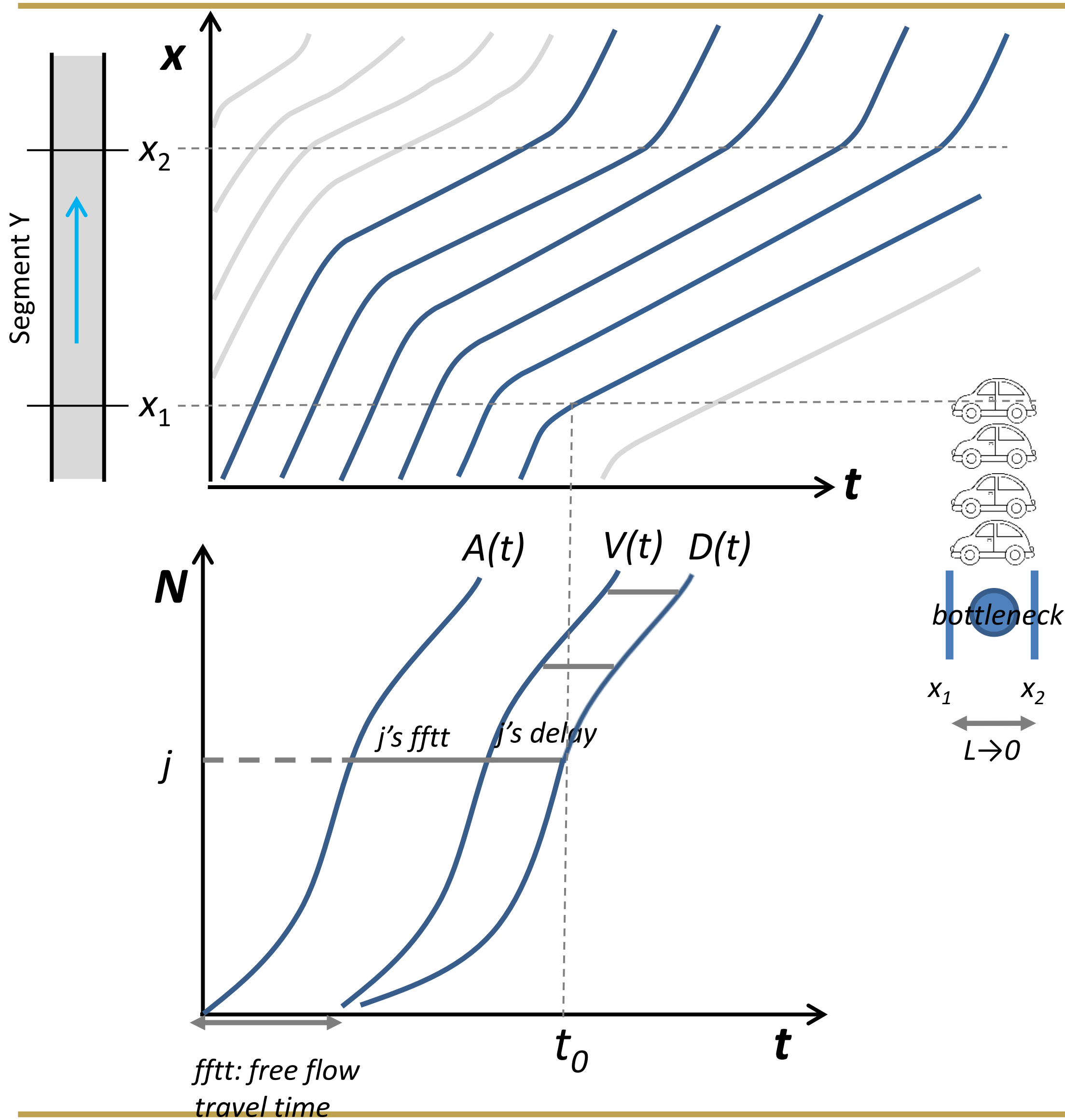
ARRIVAL-DEPARTURE CURVES
INPUT-OUTPUT CURVES

Arrival and Departure Curves

- A cumulative vehicle count curve (*N*-curve) shows the cumulative number of vehicles that pass a certain location *x* by time *t.*
- No entering/leaving traffic
- First-in-First-Out (FIFO)
- Introduce a virtual arrival curve by shifting the arrival curve to the right by the free-flow travel time to analyse queues and delays.



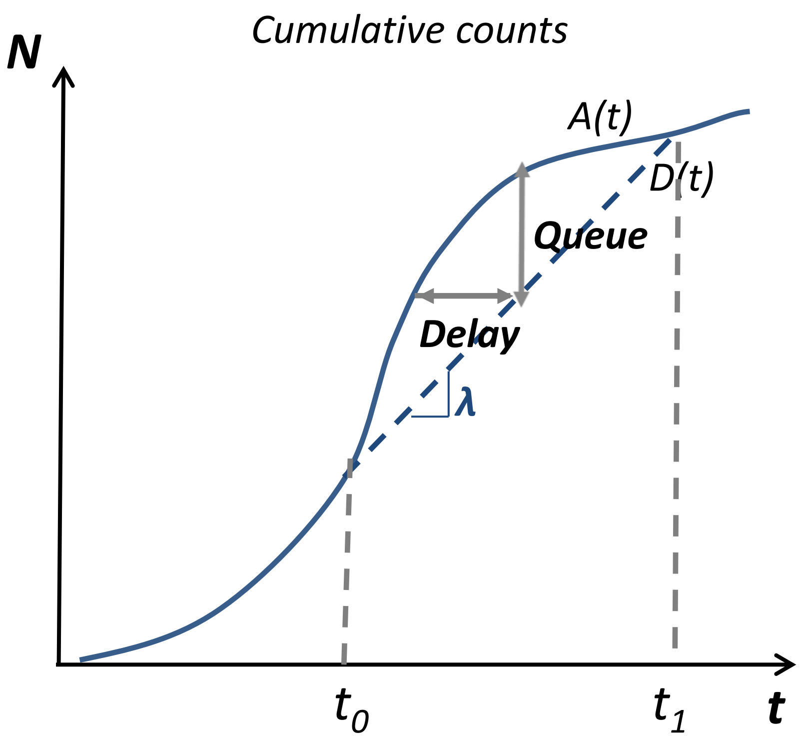
A, V and D curves



- When the *N*-curve is drawn on larger scale, the step function will look like a smooth function.
- If L → large, A(t)≠ V(t)
- If L → 0, A(t)≈V(t)
  - **A(t) ≠ D(t), point queue**
  - **V(t): number of vehicles that would arrive if there was no congestion**

Construct A-D curves (L~small)

Single Bottleneck



- At t<t0

$$A(t) = D(t)$$
$$\dot{D}(t) = \dot{A}(t) < \lambda$$

- At t0<t<t1

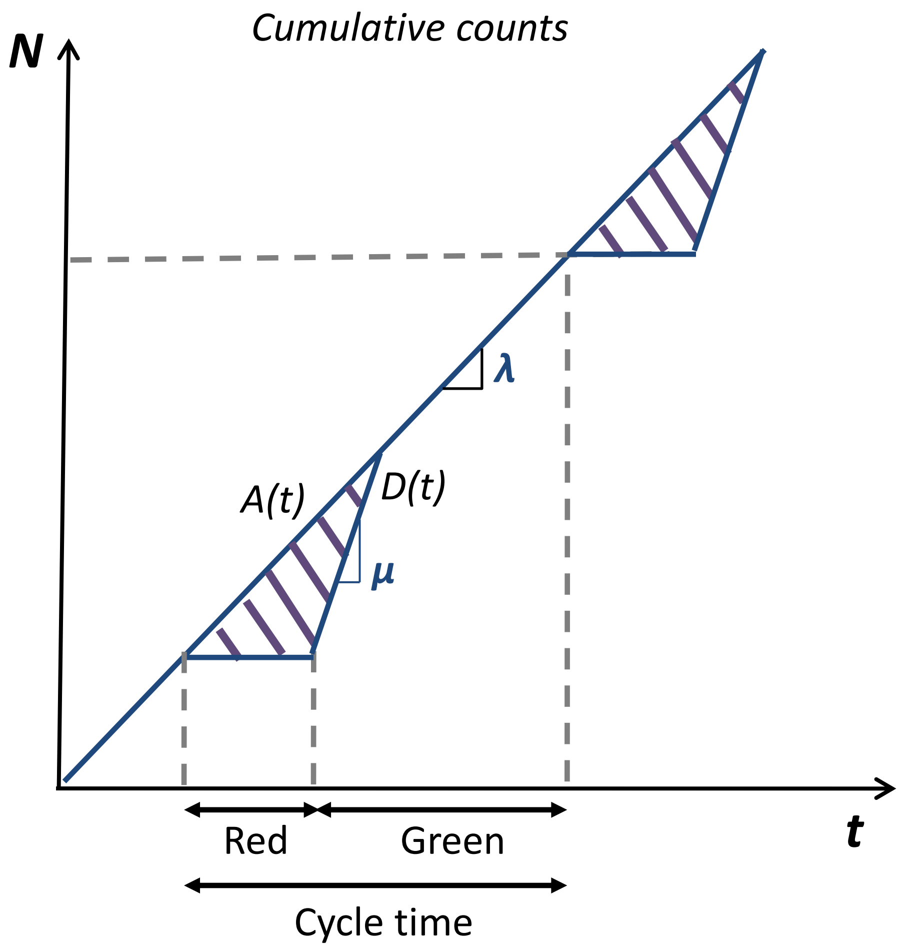
$$A(t) > D(t)$$
$$\dot{D}(t) = \lambda \textbf{ (service rate)}$$

Delay and Queue

- At t>t1

$$A(t) = D(t)$$
$$\dot{D}(t) = \dot{A}(t)$$

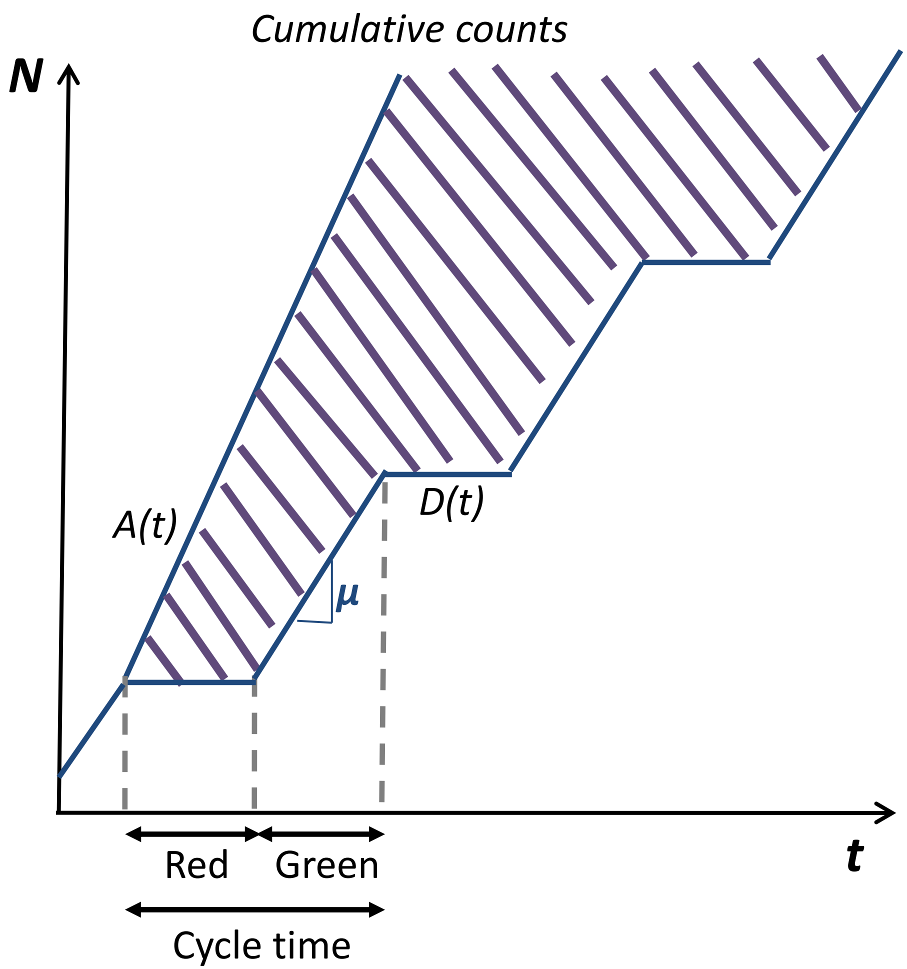
A-D (Traffic signal)



- Delay because of a traffic signal
- Traffic flows at rate λ
- Saturation flow is μ
- **λ<<μ**

- What if queue cannot be emptied?

A-D (Traffic signal)

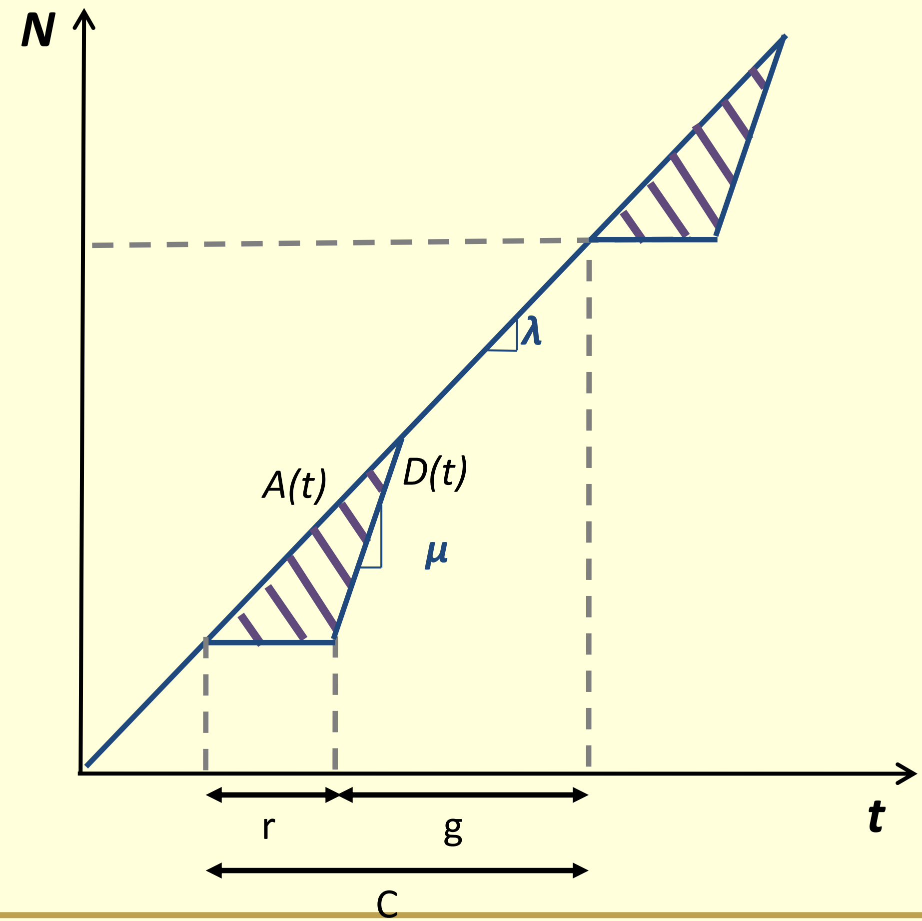


- Delay because of a traffic signal
- Traffic flows at rate λ
- Saturation flow is μ
- **λ>>μ**
- What if queue cannot be emptied?
- **SPILLBACK!!**

Question

Queue clearance

Traffic flows at rate λ, while saturation flow is μ.
Assume red phase length is r, green phase length is g, and cycle time is C, which one of the following conditions is necessary for the queue to be emptied at the end of the cycle?



A. $\lambda C \leq \mu r$

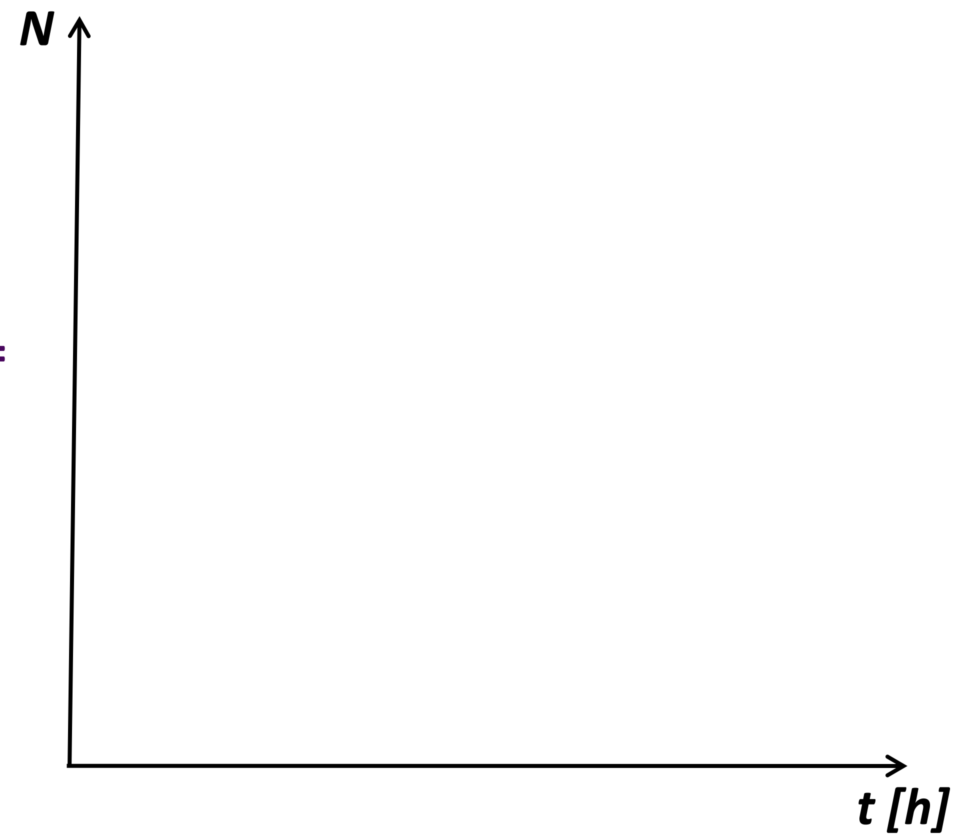
B. $\lambda C \leq \mu g$

C. $\lambda r \geq \mu C$

D. $\lambda g \geq \mu g$

Exercise (A-D)

- A freeway with 3 lanes and maximum capacity C = 6000 veh/h
- Demand = ¾*C
- Incident at $t_0 = 0$, Response time = 6 min, Clearance time = 42 min
- One lane remains open during response
- Two lanes remain open during clearance



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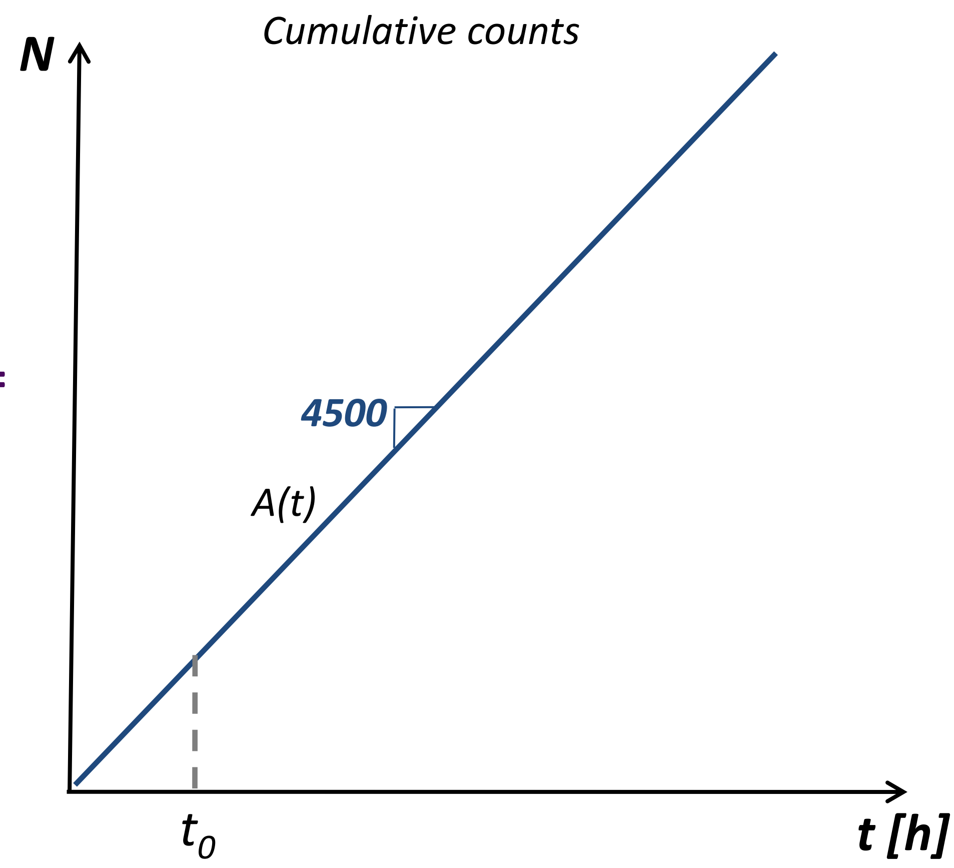
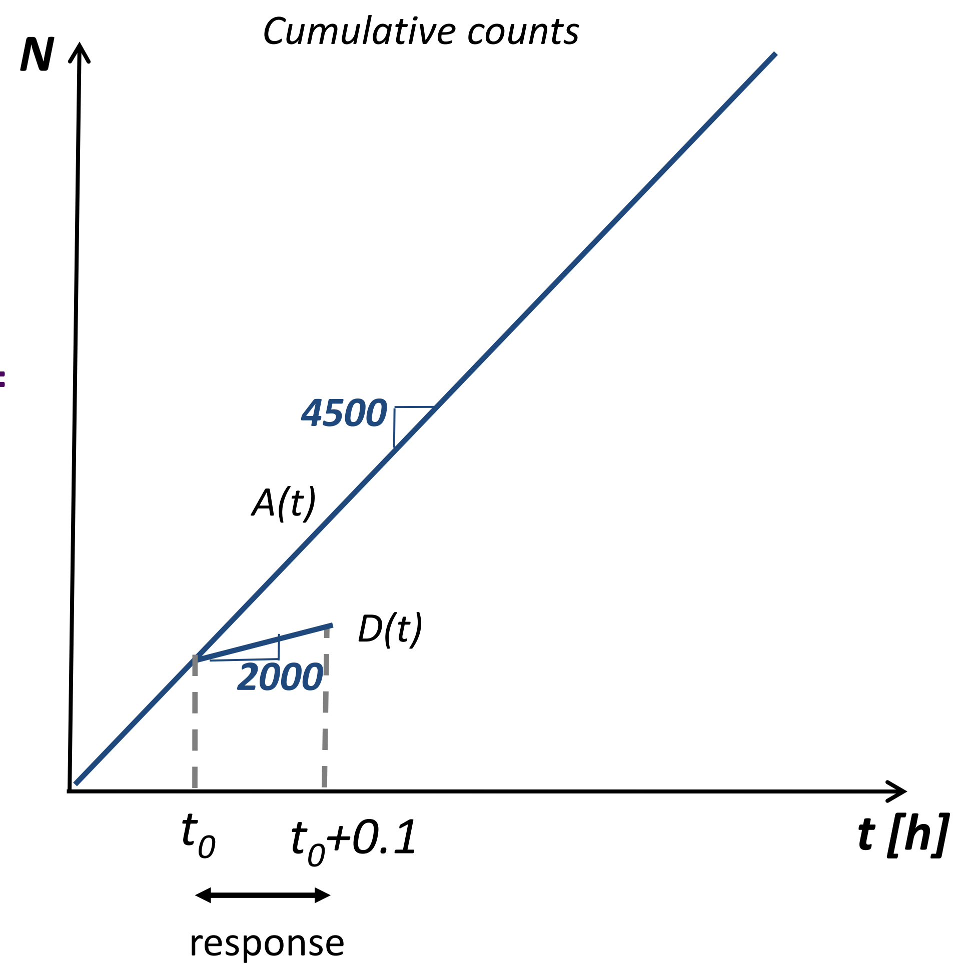


Exercise (A-D)

- **A freeway with 3 lanes and maximum capacity C = 6000 veh/h**
- **Demand = ¾*C**
- **Incident at t_0 = 0, Response time = 6 min, Clearance time = 42 min**
- **One lane remains open during response**
- **Two lanes remain open during clearance**
- **Between t_0 and t_0+0.1**

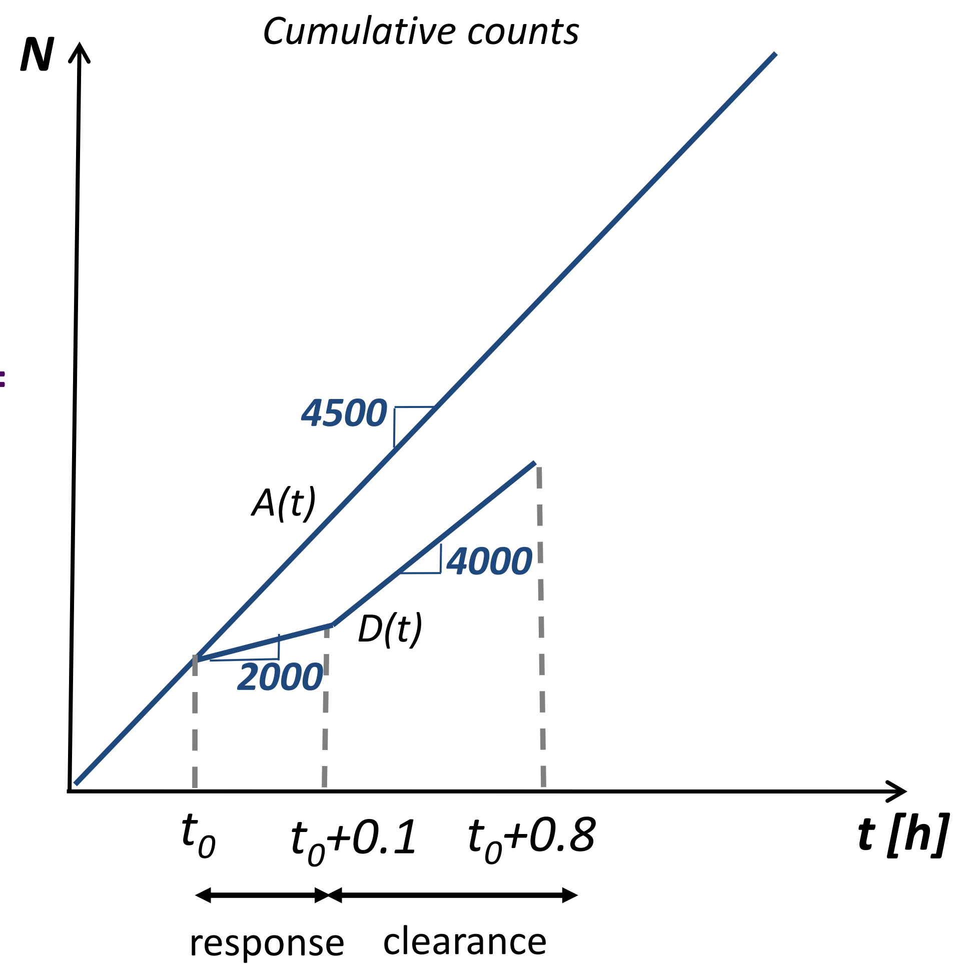
$$\dot{D}(t) = \frac{C}{3} = 2000 \text{ veh/h}$$



Exercise (A-D)

- A freeway with 3 lanes and maximum capacity C = 6000 veh/h
- Demand = ¾*C
- Incident at $t_0 = 0$, Response time = 6 min, Clearance time = 42 min
- One lane remains open during response
- Two lanes remain open during clearance
- Between $t_0+0.1$ and $t_0+0.8$

$$\dot{D}(t) = \frac{2C}{3} = 4000 \text{ veh/h}$$



Exercise (A-D)

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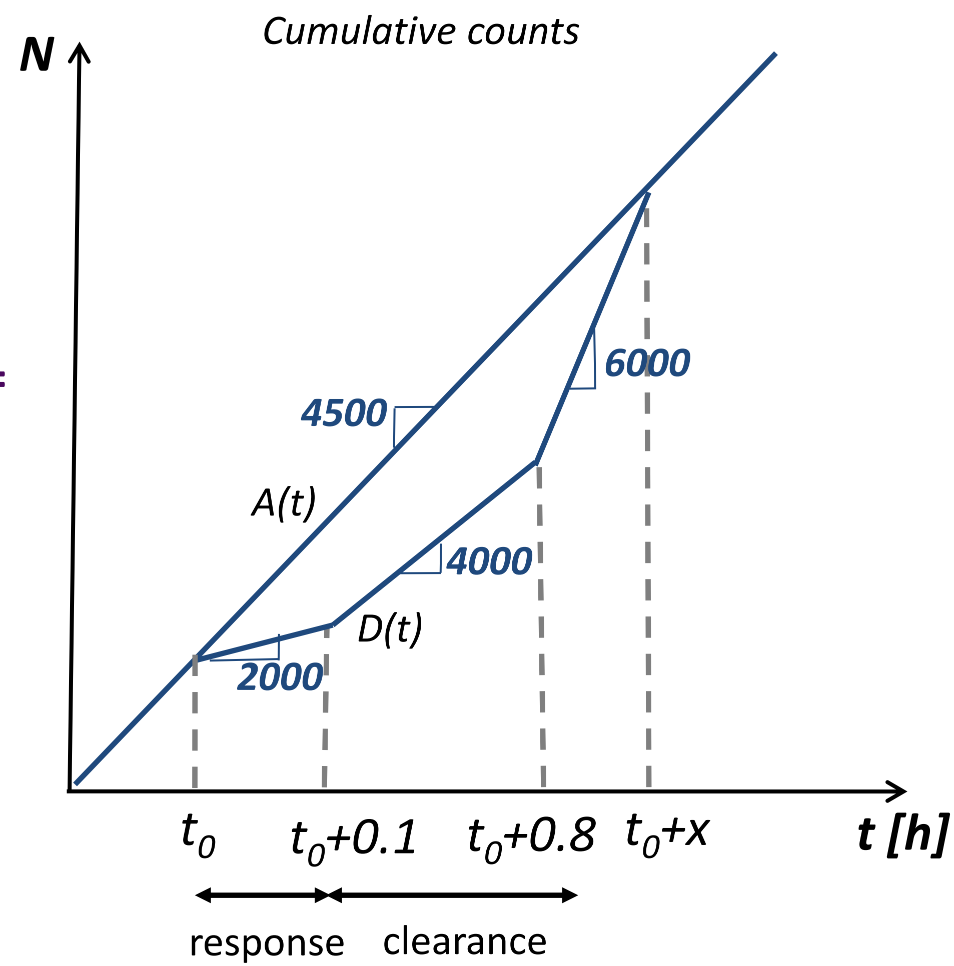


Exercise (A-D)

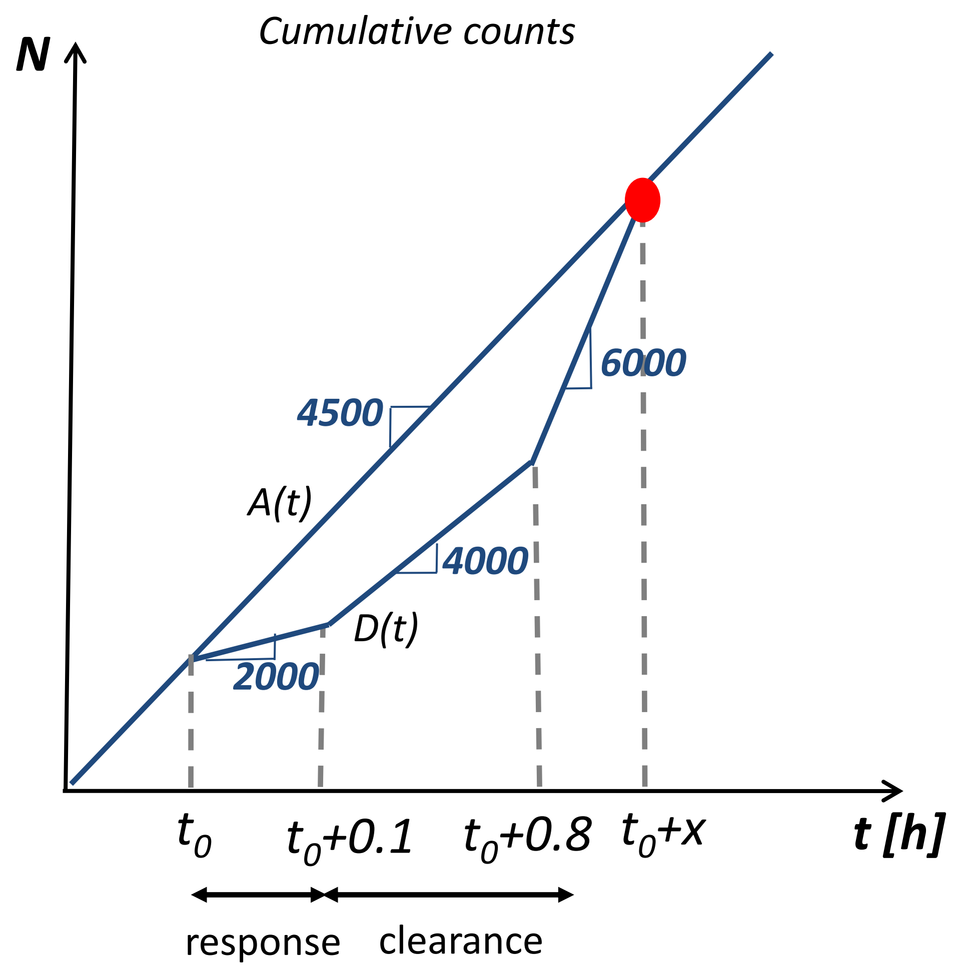
Duration of congestion

x=?

4500*x =

2000*0.1 + 4000*0.7 + 6000*(x-0.8)

x = 1.2 h



Exercise (A-D)

Total delay = w_{tot} = A1 + A2 + A3

n1 = (4500*0.1-2000*0.1) = 250 veh

n2 = (4500*0.8-2000*0.1-4000*0.7)

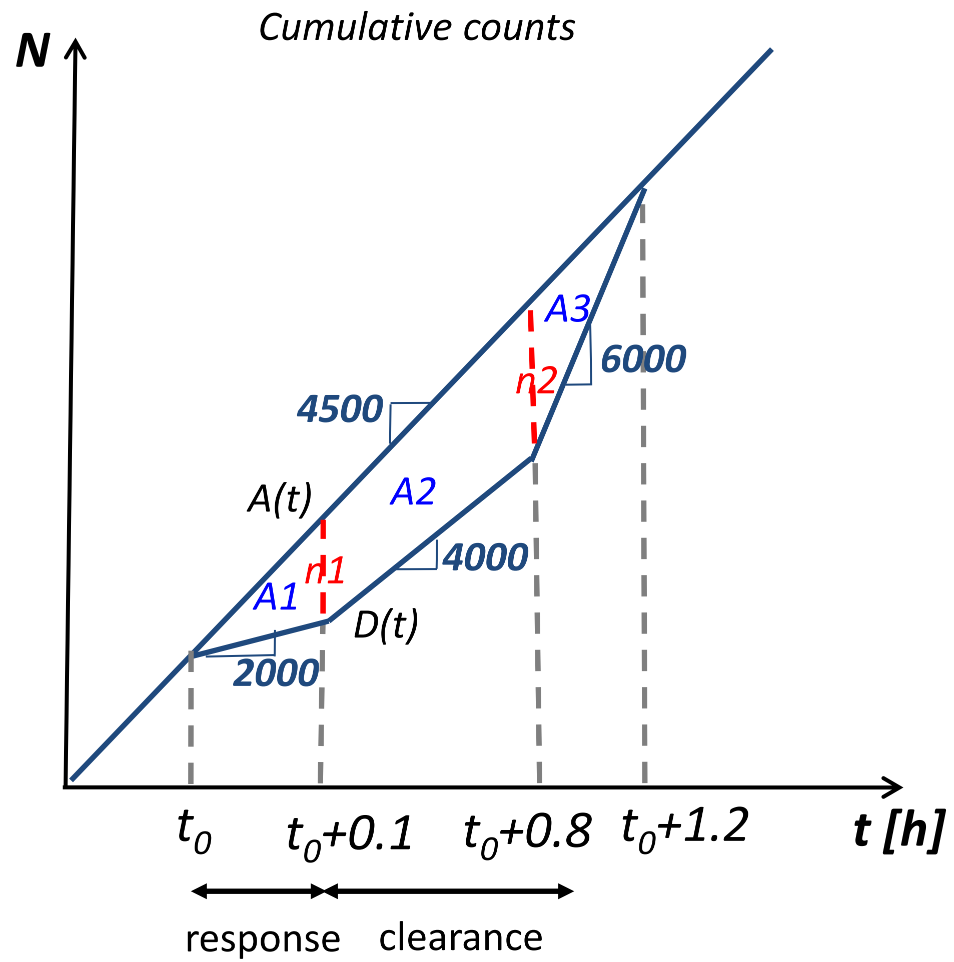
= 600 veh

A1 = n1*0.1/2 = 12.5 veh.h

A2 = (n1+n2)/2*0.7 = 297.5 veh.h

A3 = n2*0.4/2 = 120 veh.h

w_{tot} = 430 veh.h



Exercise (A-D)

Maximum delay = w_{max}?

Maximum queue = N_{max}?

N_{max} = n2 = 600 veh

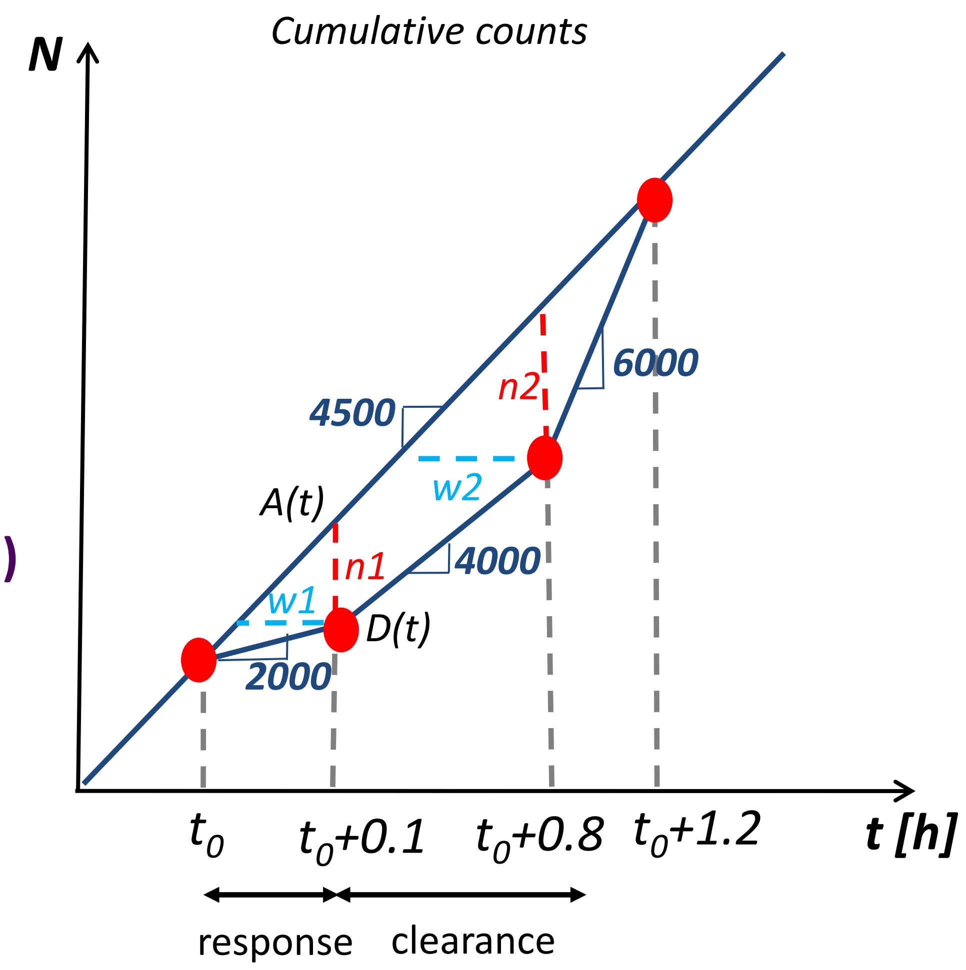
2000*0.1 = 4500(0.1-w1)

w1 = 0.055 h

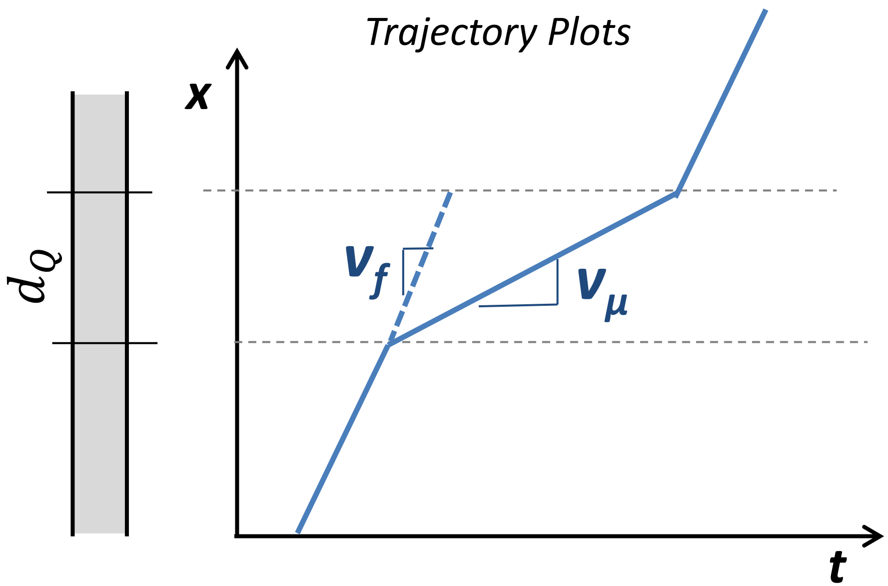
2000*0.1 + 4000*0.7 = 4500(0.8-w2)

w2 = 0.133 h

w_{max}= w2 = 0.133 h



End of queue



- Distance travelled while in the queue

$$w = \frac{d_Q}{v_\mu} - \frac{d_Q}{v_f}$$

$$d_Q = \frac{w}{\frac{1}{v_\mu} - \frac{1}{v_f}}$$

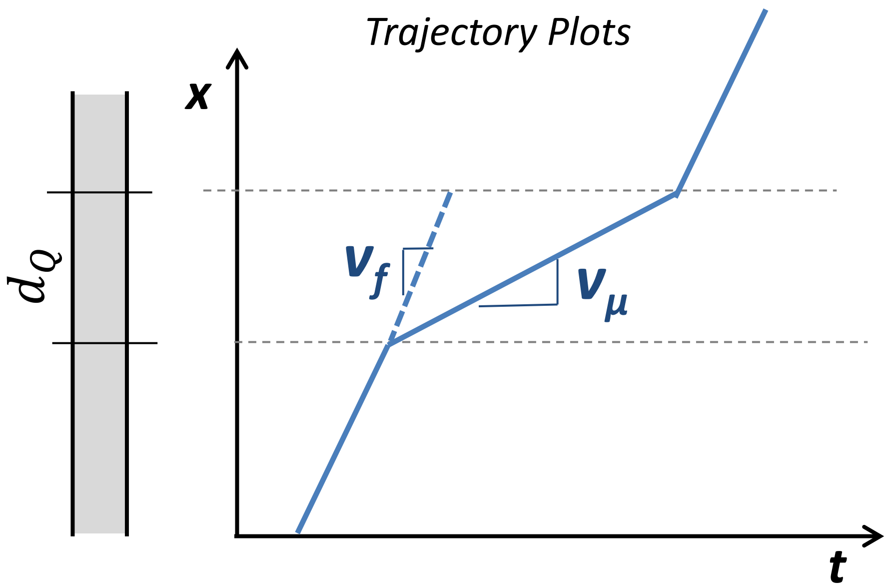
d_Q= distance of queue [km]

v_f= free flow speed [km/h]

v_μ= actual speed [km/h]

w=delay [h]

End of queue



- Time spent in the queue

$$t_Q = \frac{d_Q}{v_\mu} = \frac{d_Q}{v_f} + w$$

$$t_Q = \frac{w}{\frac{1}{v_\mu} - \frac{1}{v_f}} \frac{1}{v_f} + w$$

$$t_Q = \frac{w.v_\mu}{v_f - v_\mu} + \text{w}$$

$$t_Q = \frac{w}{1 - \frac{v_\mu}{v_f}}$$

Question 2

Input-Output

Calculate maximum length of the queue and maximum time spent in the queue based on the below A-D curve?

n1=250 veh, n2=600 veh

w1 = 0.055 h, w2=0.133 h

v_{μ}=28.57 km/h, v_f=100 km/h

A. 5.00km, 0.25 h

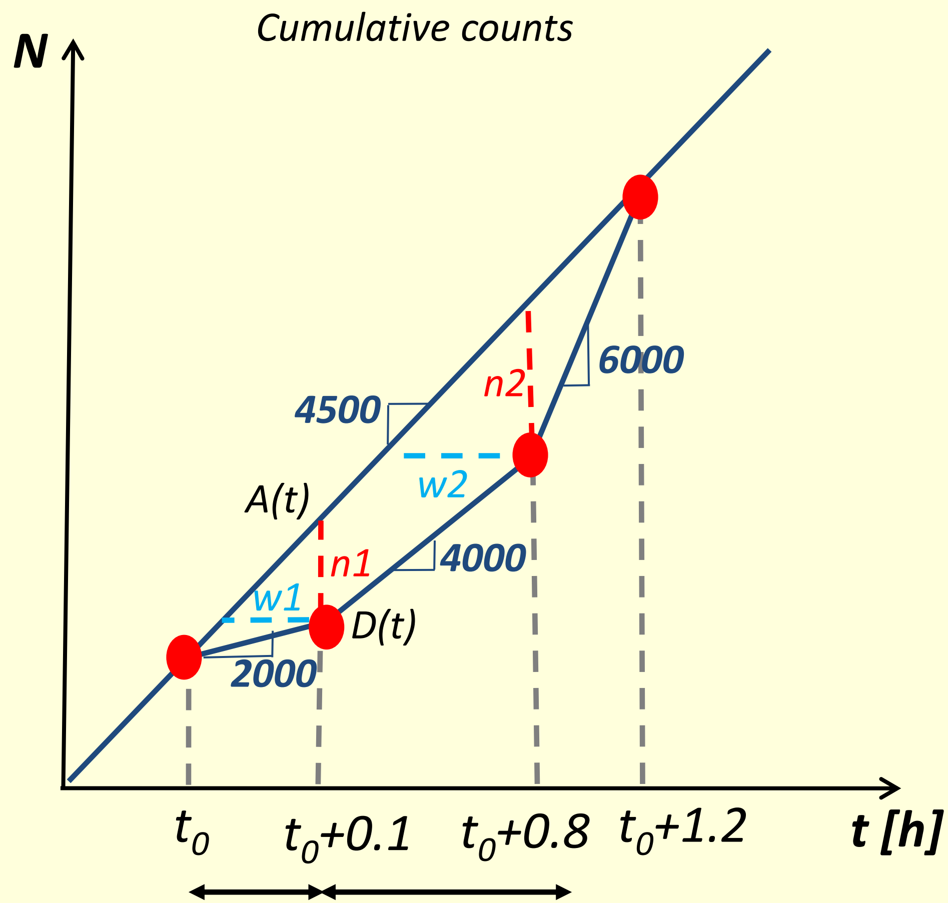
B. 4.25km, 0.10 h

C. 3.80km, 0.35 h

D. 5.32km, 0.19 h

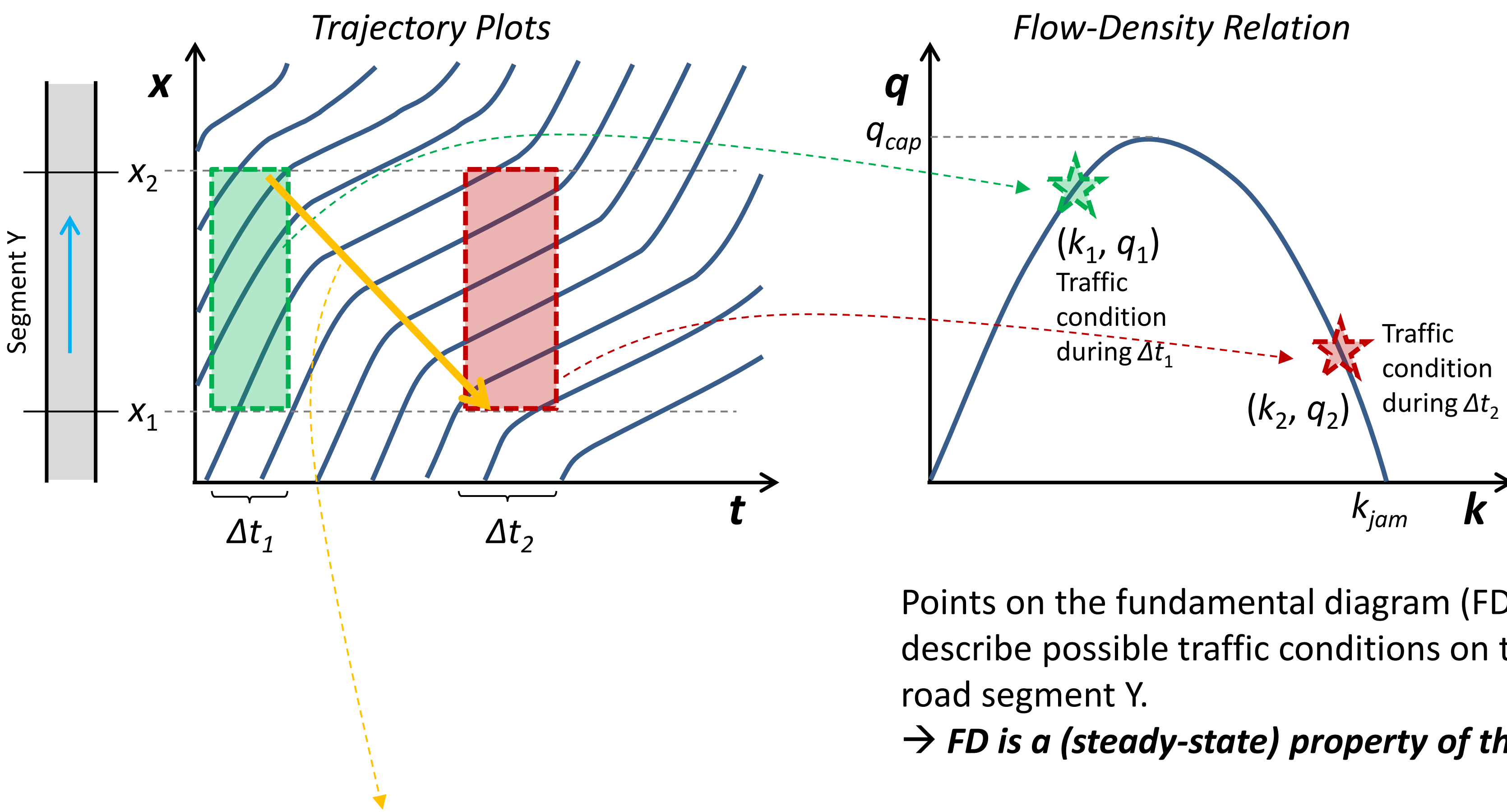
E. 4.70km, 0.35 h

$$d_Q = \frac{w}{\frac{1}{v_{\mu}} - \frac{1}{v_f}} \qquad t_Q = \frac{w}{1 - \frac{v_{\mu}}{v_f}}$$



SHOCKWAVES

Space-Time Diagram (x-t plot) vs. Fundamental Diagram (q-k plot)



Points on the fundamental diagram (FD) describe possible traffic conditions on the road segment Y.

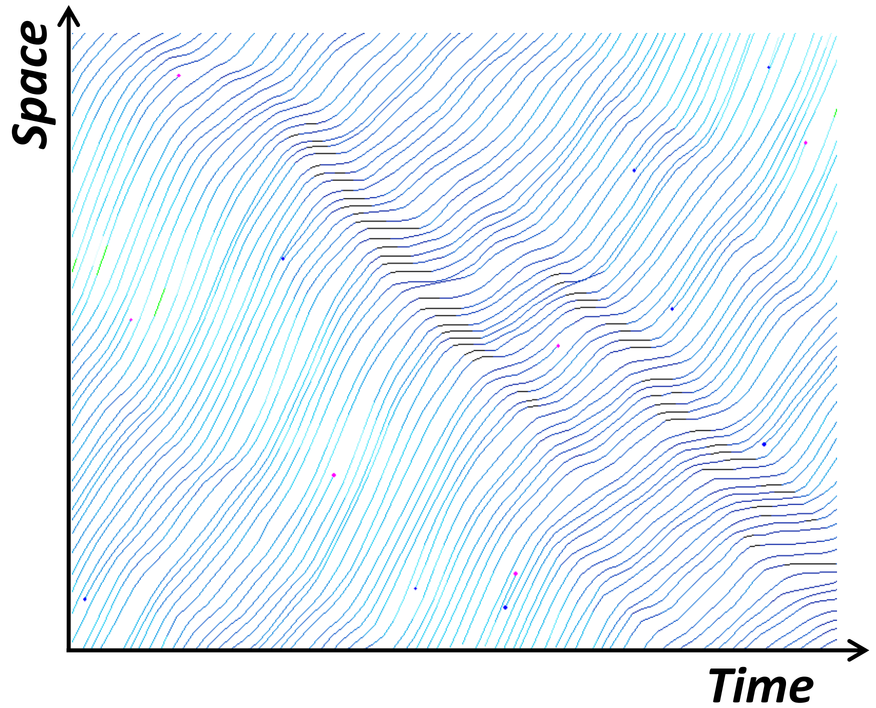
→ ***FD is a (steady-state) property of the road***.

Transition between state 1 and 2?

→ ***Shockwave***

Shockwave

- "*Shockwave*" – a motion or propagation of change in traffic conditions (flow, density, and speed)
  - Whenever a stream of traffic under certain flow conditions (q_1, k_1 and v_1) meets another stream under different conditions (q_2, k_2 and v_2), a shock wave is formed.
- A moving boundary between two different traffic states
  - a boundary in the space-time domain that shows a discontinuity in traffic stream conditions.
- The shock front may move forward or backward or stay at the same place with respect to the road.

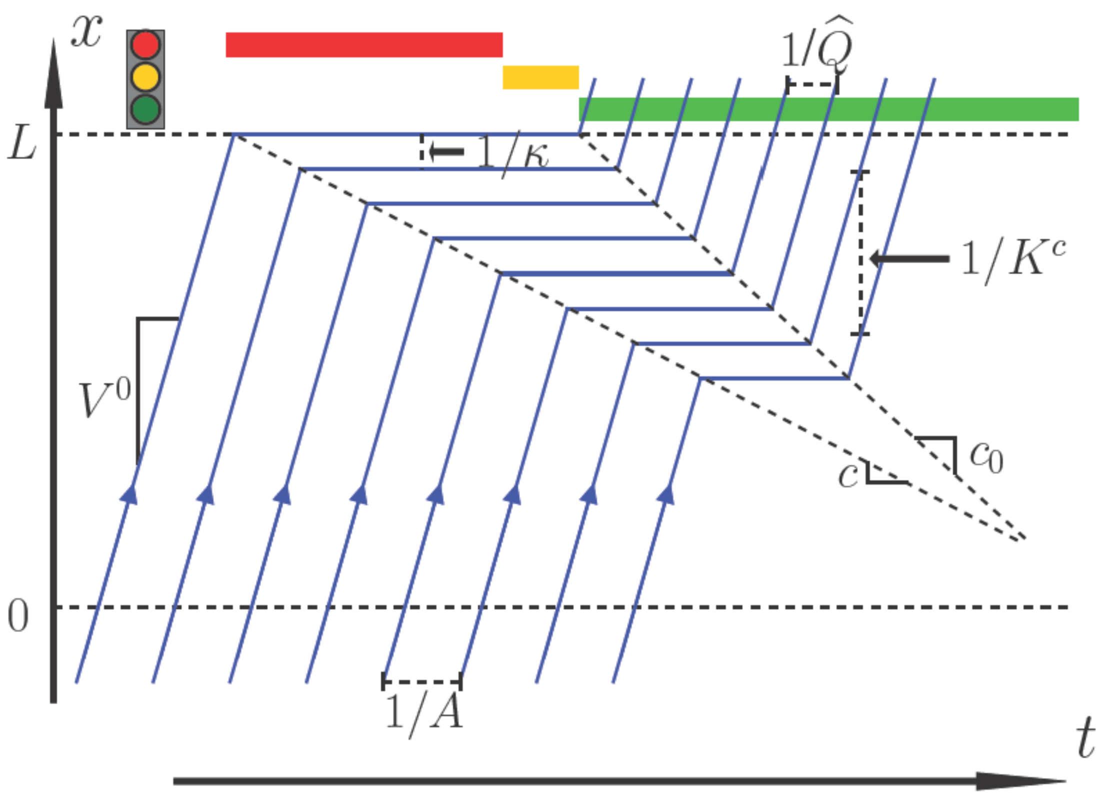


Shockwaves in Traffic Streams

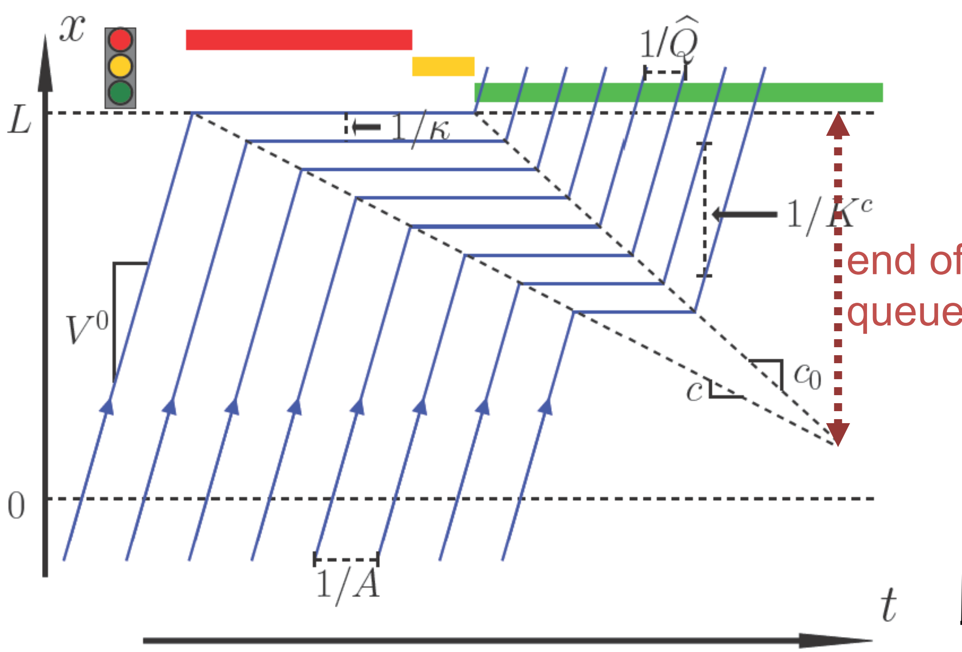
- **Phantom traffic jams or stop-and-go waves**
  – https://www.youtube.com/watch?v=goVjVVaLe10
- **How we can use this information**
  – https://www.youtube.com/watch?v=rdJFeUPc2gg
- **Simulation tools**
  – https://setosa.io/blog/2014/09/02/gridlock/
  – http://www.traffic-simulation.de/index.html

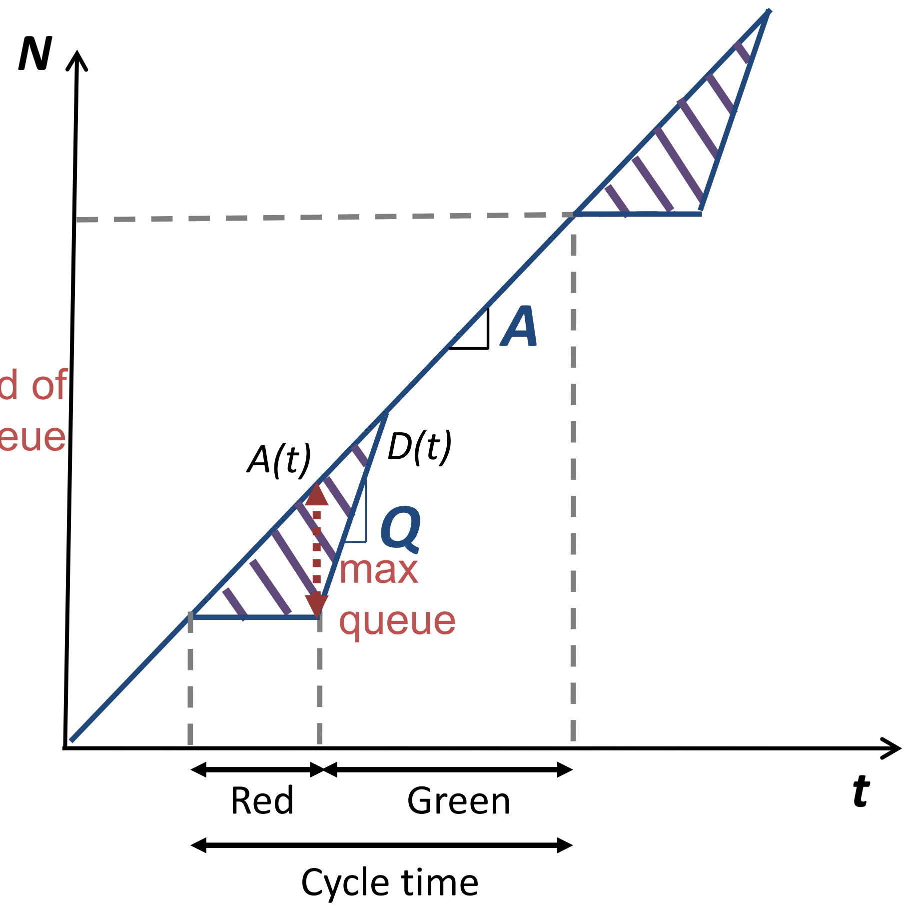
Traffic signal

- "*Shockwave*" – caused by red and green phases
  - Immediate change of speeds along the shockwaves
  - Infinite acceleration – limitation of modelling approach



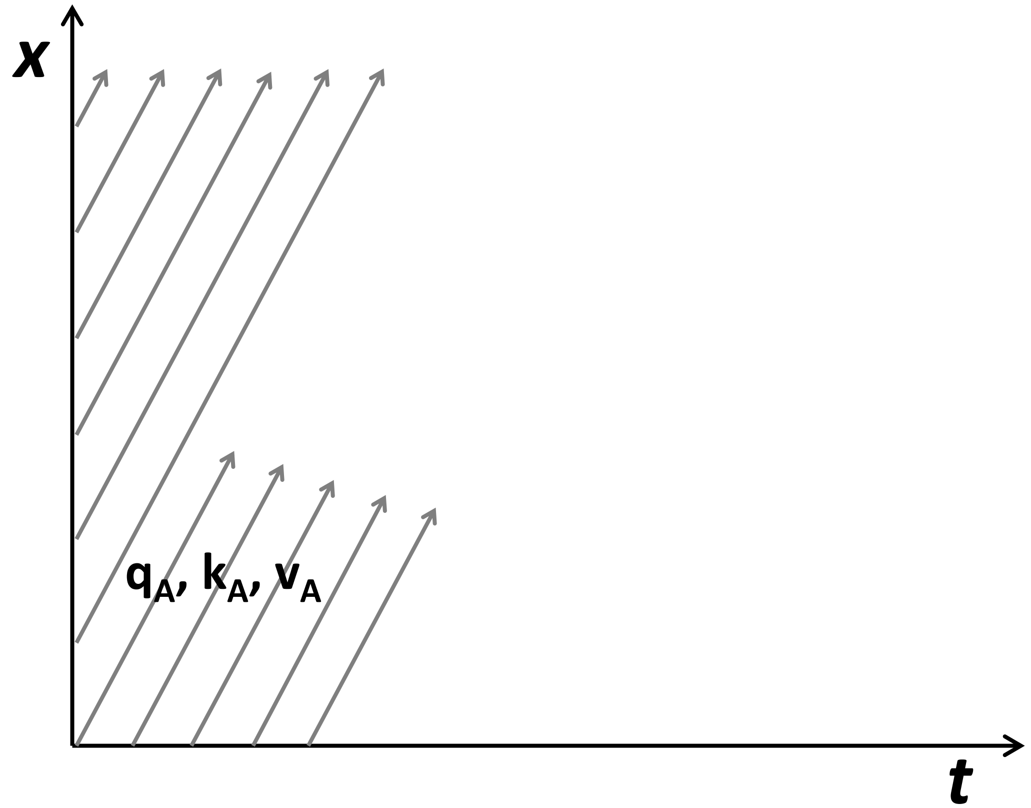
Traffic signal





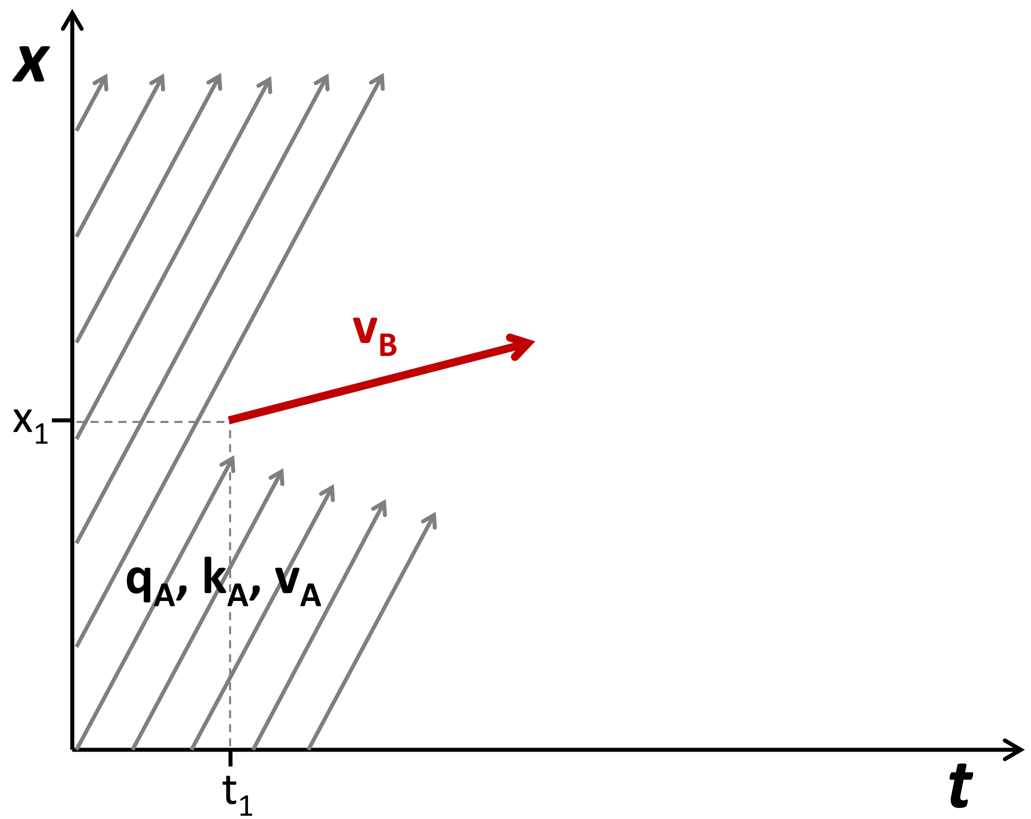
Shockwave Example

- **Formation and Propagation Scenarios [1/6]**
  - Traffic stream moving at flow q_A, density k_A, and speed v_A.



Shockwave Example

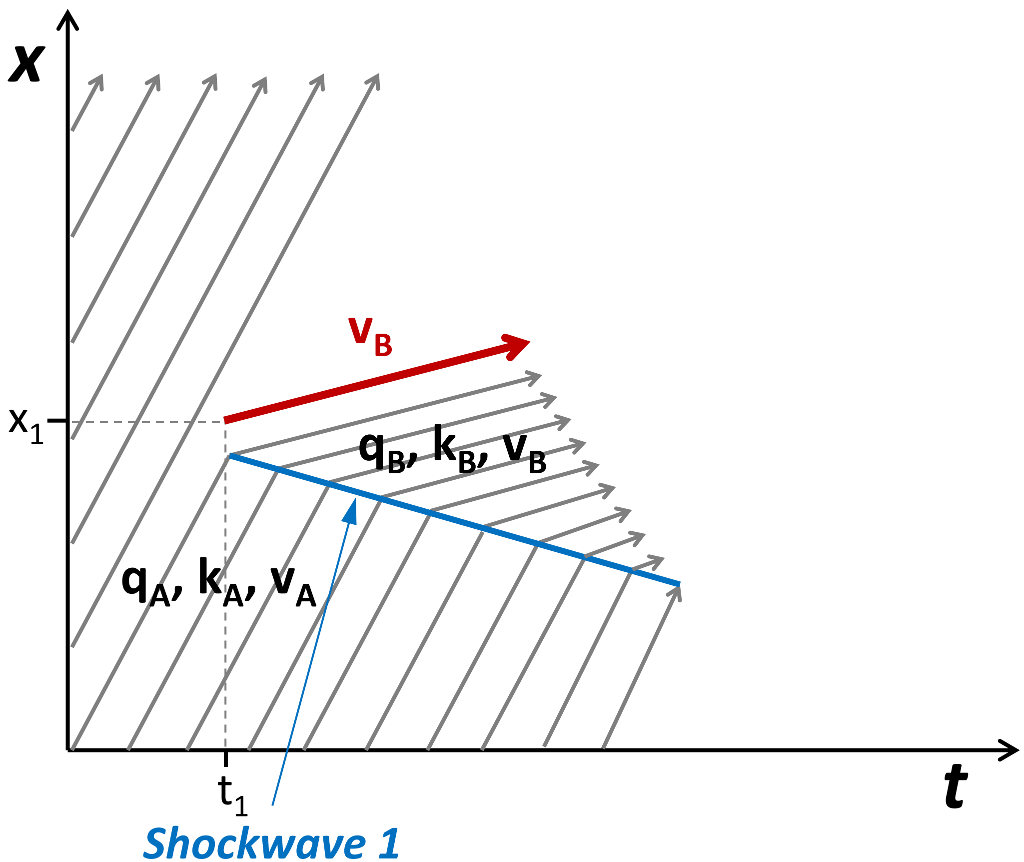
- **Formation and Propagation Scenarios [2/6]**
  - Traffic stream moving at flow **q_A**, density **k_A**, and speed **v_A**.
  - At time t_1 and point x_1, a slow moving vehicle (red trajectory) with speed **v_B** enters a traffic stream.



Shockwave Example

Formation and Propagation Scenarios [3/6]

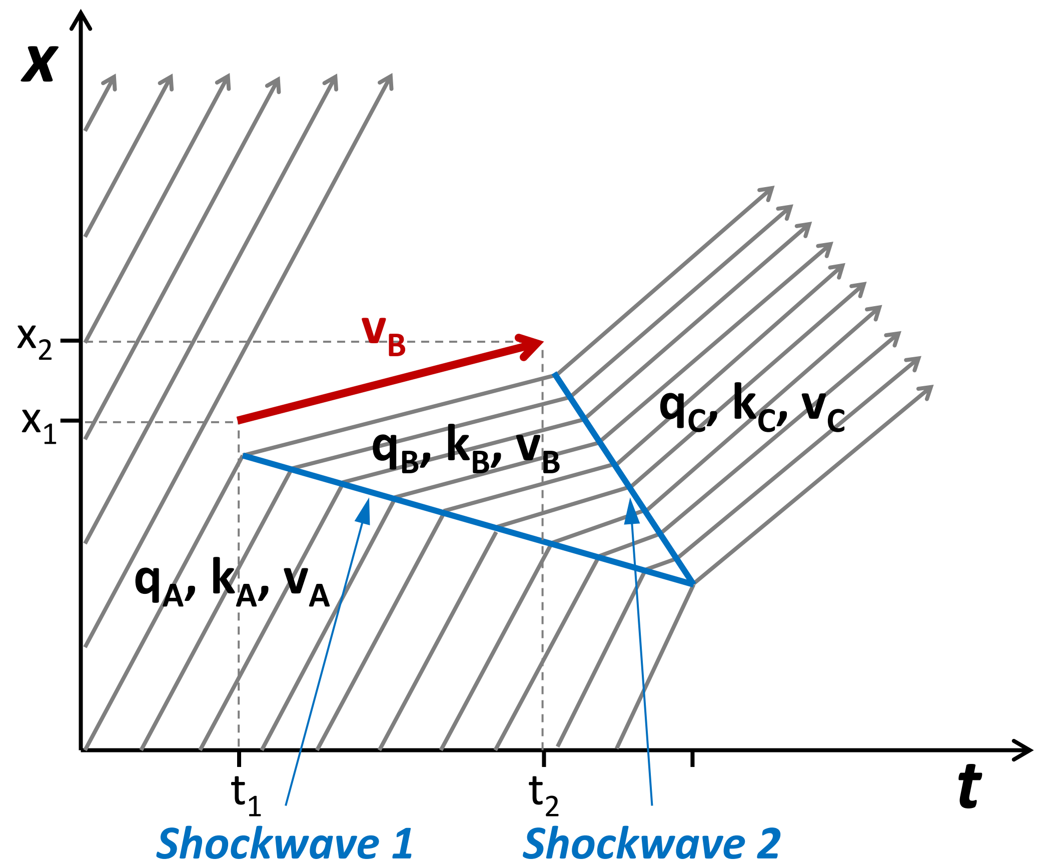
- Traffic stream moving at flow $\mathbf{q_A}$, density $\mathbf{k_A}$, and speed $\mathbf{v_A}$.
- At time t_1 and point x_1, a slow moving vehicle (red trajectory) with speed $\mathbf{v_B}$ enters a traffic stream.
- The slow moving vehicle slows the traffic and creates another stream condition $\mathbf{q_B}$, $\mathbf{k_B}$, and $\mathbf{v_B}$.
- When the stream under A meets the stream under B, ***shockwave 1*** is formed and propagates upstream.



Shockwave Example

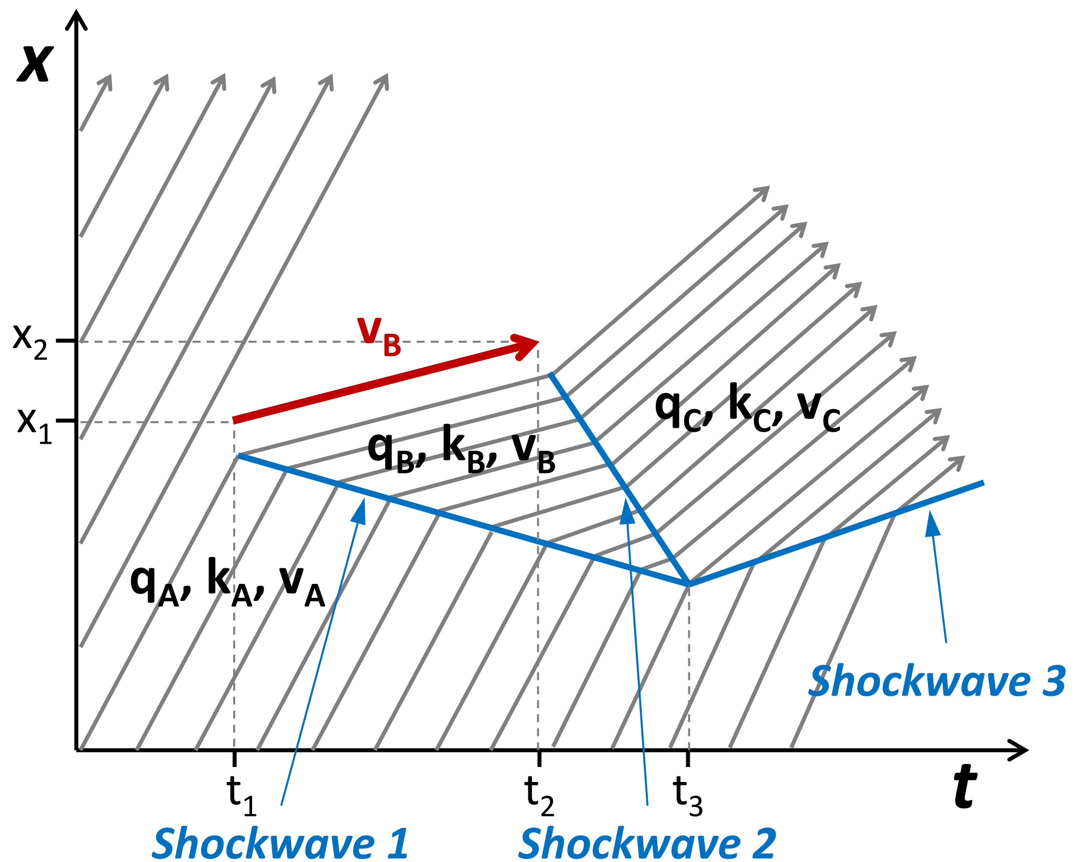
▪ Formation and Propagation Scenarios [4/6]

- At time t_2 and point x_2, a slow moving vehicle (red trajectory) leaves the traffic stream and the congested conditions is released at some other stream condition **q_C, k_C,** and **v_C**.
- When the stream under B meets the stream under C, ***shockwave 2*** is formed and propagates upstream.



Shockwave Example

- **Formation and Propagation Scenarios [5/6]**
  - At time t_3, these two shock waves (shockwaves 1 and 2) meet and the traffic condition B ends.
  - At this point, the traffic condition A meets condition C, causing ***shockwave 3***.



Shockwave Example

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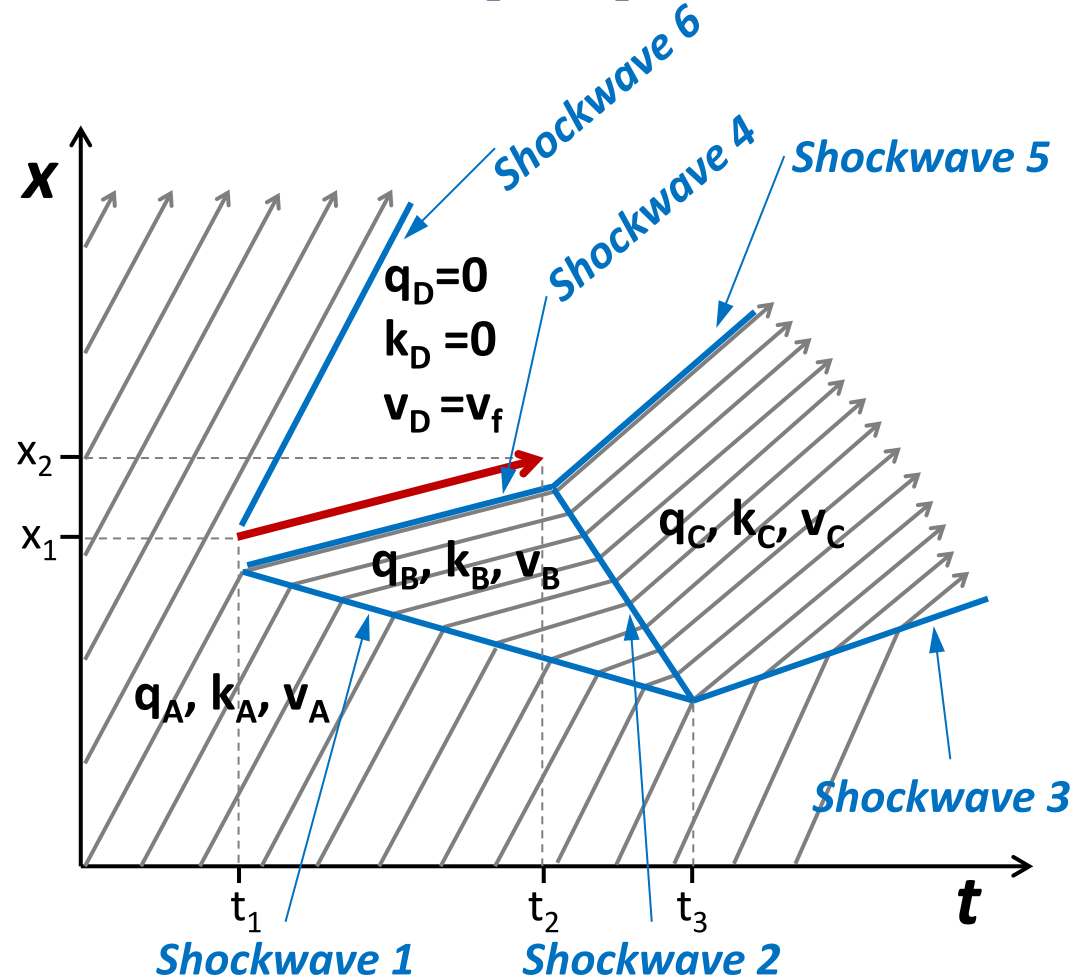
▪ Formation and Propagation Scenarios [6/6]

- Area D indicates a stream with the free-flow conditions: q_D=0, k_D=0, and $v_D = v_f$
- ***Shockwave 4*** is formed at the boundary between conditions B and D and moves with the slow moving vehicle.
- ***Shockwave 5*** is formed at the boundary between conditions C and D and moves with the first vehicle that is released.
- ***Shockwave 6*** is formed at the boundary between conditions A and D and moves with the last vehicle that avoided slow moving vehicle.



Shockwave Analysis

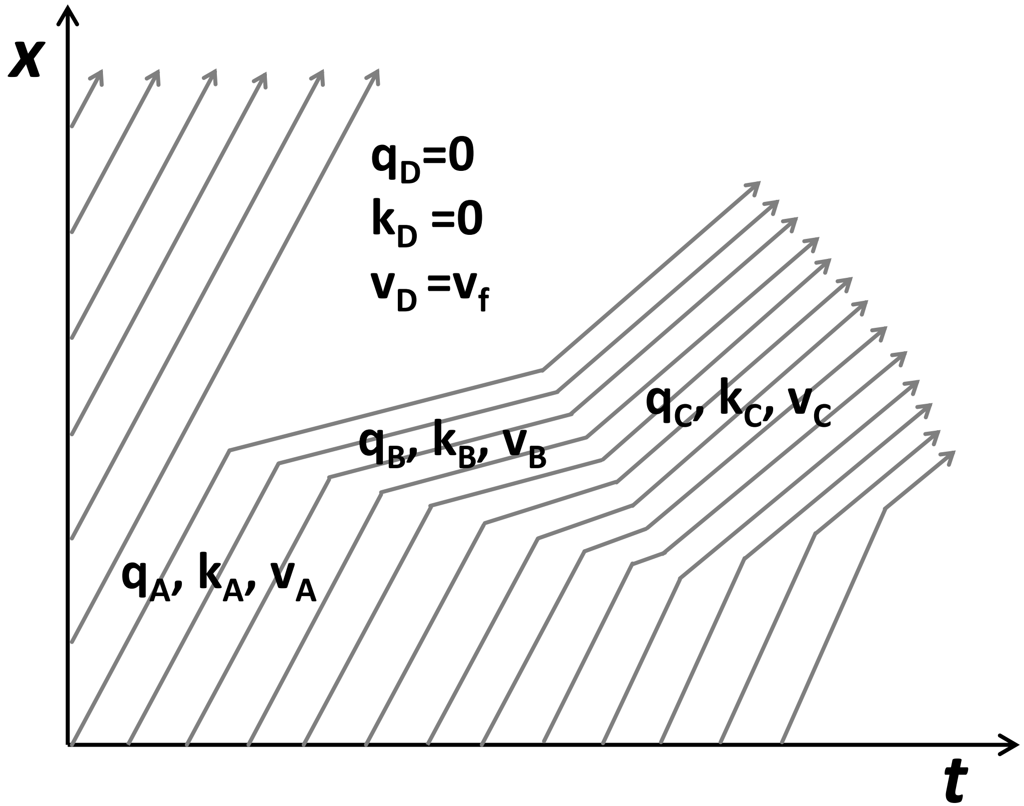
▪ Key Parameters of a Shock Wave

– Identify traffic states

– the point at which it starts,

– the point at which it ends, and

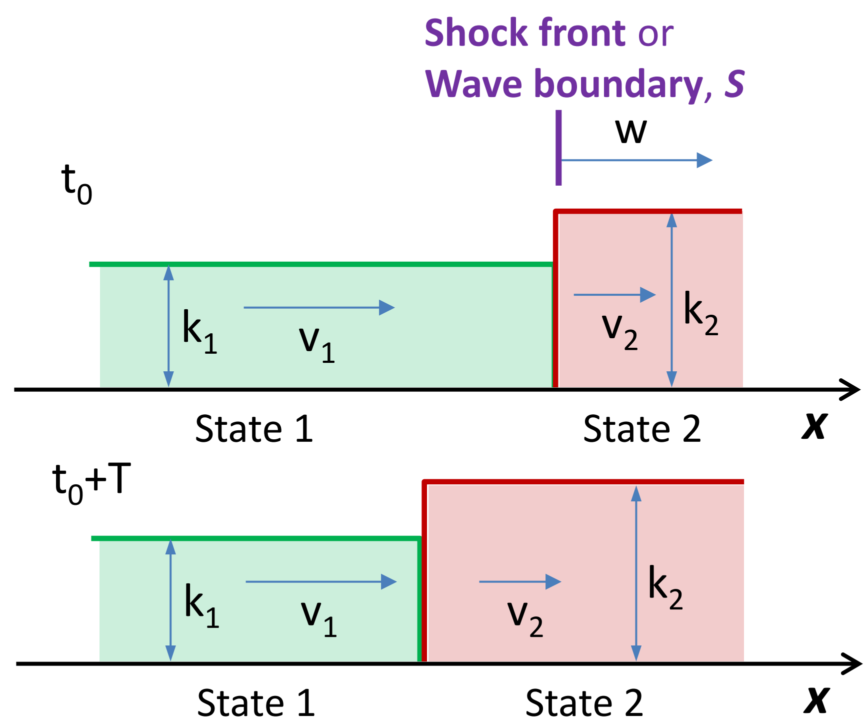
– **the velocity of the shock wave** (speed + direction).



Velocity of Shockwaves

- **Shockwave Velocity (w)**
  - Expansion of the state 2 during T in [veh]:
    - $(q_1T - q_2T - w \times T \times k_1)$
    - q_1 : inflow, q_2 : outflow,
    - $- w \times T$: space taken over by state 2
  - This is the same as;
    - $- w \times T \times k_2$
    - $(q_1 - q_2 - w \times k_1) \times T = - w \times k_2 \times T$

$$w = \frac{q_1 - q_2}{k_1 - k_2} = \frac{q_2 - q_1}{k_2 - k_1}$$ [km/hr]



Velocity of Shockwaves

- **Shockwave Velocity (w)**

$$\boxed{w = \frac{q_2 - q_1}{k_2 - k_1}}$$

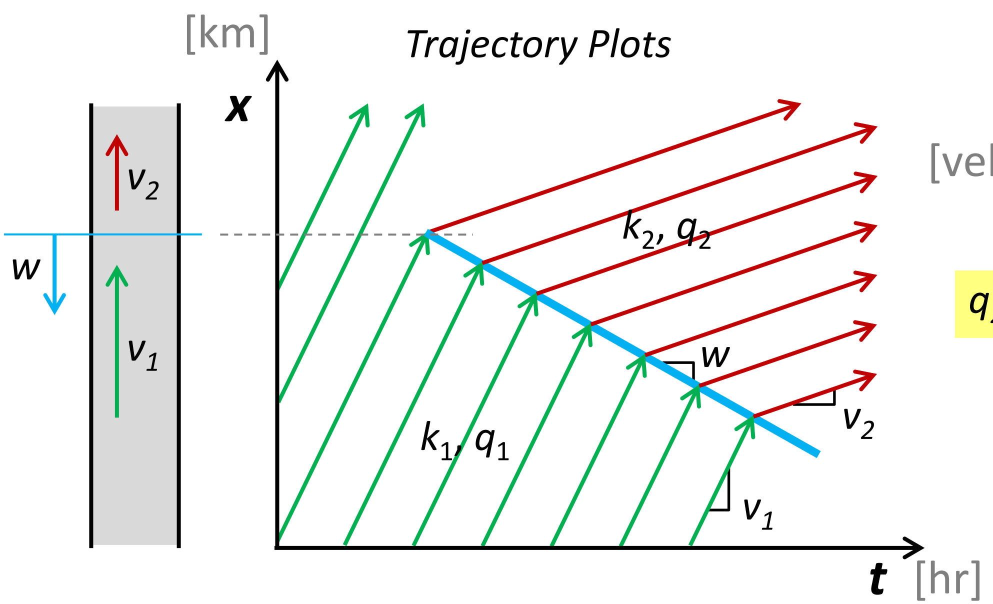
  - **is the ratio of the *change in flow* over the shock front *S* to the *change in density* over the shock front *S*.**
  - **w > 0 (+) : Shock wave is travelling in same direction as traffic stream.**
  - **w < 0 (-) : Shock wave is travelling upstream or against the traffic stream.**

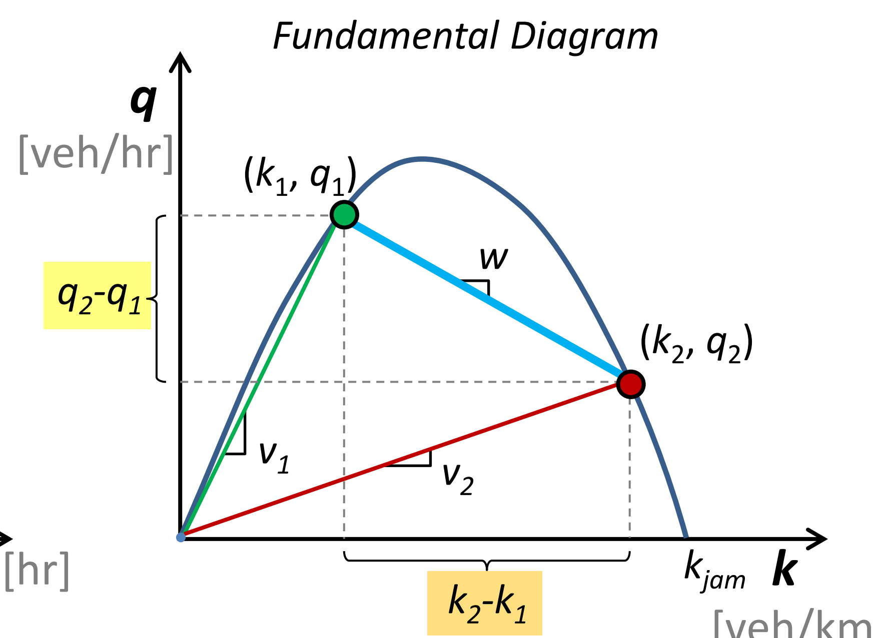
Velocity of Shockwaves

- **Shockwave Velocity (w)**

$$w = \frac{q_2 - q_1}{k_2 - k_1} \quad \text{[km/hr]}$$

- **Geometric Interpretation on Fundamental Diagram**





Question

Shockwave Velocity

The traffic flow on a highway is q=2100 veh/h with speed of v=70 km/h. As a result of an accident, the road is blocked and vehicles arriving after the accident must stop. The density in the queue is k=240 veh/km (jam density). **What is the speed of shockwave, w (km/h), generated by this accident?**

A. 10 km/h
B. 0 km/h
C. - 5 km/h
D. - 8 km/h
E. - 10 km/h

$$w = \frac{q_2 - q_1}{k_2 - k_1}$$

Queue growth rate

Traffic conditions within the impacted area are k_2 = 240 veh/km and v_2 = 0 veh/km. The capacity of the system is 2800 veh/h, free flow speed is 70 km/h, and the fundamental diagram is triangular. The accident is cleared after half an hour. **What is the speed of second shockwave due to incident clearance?**

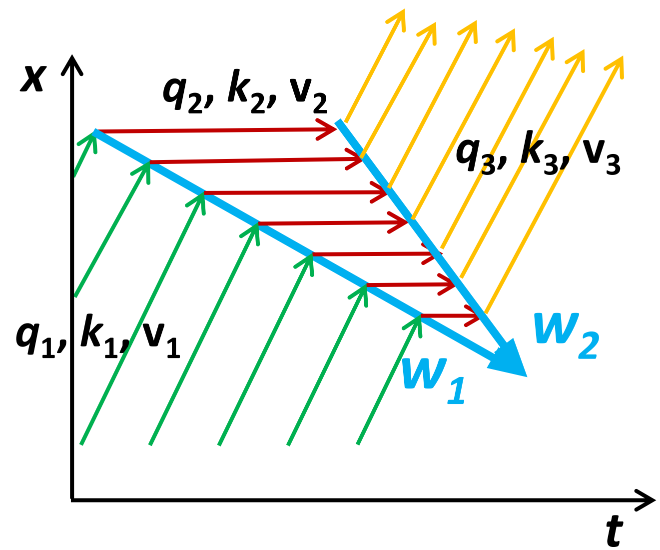
[Solution]

q_3 = 2800 veh/h, v_3 = 70 km/h

$\rightarrow$ k_3 = 2800/70 = 40 veh/km

The shockwave speed

$$w_2 = \frac{q_3 - q_2}{k_3 - k_2} = \frac{2800 - 0}{40 - 240} = \frac{2800}{-200} = -14 \text{ km/h}$$



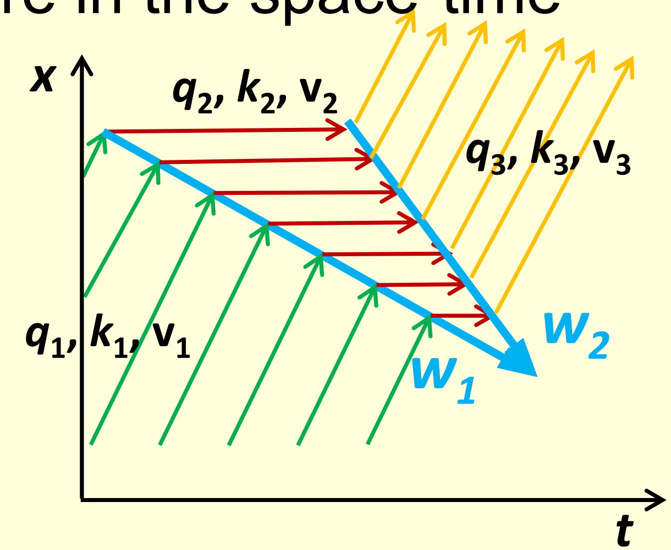
→ The shock front moves upstream at 14km/h.

→ The queue grows against traffic.

Question

Queue Growth Rate

In total, how many shockwaves are there in the space-time diagram?



A. 2

B. 3

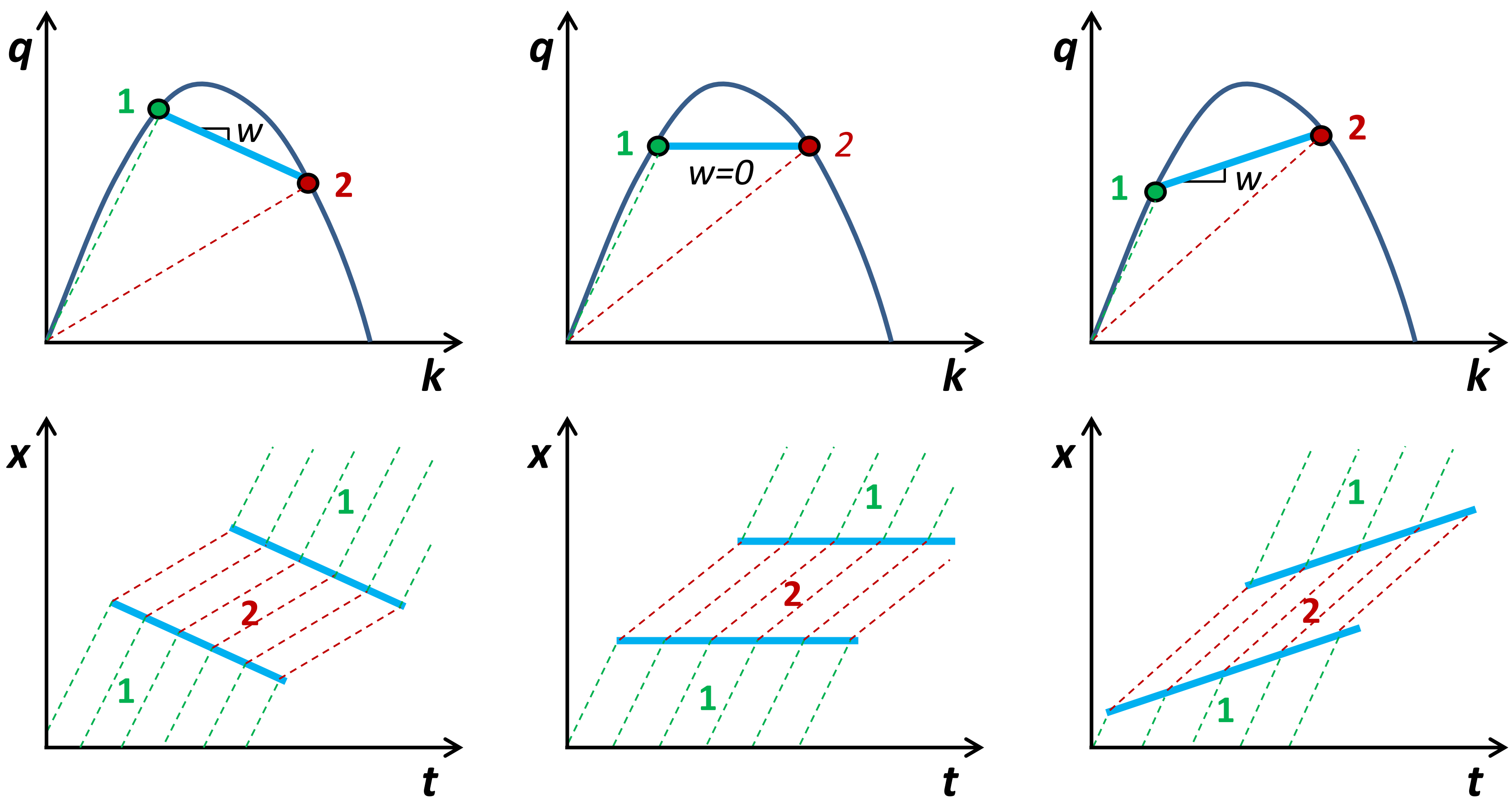
C. 4

D. 5

E. 6

Types of Shockwaves

Backward moving shock w < 0 (negative)	Stationary shockwave w = 0	Forward moving shock w > 0 (positive)



Next

This week
- Tutorial – A/D curves

CIVL6415, Semester 2, 2024

TRAFFIC ANALYSIS AND SIMULATION

MODULE 4

Macroscopic traffic models

Lecture Week 8

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Outline

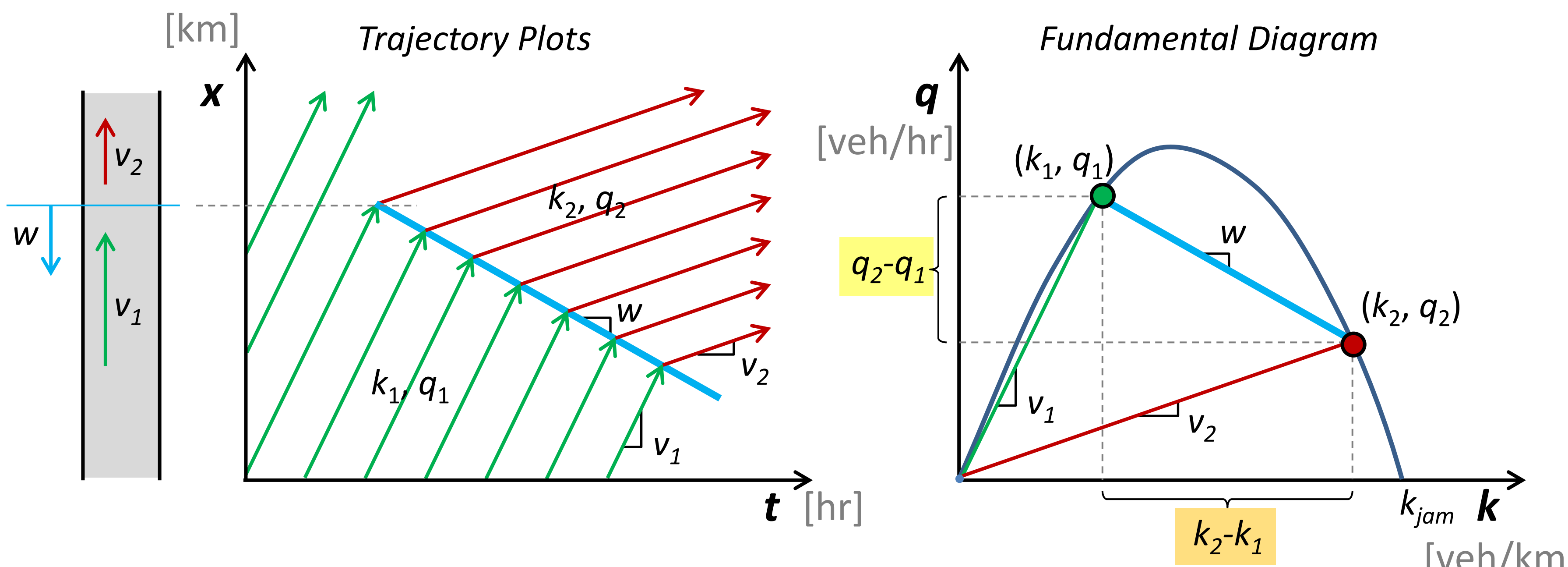
- Arrival / Departure or Input / Output curves
- Shockwaves
- Shockwave velocity equation
- **Types of bottlenecks**
  - Temporary bottlenecks
  - Stationary bottlenecks
  - Moving bottlenecks
- Flow conservation law

Velocity of Shockwaves

- **Shockwave Velocity (w)**

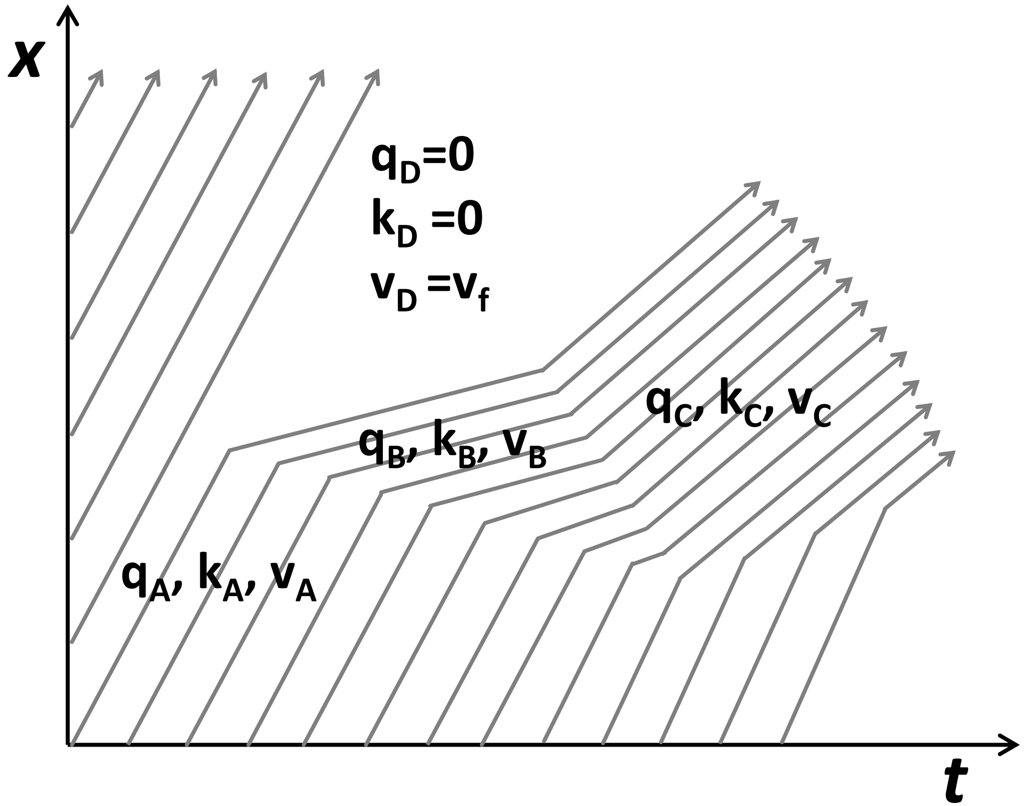
$$\boxed{w = \frac{q_2 - q_1}{k_2 - k_1}}$$ [km/hr]

- **Geometric Interpretation on Fundamental Diagram**



Shockwave Analysis

- **Key Parameters of a Shock Wave**
  - Identify traffic states
  - the point at which it starts,
  - the point at which it ends, and
  - **the velocity of the shock wave** (speed + direction).



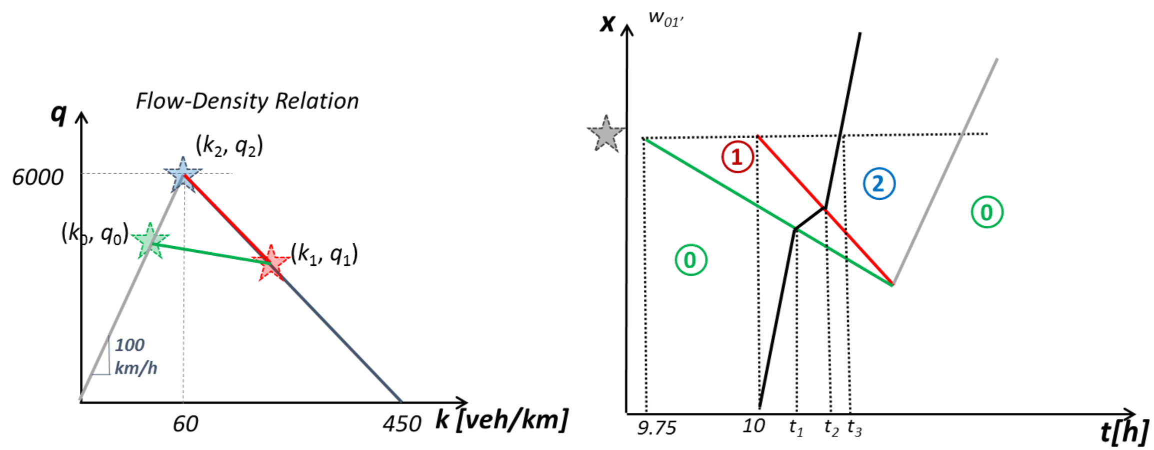
Temporary Bottlenecks

Exercise

- Traffic on a three-lane freeway observes a triangular flow-density relationship with free-flow speed u_f=100km/h, capacity c=2000veh/h/lane, jam density k_j=150veh/km/lane. Flow in the freeway from upstream is 5000veh/h, while the ramp flow is negligible (almost zero).
- At 9:45am, an accident occurs at point A. The vehicles involved close one lane of traffic (total capacity reduces to 4000veh/h) for 15min and afterwards the scene is clear (capacity returns to 6000veh/h). Lucy enters the freeway via the ramp at point B (upstream of point A) at 10:00am. Distance AB is equal to 6.5km.

- Space-time diagram with all the shockwaves
- Lucy's trajectory

Exercise



$q_0 = 5000veh/h$, $k_0 = 5000/100 = 50veh/km$, $v_0 = 100km/h$

$q_2 = 6000veh/h$, $k_2 = 6000/100 = 60veh/km$, $v_2 = 100km/h$

$q_1 = 4000veh/h$

Using similar triangles; $k_1 = 60 + \frac{450-60}{6000} * (6000 - 4000) = 190veh/km$

$v_1 = \frac{4000}{190} = 21.05\ km/h$

$$w_{0-1} = \frac{q_0 - q_1}{k_0 - k_1} = \frac{5000 - 4000}{50 - 190} = -7.14\ \text{km/h}$$

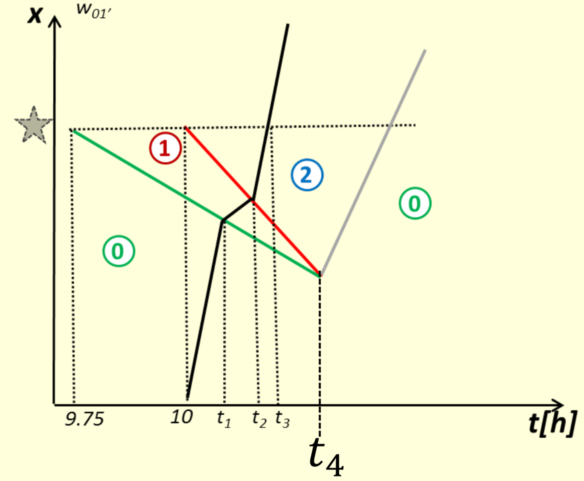
$$w_{1-2} = \frac{q_1 - q_2}{k_1 - k_2} = \frac{4000 - 6000}{190 - 60} = -15.38\ \text{km/h}$$

Question

q-k Curves at Lane-Drop Bottleneck

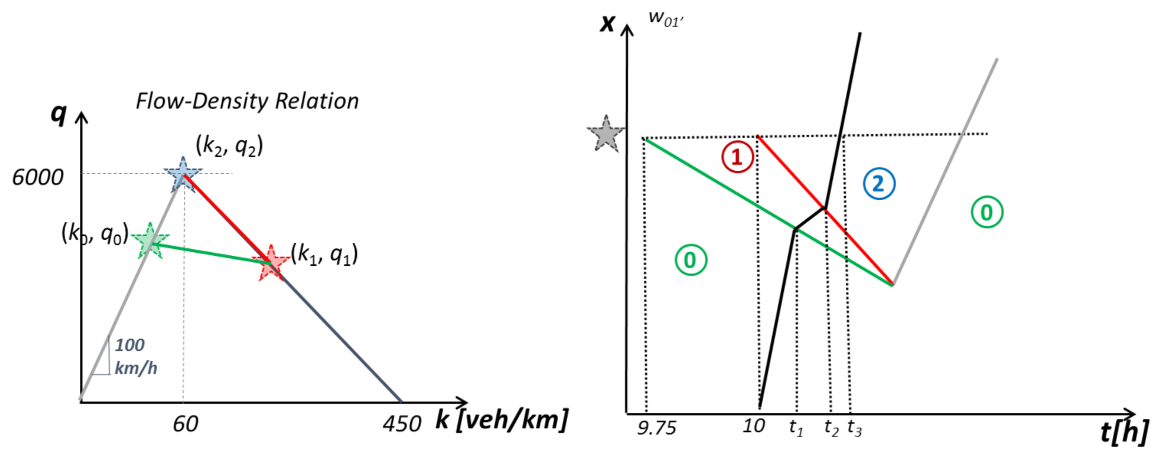
Calculate the time when the congestion ends, t_4[h].

$$w_{0-1} = -7.14 \text{ km/h}$$
$$w_{1-2} = -15.38 \text{ km/h}$$



a. 10.111
b. 10.150
c. 10.166
d. 10.200
e. 10.217

Exercise

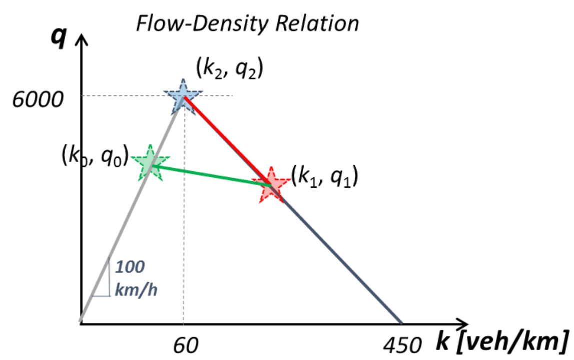


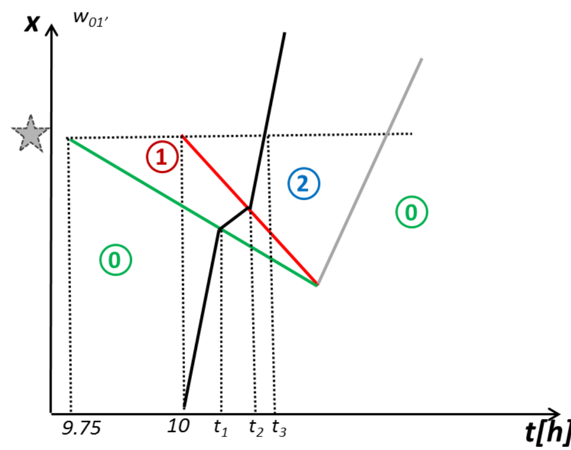
When does Lucy hit the queue? Point t_1 in the space-time diagram.

$$v_0 * (t_1 - 10) = 6.5 + w_{0-1} * (t_1 - 9.75)$$
$$100 * (t_1 - 10) = 6.5 - 7.14 * (t_1 - 9.75)$$
$$\rightarrow t_1 = 10.044\ h$$

Make sure t_1<t_4.

Exercise





Note that by the time Lucy crosses the accident location, the scene is clear

When does Lucy leave the queue? Point t_2 in the space-time diagram.

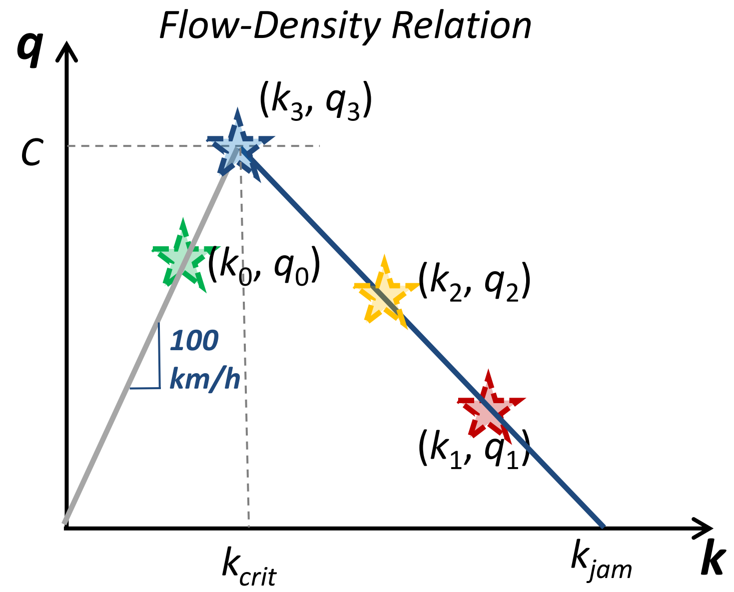
$$v_0 * (t_1 - 10) + v_1 * (t_2 - t_1) = 6.5 + w_{1-2} * (t_2 - 10)$$
$$100 * 0.044 + 21.05 * (t_2 - 10.044) = 6.5 - 15.38 * (t_2 - 10)$$
$$\rightarrow t_2 = 10.083\ h$$

When does Lucy cross the accident scene? Point t_3 in the space-time diagram.

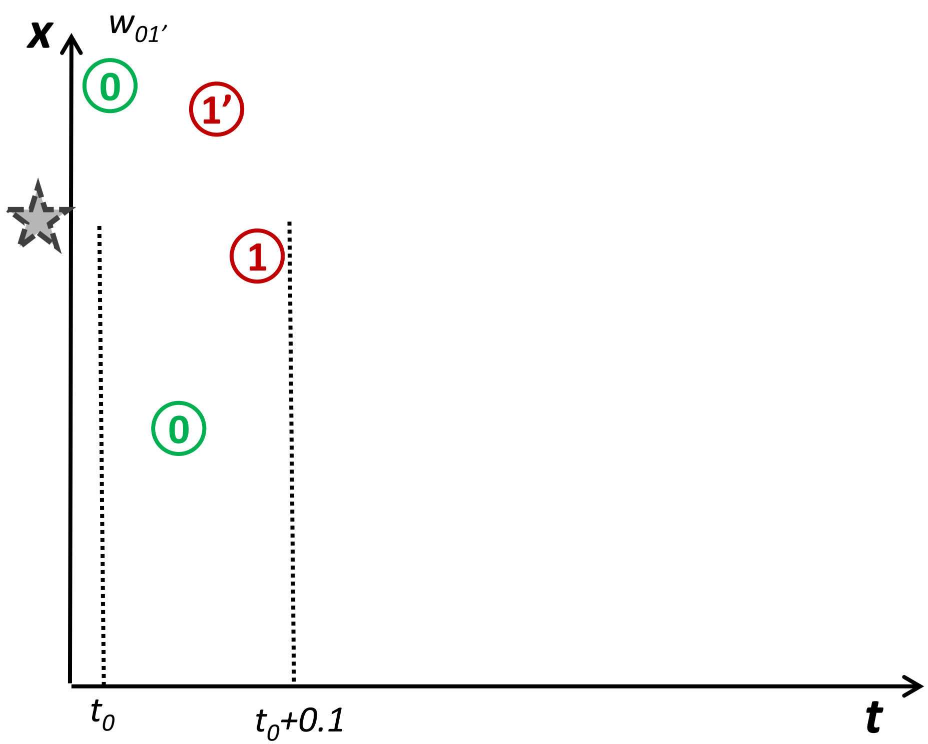
$$v_0 * (t_1 - 10) + v_1 * (t_2 - t_1) + v_2 * (t_3 - t_2) = 6.5$$
$$100 * 0.044 + 21.05 * 0.039 + 100 * (t_3 - 10.083) = 6.5$$
$$\rightarrow t_3 = 10.096\ h$$

Exercise

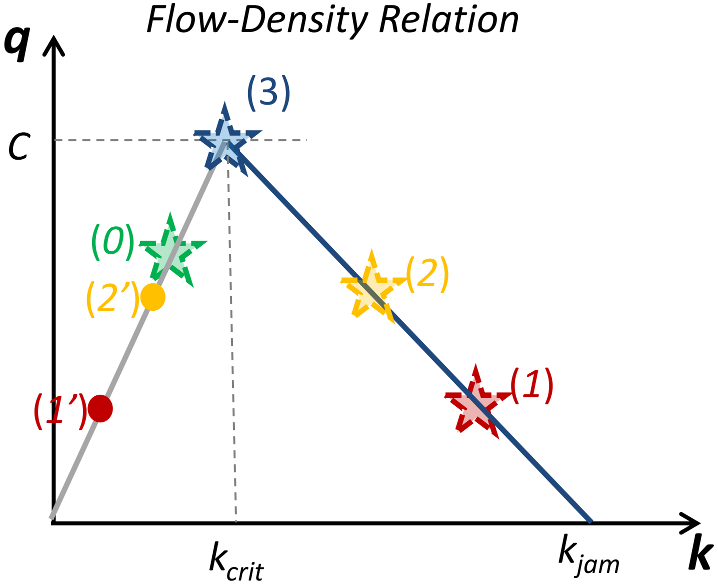
- A freeway with 3 lanes and maximum capacity C = 6000 veh/h
- Triangular fundamental diagram, where v_f = 100 km/h, $k_{jam} = k_{crit} * 5$
- Demand = ¾*C
- Incident at $t_0 = 0$, Response time = 6 min, Clearance time = 42 min
- One lane remains open during response
- Two lanes remain open during clearance



Exercise

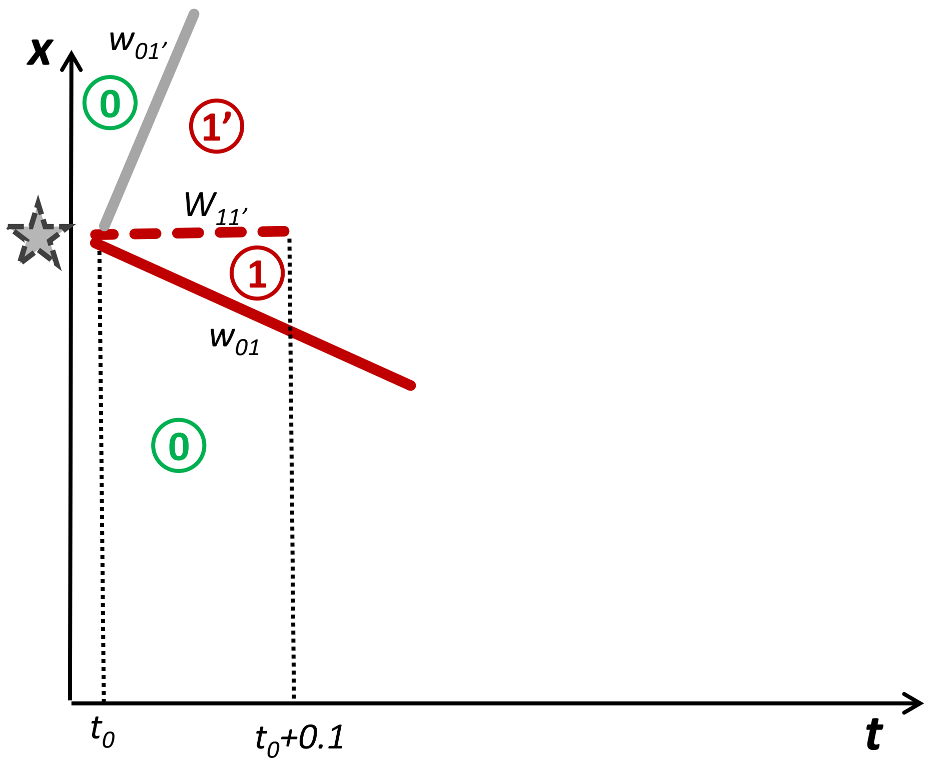


x
$w_{01'}$
0
1'
1
0
t_0
t_0+0.1
t

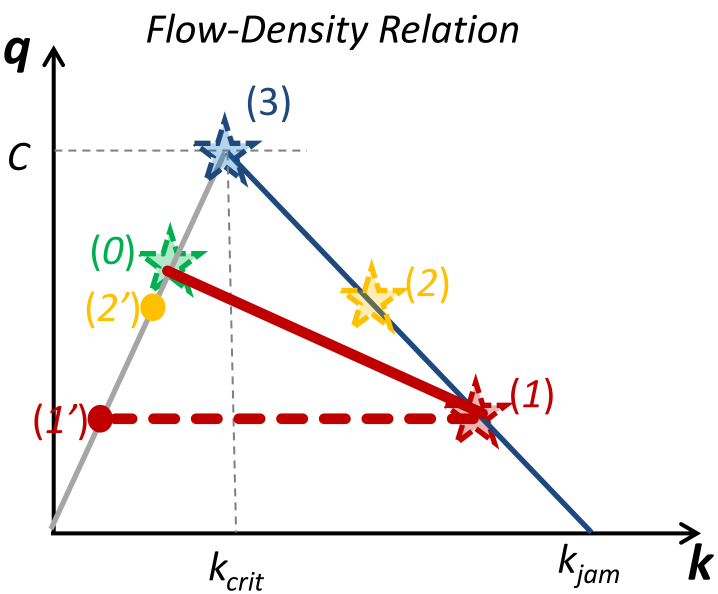


Flow-Density Relation
q
C
(3)
(0)
(2')
(1')
(2)
(1)
k_{crit}
k_{jam}
k

Exercise

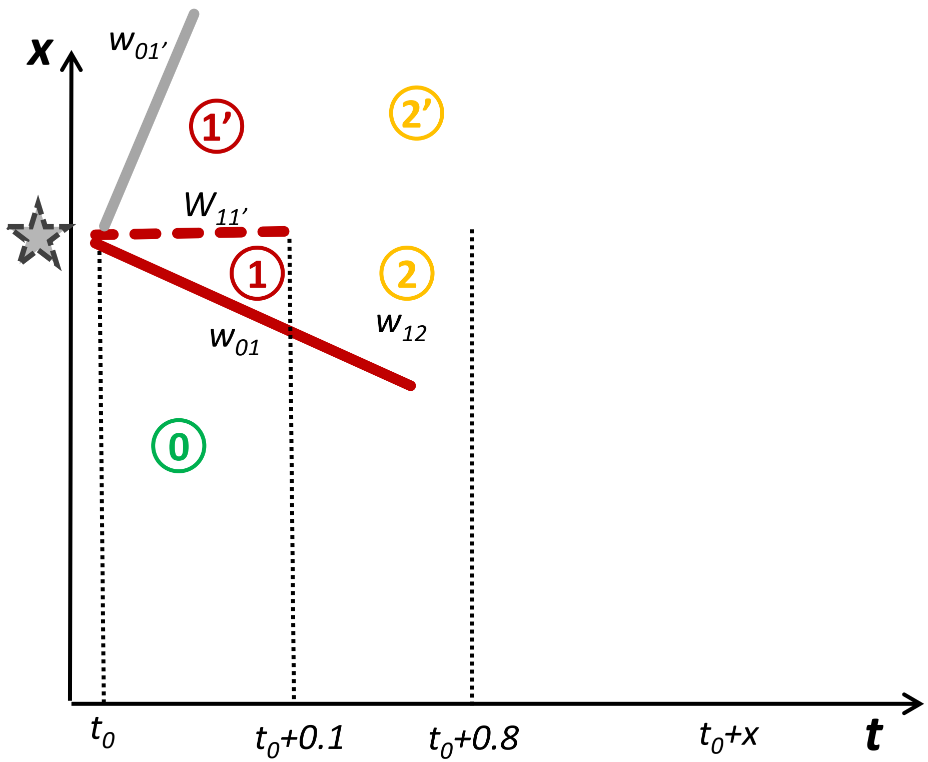


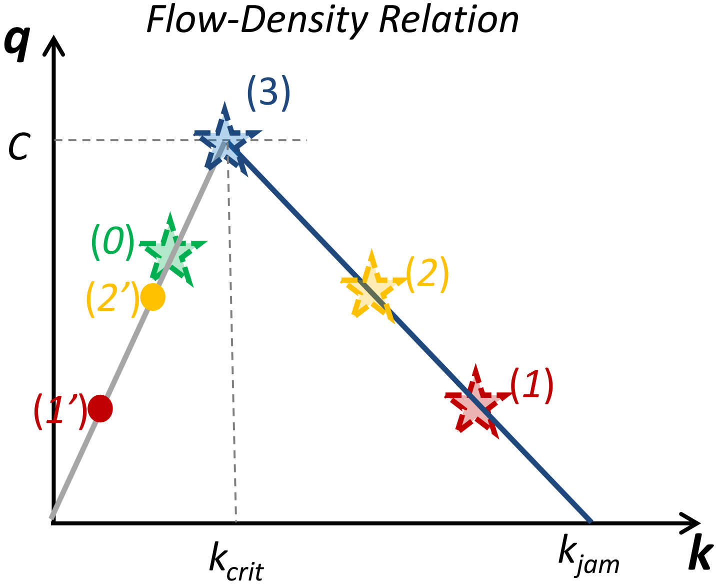
x
$w_{01'}$
0
1'
$W_{11'}$
1
w_{01}
0
t_0
t_0+0.1
t



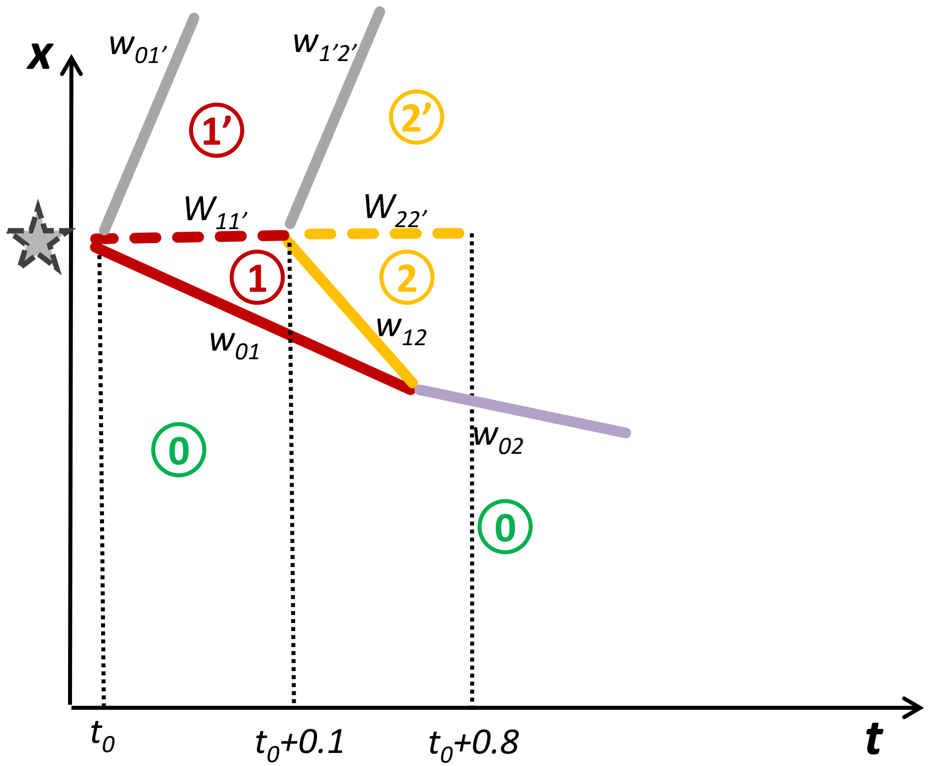
Flow-Density Relation
q
(3)
C
(0)
(2')
(2)
(1')
(1)
k_{crit}
k_{jam}
k

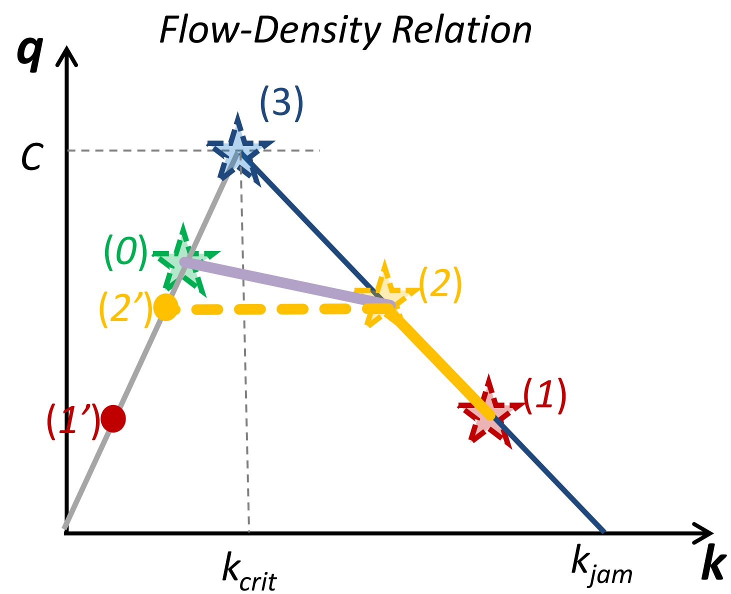
Exercise


x
w01′
1′
2′
W11′
1
2
w01
w12
0
t0
t0+0.1
t0+0.8
t0+x
t

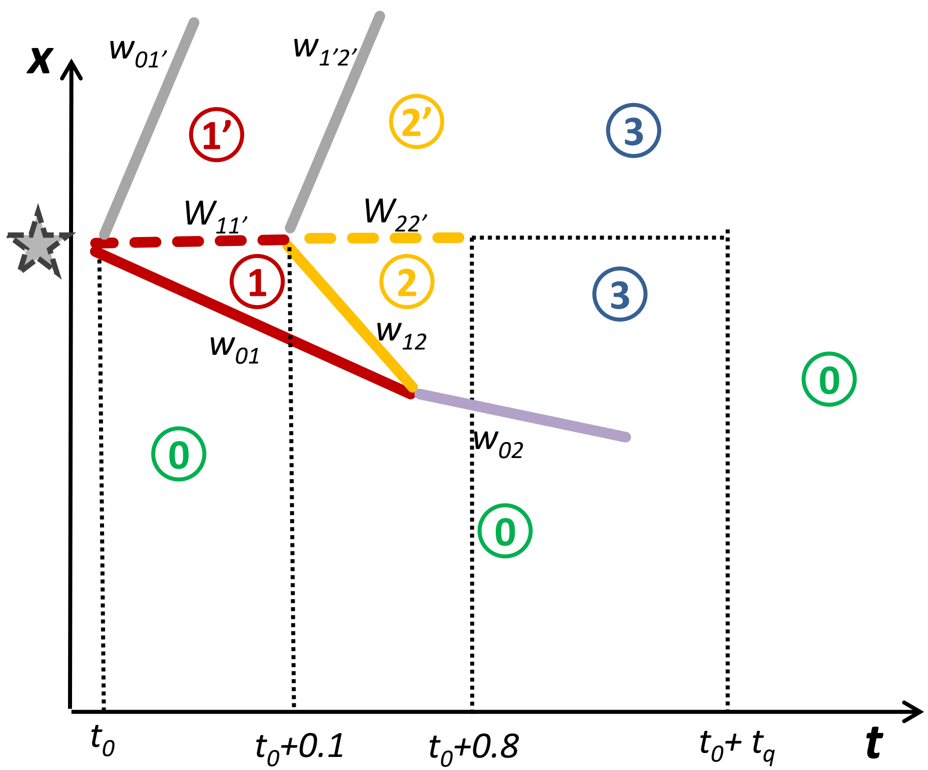

Flow-Density Relation
q
(3)
C
(0)
(2′)
(2)
(1′)
(1)
kcrit
kjam
k

Exercise

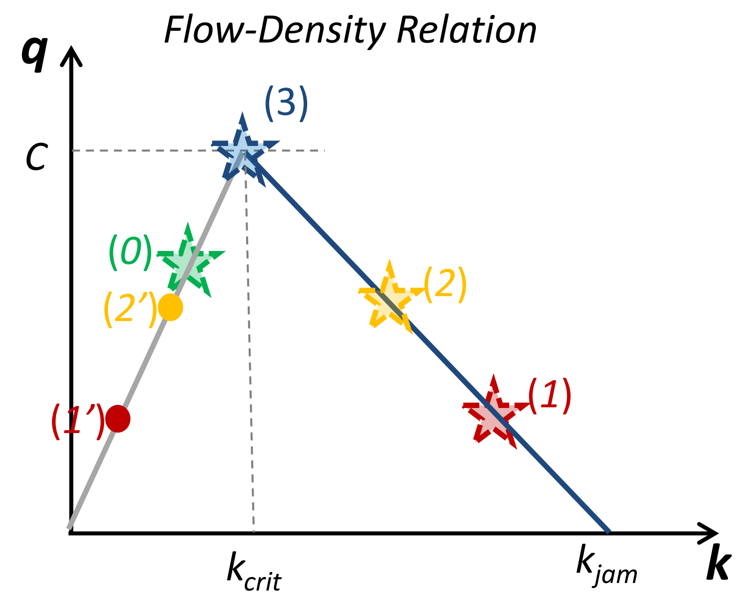

x
t
$w_{01'}$
$w_{1'2'}$
1'
2'
$W_{11'}$
$W_{22'}$
1
2
w_{12}
w_{01}
w_{02}
0
0
t_0
t_0+0.1
t_0+0.8


Flow-Density Relation
q
(3)
C
(0)
(2)
(2')
(1)
(1')
k_{crit}
k_{jam}
k

Exercise

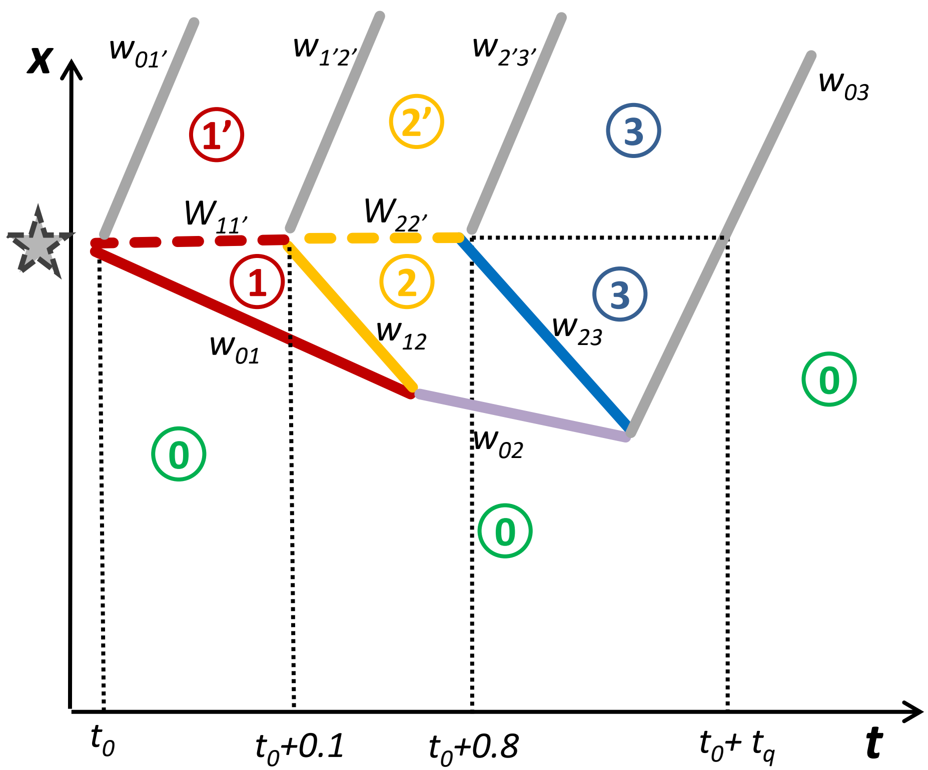


x
$w_{01'}$
$w_{1'2'}$
1'
2'
3
$W_{11'}$
$W_{22'}$
1
2
3
w_{12}
w_{01}
0
w_{02}
0
0
t_0
t_0+0.1
t_0+0.8
t_0+t_q
t

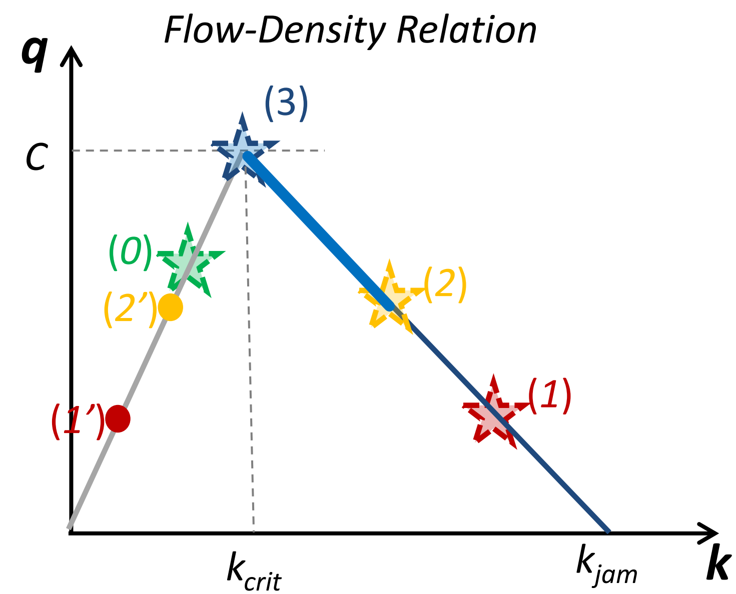


Flow-Density Relation
q
(3)
C
(0)
(2')
(2)
(1')
(1)
k_{crit}
k_{jam}
k

Exercise



x
t
$w_{01'}$
$w_{1'2'}$
$w_{2'3'}$
w_{03}
1'
2'
3
$W_{11'}$
$W_{22'}$
1
2
3
w_{12}
w_{23}
w_{01}
w_{02}
0
0
0
t_0
t_0+0.1
t_0+0.8
t_0+ t_q



Flow-Density Relation
q
k
C
(3)
(0)
(2')
(2)
(1')
(1)
k_{crit}
k_{jam}

Exercise

3 lanes, C = 6000 veh/h, Demand = ¾*C

v_f = 100 km/h, $k_{jam} = k_{crit} * 5$

k_{crit} = 6000/100 =60 veh/km

k_{jam} = 300 veh/km

q_2 = 2/3*C = 4000 veh/h

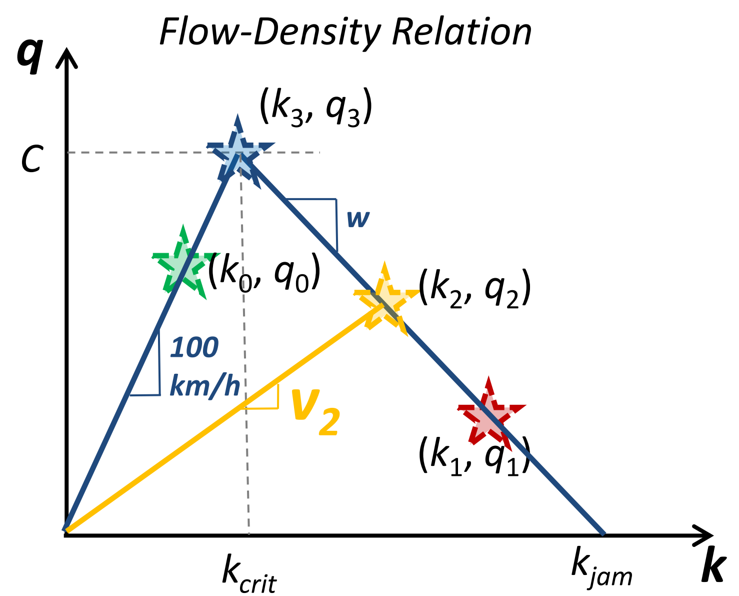
$w = (q_{jam} - q_{crit}) / (k_{jam} - k_{crit})$ = -25km/h

$w = (q_2 - q_{crit}) / (k_2 - k_{crit})$

$k_2 = k_{crit} + 1/w*(q_2 - q_{crit})$

k_2 = 60 - 1/25*(4000-6000) = 140 veh/km

$v_2 = q_2/k_2$ = 4000/140 = 28.57 km/h



q_0 = 4500 veh/h, k_0 = 45 veh/km

q_1 = 2000 veh/h, k_1 = 220 veh/km

q_2 = 4000 veh/h, k_2 = 140 veh/km

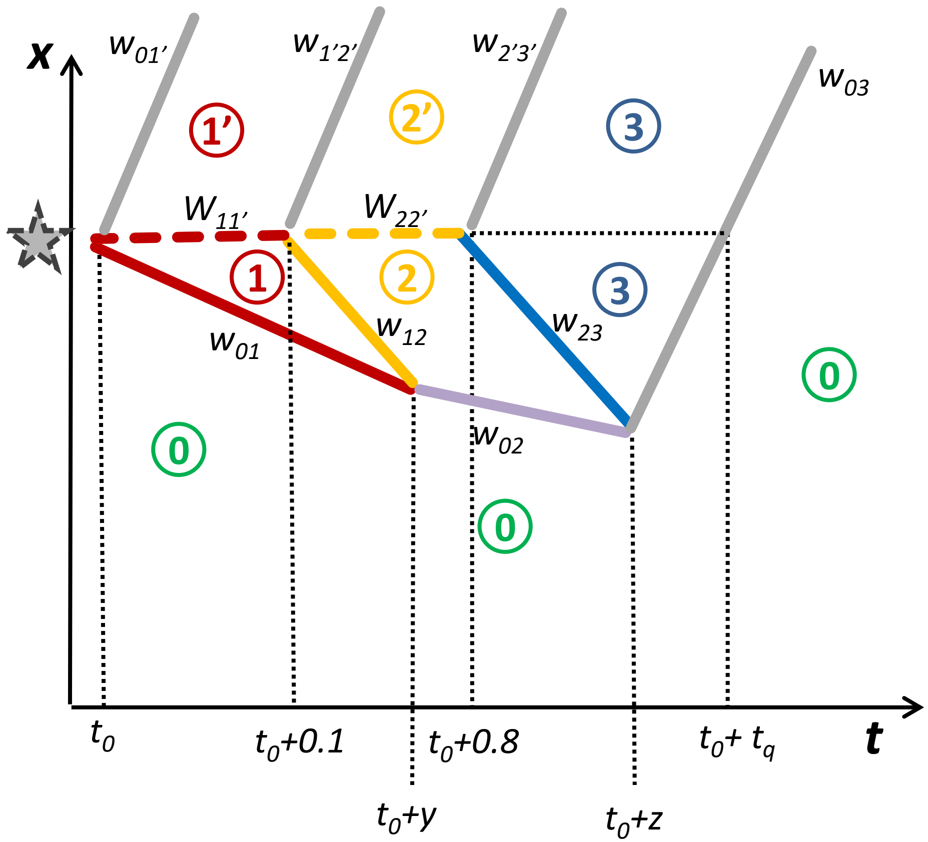
q_3 = 6000 veh/h, k_3 = 300 veh/km

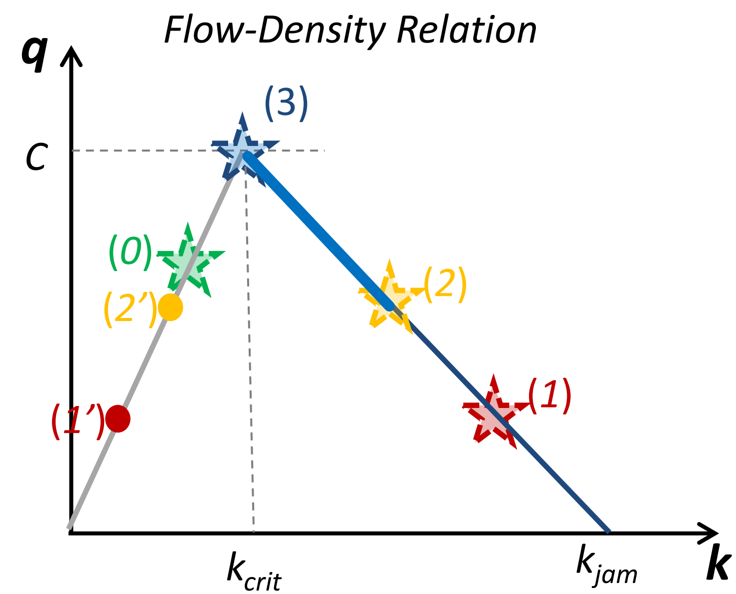
Exercise

$w_{01} = (2000\text{-}4500)/(220\text{-}45) = -14.29$ km/h

$w_{02} = (4000\text{-}4500)/(140\text{-}45) = -5.26$km/h

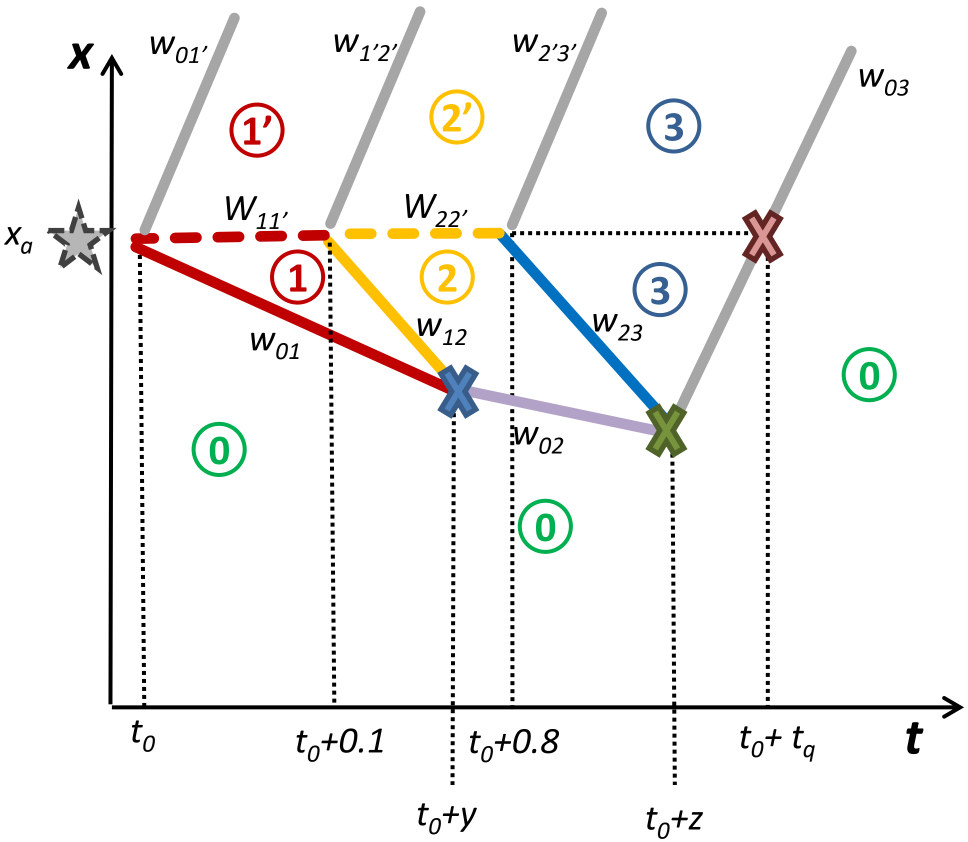
$w_{12} = w_{23} = 6000/240 = -25$ km/h





$v_f = w_{01'} = w_{1'2'} = w_{2'3'} = w_{03} = 100$km/h

Exercise



$x_a + w_{01}*y = x_a + w_{12}*(y-0.1)$

$\rightarrow$ y = 0.23h

$x_a + w_{12}*(y-0.1) + w_{02}*(z-y) = x_a + w_{23}*(z-0.8)$

$\rightarrow$ z = 1.12h

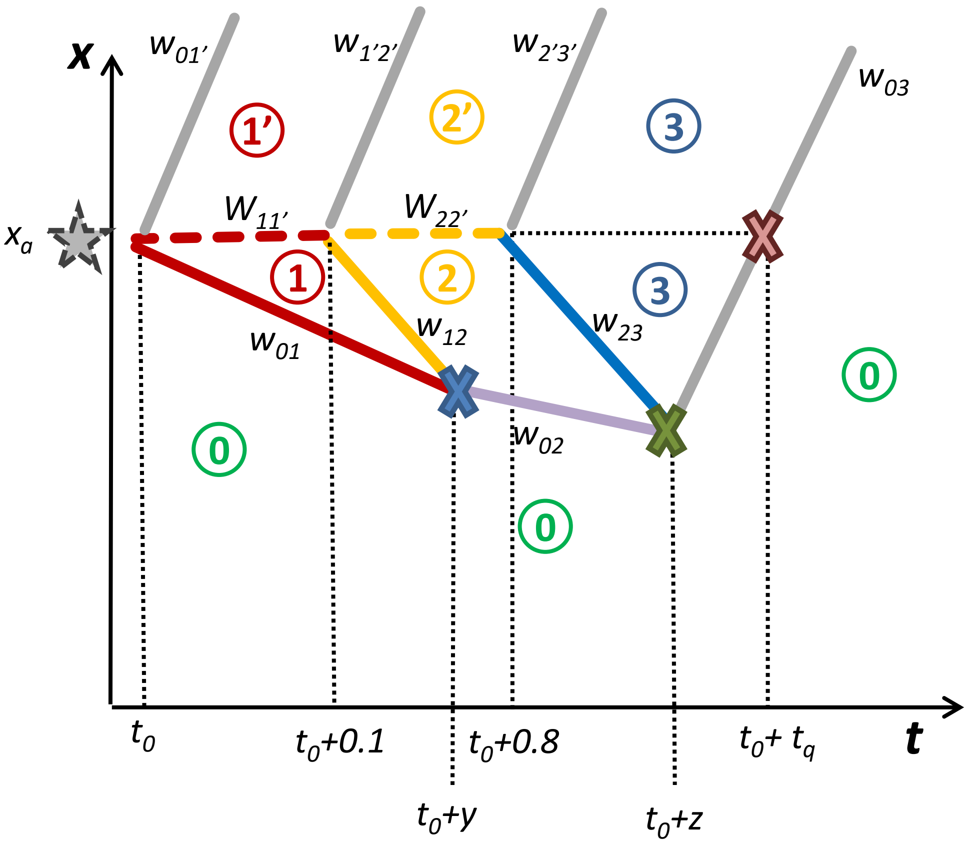
w_{01} = -14.29 km/h

w_{02} = -5.26km/h

$w_{12} = w_{23}$ = -25 km/h

$v_f = w_{01'} = w_{1'2'} = w_{2'3'} = w_{03}$ = 100km/h

Exercise



$$x_a + w_{23}*(z-0.8) + v_f*(t_a-z) = x_a$$

$$-25*(1.12-0.8) + 100 *(t_a-1.12) = 0$$

$$\rightarrow t_a = 1.20h$$

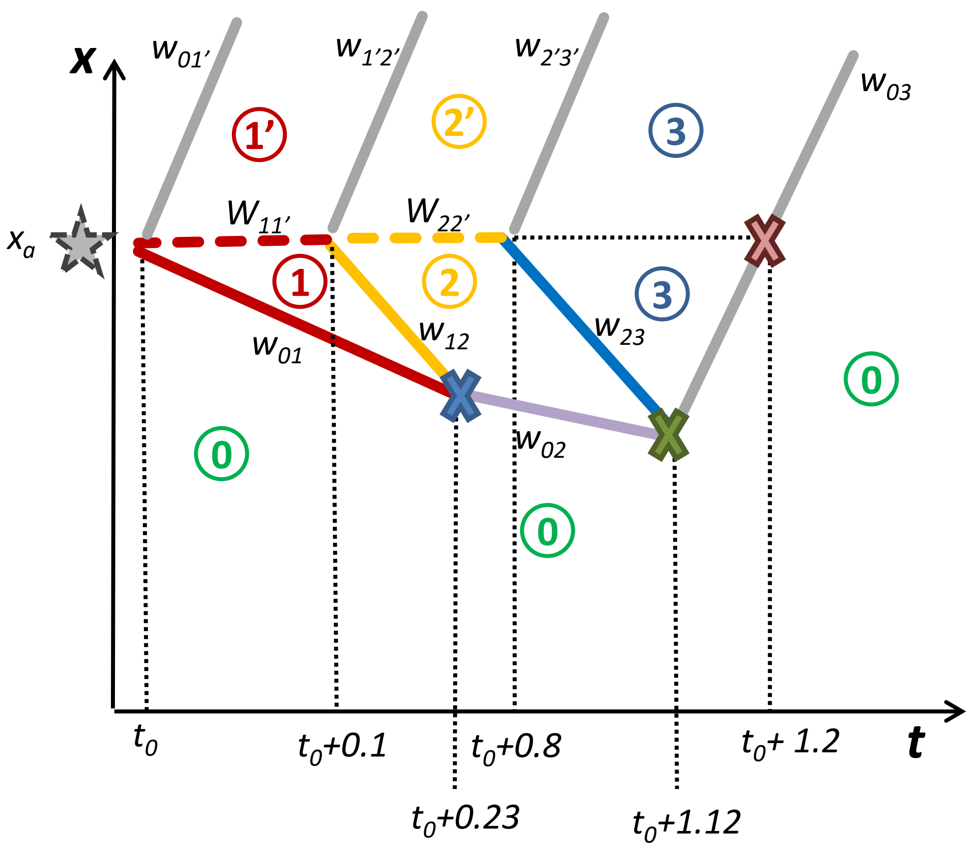
w_{01} = -14.29 km/h

w_{02} = -5.26km/h

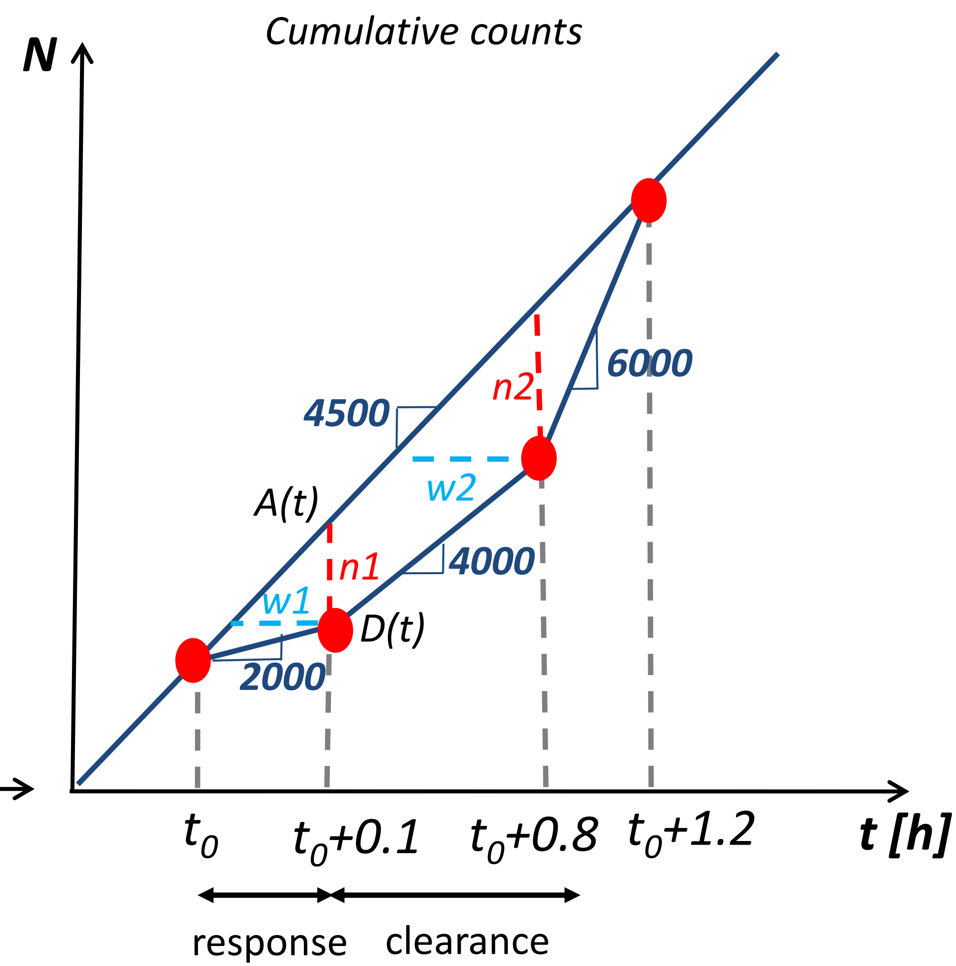
$w_{12} = w_{23}$ = -25 km/h

$v_f = w_{01'} = w_{1'2'} = w_{2'3'} = w_{03}$ = 100km/h

Exercise

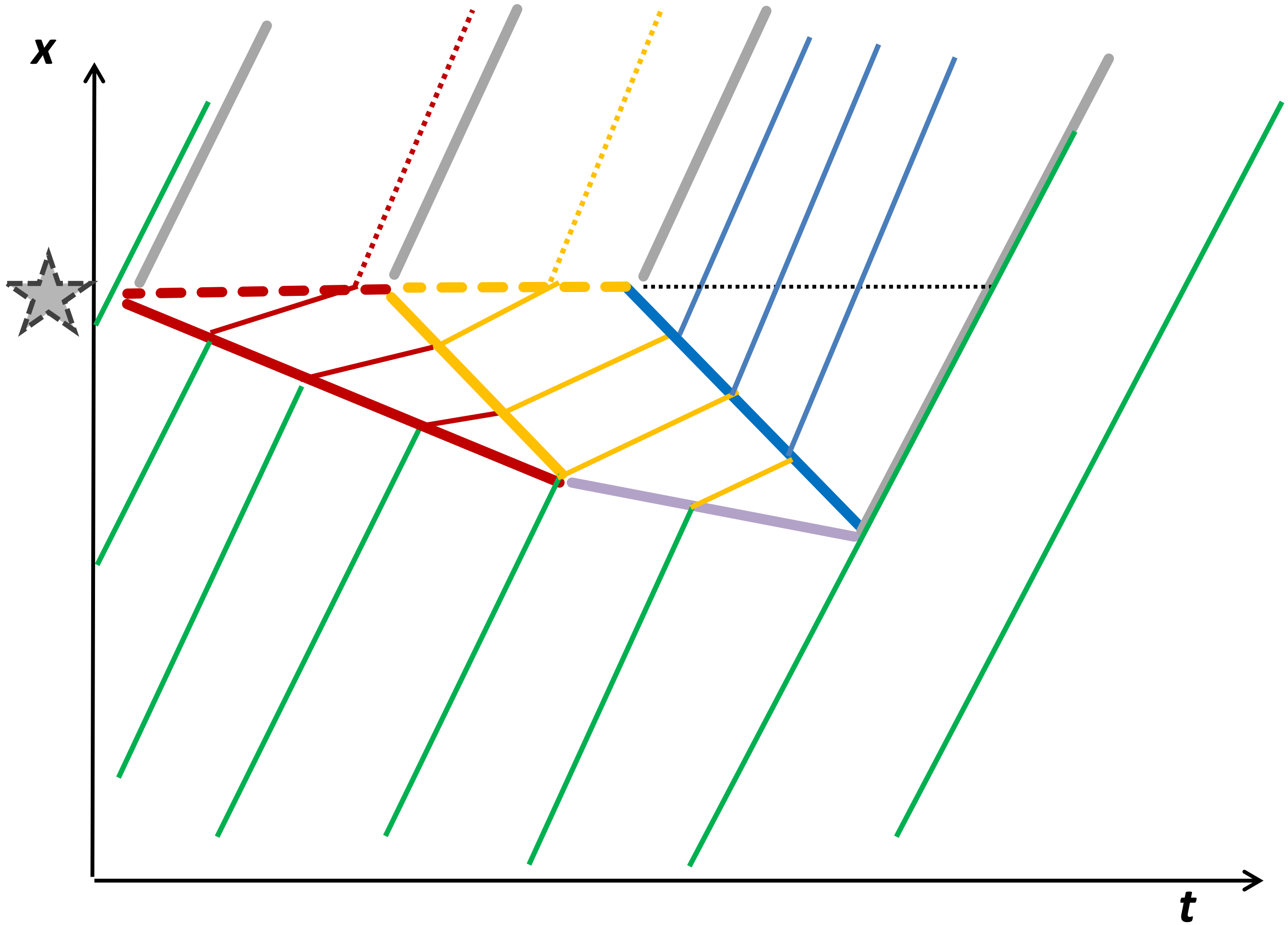


x
$w_{01'}$
$w_{1'2'}$
$w_{2'3'}$
w_{03}
1'
2'
3
x_a
$W_{11'}$
$W_{22'}$
1
2
3
w_{12}
w_{23}
w_{01}
0
w_{02}
0
0
0
t_0
t_0+0.1
t_0+0.8
t_0+ 1.2
t
t_0+0.23
t_0+1.12



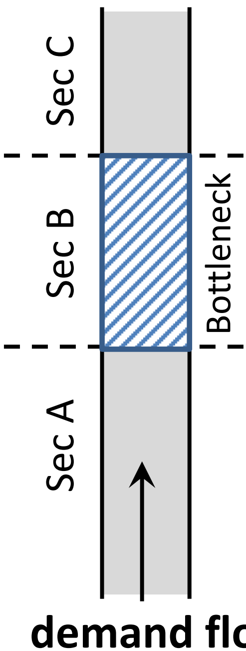
Cumulative counts
N
4500
6000
n2
w2
A(t)
n1
4000
w1
D(t)
2000
t_0
t_0+0.1
t_0+0.8
t_0+1.2
t [h]
response
clearance

Exercise

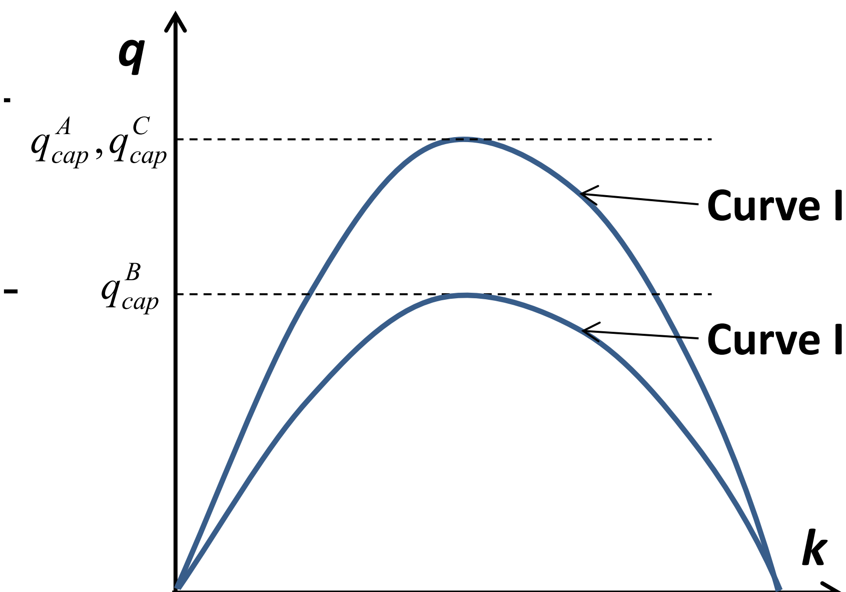


Stationary Bottlenecks

- **Different Flow-Density Curves observed near Classical Flow-Conserving Bottleneck**
  - Consider road sections A, B and C, where Section B is a classical flow-conserving bottleneck that has a lower capacity than sections A and C.
  - Two flow-density diagrams exist: Curve I defining Sections A and C, and Curve II defining Section B.



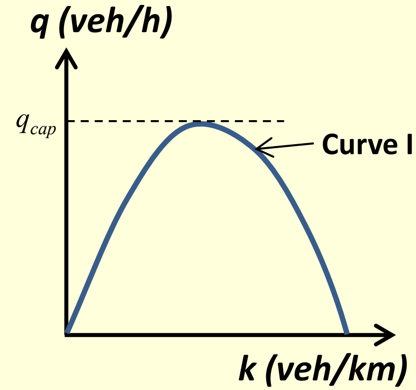
q-k curve	capacity (veh/h)
Curve I	q_{cap}^{C}
Curve II	q_{cap}^{B}
Curve I	q_{cap}^{A}

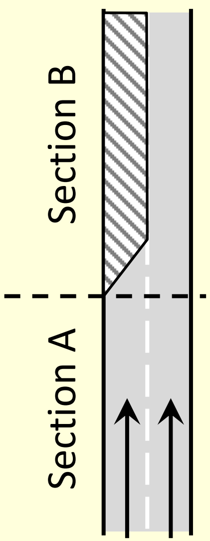


Question

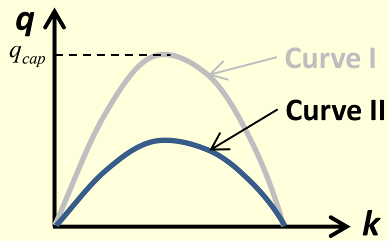
q-k Curves at Lane-Drop Bottleneck

A two-lane highway has the flow-density diagram defined by Curve I.
If one lane is closed in Section B, **what would be the q-k curve like for the bottleneck section B (Curve II) ?**

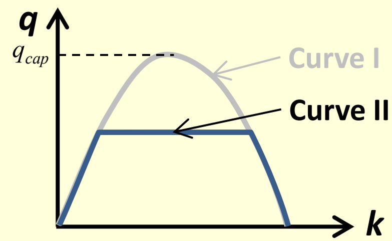




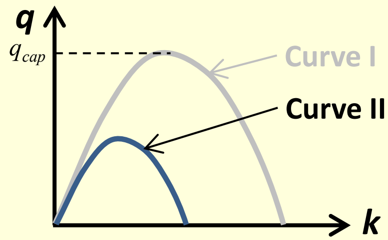
A.



C.



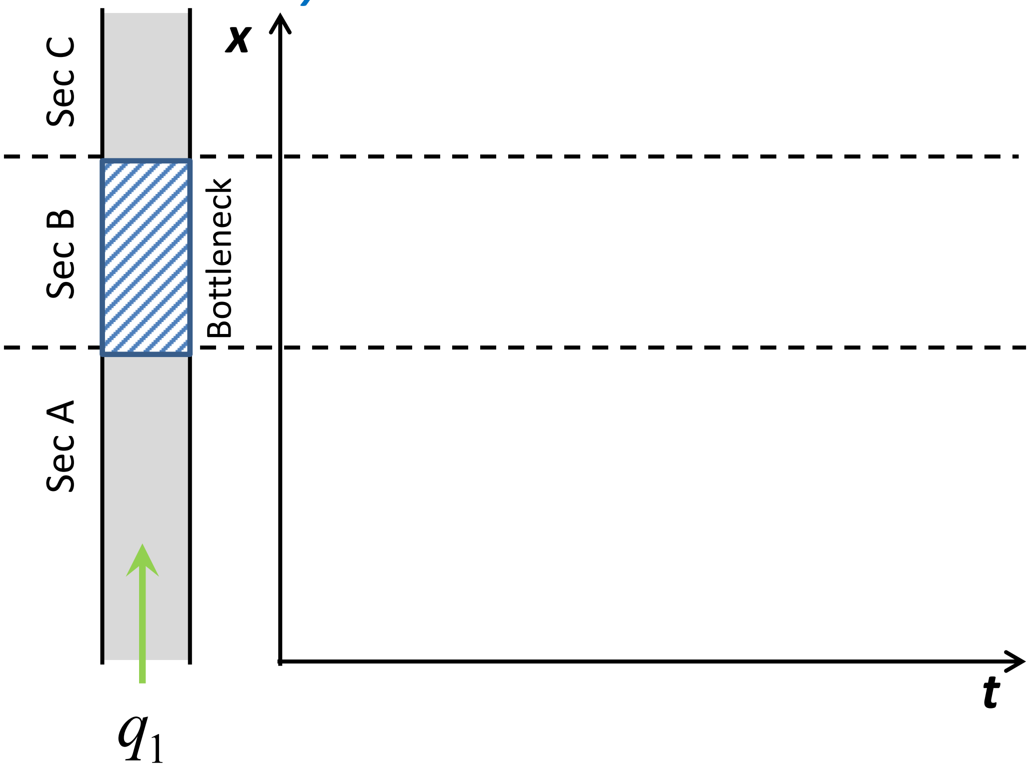
B.

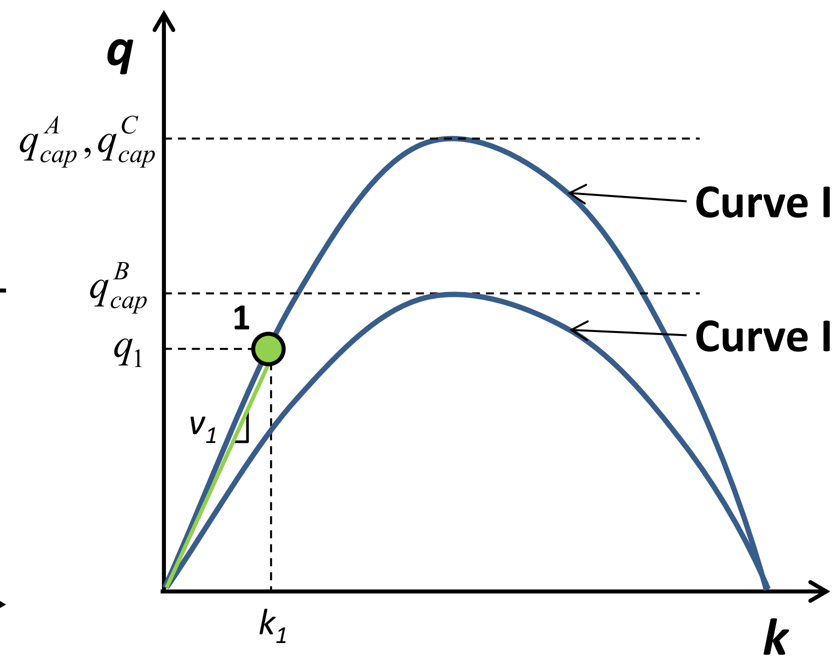


D. None of the Above

Shockwaves at Stationary Bottleneck

- **The demand flow rate q_1 is less than the lower capacity:** $q_1 < q_{cap}^B$
  - Sections A and C have k_1 and v_1 defined by Curve I.
  - Section B has k_2 and v_2 defined by Curve II at the given demand q_1.
  - At each section, the demand is served and flows on into the next section.
  - ***Stationary shockwaves***.





Shockwaves at Stationary Bottleneck

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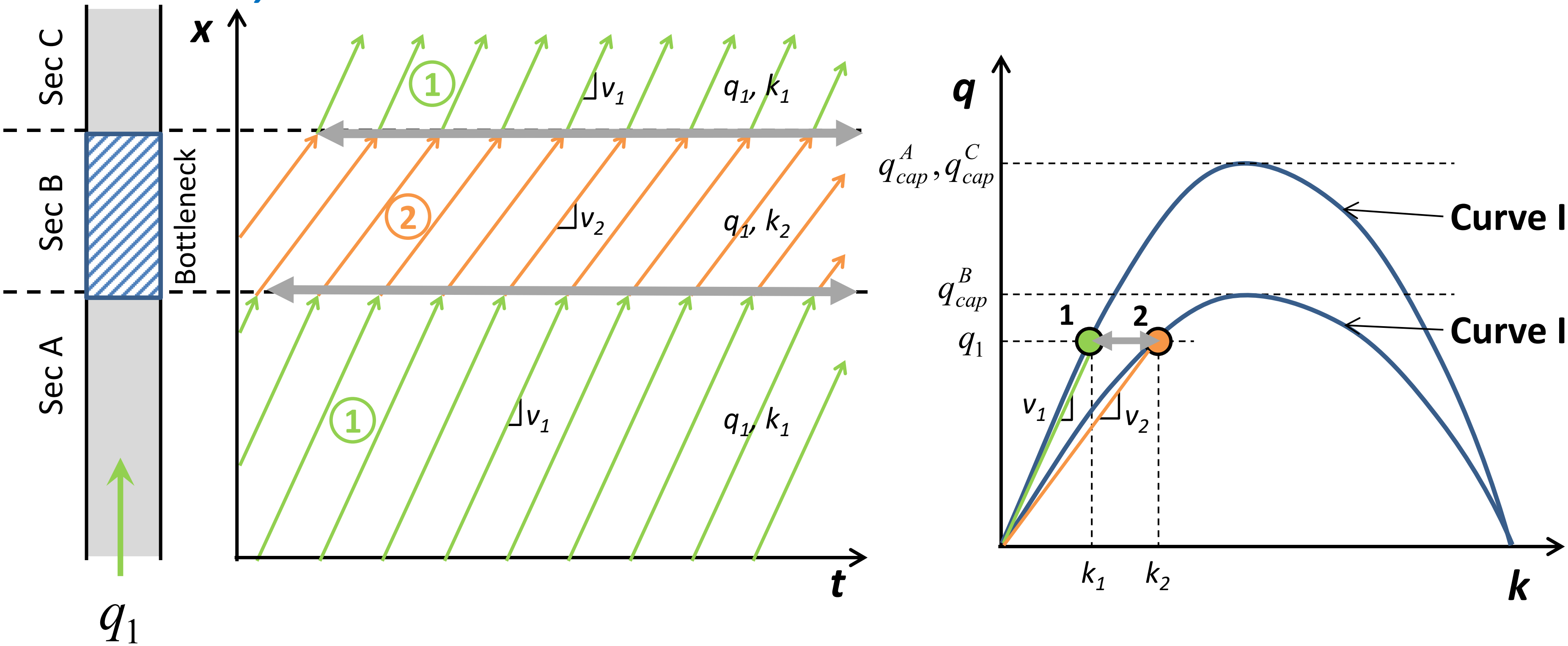
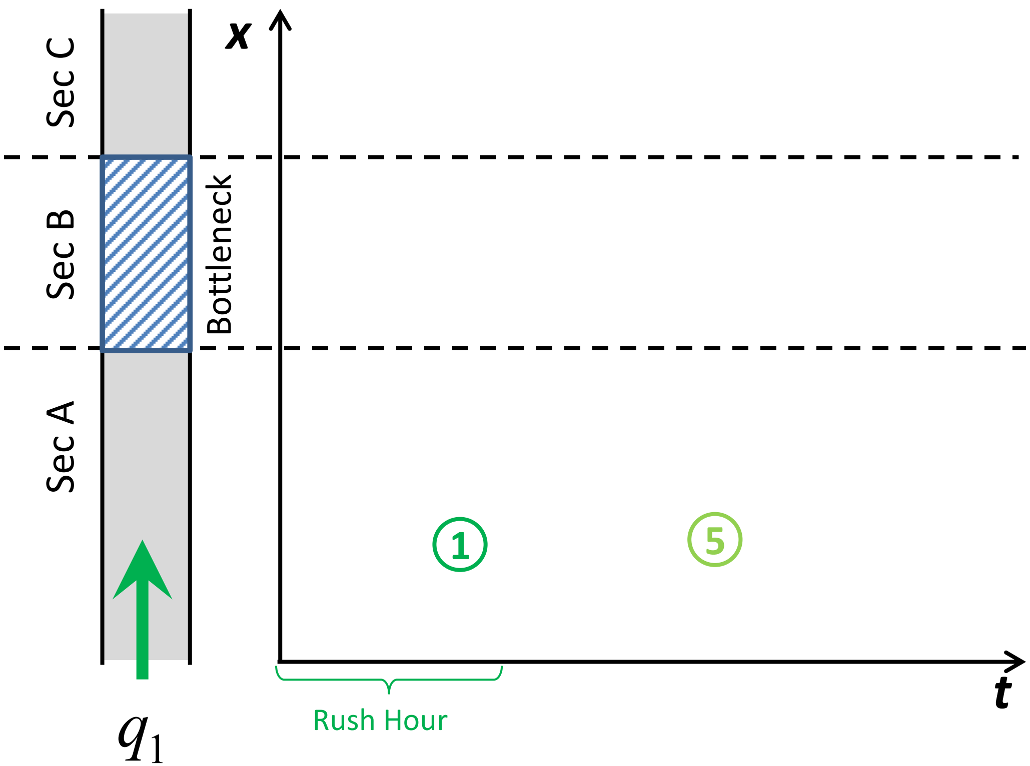
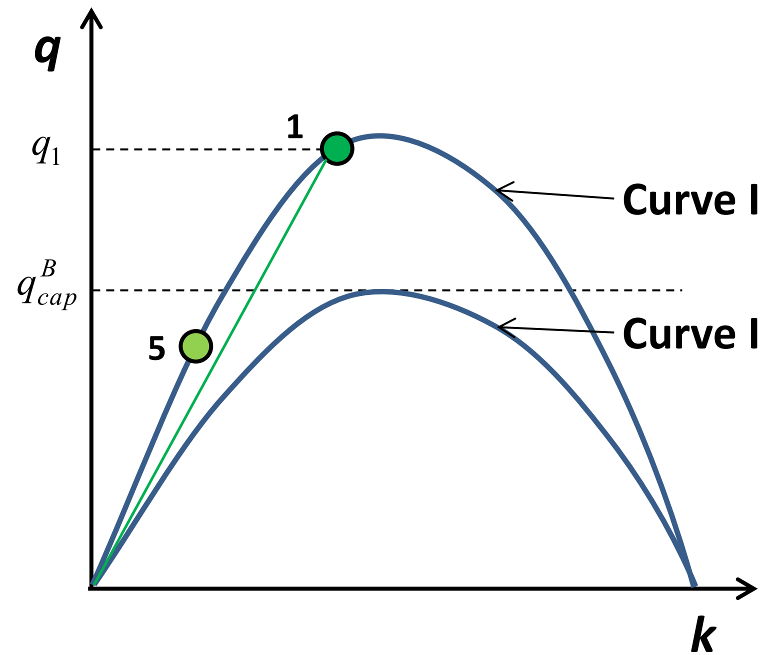


Shockwaves at Stationary Bottleneck

- **The demand flow rate q_1 is greater than Sec B's capacity: $q_1 > q_{cap}^B$**
  - The demand flow rate q_1 can be served at some point in Sec A operating at **Point 1** in Curve I.
  - The demand q_1 cannot be passed to Sec B; the most that can be passed is $q_2 = q_{cap}^B$.
  - The difference q_1-q_2 is stored upstream of Sec B; Sec A operates at **Point 2** on Curve I.
  - In Sec A, a jump from 2 to 1 on Curve I creates **a shockwave traveling upstream at velocity w_{12}**.





Shockwaves at Stationary Bottleneck

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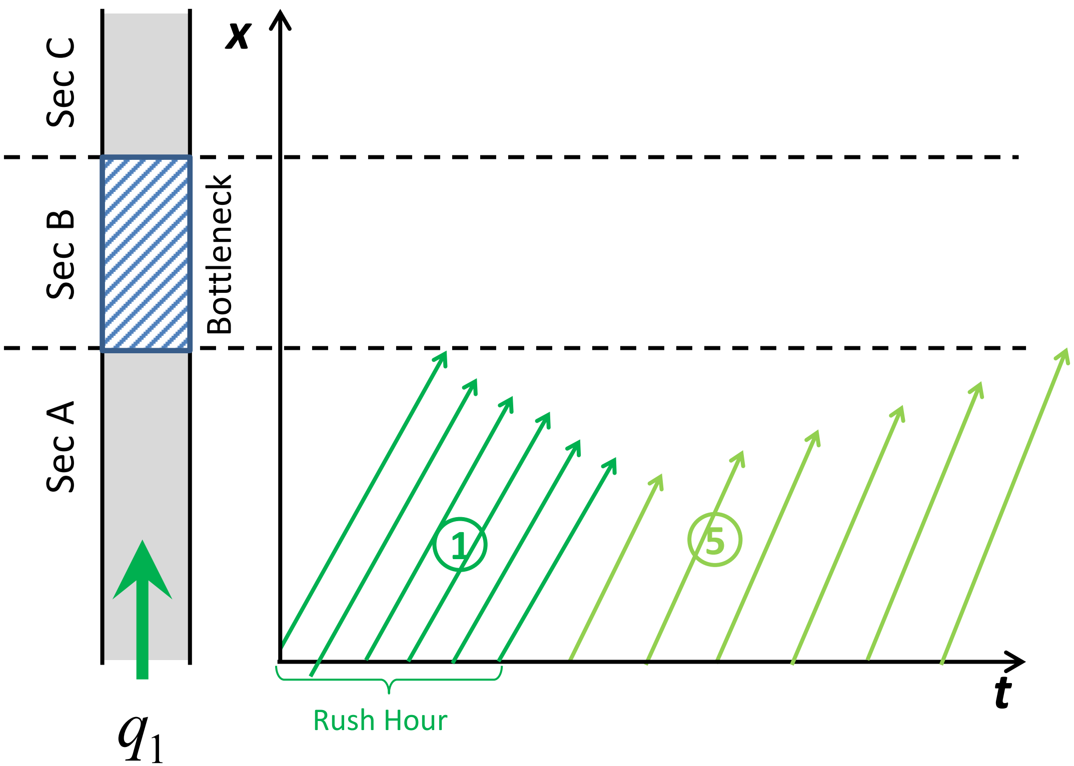
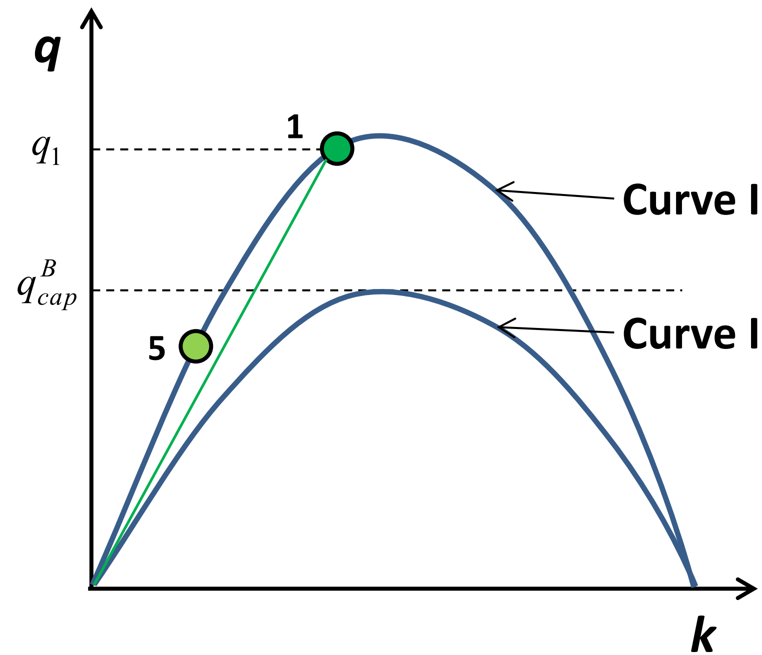
  - In Sec A, a jump from 2 to 1 on Curve I creates **a shockwave traveling upstream at velocity w_{12}**.





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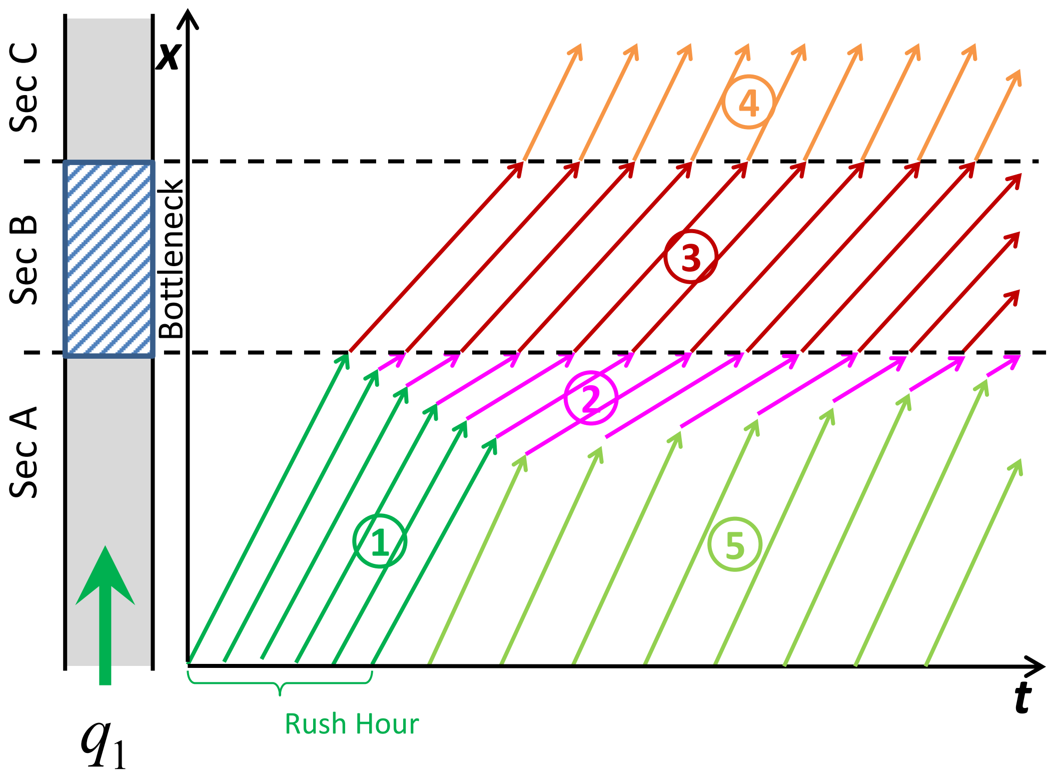
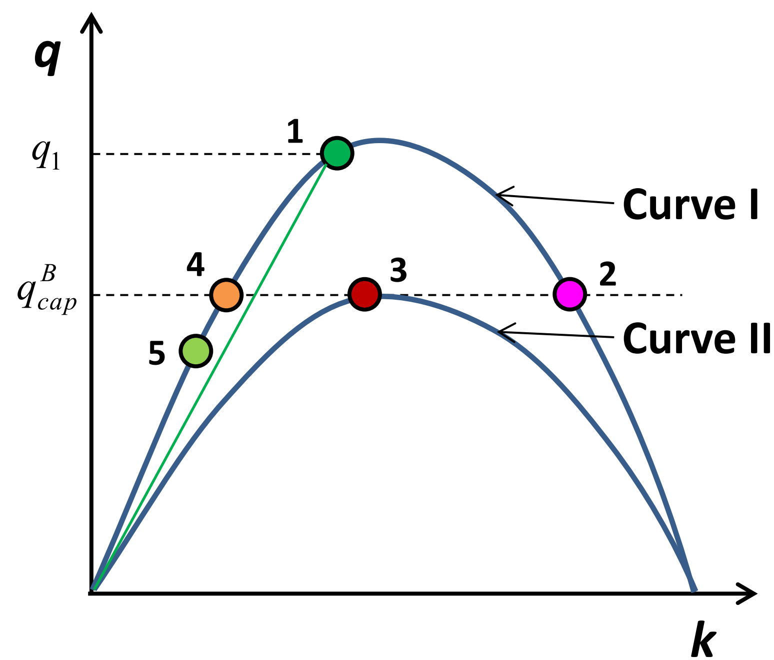
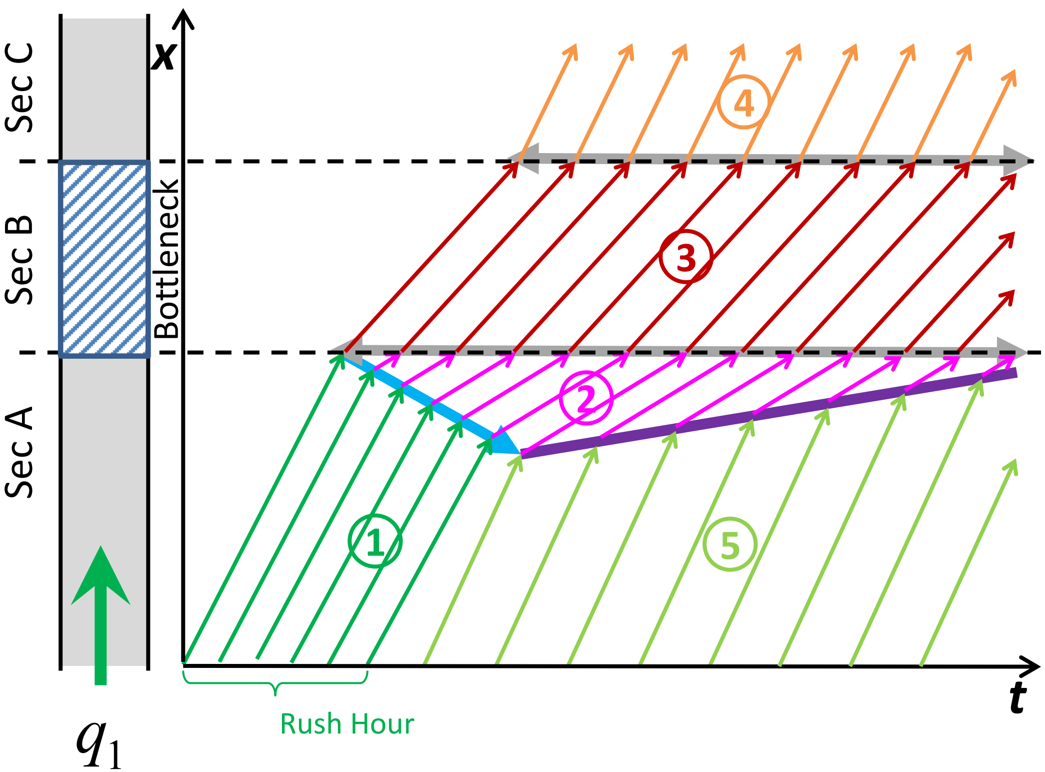
  - In Sec A, a jump from 2 to 1 on Curve I creates **a shockwave traveling upstream at velocity w_{12}**.

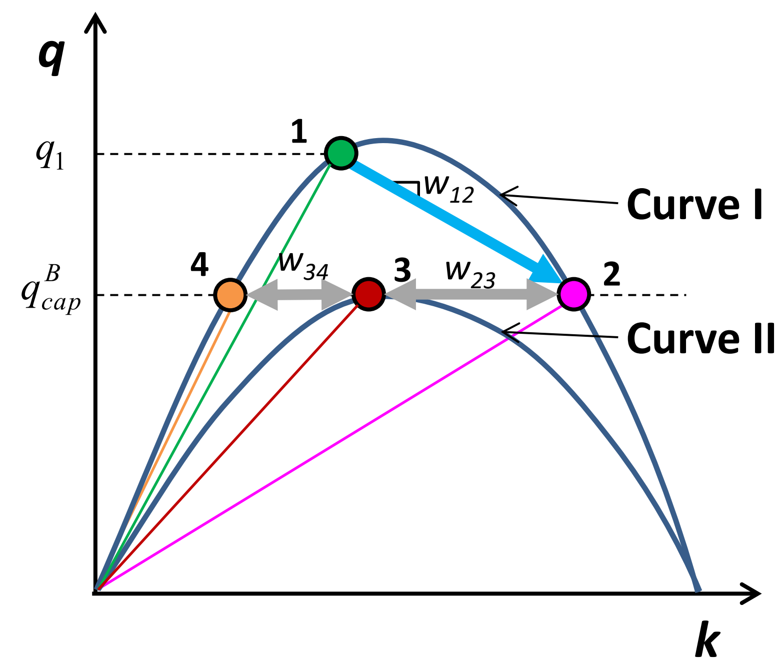




Shockwaves at Stationary Bottleneck

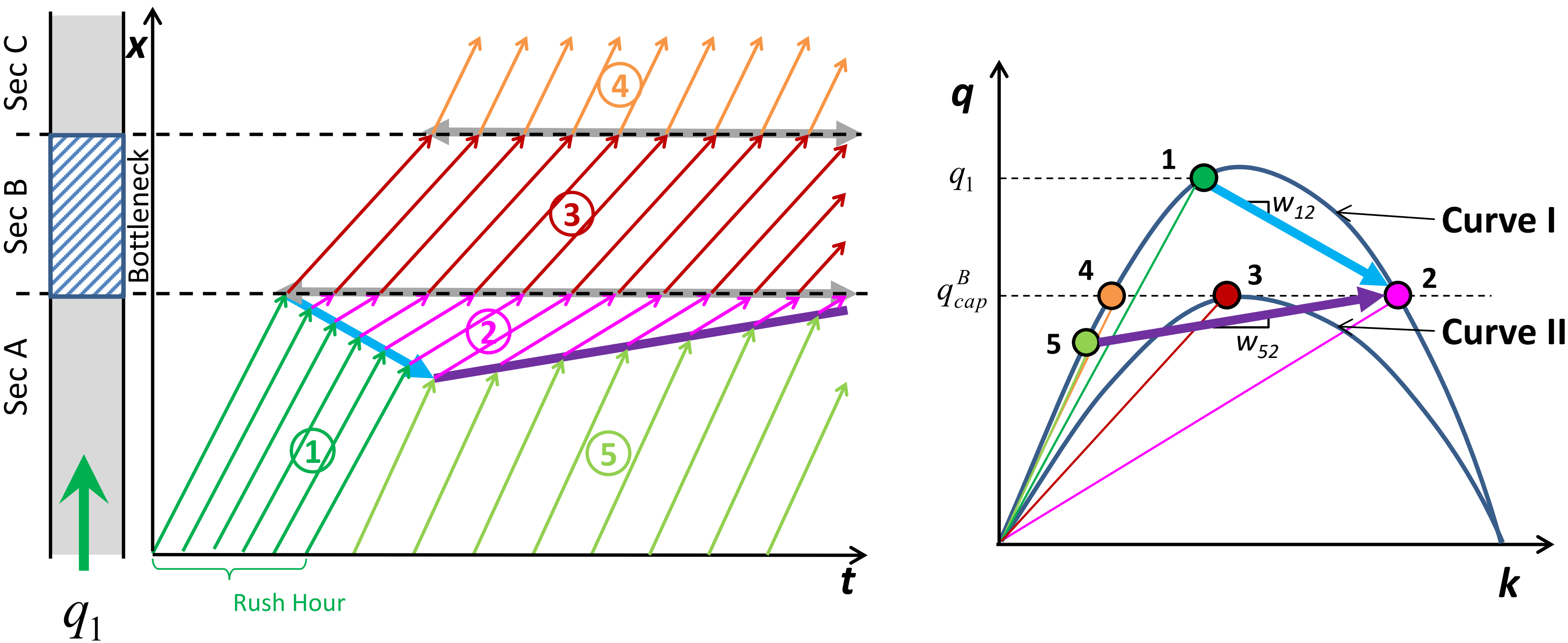
- **The demand flow rate q_1 is greater than Sec B's capacity:** $q_1 > q_{cap}^B$
  - In Sec B, the maximum flow (q_{cap}^B) is served and the section operates at its capacity (**Point 3** on Curve II).
  - Only a flow level q_2 can be passed to Sec C, which operates at **Point 4** on Curve I.





Shockwaves at Stationary Bottleneck

- **The demand flow rate q_1 is greater than Sec B's capacity:** $q_1 > q_{cap}^{B}$
  - In Sec A, the queue under condition 2 grows until the demand decreases to $q_5 < q_{cap}^{B}$. Sec A now operates at Point 5 on Curve I.
  - When the new traffic stream under condition 5 meets condition 2, **a shockwave is formed and propagates downstream at velocity w_{52}**, decreasing the size of queue.



Moving Bottlenecks

Shockwaves at Moving Bottleneck

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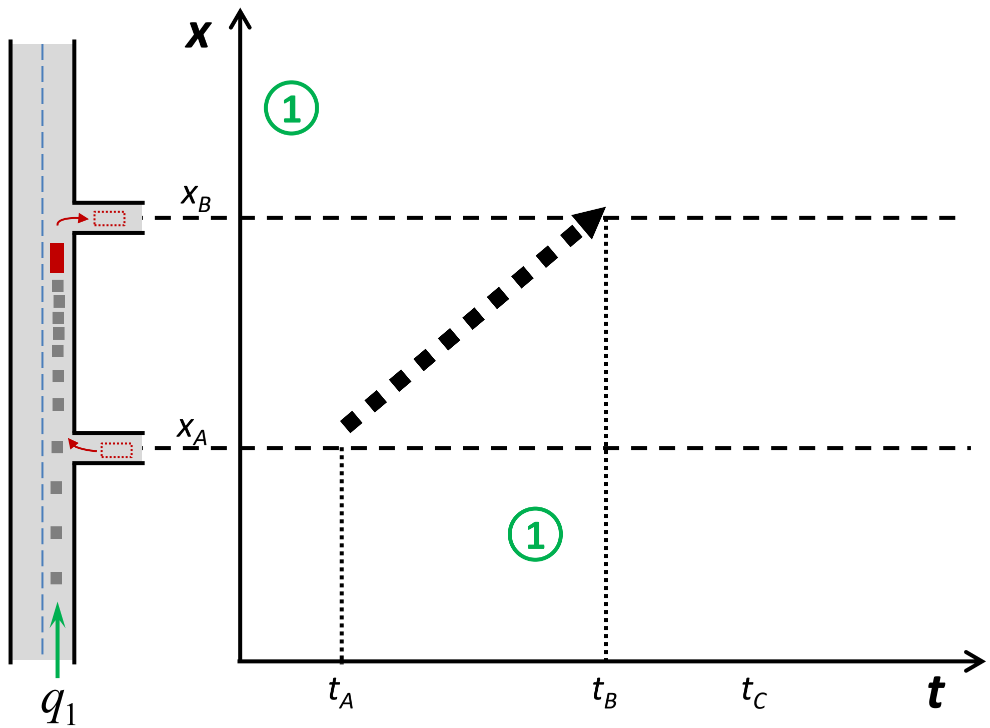
- **Moving Bottleneck**
  - A slow-moving obstruction on a traffic stream; a slowly moving vehicle that represents a bottleneck on a road.
  - Examples
    - A tractor or heavy trucks with a speed of 10-20 km/h on a two-lane road.
    - "rolling roadblocks" or "traffic pacing": a method of temporary traffic control; a police cruiser travelling at a reduced speed (while keeping traffic blocked behind it) to pace the traffic.

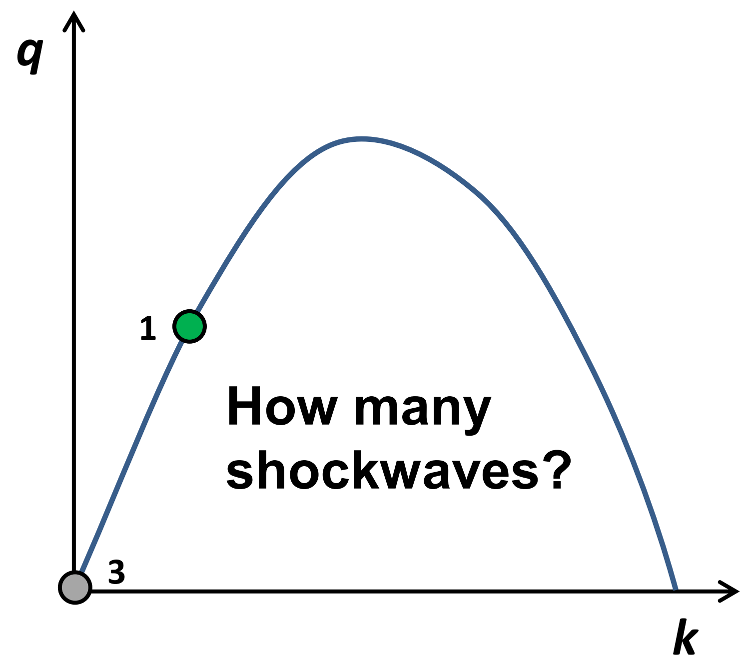
[source: wikipedia]

[source: https://www.flickr.com/photos/inverness_trucker/8466083182]

Shockwaves at Moving Bottleneck

- **Slow-moving Truck Example:**
  - At time t_A, a truck joins a traffic stream operating at q_1, k_1, and v_1 and travels at a much lower speed v_2 until it leaves the road at t_B. Vehicles cannot pass the truck and form a queue behind the truck.





Shockwaves at Moving Bottleneck

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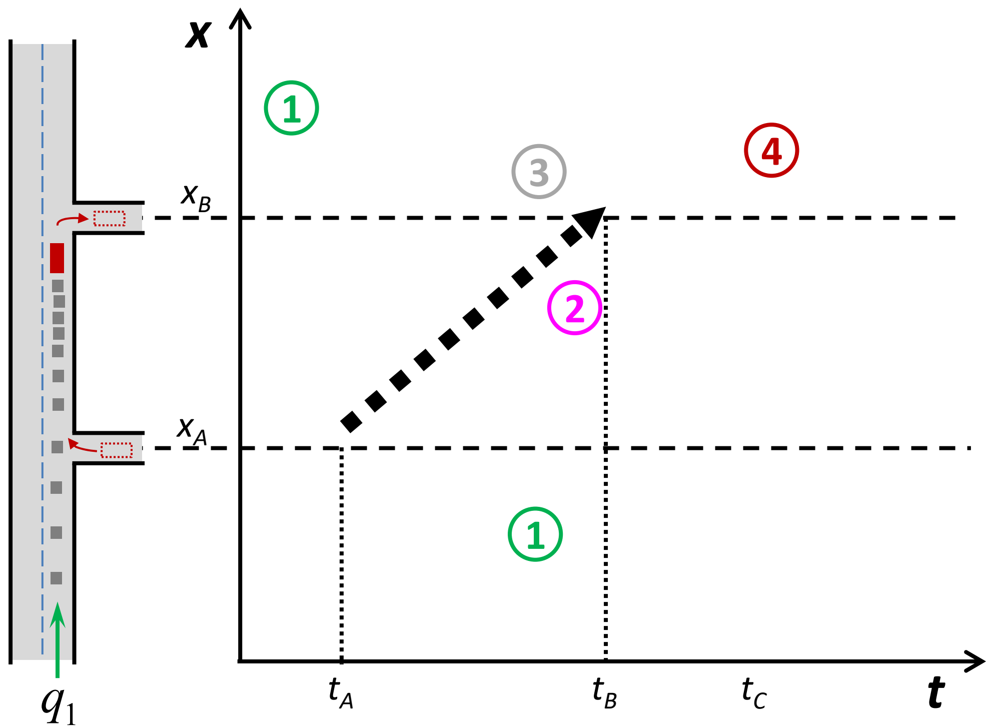
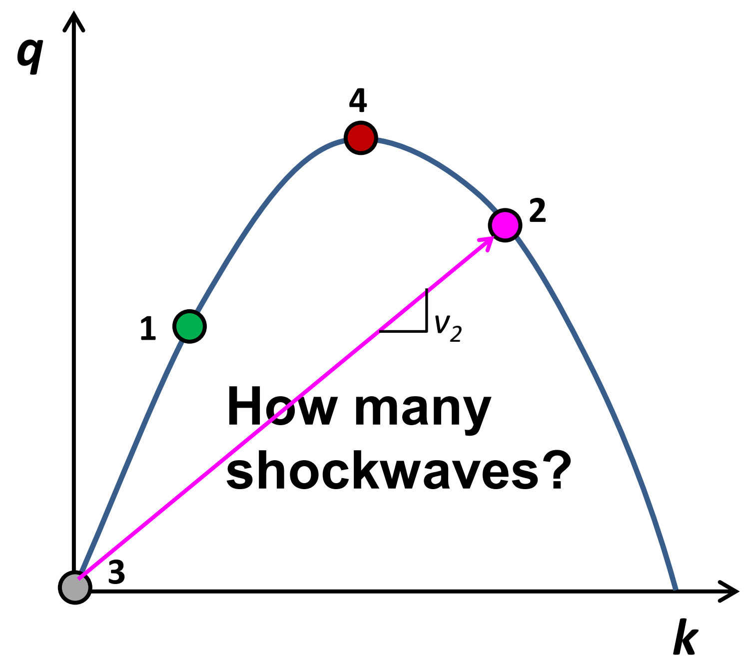
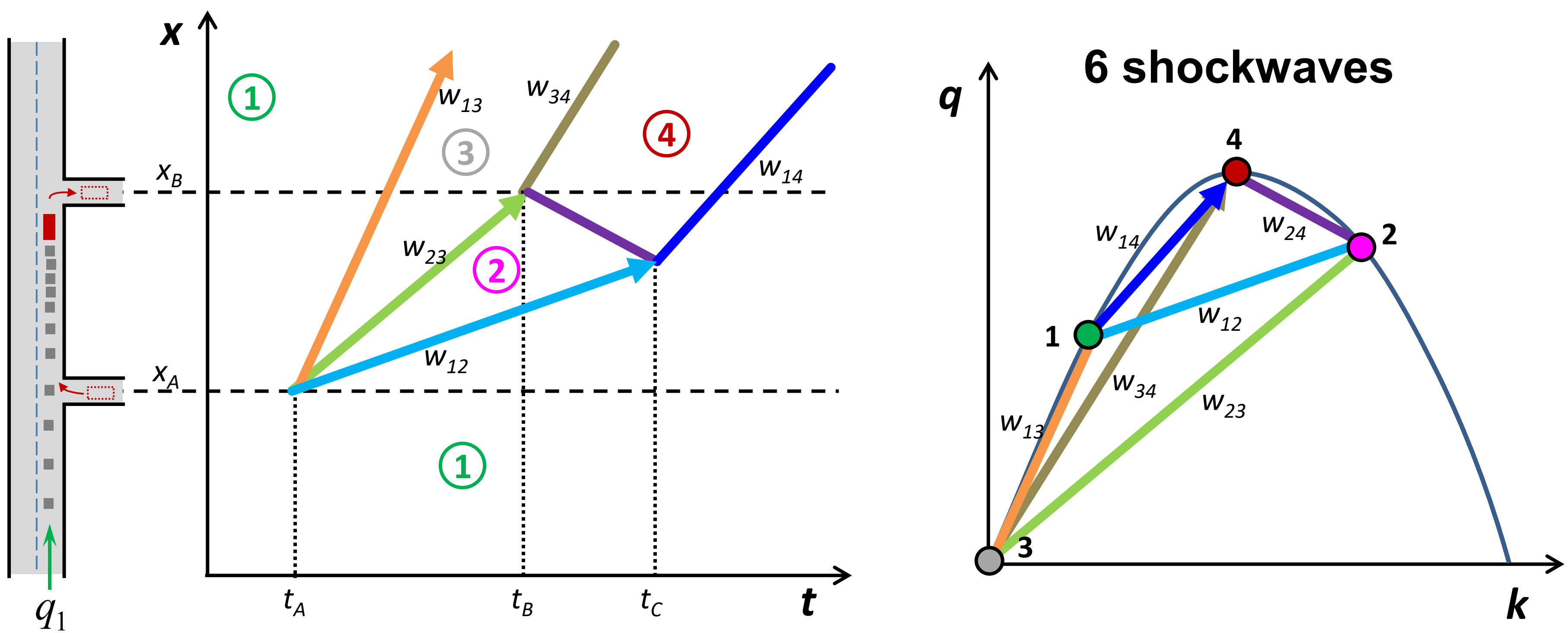




Shockwaves at Moving Bottleneck

- **At time t_A, the truck enters the road:**
  - Vehicles behind the truck slow down to v_2 + upstream traffic in 1 → shockwave w_{12}
  - Vehicles ahead of the truck continue to travel at v_1 + no vehicle in 3 → shockwave w_{13}
  - The truck and platoon moving under 2 + no vehicle in 3 → shockwave w_{23}



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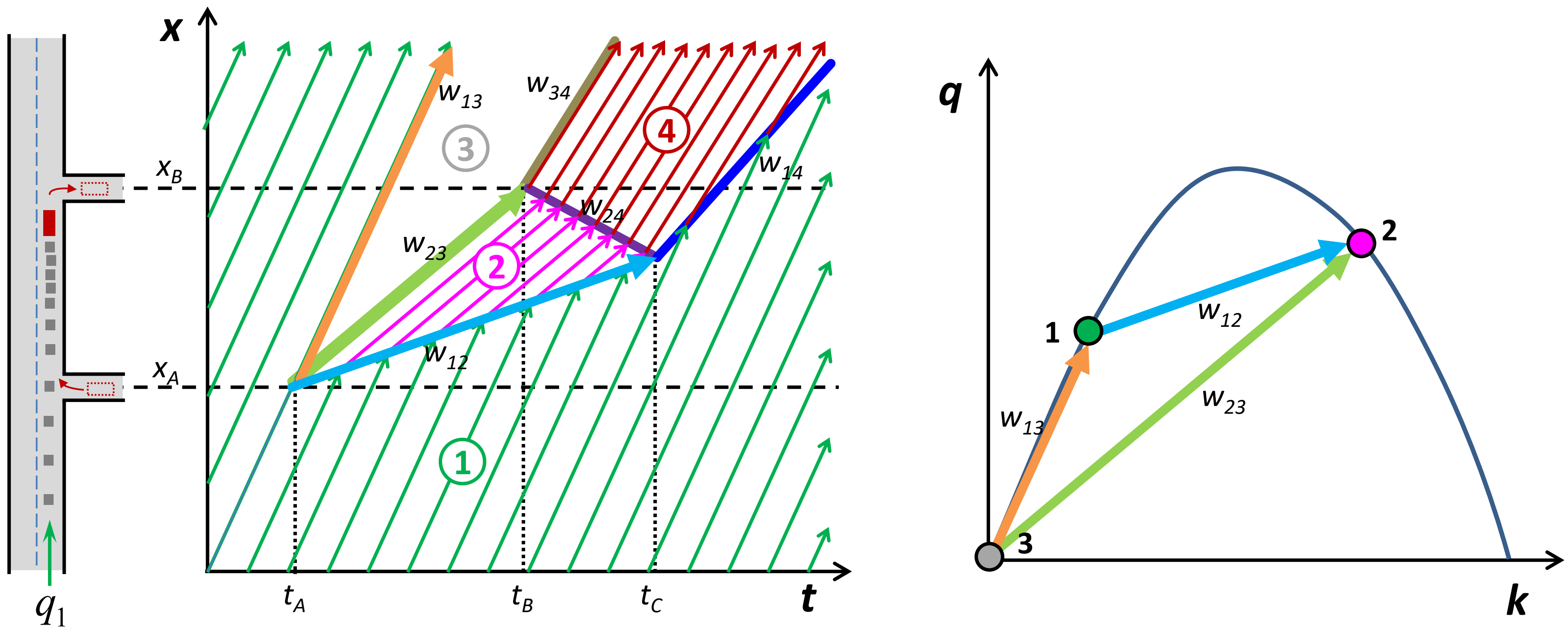
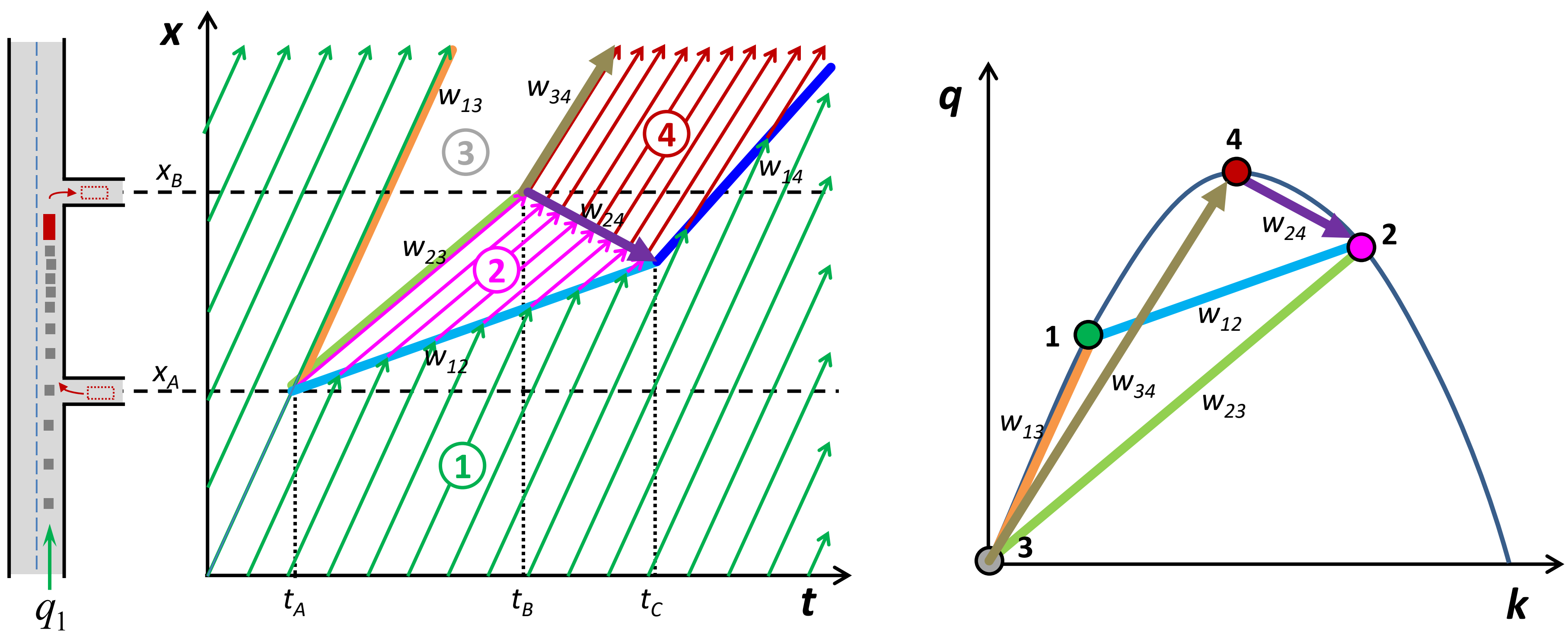


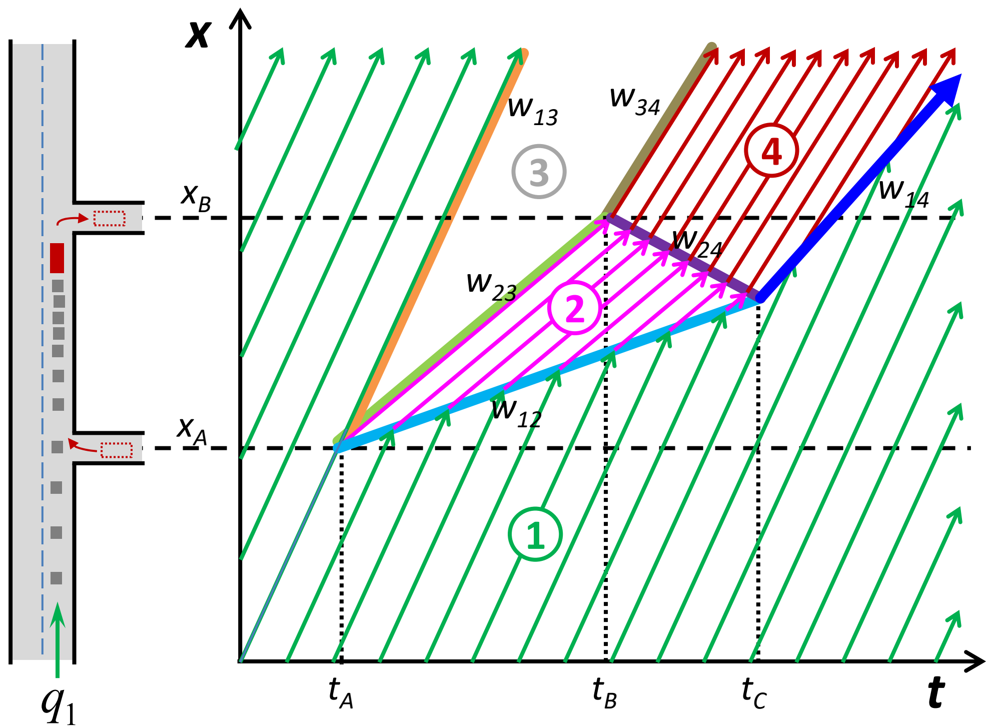
Shockwaves at Moving Bottleneck

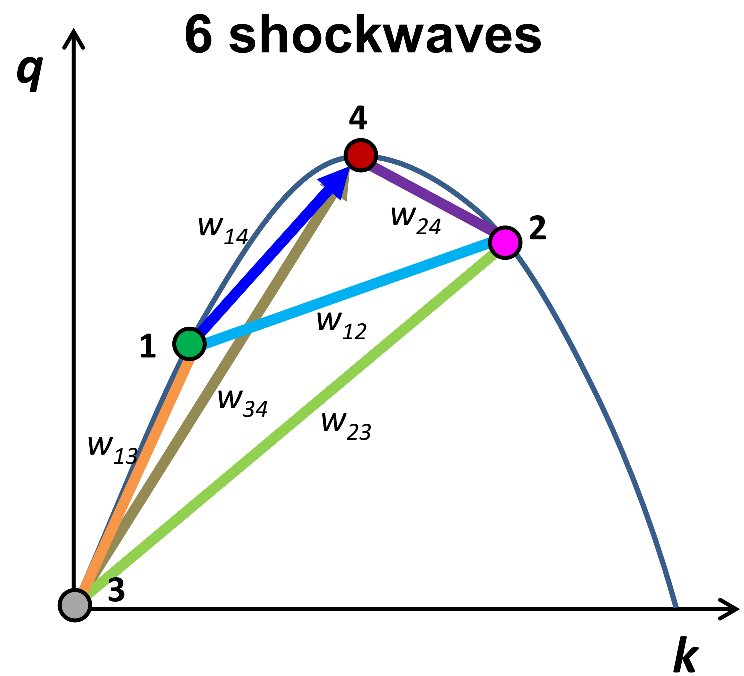
- **At time t_B, the truck leaves the road:**
  - **The flow will be increased to the capacity of the road with condition 4**
    - **creates a forward moving shockwave w_{34}**
    - **creates a backward moving shockwave w_{24}**



Shockwaves at Moving Bottleneck

- At time t_C, two shockwaves w_{12} and w_{24} coincide:
  - The queued condition 2 ends.
  - A new forward moving shockwave w_{14} is formed.





Exercise

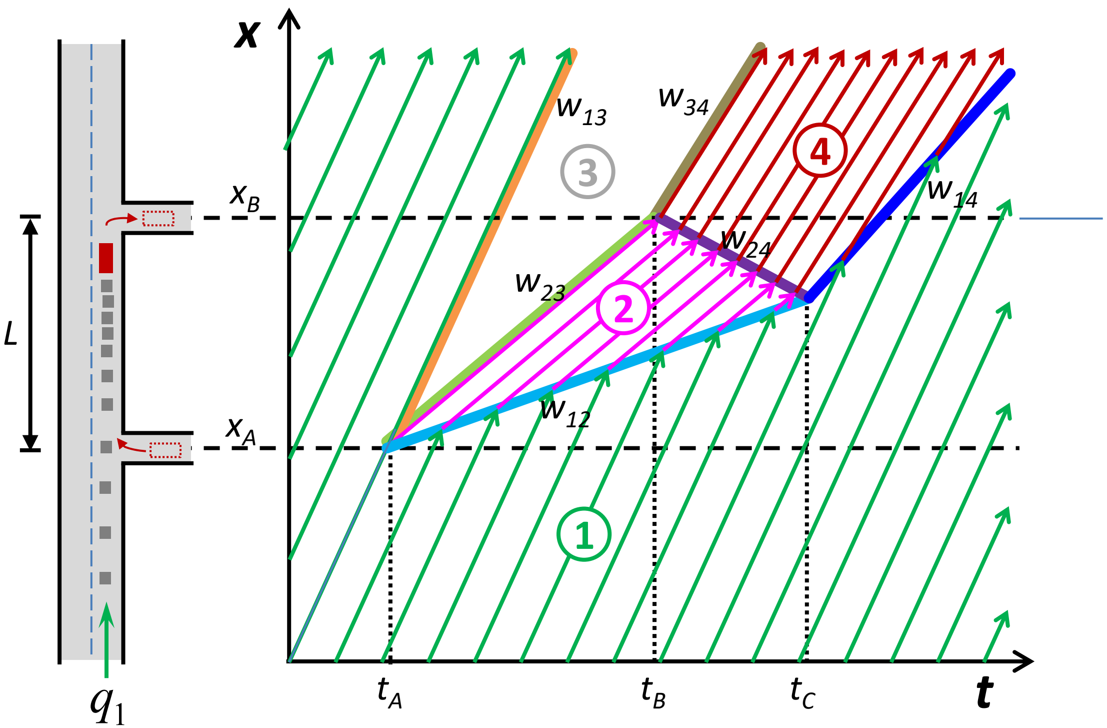
Original traffic condition: q_1 = 1500 veh/h, k_1 = 25 veh/km,

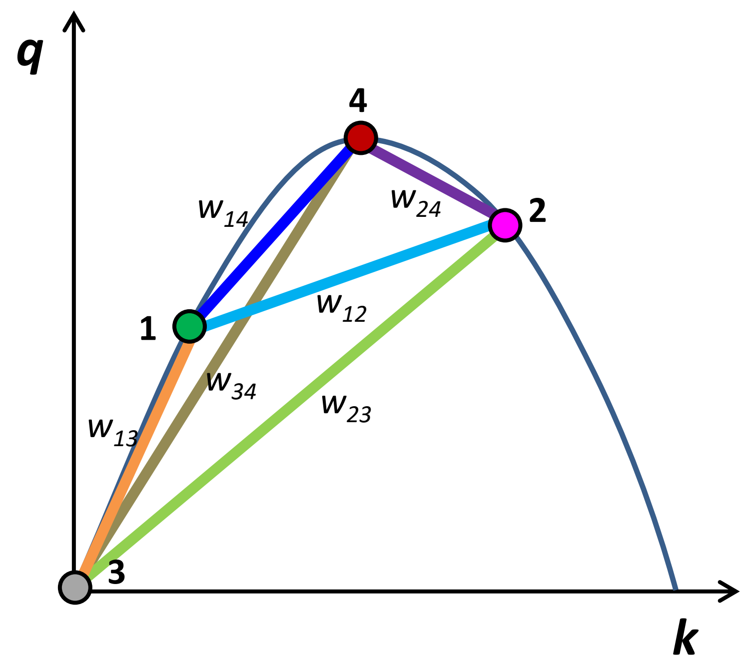
Truck: speed (v_2) = 20 km/h, travel distance L = 2.5 km

Platoon: density k_2 = 100 veh/km

(a) What is the rate at which the platoon grows, in units of vehicles per hour?

(b) How many vehicles will be in the platoon by the time the truck leaves the road?





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(a) What is the rate at which the platoon grows, in units of vehicles per hour?

(b) How many vehicles will be in the platoon by the time the truck leaves the road?

[Solution to (a)]

The velocity of a shockwave upstream of the platoon is:

$$w_{12} = \frac{q_2 - q_1}{k_2 - k_1} = \frac{(100 \times 20) - 1500}{100 - 25} = \frac{500}{75} = 6.67 \text{ km/h}$$

Since the platoon is moving at 20km/km and the shock front moves forward relative to the road at 6.67 km/h, the platoon growth speed is: 20 - (6.67)=13.33 km/h

The rate at which the queue grows, in units of veh/h is:
platoon density * growth speed = 100 * 13.33 = $\boxed{\textbf{1333 veh/h}}$
(or q_1-k_1w_{12} = q_2-k_2w_{12} = 1333 veh/h)

Exercise

Original traffic condition: q_1 = 1500 veh/h, k_1 = 25 veh/km,

Truck: speed (v_2) = 20 km/h, travel distance L = 2.5 km

Platoon: density k_2 = 100 veh/km

(a) What is the rate at which the platoon grows, in units of vehicles per hour?

(b) How many vehicles will be in the platoon by the time the truck leaves the road?

[Solution to (b)]

The time spent by the truck on the road = 2.5[km] / 20[km/h] = 0.125 h

The number of vehicles in platoon after 0.125 h is
= the platoon growth rate * 0.125 = 1333 veh/h * 0.125 h = **167 veh**

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Next

This week

- Tutorial – Shockwaves

CIVL6415, Semester 2, 2024

TRAFFIC ANALYSIS AND SIMULATION

MODULE 4

Macroscopic traffic models

Lecture Week 9

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Outline

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Flow conservation law

First-order models - Cell Tranmission Model (CTM)

Second-order macroscopic models

Ramp Metering

The conservation law

- **An equation relating point flows, i.e. q(t, x), and densities, i.e. k(t, x), in a road section:**
  - **Theory of traffic flow that can be used to predict traffic conditions in the network**
- **No entering or exiting flow to the physical road section**

$$\frac{\partial q}{\partial x} + \frac{\partial k}{\partial t} = 0$$

"it ensures that flow and density variation in space and time is consistent with no entering / leaving traffic hypothesis."

"entering and leaving traffic is equal in the space-time context"

Conservation law (visual)

entering and leaving traffic is equal

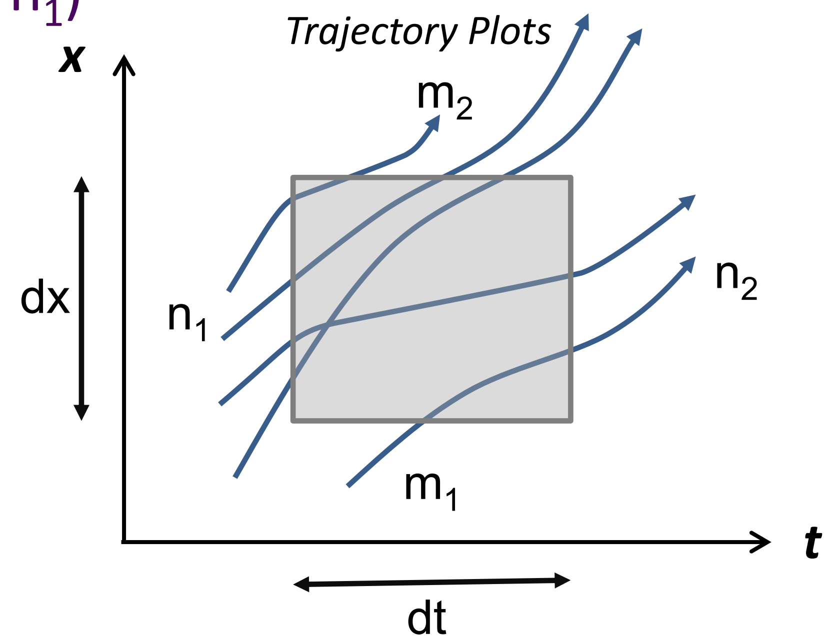
number of trajectories leaving: (m_2+n_2)

number of trajectories entering: (m_1+n_1)

$$(m_2 + n_2) = (m_1 + n_1)$$

$$(m_2 - m_1) + (n_2 - n_1) = 0$$

$$(3-1) + (2-4) = 0$$



Conservation law (visual)

entering and leaving traffic is equal

number of trajectories leaving: (m_2+n_2)

number of trajectories entering: (m_1+n_1)

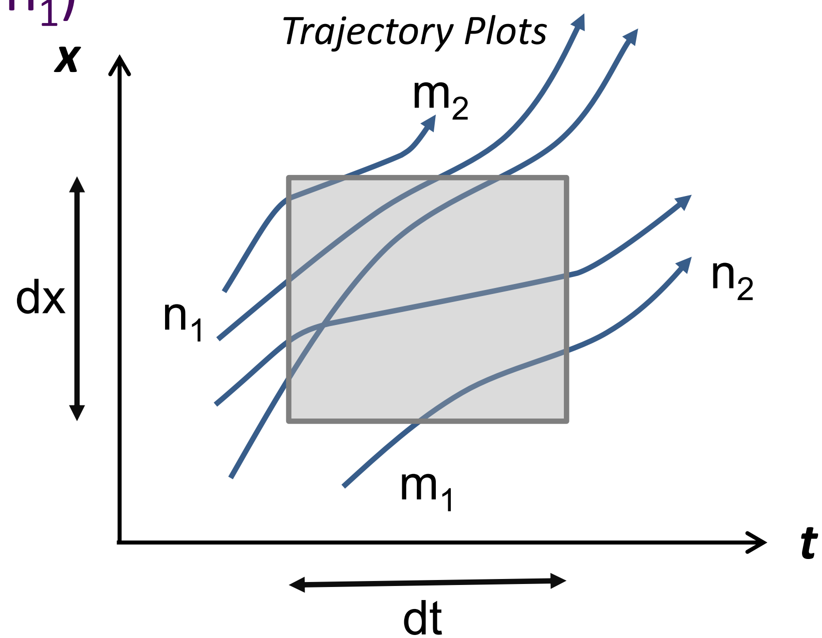
$$(m_2 - m_1) + (n_2 - n_1) = 0$$

(3-1) + (2-4) = 0

$$\frac{(m_2 - m_1)}{(dxdt)} + \frac{(n_2 - n_1)}{(dxdt)} = 0$$

$$\frac{q_{2-}q_1}{dx} + \frac{k_{2-}k_1}{dt} = 0$$

$$\frac{\partial q}{\partial x} + \frac{\partial k}{\partial t} = 0$$



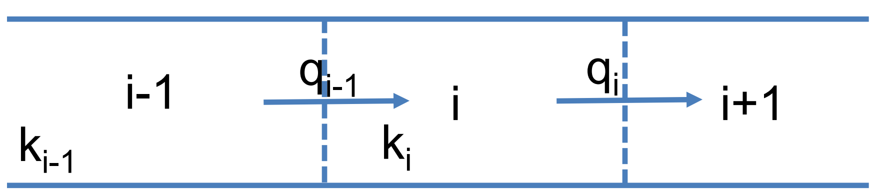
The conservation law

- **Discretised formula**
- **T: length of time step, Δ_i: length of section *i***

$$\frac{\partial q}{\partial x} + \frac{\partial k}{\partial t} = 0$$

$$\frac{q_i(t) - q_{i-1}(t)}{\Delta_i} + \frac{k_i(t+1) - k_i(t)}{T} = 0$$

$$k_i(t+1) = k_i(t) + \frac{T}{\Delta_i}[q_{i-1}(t) - q_i(t)]$$



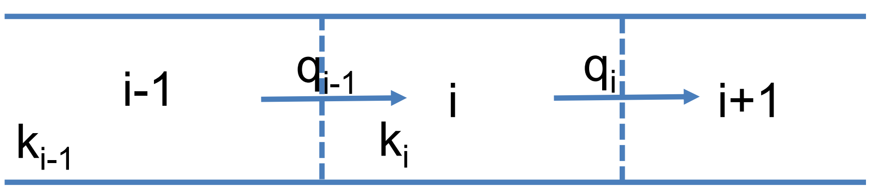
The conservation law

- **Discretised formula**
- **T: length of time step, Δ_i: length of section *i***

$$k_i(t+1) = k_i(t) + \frac{T}{\Delta_i}\underbrace{[q_{i-1}(t) - q_i(t)]}_{\text{Inflow-Outflow}}$$

Density at next time step

Density at current time step



Question

Flow conservation

Which one of the statements below is TRUE?

A. **The number of vehicles entering and leaving a physical roadway section in a given amount of time is always the same.**

B. **Flow conservation law holds only if vehicles do not overtake each other**

C. **Entering traffic is equal to leaving traffic in space-time context**

D. **Flow conservation law cannot be used to predict traffic conditions**

E. **None**

Cell Transmission Model (CTM)

CTM – Numerical scheme

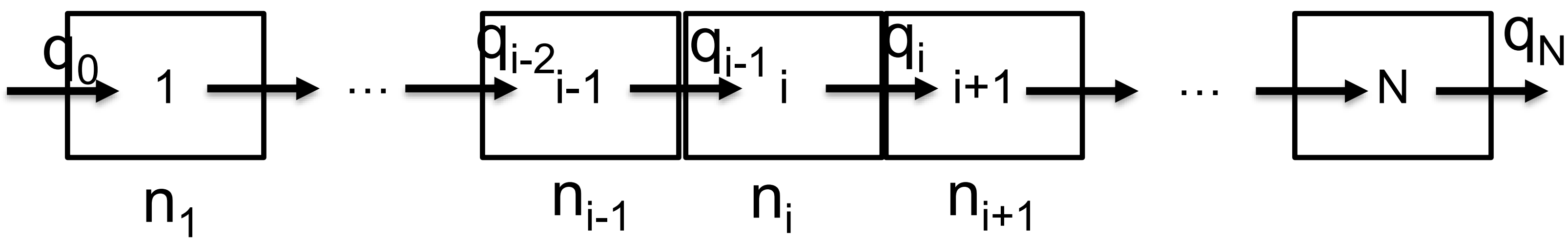
- Analytical solution to kinematic wave model is exact
- Only applicable in relatively simple situations, e.g. with respect to upstream traffic demand, off-ramps and on-ramps, etc.
- What to do when demand on main-road and on-ramps is changing dynamically? Use numerical approximations!

- Discrete difference equations
- First order discrete Godunov approximation of LWR

Daganzo, Carlos F. "The cell transmission model: A dynamic representation of highway traffic consistent with the hydrodynamic theory." *Transportation Research Part B: Methodological* 28.4 (1994): 269-287.

Daganzo, Carlos F. "The cell transmission model, part II: network traffic." Transportation Research Part B: Methodological 29.2 (1995): 79-93.

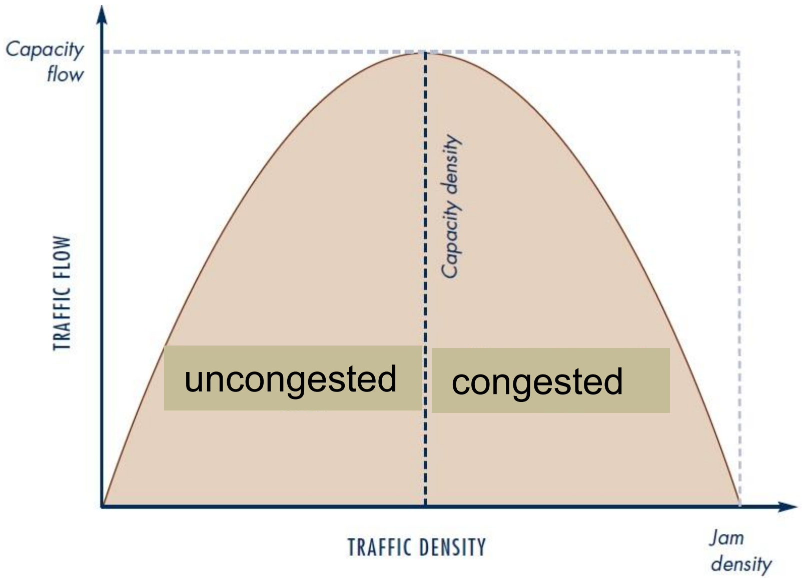
CTM assumptions



- Divide roadway into cells i, length dx
- Divide time into steps with length dt

- Cells are homogeneous
- Within the time step, traffic is stationary!

Fundamental diagram



The steady state relationship between flow and density

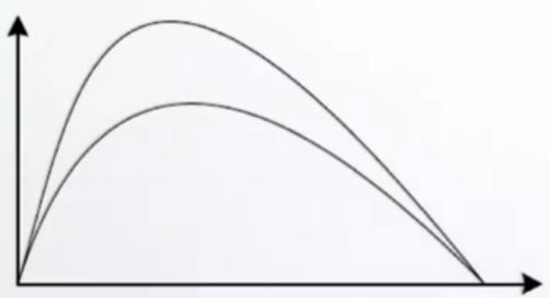
$$V(\rho) = v_f \cdot \exp\left(-\frac{1}{2}\left(\frac{\rho}{\rho_{cr}}\right)^2\right)$$

$$Q(\rho) = v_f \cdot \rho \cdot \exp\left(-\frac{1}{2}\left(\frac{\rho}{\rho_{cr}}\right)^2\right)$$

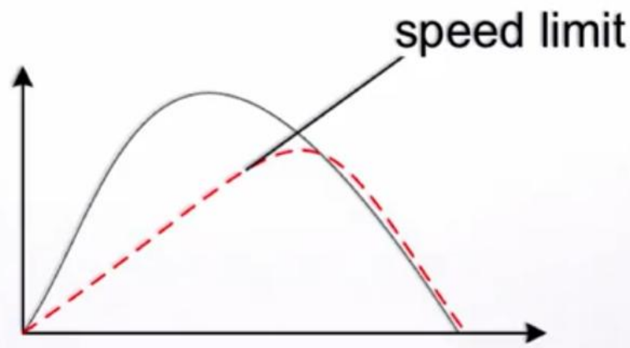
- Macroscopic relationship: $Q(\rho) = \rho \cdot V(\rho)$
- It does not violate the conservation equation!

FD - examples

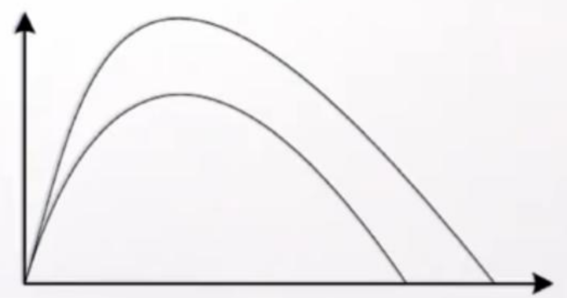
Uphill, curvature



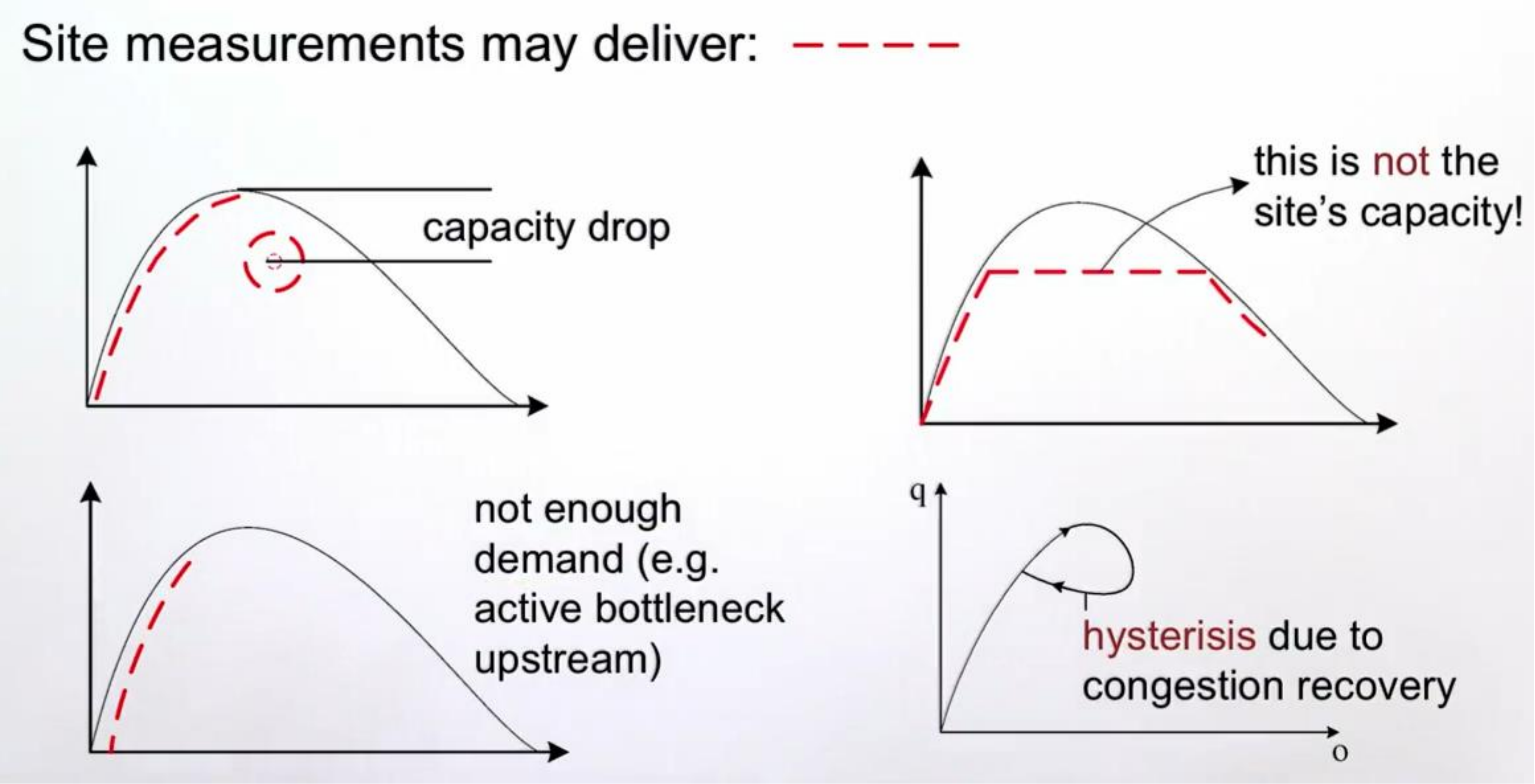
Variable speed limit (VSL)



Lane drop



FD - limitations



Demand / Supply FDs

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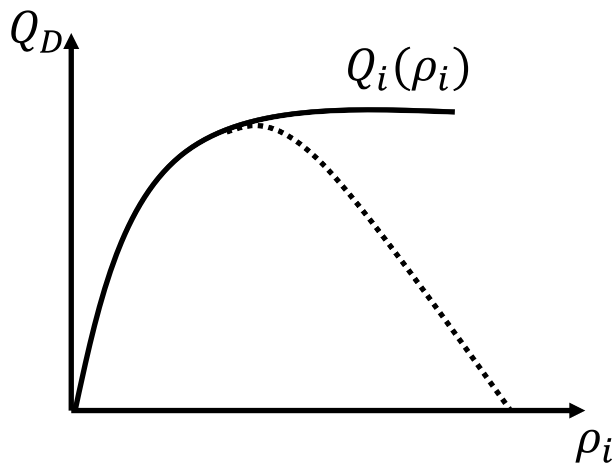

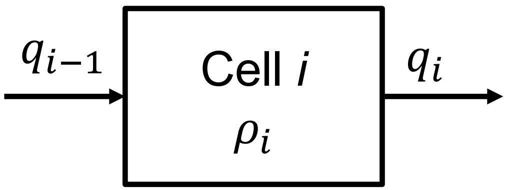

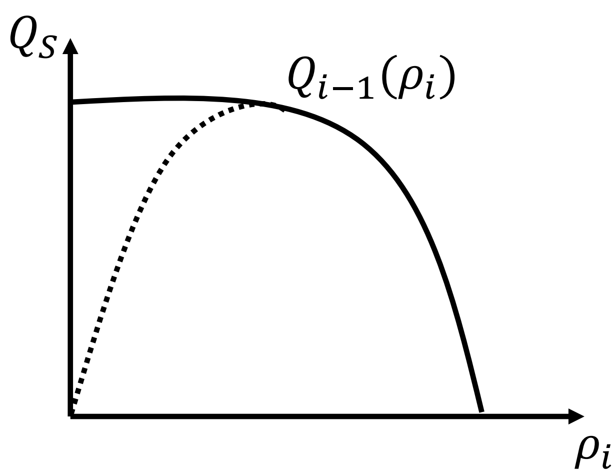

The flows from one cell to the other restricted by

- – Demand capacity
- – Supply capacity

Traffic variables

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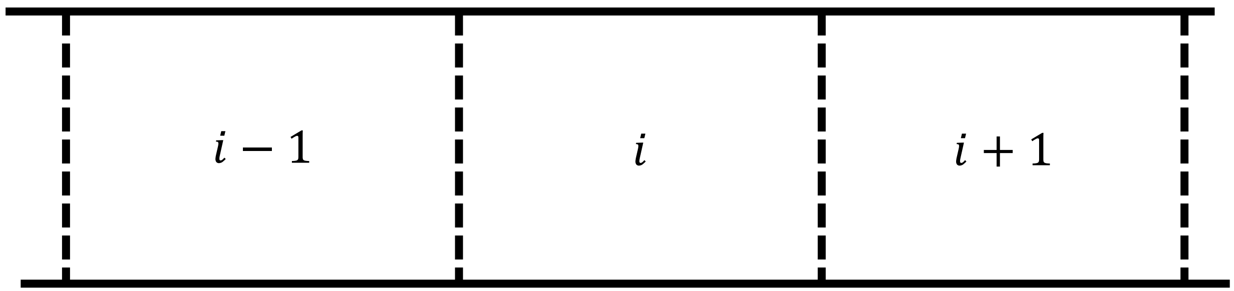
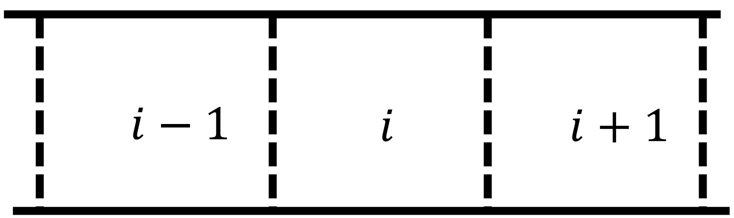


- Density $\rho_i(k)$: density in cell i at time step k or $t = kT$
- Speed $v_i(k)$: speed in cell i at time step k or $t = kT$
- Flow $q_i(k)$: number of vehicles leaving cell i between time step $k-1$ and k or $[(k-1)T, kT]$

$$q_i(k)=\rho_i(k) \cdot v_i(k)$$

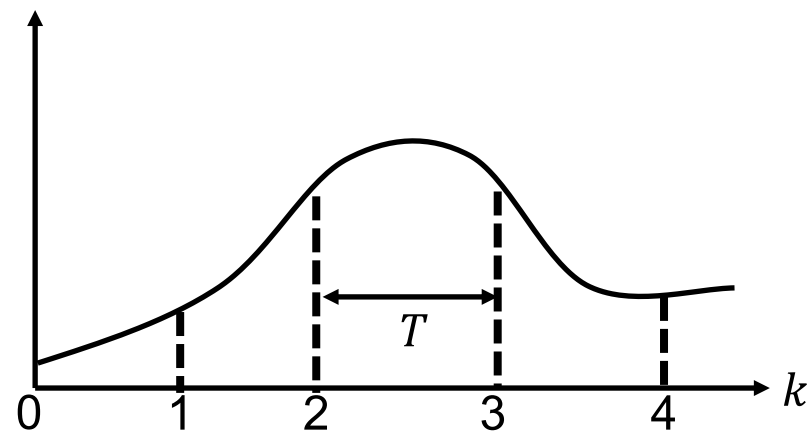
Spatial and temporal modelling

Spatial discretisation



Δ_i ≈10-500m

Temporal discretisation



T ≈0.2-15s

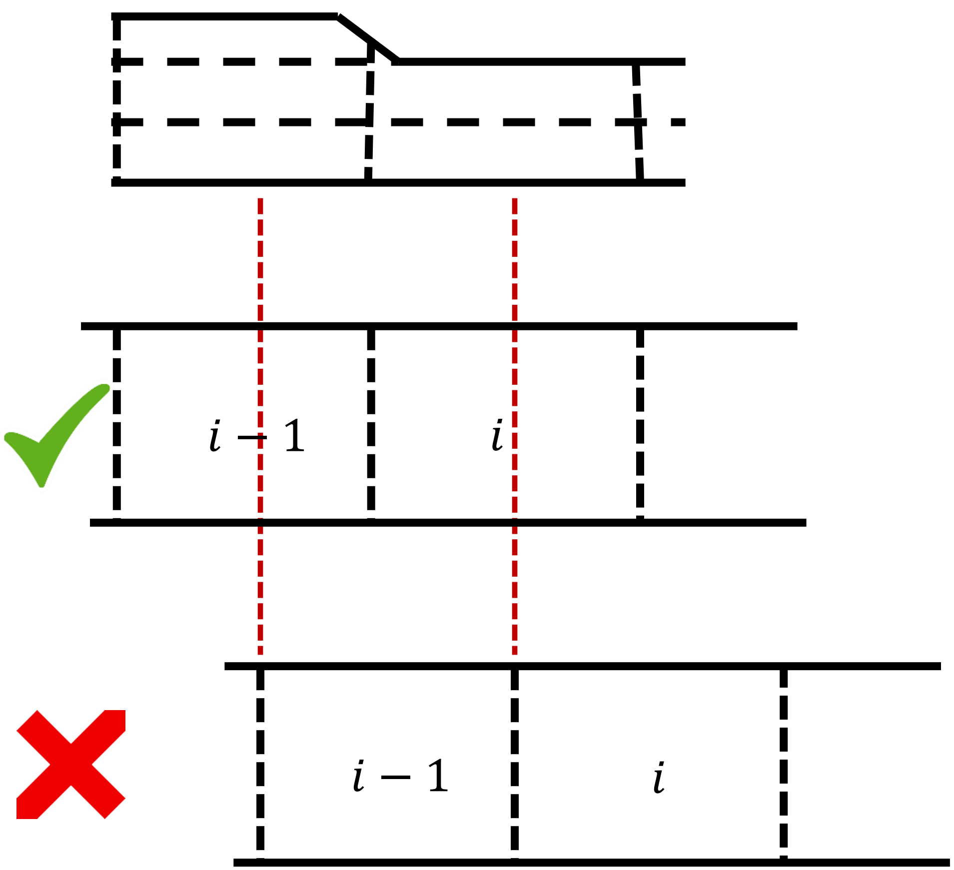
CFL conditions! $T < \dfrac{\Delta_i}{v_f} \quad \forall i$

- In mathematics, the Courant–Friedrichs–Lewy (CFL) condition is a necessary condition for convergence while solving certain partial differential equations (usually hyperbolic PDEs) numerically.

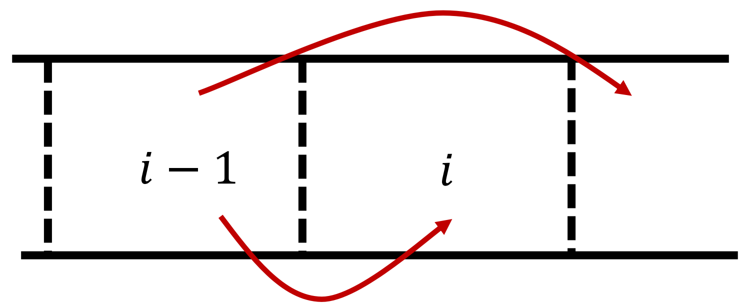
Courant, R.; Friedrichs, K.; Lewy, H. (1928), "Über die partiellen Differenzengleichungen der mathematischen Physik", Mathematische Annalen (in German), 100 (1): 32–74.

Spatial and temporal modelling

Spatial discretisation



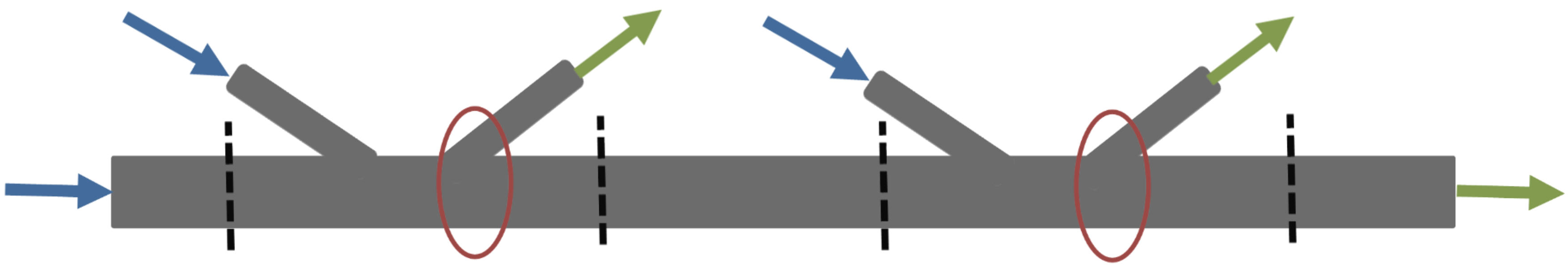
Temporal discretisation



CFL conditions! $T < \frac{\Delta_i}{v_f} \quad \forall i$

Vehicles should not be able to traverse more than one cell in one time step

Freeway structure



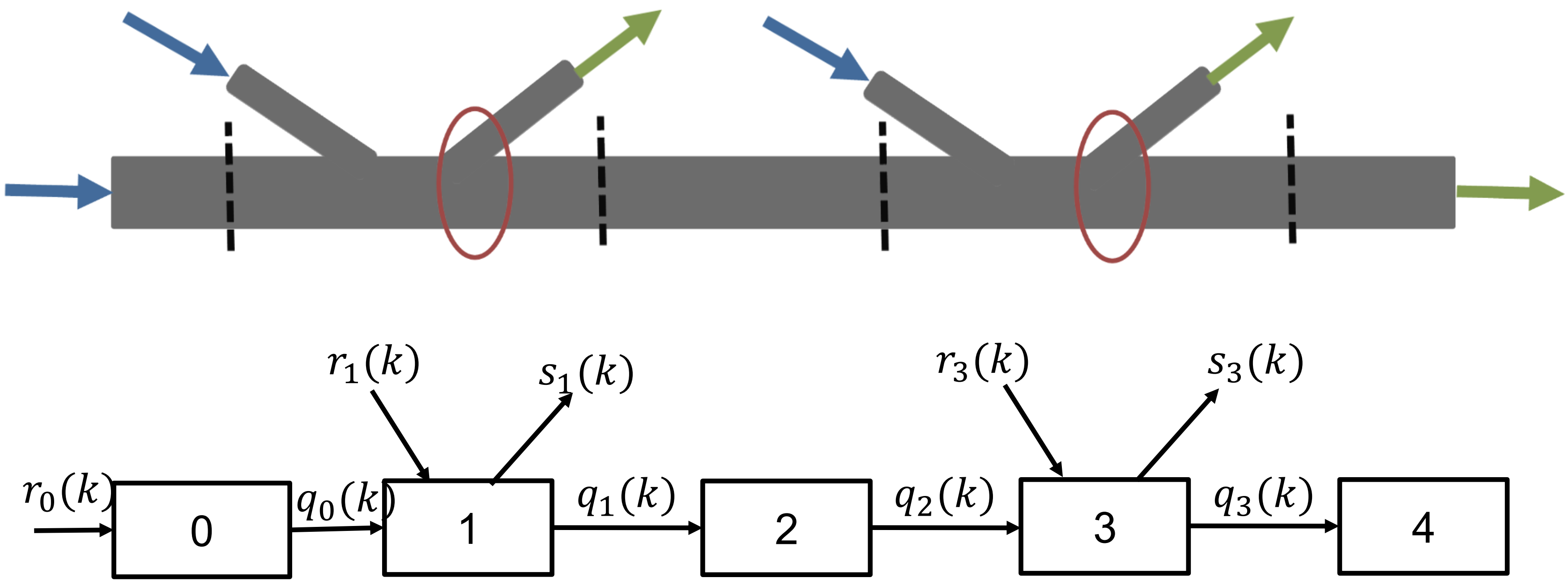
- Three requirements

Known demands

Known split ratios

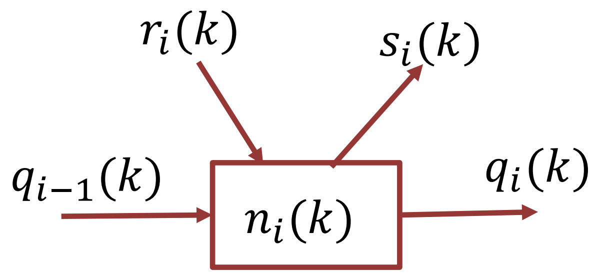
Unobstructed outflows

Freeway structure



- A cell has at most one on-ramp and one off-ramp
- On-ramps have known inflows
- Off-ramps have known split-ratios

CTM - Interflows



Split ratios

$$\beta_i(k) \in [0,1]$$
$$\overline{\beta}_i(k) = 1 - \beta_i(\text{k})$$

$$n_i(k+1) = n_i(k) + T\,[q_{i-1}(k) + r_i(k) - q_i(k) - s_i(k)]$$

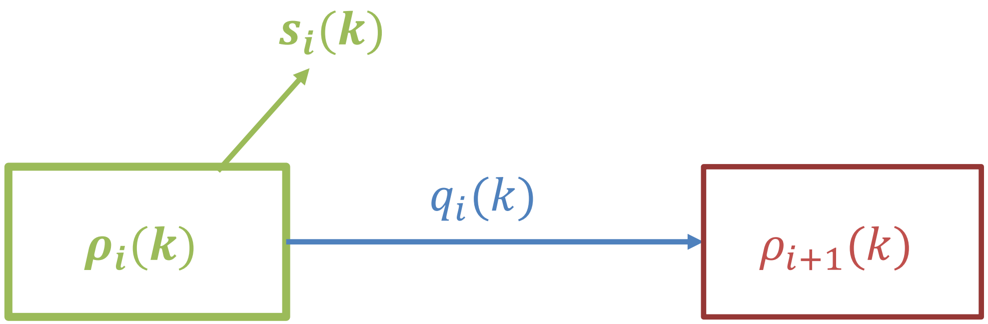
$$\rho_i(k+1) = \rho_i(k) + \frac{T}{\Delta_i}\,[q_{i-1}(k) + r_i(k) - q_i(k) - s_i(k)]$$

Off-ramp flows

$$s_i(k) = \beta_i(k)[s_i(k) + q_i(k)]$$

$$s_i(k) = \frac{\beta_i(k)}{\overline{\beta}_i(k)}\,q_i(k)$$

CTM - Interflows



Demand for space
$v_i\rho_i(k) - s_i(k)$

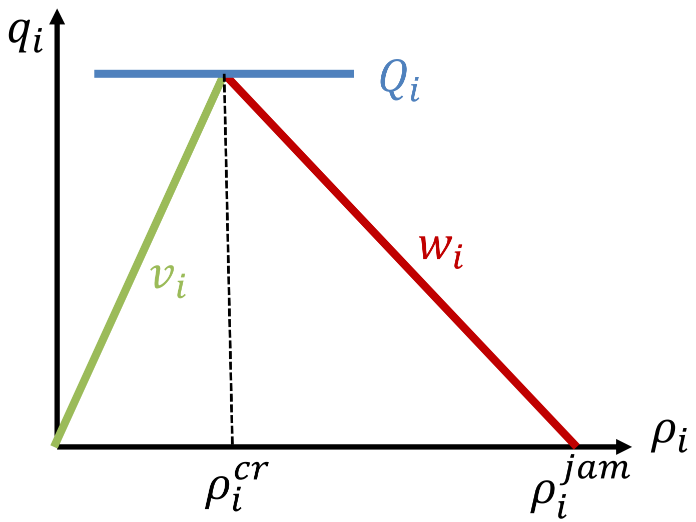
Capacity
Q_i

Supply for space
$w_{i+1}\left(\rho_{i+1}^{jam} - \rho_{i+1}(k)\right)$

$$q_i(k) = \min\left(\bar{\beta}_i(k)v_i\rho_i(k), w_{i+1}\left(\rho_{i+1}^{jam} - \rho_{i+1}(k)\right), Q_i\right)$$

CTM - FD

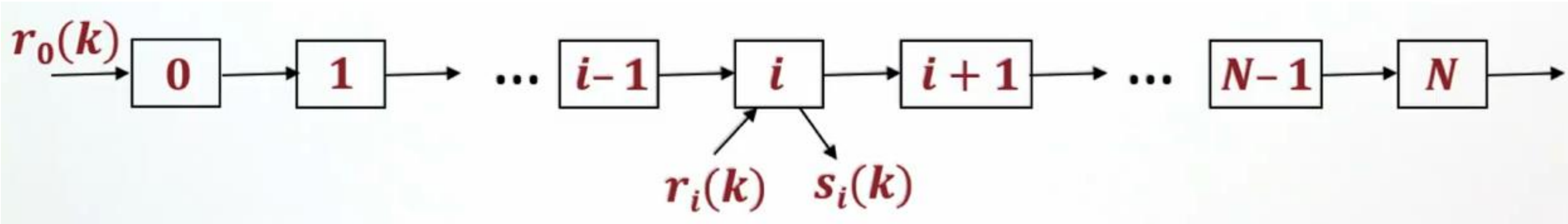
$$q_i(k) = \min\left(\bar{\beta}_i(k) v_i \rho_i(k), w_{i+1}\left(\rho_{i+1}^{jam} - \rho_{i+1}(k)\right), Q_i\right)$$



- Triangular FD implies, at ρ_i=ρ_i^{cr}

$$Q_i = \bar{\beta}_i(k)\, v_i\ \rho_i^{cr} = w_{i+1}\left(\rho_{i+1}^{jam} - \rho_i^{cr}\right)$$

CTM – Putting it together



$$q_i(k) = \min\left(\overline{\beta}_i(k) v_i \rho_i(k), w_{i+1}\left(\rho_{i+1}^{jam} - \rho_{i+1}(k)\right), Q_i\right), \qquad 0 \le i \le N-1$$

$$q_N(k) = \min\left(\overline{\beta_N}(k) v_N \rho_N(k), Q_N\right)$$

$$\rho_i(k+1) = \rho_i(k) + \frac{T}{\Delta_i}\left[q_{i-1}(k) + r_i(k) - \frac{q_i(k)}{\overline{\beta_i(k)}}\right], \qquad 1 \le i \le N$$

$$\rho_0(k+1) = \rho_0(k) + \frac{T}{\Delta_i} T \left[r_0(k) - \frac{q_0(k)}{\overline{\beta_0(k)}}\right]$$

What if congested?

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Density in cell *i* cannot exceed the jam density

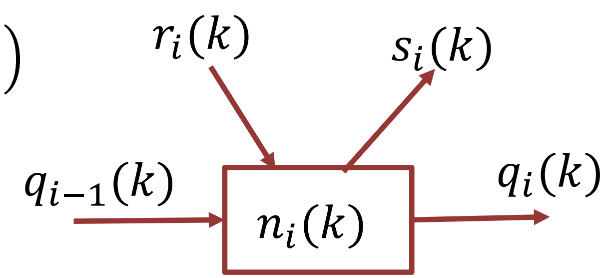
1. Define a virtual queue for the on-ramp with $N_i(k)$ denoting the number of vehicles at time kT and $N_i(0)$=0 .

2. At every time-step, calculate the two inflows that want to enter the cell as;

$$\overline{q_{i-1}}(k) = \min\left(\overline{\beta_{i-1}}(k) v_{i-1}\rho_{i-1}(k), w_i(k)\left(\rho_i^{jam} - \rho_i(k)\right), Q_{i-1}\right)$$

$$\overline{r_i}(k) = d_i(k) + \frac{N_i(k)}{T}$$

where $d_i(k)$ is the demand



3. If $\overline{q_{i-1}}(k) + \overline{r_i}(k) \leq w_i(k)\left(\rho_i^{jam} - \rho_i(k)\right)$,

then you can apply these inflows,

i.e. $q_{i-1}(k) = \overline{q_{i-1}}(k)$ and $r_i(k) = \overline{r_i}(k)$.

What if congested?

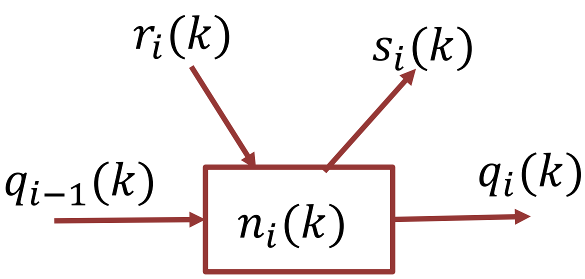
4. Else, the actual inflows are calculated

$$q_{i-1}(k) = \frac{\overline{q_{i-1}}(k)}{\overline{q_{i-1}}(k) + \overline{r_i}(k)} \cdot w_i(k) \left(\rho_i^{jam} - \rho_i(k)\right)$$

$$r_i(k) = \frac{\overline{r_i}(k)}{\overline{q_{i-1}}(k) + \overline{r_i}(k)} \cdot w_i(k) \left(\rho_i^{jam} - \rho_i(k)\right)$$

5. In both cases, the virtual queue is updated by

$$N_i(k+1) = N_i(k) + [d_i(k) - r_i(k)] \cdot T$$



Second-Order Models

Second-Order Models

CTM: a first-order traffic model

Issues with the first-order models

- Instantaneous change of speeds – Infinite acceleration
- Motion through shockwaves
- Driver anticipation not considered

Higher-order models

- Attempt to represent the acceleration of the traffic stream

Acceleration = *f(traffic ahead, steady-state speed, reaction time,…)*

Driver anticipation

Anticipation based on the density at $x + \Delta x$

$$v(x,t) = V\big(\rho(x+\Delta x, t)\big)$$

Δx>0

$$V(\rho) = v_f \cdot \exp\left(-\frac{1}{2}\left(\frac{\rho}{\rho_{cr}}\right)^2\right)$$

$v(x,t)$: Speed at location *x* and time *t*

Taylor series expansion;

$$v(x,t) = V\big(\rho(x,t)\big) + \left[\frac{\partial V}{\partial \rho}\frac{\partial \rho}{\partial x}\right]_{x,t} \cdot \Delta x$$

with $\Delta x \sim 1/\ \rho$ (1 vehicle ahead) – e.g., $A/\ \rho$ (A vehicles ahead)

Assume an anticipation constant; $\upsilon = -A\,\partial V/\partial \rho > 0$: constant parameter

$$v(x,t) = V\big(\rho(x,t)\big) - \frac{\upsilon}{\rho(x,t)}\frac{\partial \rho}{\partial x}$$

Discretised version;

$$v_i(k) = V(\rho_i(k)) - \frac{\upsilon}{\Delta_i}\frac{\rho_{i+1}(k) - \rho_i(k)}{\rho_i(k) + \kappa}$$

A constant for the network
Gives the flexibility to calibrate

Reaction time

Reaction time, τ

$$v(x, t+\tau) = V\big(\rho(x+\Delta x, t)\big)$$

$$v + \tau \cdot \frac{dv}{dt} = V(\rho) - \frac{v}{\rho}\frac{\partial \rho}{\partial x}$$

Taylor series expansion

with Chain rule

$$\frac{dv}{dt} = \frac{\partial v}{\partial t} + \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial t} = \frac{\partial v}{\partial t} + v\frac{\partial v}{\partial x}$$

We obtain

$$\frac{\partial v}{\partial t} = \frac{dv}{dt} - v\frac{\partial v}{\partial x} = \left[V(\rho) - \frac{v}{\rho}\frac{\partial \rho}{\partial x} - v\right] \cdot \frac{1}{\tau} - v\frac{\partial v}{\partial x}$$

Discretised version: $\frac{\partial v}{\partial t} = \frac{v_i(k+1)-v_i(k)}{T}$ and $\frac{\partial v}{\partial x} = \frac{v_i(k)-v_{i-1}(k)}{\Delta_i}$

Dynamic equation

$$v_i(k+1) = v_i(k)$$

$$+\frac{T}{\tau}[V(\rho_i(k)) - v_i(k)]$$

Relaxation (to the steady-state)

$$+\frac{T}{\Delta_i} v_i(k)[v_{i-1}(k) - v_i(k)]$$

Convection (tendency to travel with current speed)

$$-\frac{\upsilon T}{\tau \Delta_i} \frac{\rho_{i+1}(k) - \rho_i(k)}{\rho_i(k) + \kappa}$$

Anticipation (impact of density ahead)

Parameters to calibrate: τ, υ, κ

Second-order models

Advantages:

- potential improvement of accuracy,
- fundamental diagram “noise” and capacity drop,
- traffic instability,
- congestion outflow

Limitations

- Inaccurate results with low traffic volumes
- lane-use restrictions, off-ramp spillback, dedicated lanes

No need to memorise the equations!!

Question

Macroscopic models

Which one of the statements below is NOT TRUE?

A. **CTM is a numerical approximation of LWR theory**
B. **CTM is a first-order model, as it does not account for the acceleration of the traffic stream**
C. **Second-order models can produce capacity drop**
D. **CTM accounts for driver anticipation**
E. **None**

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Freeway Control:
Ramp Metering

Freeway / Motorway / Highway

- Originally…
  - sufficient capacity for virtually unlimited mobility (without the need for control)

- Ever increasing demand…
  - Congestions
  - Degraded infrastructure use

- Congestion: recurrent and non-recurrent
- Goal: Operate freeway networks optimally (as a **controllable** system)

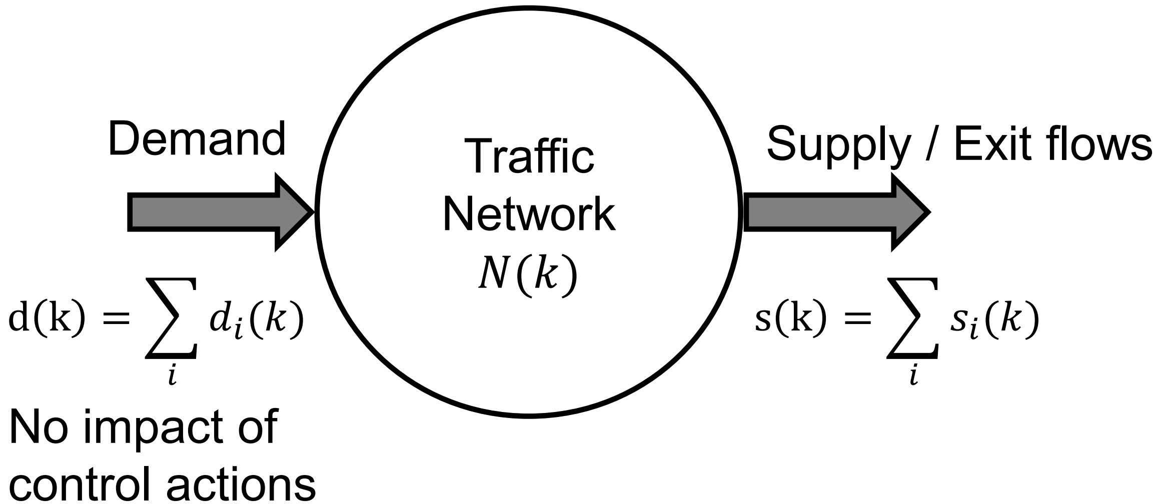
Control Techniques

- Ramp Metering
  - (direct impact on density ⇒ congestion)
- Link Control
  - (variable speed limits, keep lane, lane control, congestion warning, tidal flow, environmental conditions: fog, ice, rain)
- Driver Information and Guidance
  - VMS Information
  - VMS Route Recommendation
  - Individual Route Guidance

https://www.youtube.com/watch?v=8G7ViTTuwno.

https://youtu.be/QDMYODIgLcs?t=73

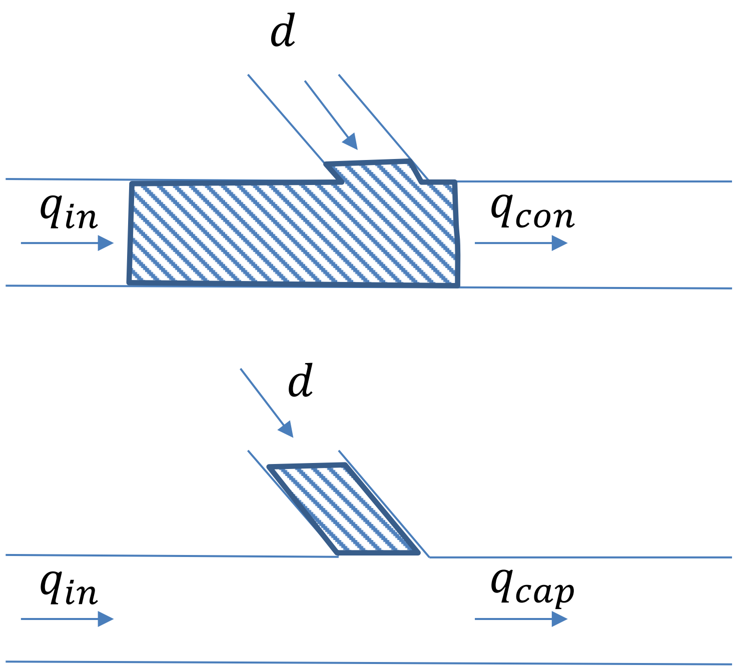
Traffic Control



- Total travel time or Total time spent $TTS = T\sum_{k=0}^{K} N(k)$
- How to minimise total time spent [veh.h]?
- $\min TTS,\ \max \sum_{k=0}^{K}(K-k)\,s(k)$

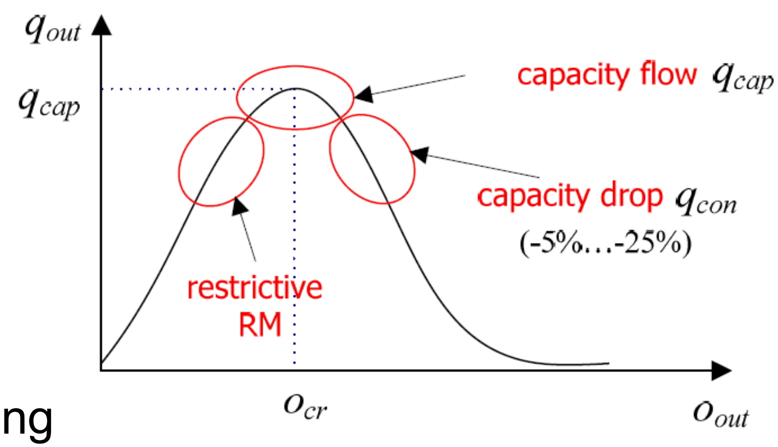
Ramp Metering

- Why useful? ***1st answer***



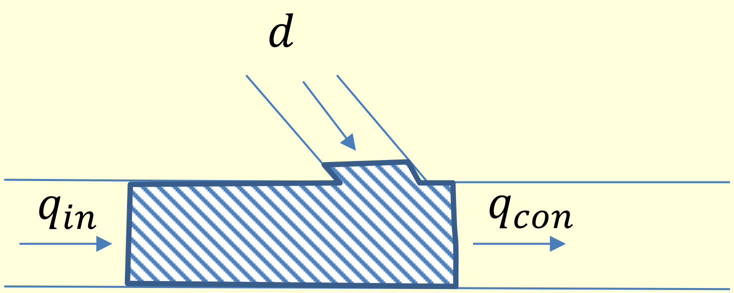
No Control

Ramp Metering

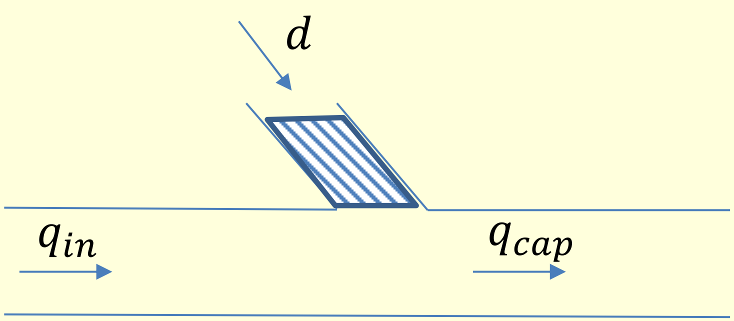


Ramp metering

- Calculate travel time savings in terms of the variables given below.



No Control

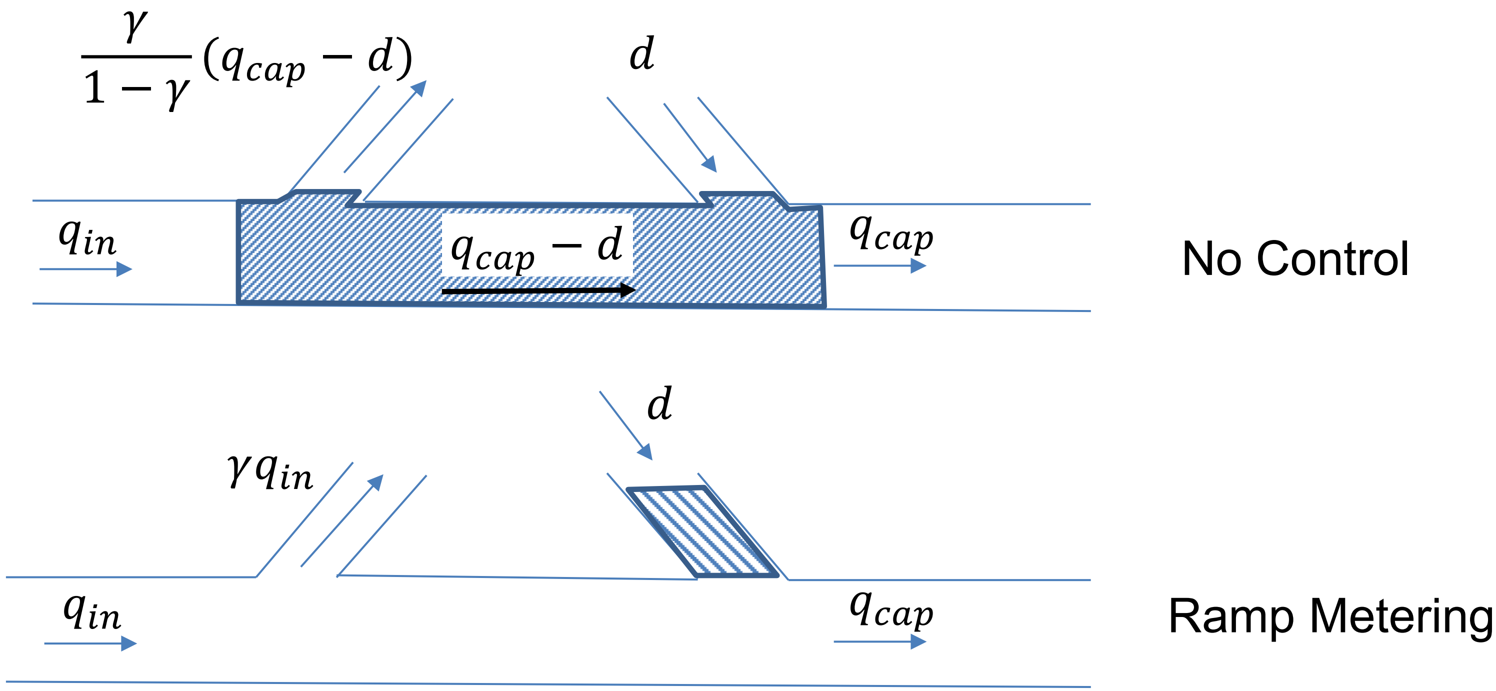


Ramp Metering

$$\Delta TTS = \frac{TTS^{nc} - TTS^{rm}}{TTS^{nc}} = ?$$

Ramp Metering

- Why useful? ***2nd answer***

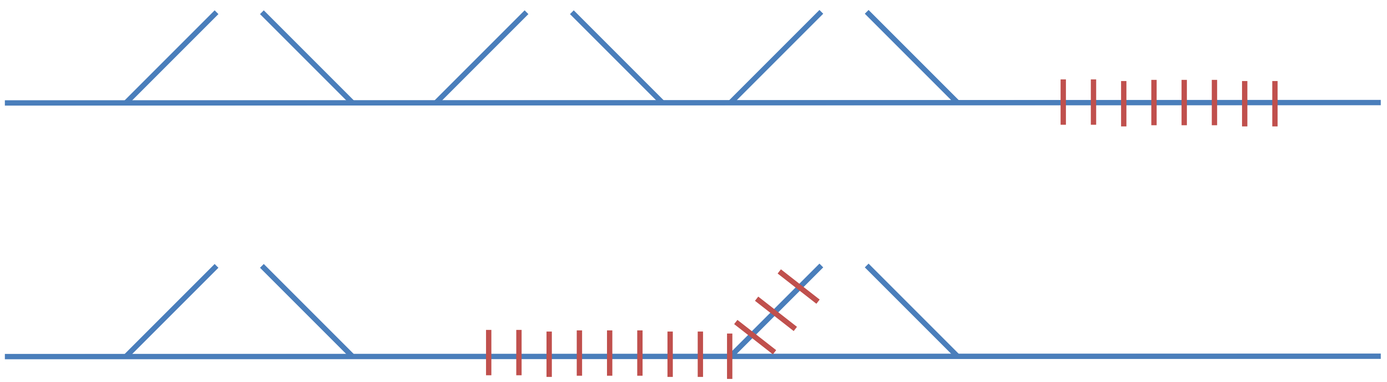


- Travel time savings

$$\Delta TTS = \frac{TTS^{nc} - TTS^{rm}}{TTS^{nc}} = \gamma$$

Ramp metering

- When not useful?

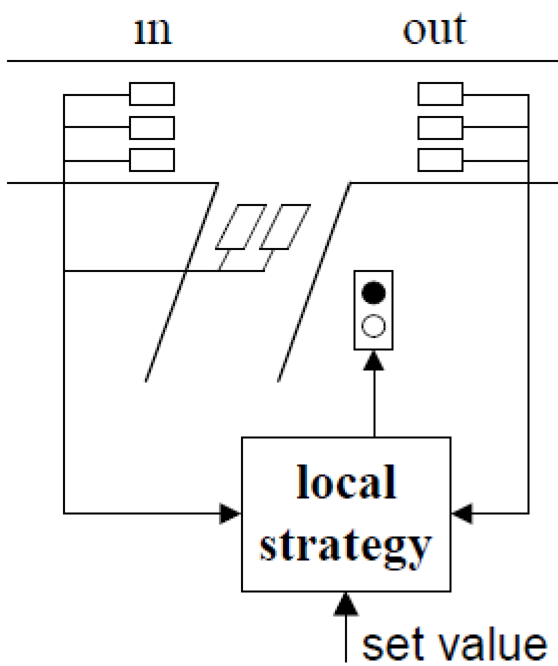


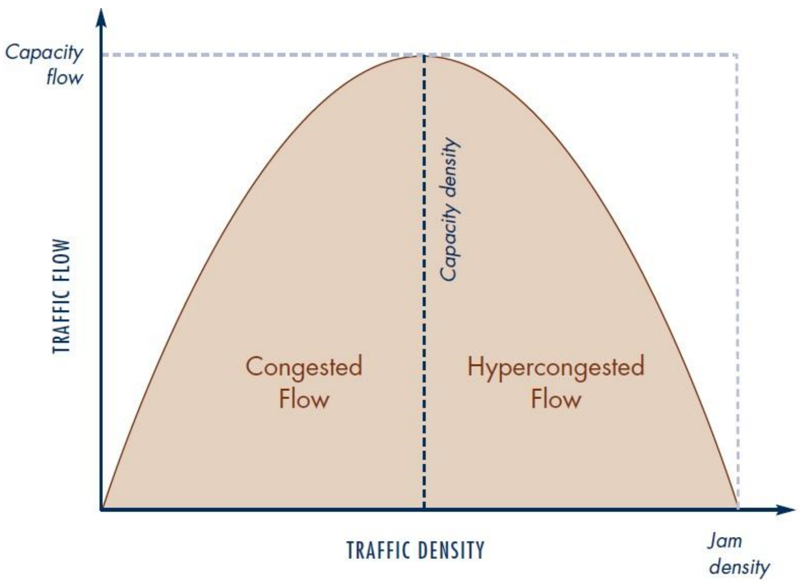
- Congestion further downstream
- Exit flow problems

Note: On-ramp queue should not interfere with surface street traffic.

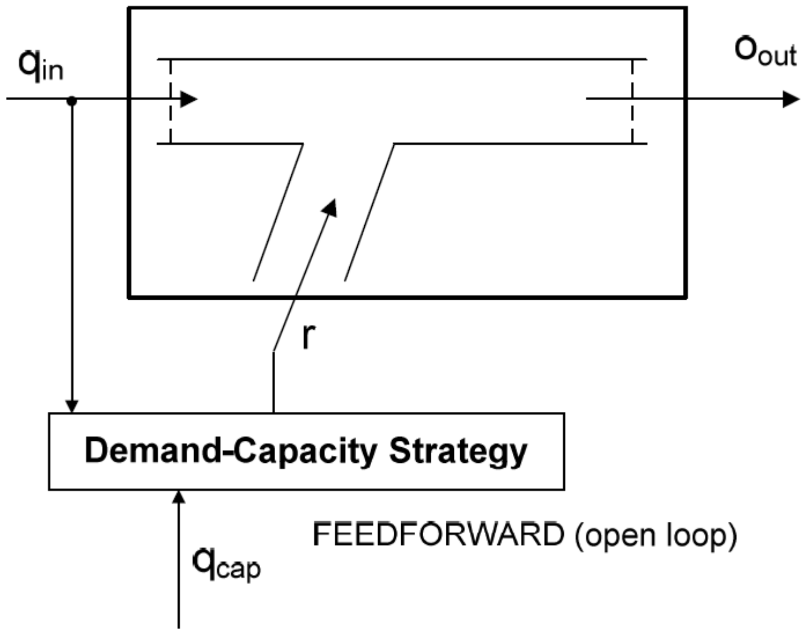
Regulator approaches

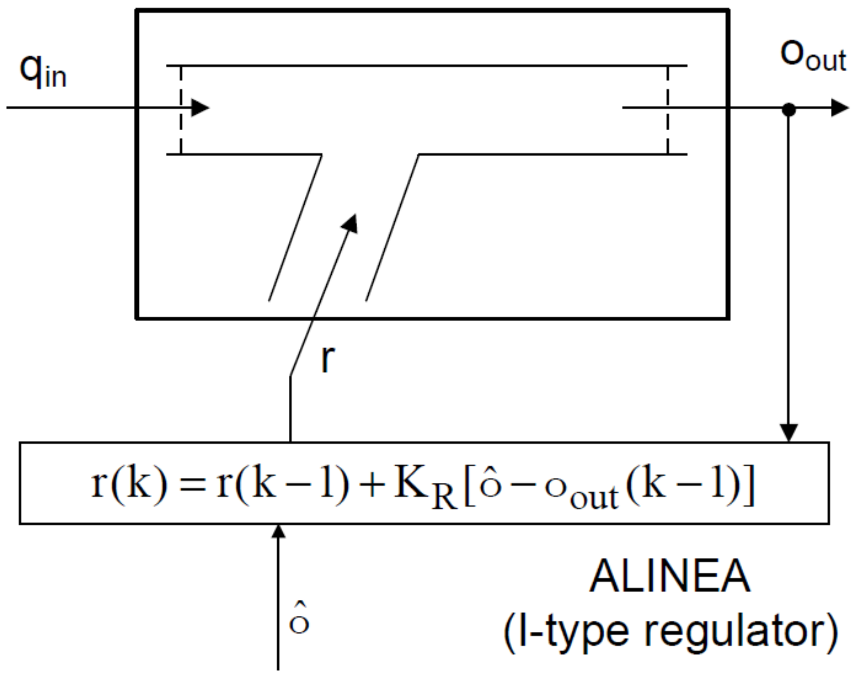
- Local ramp metering





Regulator approaches





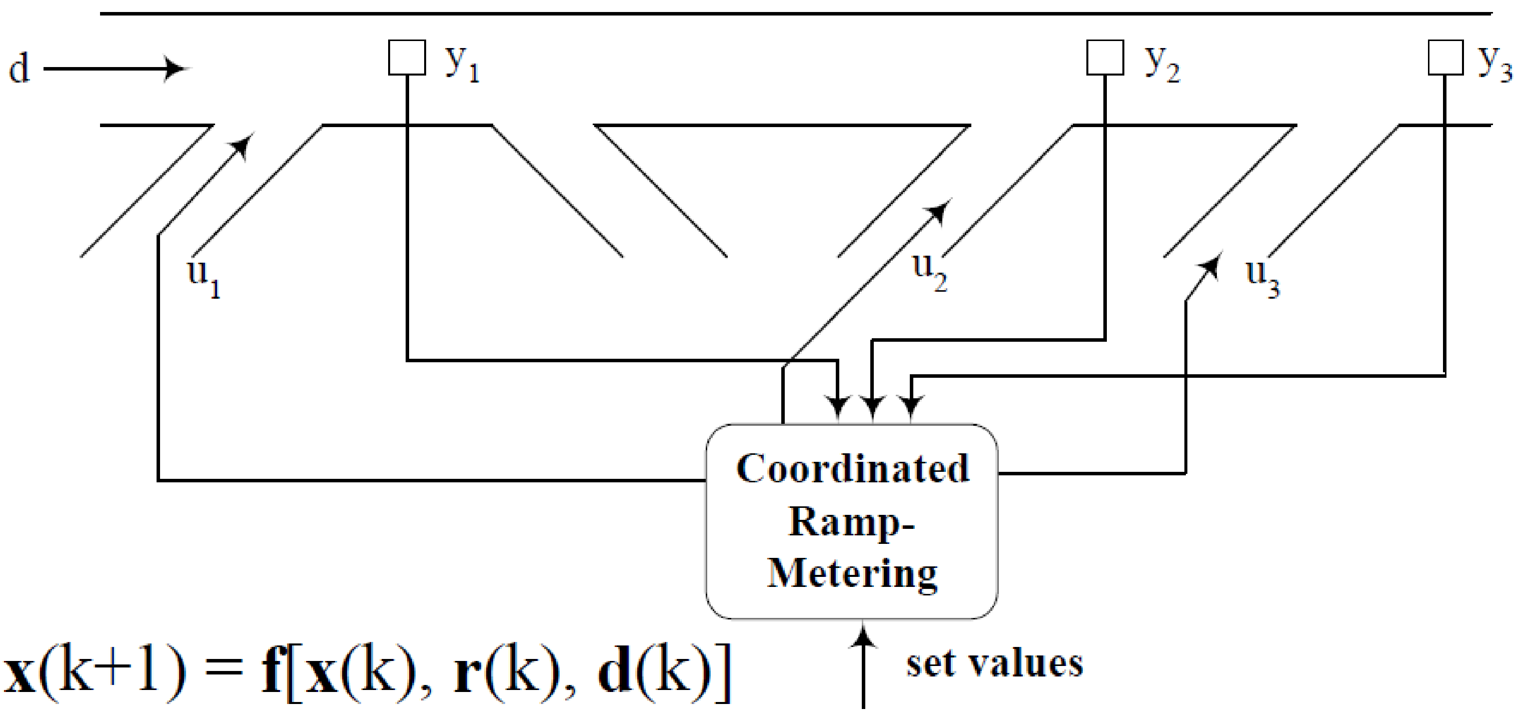
$$r + q_{in} = q_{cap}$$

Metering implementation

- Traffic lights
  – One car per green (adjust red phase wrt control flows)
  – Traffic cycles
- Intermediate applications
  – n cars at a time (n=2,3,…)
  – Short traffic cycles
- Ramp capacity
  – Usually exceeded
  – Important for overall efficiency

Coordinated ramp metering

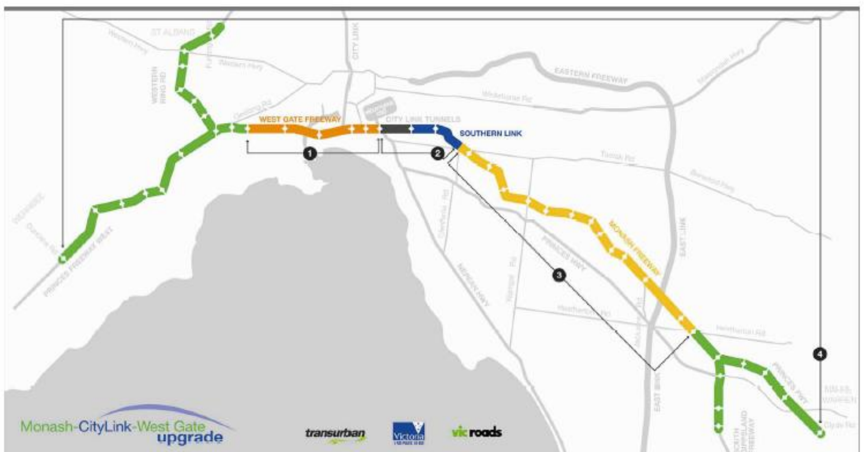
- Improves efficiency and equity across the ramps
- Crucial when limited storage capacity



$$\mathbf{x}(k+1) = \mathbf{f}[\mathbf{x}(k), \mathbf{r}(k), \mathbf{d}(k)]$$

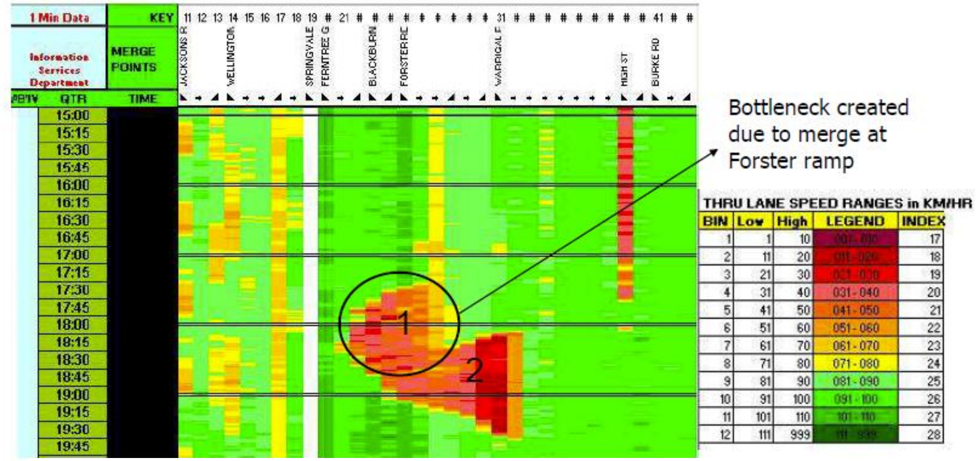
Coordinated ramp metering

- HERO (HEuristic Ramp metering cOordination)
- Monash freeway, Melbourne

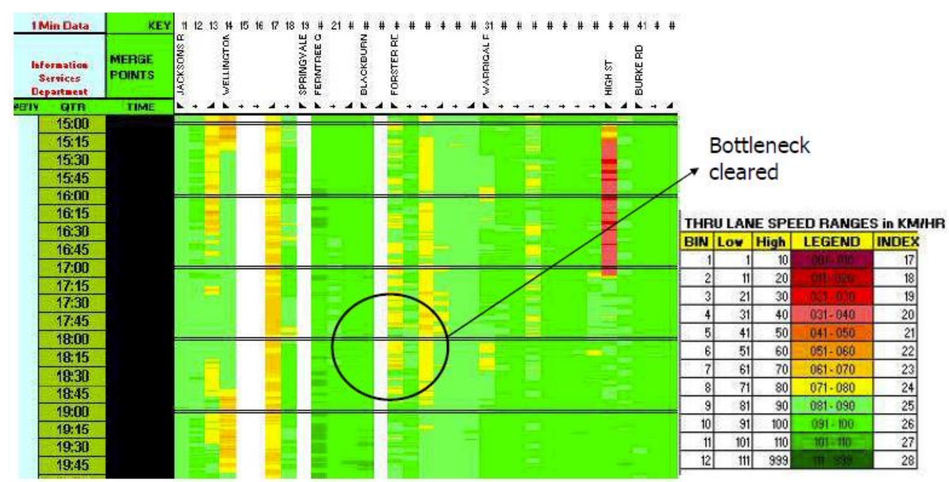


Papamichail, Ioannis, et al. "Heuristic ramp-metering coordination strategy implemented at monash freeway, australia." Transportation Research Record 2178.1 (2010): 10-20.

PM PEAK Typical day (No RM)



PM PEAK Typical day (ALINEA/HERO)



Next

This week
- Tutorial – Python

Computer exercise will build on this week's tutorial

CIVL6415, Semester 2, 2025

TRAFFIC ANALYSIS AND SIMULATION

MODULE 5

Traffic Assignment

Lecture Week 11

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Outline

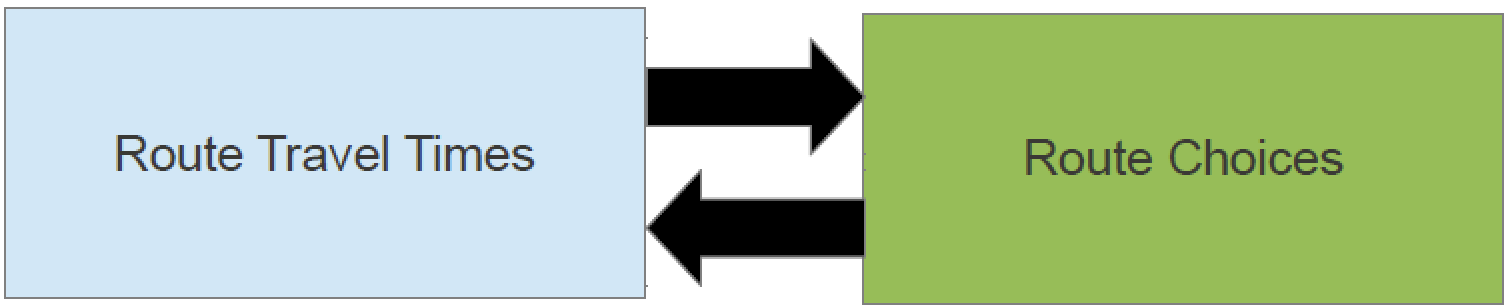
- Introduction to traffic assignment / equilibrium
- Path choice models
  - Deterministic vs. Probabilistic
- Static traffic assignment
  - User Equilibrium (UE)
  - System Optimum (SO)
  - Methods to reach equilibrium
- Dynamic traffic assignment
  - Dynamic User Equilibrium (DUE), Dynamic System Optimum (DSO)
  - Method of Successive Averages (MSA)
- Departure time choice

Introduction

- Transportation systems involve interactions among multiple agents. The basic facts are:
  - Although travel choices are made individually, they impact others as well (congestion).
  - The impacts of others' travel choices are important as you make your choices.

It's the evening peak period, so the freeway will probably be congested. I'll take another route." (But isn't everyone else thinking the same way?)

Introduction



4-step modelling

- ***4-Step Modelling***
  - Trip generation (Estimate trips produced and attracted in each zone)
  - Trip distribution (Find O-D flows)
  - Mode split (Consider alternative transport modes)
  - **Traffic assignment (Route choice for O-D car flows)**
    - **Equilibrium**

Introduction

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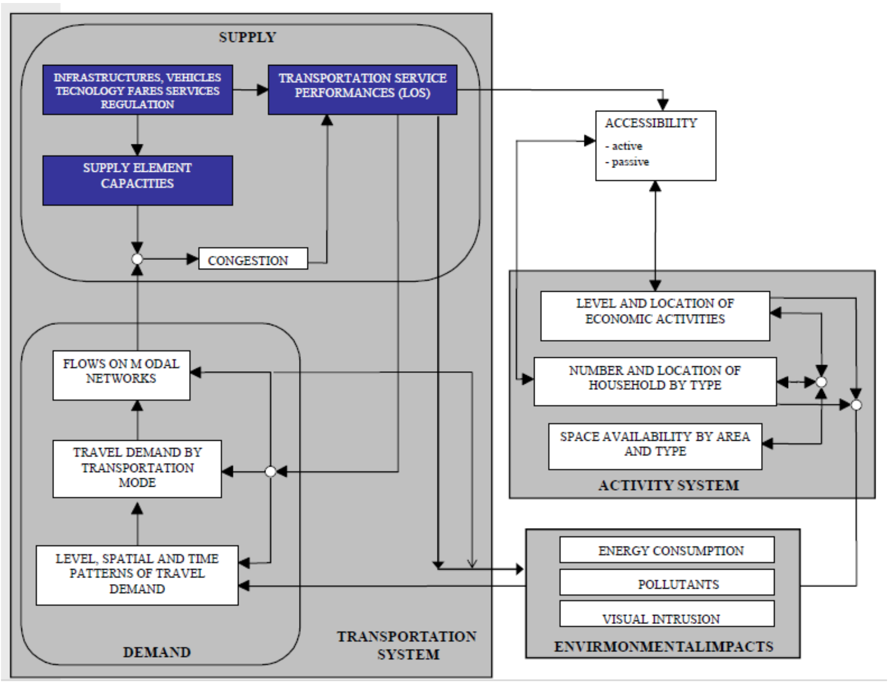
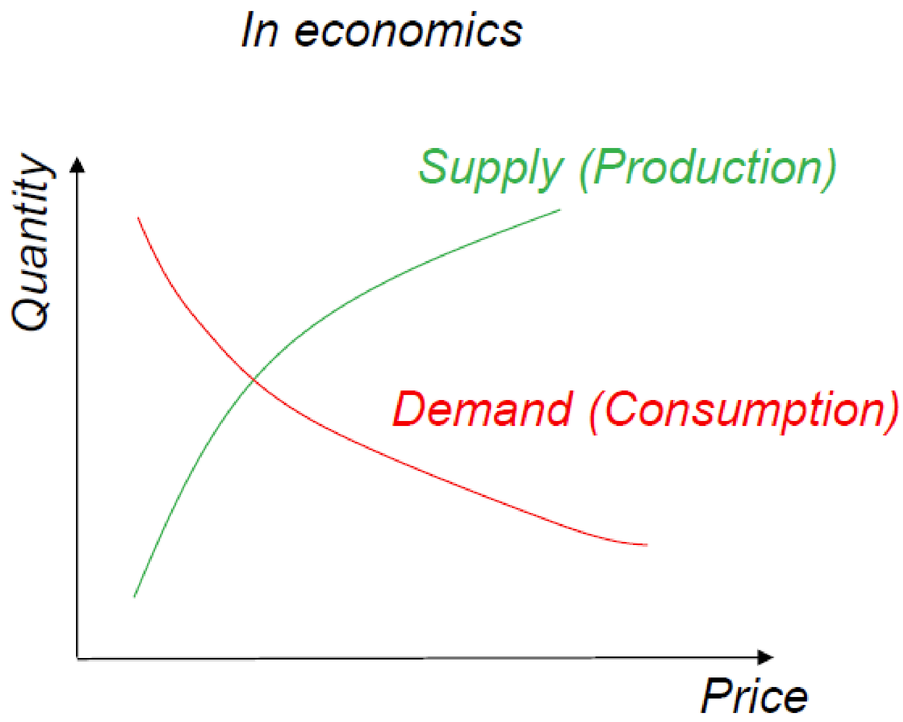
Simulate the way in which demand and supply interact in transportation systems, resulting in flows and performance levels on network elements.



Supply / Demand Interactions

- Demand curve represents the amount of some goods that buyers are willing and able to purchase at various prices
- Supply curve represents the amount of some goods that producers are willing to provide at various prices

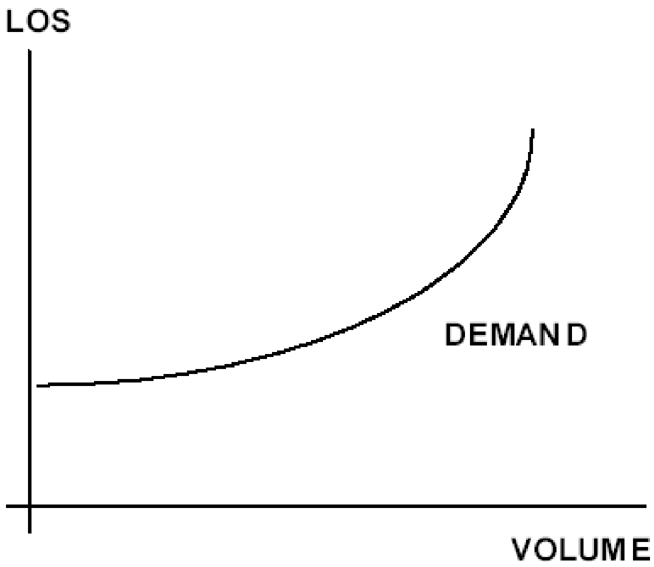
- This is simply a micro-economic concept. For example, if a movie theater doubles its price, therefore making its service less attractive -- in this case, more expensive -- fewer people will go to that movie theater. If a movie theater halves its price, more people will go.



Demand curve

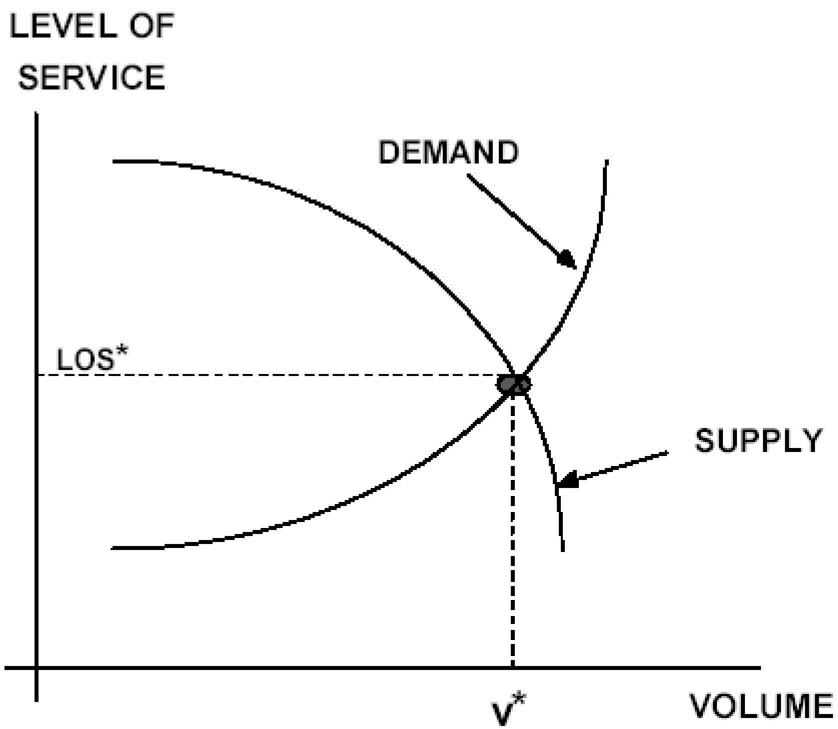
- The volume that a transportation service attracts is a function of the level-of-service provided to customers. If the level-of-service deteriorates, less people will want to use the service.

- LOS similar to utility, a measure of benefit or satisfaction. Might be linear sum of various LOS variables, such as travel time, access time, waiting time, fare and comfort. Inverse relationship between travel time and LOS



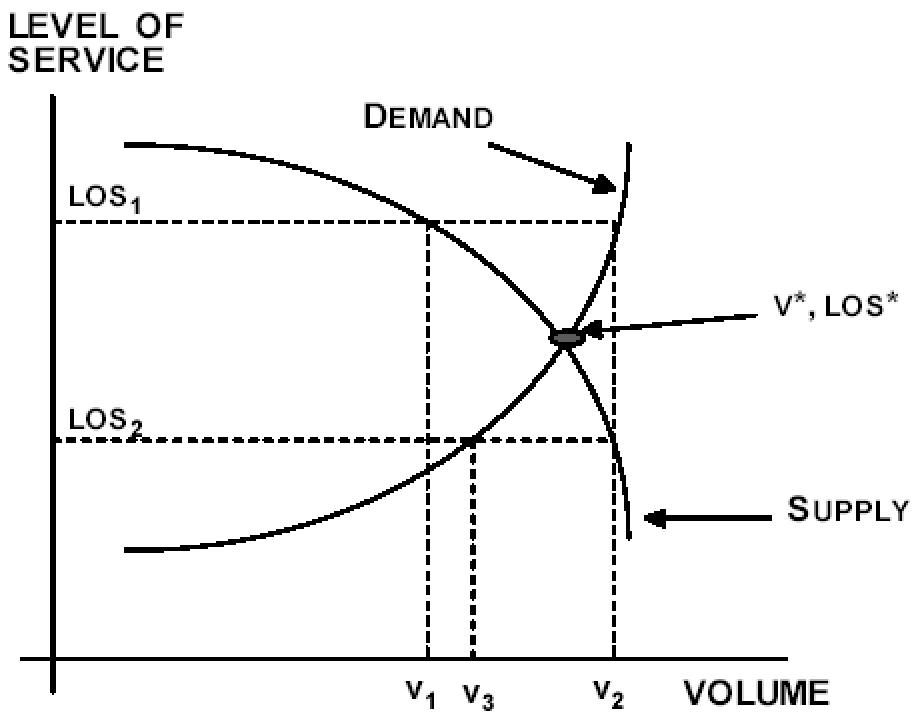
Supply curve

- Level-of-service as a function of volume.
- As a facility gets congested, the level-of-service for customers deteriorates
- **At equilibrium:** Quantity produced = Quantity demanded



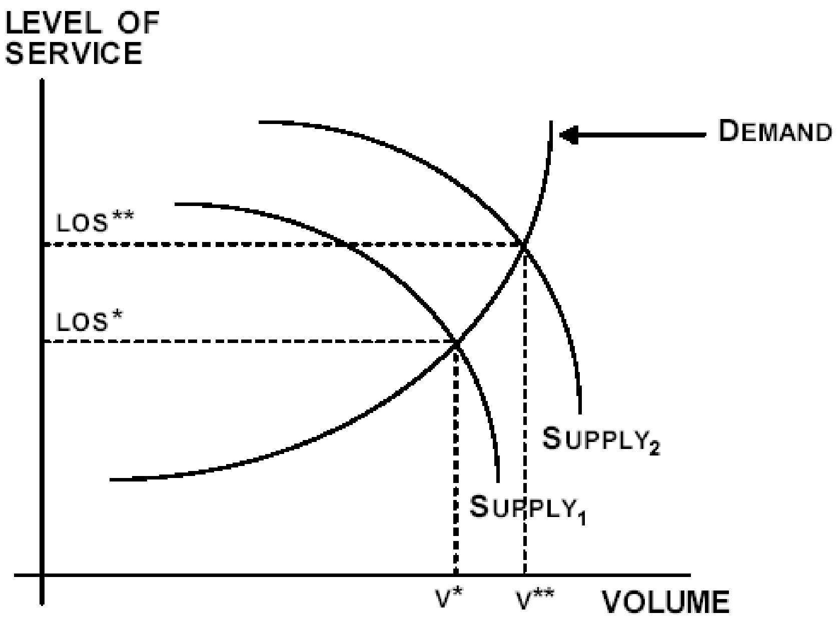
Iteration to find equilibrium

v_1 --first guess at volume
LOS_1 implies demand v_2
v_2 implies LOS_2
LOS_2 implies v_3 , etc., until v*, LOS* are found.



Measures to increase supply

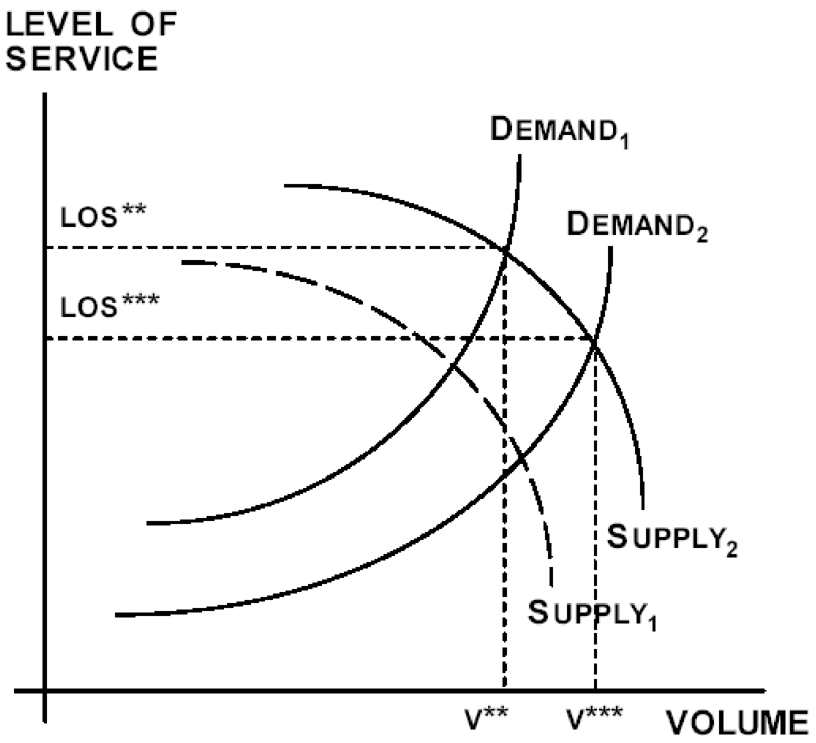
- General street traffic engineering improvements
  – minor design improvements, improved
  – signal timing
- Freeway traffic management – ramp metering
- From Supply 1 to Supply 2



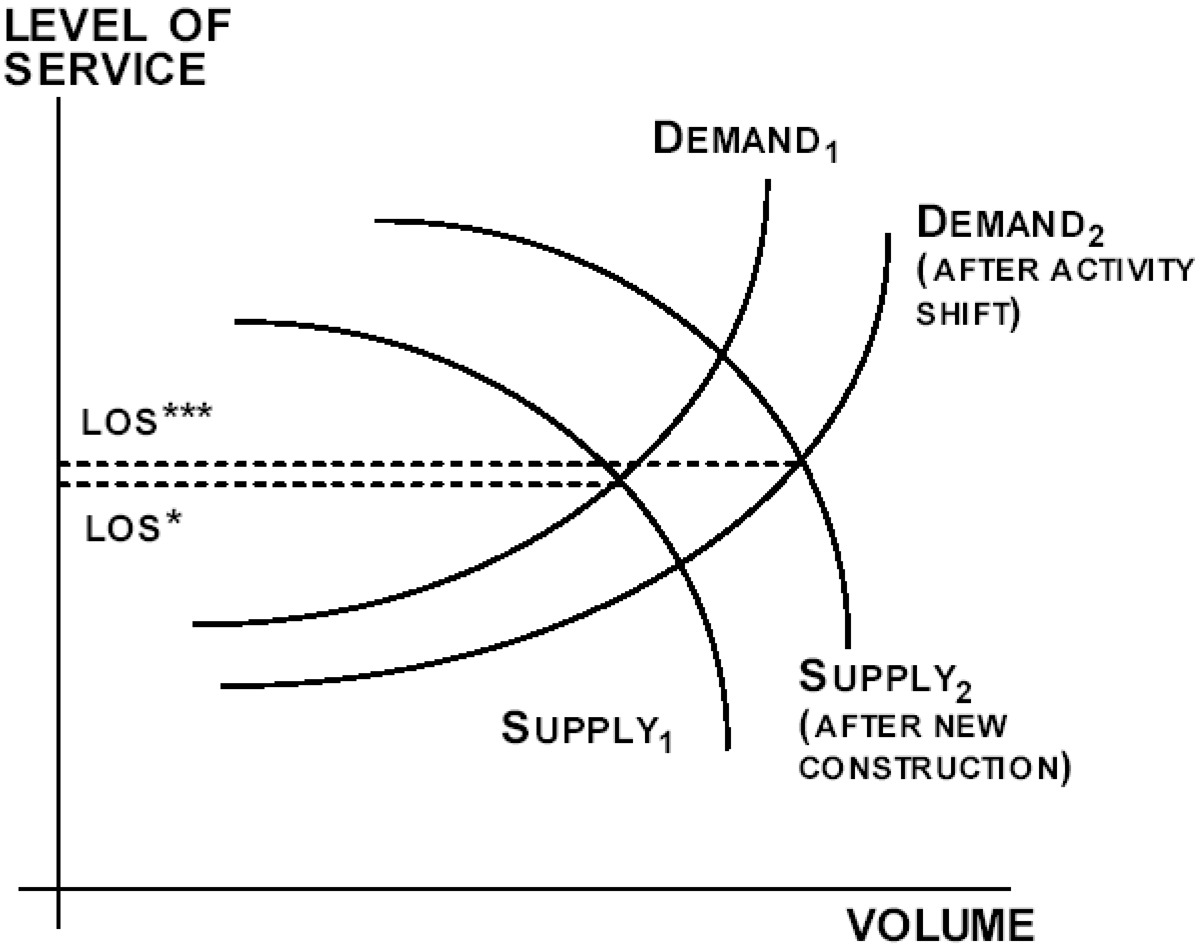
Measures to reduce demand

- Ridesharing, carpooling
- Park-and-ride
- Telecommuting
- Promoting off-peak travel

- From Demand 2 to Demand 1

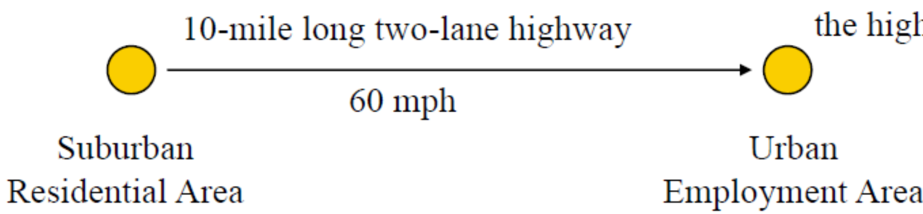


Induced demand



Exercise

Consider



(without intermediate access to the highway).

Exercise

Assume a simple, linear model specification for Supply Function:

$$t = m + n*V$$

where:

t = travel time (min.) a measure of LOS

V = volume (vehicle per hour, vph)

m, n = parameters defining the Transportation System 'T' = {m,n}

Let m = 10 min. (corresponding to a mean speed of 60 mph over the 10 mile highway)

n = 0.01 min. per vph (the sensitivity of travel time to increases in volume)

Thus:

$$t = 10 + 0.01V$$

Exercise

Again, assume a simple, linear model specification for Demand Function:

$$V = a + b*t$$

where:

a, b = parameters defining the Activity System 'A' = {a,b}

Let a = 5000 vph (corresponding to a potential, or latent, demand for the 10 mile highway with zero travel time)

b = -100 vph per minute (the sensitivity of demand to increases in travel time)

Thus:

$$V = 5000 – 100t$$

Exercise

The Flow Pattern ‘F’ defined by ‘V’ and ‘S’ is the resultant of the equilibration of the Service Function ‘J’ and the Demand Function ‘D’.

For this simple example, ‘F’ may be determined either graphically or by solving the two functions simultaneously.

Travel Market Equilibration

The equilibrium flow pattern is:

$t = 10 + 0.01V$

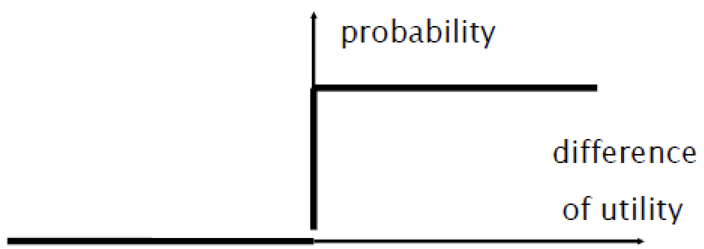
$V = 5000 – 100t$ => F1 = {2000 vph, 30.0 minutes}

Path choice

Path choice models

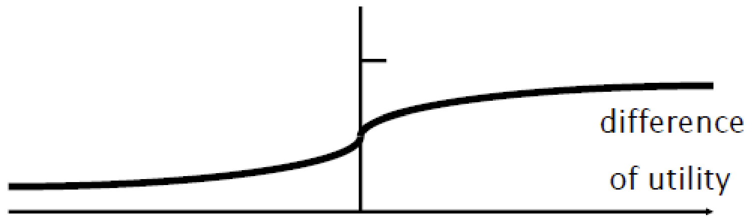
Deterministic

Users choose only the minimum cost alternative



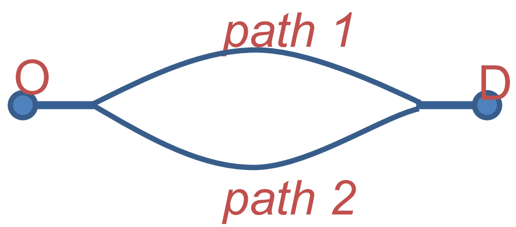
Probabilistic

Users may choose any available alternative with a probability depending on costs



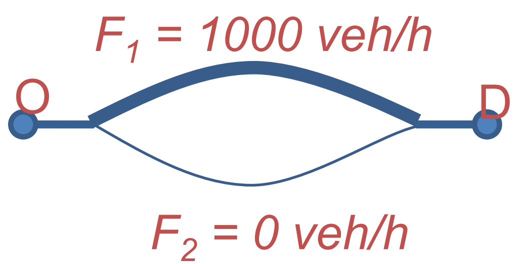
Example

Deterministic vs. Probabilistic path choice



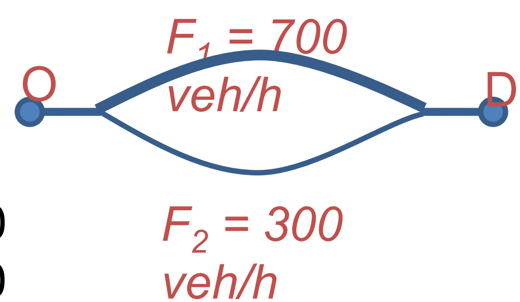
d_{OD} = 1000 veh/h
Path 1: t_1 = 10 min
Path 2: t_2 = 15 min

Deterministic



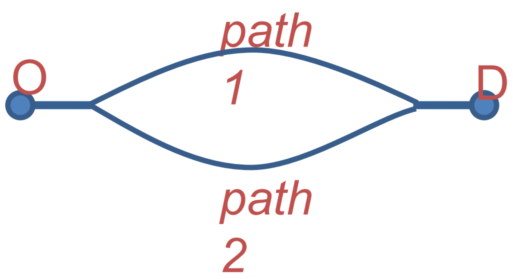
Probabilistic

$V_1 = \alpha \cdot t_1$
$V_2 = \alpha \cdot t_2$
$\alpha = -5.90$
Prob[1] = 0.70
Prob[2] = 0.30

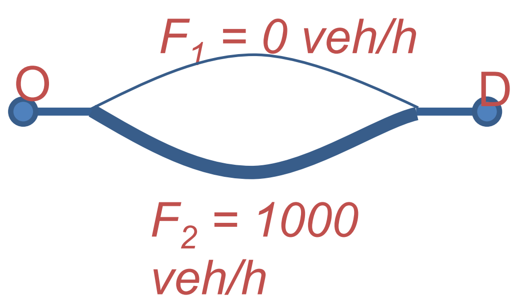


Example

New scenario: t_2 = 15 min → 8 min

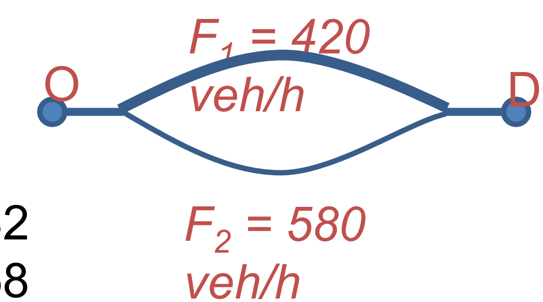


Deterministic



Probabilistic

$V_1 = \alpha \cdot t_1$
$V_2 = \alpha \cdot t_2$
$\alpha = -5.90$
Prob[1] = 0.42
Prob[2] = 0.58



Deterministic vs. Probabilistic

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DETERMINISTIC MODELS

ADVANTAGES

- computational

DISADVANTAGES

- lack of realism
- path flow map is a multi-valued application

PROBABILISTIC MODELS

ADVANTAGES

- more realistic
- wider modelling flexibility
- path flow map is a continuous function

DISADVANTAGES

- computational

Probabilistic choice

Random utility model

$$U_i = V_i(\text{attributes of } i; \text{parameters}) + \varepsilon_i$$

What is in ε?

Analysts' imperfect knowledge:

– Unobserved attributes

– Unobserved taste variations

– Measurement errors

– Use of proxy variables

$$\begin{aligned} U(\text{bus}) &= \beta_1 \times (\text{walk\_timebus}) + \beta_2 \times (\text{in\_vehicle\_timebus}) \\ &\quad + \beta_3 \times (\text{farebus} / \text{income}) + \varepsilon_{bus} \\ &= V(\text{bus}) + \varepsilon_{bus} \end{aligned}$$

Systematic utility (V)

- Attributes: describing the alternative
  - Generic vs. Specific
    - Examples: bus fare vs. travel time
  - Quantitative vs. Qualitative
    - Examples: reliability vs. comfort
  - Perception
  - Data availability

- Characteristics: describing the decision-maker
  - Socio-economic variables
    - Examples: income, gender, education

Random term (ε)

- Distribution of *epsilons* (ε)
  - Uniform → Linear
  - …
  - Normal → Probit
  - Extreme Value or Gumbel → Logit

- Variance/covariance structure
  - Correlation between alternatives

Random utility

Decision rule: Utility maximization

Decision maker *n* selects the alternative *i* with the highest utility U_{in} in the choice set C_n.

$$U_{in} = V_{in} + \varepsilon_{in}$$

V_{in}: Systematic utility expressed as a function of *k* observable variables, e.g. a linear-in-parameter specification. V_{in} is the deterministic part of the utility that the researcher can capture with observable attributes and their weights

ε_{in}: Random utility component

Random utility

- Multinomial choice probability:

$$P(i|C_n) \;=\; P(U_{in} \;\geq U_{jn}, \forall j \;\in C_n)$$
$$P(i|C_n) \;=\; P(U_{in} - U_{jn} \geq 0, \forall j \;\in C_n)$$
$$P(i|C_n) \;=\; P(U_{in} = max_j\, U_{jn}, \forall j \;\in C_n)$$

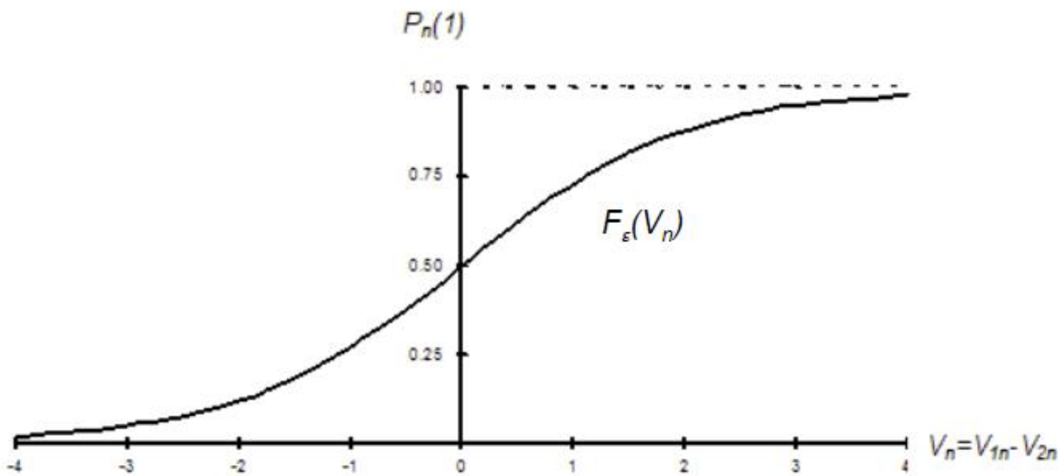
- Binary choice – Special case

$$P_n(1) = P(U_{1n} \geq U_{2n})$$
$$P_n(1) = P(U_{1n} - U_{2n} \geq 0)$$

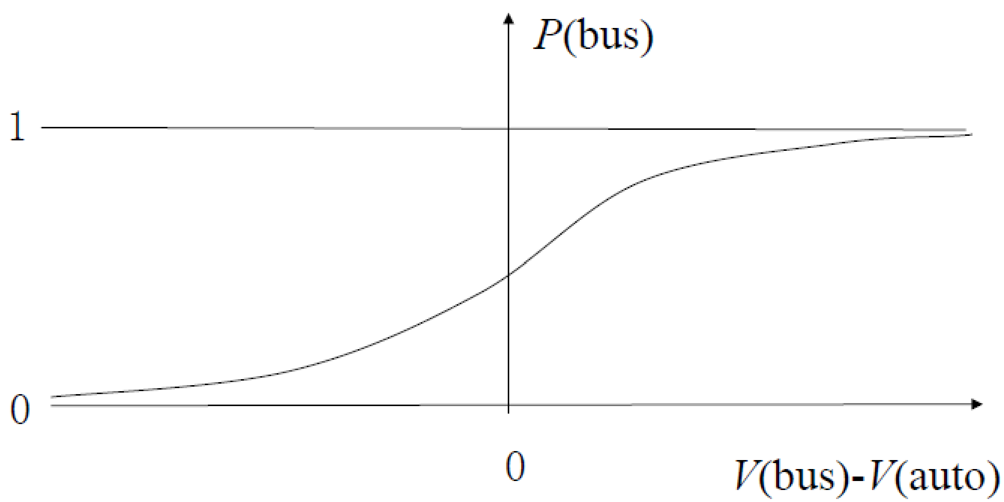
Binary choice

- Choice set $C_n = \{1,2\}\ \forall n$

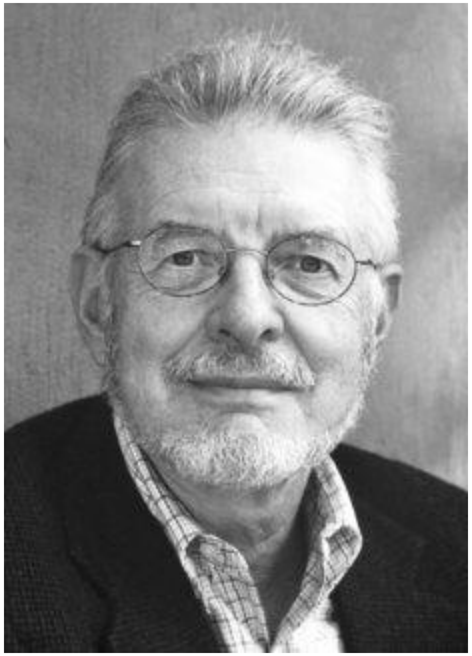
$$
\begin{aligned}
P_n(1) = P(1|C_n) &= P(U_{1n} \geq U_{2n}) \\
&= P(V_{1n} + \varepsilon_{1n} \geq V_{2n} + \varepsilon_{2n}) \\
&= P(V_{1n} - V_{2n} \geq \varepsilon_{2n} - \varepsilon_{1n}) \\
&= P(V_{1n} - V_{2n} \geq \varepsilon_n) \\
&= P(V_n \geq \varepsilon_n) = F_\varepsilon(V_n)
\end{aligned}
$$



Probabilistic choice



Discrete choice models



Daniel McFadden

- 2000 Nobel Prize in Economic sciences
- His research on choice behaviour and the problem of linking economic theory and measurement.

Why the Gumbel distribution became popular (and the derivation of the family of models gave a Nobel prize to Daniel McFadden)?

- Gumbel distribution for the error terms is nearly the same as a normal distribution, but the choice probability is closed-form
- the difference between two error terms that are Gumbel distributed follows a logistic distribution and provides the closed-form formula

Logit models

- The analytical solution of the integral yields the closed-form choice probability of the logit model

$$P_{ni} = \frac{\exp(V_{ni})}{\exp(V_{n1}) + \exp(Vn_2) + \cdots + \exp(Vnj)}$$

- The binary logit formula;

$$P_{n1} = \frac{\exp(Vn_1)}{\exp(V_{n1}) + \exp(Vn_2)} = \frac{1}{1 + \exp(V_{n2} - V_{n1})}$$

- V_{ni} involve constants that need to be calibrated
- Maximum Likelihood estimation
- Stated Preference or Revealed Preference data

Properties of Logit

Independence from irrelevant alternatives (IIA)

Probability of option i in the choice set C_n

$$P(i|C_n) = \frac{\exp(V_{in})}{\sum_{j \in C_n} \exp(V_{jn})}$$

Odds ratio;

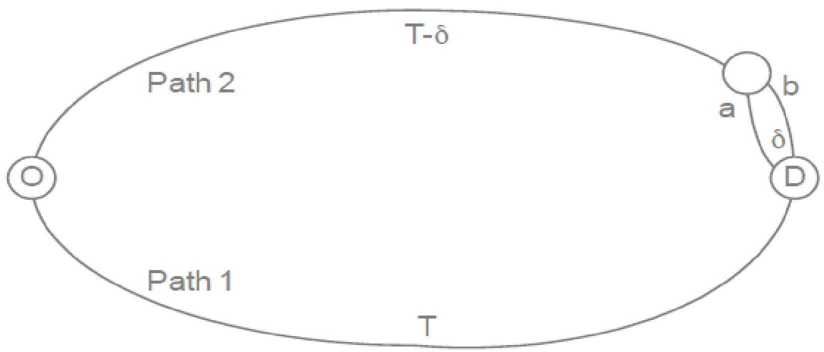
$$\frac{P(i|C_{1n})}{P(j|C_{1n})} = \frac{P(i|C_{2n})}{P(j|C_{2n})}$$

$$i, j \in C_{1n},\ i, j \in C_{2n},\ C_{1n} \subseteq C_n \text{ and } C_{2n} \subseteq C_n$$

If A is preferred to B out of the choice set {A,B}, introducing a third option X, expanding the choice set to {A,B,X}, must not make B preferable to A. Guarantees rational behaviour.

Route choice - Logit

- Route choice example



$$P(1 \mid \{1,2a,2b\}) = P(2a \mid \{1,2a,2b\}) = P(2b \mid \{1,2a,2b\}) = \frac{e^{\mu T}}{\sum_{j \in \{1,2a,2b\}} e^{\mu T}} = \frac{1}{3}$$

- Paths may have overlapping sections
- Alternative paths may be correlated due to overlapping portions

Overlapping problem

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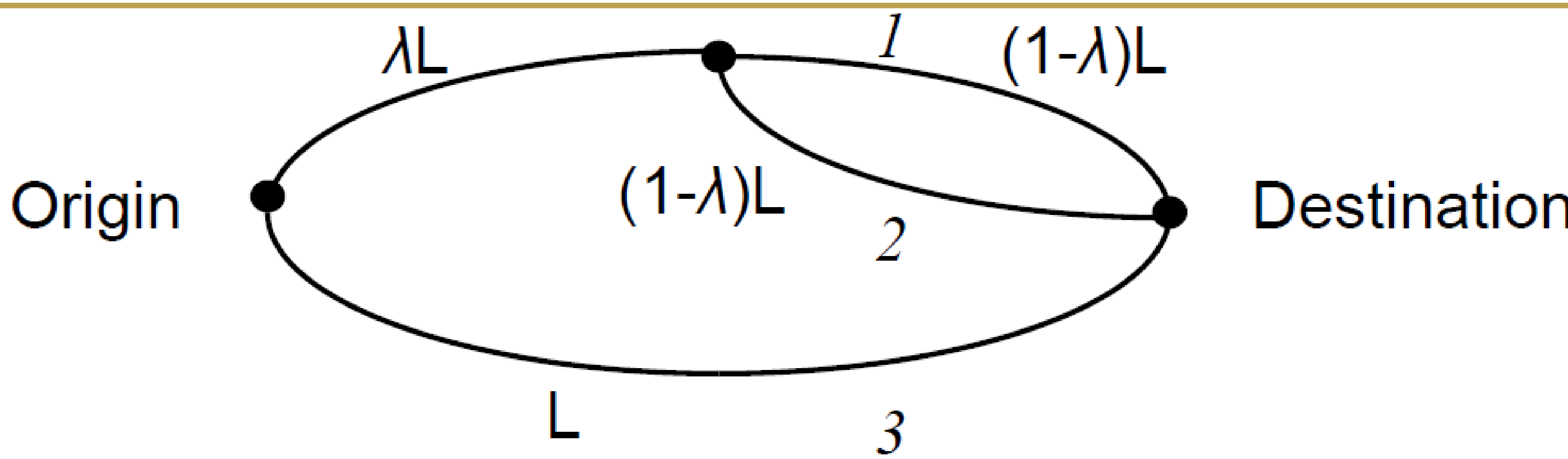

- Introduces correlation - Violates IIA - Logit is unsuitable
- Traditional example with three paths, two overlapping
  - As overlap approaches 100 percent ($\lambda \rightarrow 1$), expect close to 25/25/50 shares
  - As overlap approaches 0 percent ($\lambda \rightarrow 0$), expect close to 33/33/33 shares
- Logit predicts 33/33/33 shares for any value of λ

C-Logit

- A variation of the logit model to address the commonality problem

$$P(i|C_n) = \frac{\exp(\mu V_{in} - CF_i)}{\sum_{j \in C_n} \exp(\mu V_{jn} - CF_j)}$$

- directly proportional to the degree of overlapping of path *k* with other alternative paths. Thus, highly overlapped paths have a larger CF factor and therefore smaller utility with respect to similar paths. CF_k is calculated as follows:

Commonality Factor

$$CF_i = \beta \cdot \ln \sum_{j \epsilon C_n} \left(\frac{L_{ij}}{L_i^{1/2} L_j^{1/2}} \right)^{\gamma}$$

- L_{ij} is the cost of links common to paths *i* and *j*,
- L_i and L_j are the cost of paths *i* and *j* respectively (expressed in hours if cost is travel time)

- Depending on the two factor parameters β and γ, a greater or lesser weighting is given to the 'commonality factor'.
- Larger values of β means that the overlapping factor has greater importance with respect to the utility V_i.
- The factor γ is a positive parameter, usually taken in the range [0, 2], whose influence is smaller than β and which has the opposite effect on the choice.

Route choice in simulation

- Adaptive path choice
  - Apply path choice model repeatedly with certain frequency
  - The more frequent, the closer to equilibrium
  - The decision is based on instantaneous traffic conditions

- Utility usually consists of
  - Travel time
  - Link attractiveness
    - Some roads are preferred more than others because of a variety of reasons. This function allows the modeller to calibrate the simulation.

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Next

This week

- Tutorial - Python

CIVL6415, Semester 2, 2025

TRAFFIC ANALYSIS AND SIMULATION

MODULE 5

Traffic Assignment

Lecture Week 12

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Outline

- Introduction to traffic assignment / equilibrium
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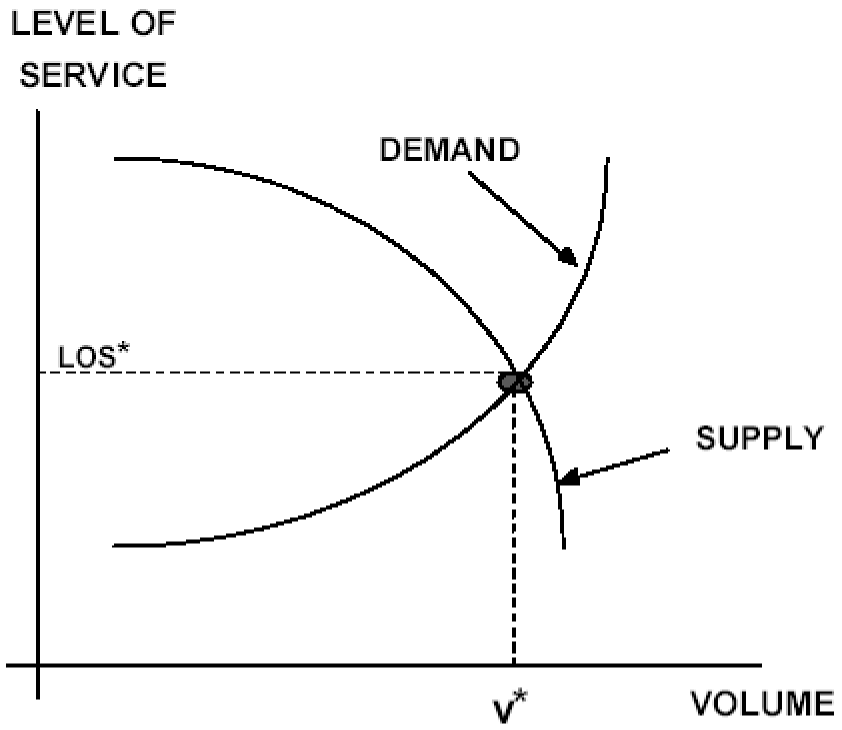
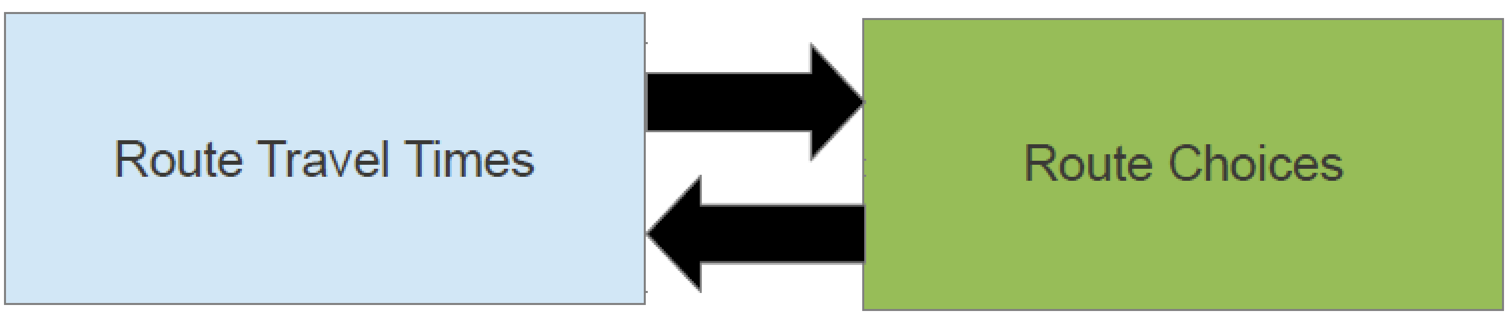
Introduction





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Static traffic assignment

Applications of Assignment models

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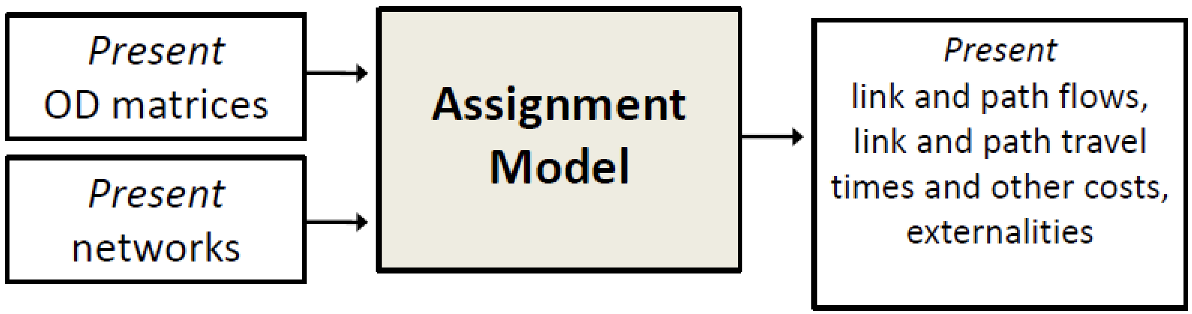
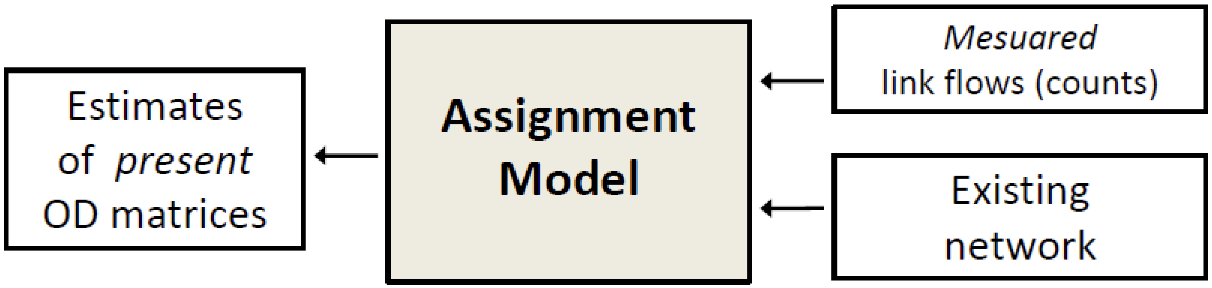
- As estimators of the current state of the transport systems



- A good match between observed and simulated traffic conditions is needed to guarantee a well calibrated simulation model
- The results from the assignment model can complement other traffic observations (e.g., link flows) as such measures are not available for all network components

Applications of Assignment models

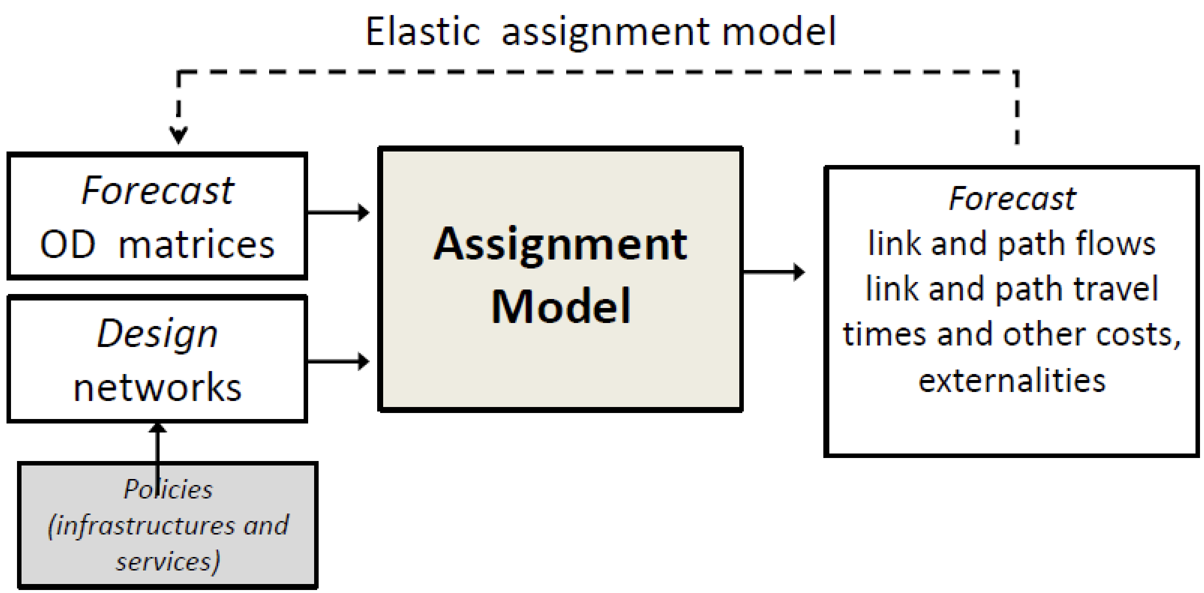
- For the estimation of current travel demand



- If prior OD matrices are not reliable, assignment models can be useful to estimate present OD matrices
- Prior ODs usually depend on historical surveys and may not reflect current traffic state

Applications of Assignment models

- For evaluating the changes in the transport systems



Traffic assignment

- Traffic assignment considers a trade-off between the demand (the number of vehicles willing to drive along a network link) and supply (the available capacity of that network link)

- This trade-off implies that there are three building blocks
  - A network of nodes and links
  - A link performance function (speed-flow or time-flow) that expresses the trade-off between traffic volume and travel time. This function may be static or dynamic depending on the application.
  - A route choice model that controls the assignment of the demand to the network links

Solving traffic assignment

- Given the demand, the network and the link performance functions
  - All or Nothing assignment with no road capacity constraints
  - User Equilibrium (UE) with capacity constraints
  - Stochastic User Equilibrium (SUE) with capacity and stochastic effects

- Dynamic extensions
  - Dynamic User Equilibrium
  - Dynamic Stochastic User Equilibrium

All or Nothing assignment

- Assumptions
  – Traffic flows freely without delays (no congestion)
  – Travellers have full knowledge of the network
  – All travellers are equal and have the same preferences

- The principle is then that all travellers choose the shortest path from their origin to their destination

- The advantage is that the assignment model is really fast, given that it is a search for shortest paths from multiple origins to multiple destinations

- The disadvantage is that the assignment model does not consider congestion, it is unrealistic and it is unusable for evaluating planning projects

Equilibrium models

- Given the demand, an equilibrium principle exists so that the system can be described by an equilibrium state
- An equilibrium model needs to be given to represent the state where demand and supply reach the equilibrium itself

- The All or Nothing assignment model is not an equilibrium
  - The model allows for infinite demand to be on one path
  - The model converges in 1 iteration (shortest path search)

Wardrop's equilibrium principles

In 1952, Wardrop stated two principles that formalize different notions of equilibrium

- 1. Principle

"Under equilibrium conditions traffic arranges itself in congested networks in such a way that no individual traveller can improve their path cost by switching routes"

- 2. Principle

"Under equilibrium conditions traffic should be arranged in congested networks in such a way that the total travel cost is minimised"

Wardrop's equilibrium principles

- Wardrop's first principle is a UE (user equilibrium)
- **Descriptive:** it describes how individuals travel through an urban transportation system given the components of that system

- Each traveller minimises his/her own travel cost, in line with the utility maximisation principle (i.e., realistically travellers are selfish)

- The UE implies mathematically that, if T^*_{ijr} is the traffic flow at equilibrium for OD pair {*i*,*j*} on a given route *r,* and c^*_{ij} is the travel cost at equilibrium for the same OD pair, then

$$T^*_{ijr} = 0 \ \& \ c_{ijr} > c^*_{ij}$$

$$T^*_{ijr} > 0 \ \& \ c_{ijr} = c^*_{ij}$$

- The UE implies that the travel times on all used routes are equal and minimal

Wardrop's equilibrium principles

- Wardrop's second principle is a SO (system optimum)
- **Normative:** attempt to determine the optimal system configuration

- Force route choices such that the total travel time on the network is minimised

- From the planners' perspective, this is the most beneficial principle although selfish travellers would change route for a better one

- As UE and SO are not always the same, planners will push the UE as close as possible to the SO with a set of instruments
  - Road pricing (video)
  - Renovated or new infrastructure
  - Public transport subsidising

System optimal

- By definition *system optimal* flows are not stable. Travelers will always be tempted to switch to non-system optimal routes.
- Therefore system optimal flows are not realistic of actual traffic.
- However, system optimal flows do provide useful comparisons with the more realistic user equilibrium traffic forecasts.

Stochastic user equilibrium

- SUE means that travellers cannot improve their perceived travel cost by unilaterally changing their routes
- All routes are used and the travel times are not equal
- Probabilistic path choice models

Wardrop's equilibrium principles

- UE formulated as a mathematical program by Beckmann, 1956

$$\text{Min } Z = \sum_a \int_{0,xa} t_a(x)dx$$

st.

$$\sum_{\mathrm{k}} f_{k,i,j} = T_{i,j}$$

$$x_a = \sum_{\mathrm{k}} f_{k,i,j}\, \delta_{a,k,i,j}$$

$$f_{k,i,j} \geq 0 \ \forall k,i,j$$

$$x_a \geq 0 \ \forall_{\mathrm{a}}$$

- Recall that UE means that the travel times of used routes are equal

Wardrop's equilibrium principles

- SO was formulated as a mathematical program by Sheffi

$$\text{Min } Z = \sum_a x_a t_a(x_a)$$

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$\sum_k f_{k,i,j} = T_{i,j}$

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$f_{k,i,j} \geq 0 \ \forall k,i,j$

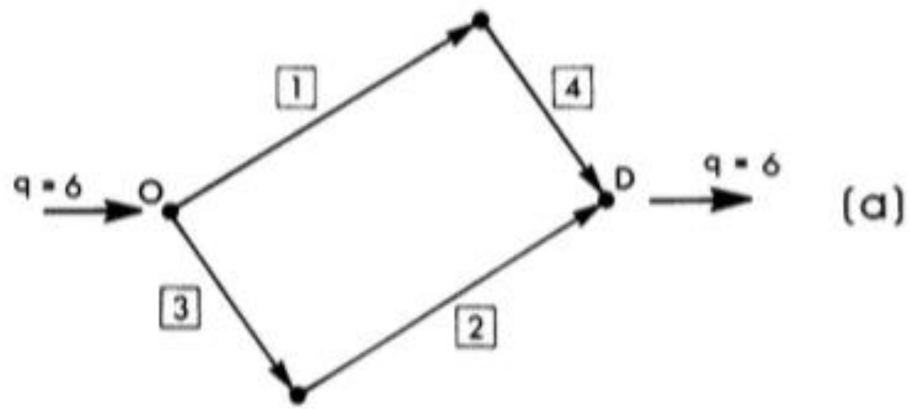
$x_a \geq 0 \ \forall a$

- Recall that SO means that the total travel time is minimum, and the total travel time is the sum of the travel time on each link multiplied by the traffic flow on that link

Exercise

- Two routes connect a city and a suburb. During the peak-hour morning commute, a total of 4500 vehicles travel from the suburb to he city. Route 1 has a 60 km/h speed limit and 6km in length. route 2 is 3 km in length with 45km/h speed limit. Studies show that the travel time on route 1 increases four minutes for every additional 1000 vehicles added. Minutes of travel time on route 2 increase with the square of the number of vehicles expressed in thousands of vehicles per hour. Determine user equilibrium travel times.

Example of UE and SO calculation



Performance Data

$t_1(x_1) = 50 + x_1$

$t_2(x_2) = 50 + x_2$

$t_3(x_3) = 10x_3$

$t_4(x_4) = 10x_4$

Path Definition

Path 1 $\boxed{1}$ $\boxed{4}$

Path 2 $\boxed{3}$ $\boxed{2}$

Example of UE calculation

- The answers to the problems are
  - The four travel times t_1, t_2, t_3, t_4
  - The four link volumes x_1, x_2, x_3, x_4

- User equilibrium means that the travel time on the two routes is equal
  - In the specific case $t_1 + t_4 = t_2 + t_3$
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  - Write the total demand (in the specific case $x_1 + x_2 = 6$)
  - This is a system of four equations in four unknowns that can be easily solved

Example of SO calculation

- The answers to the problems are
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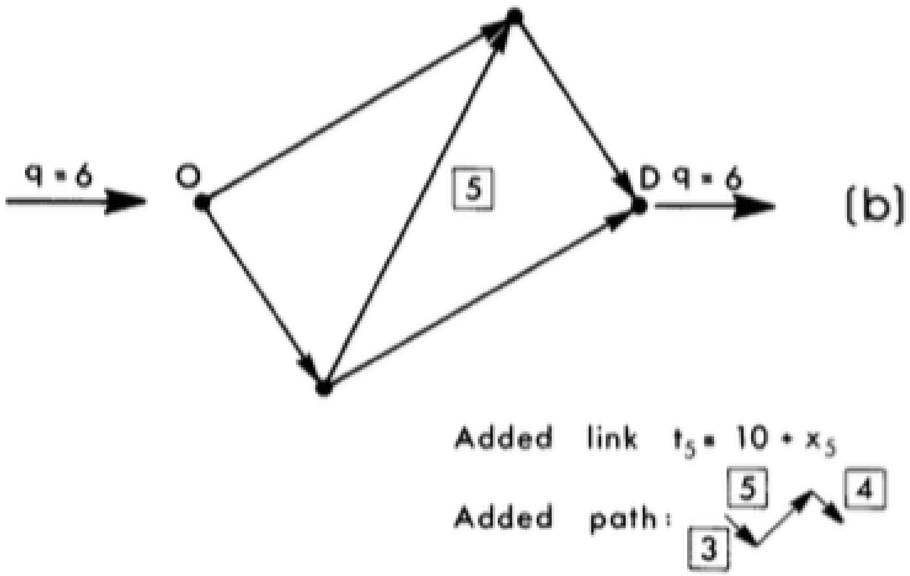

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- In the second case the total travel time in the network under UE conditions was worse even though the demand was the same and the network improved: this is the **Braess paradox** and it is related to the travellers being selfish

- From a more general perspective, **Braess paradox** emphasizes the importance of a careful and systematic analysis of investments in urban networks. Not every addition of capacity can bring about all the anticipated benefits and in some cases, the situation may be worsened.

Method of successive averages (MSA)

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- **Step 0:** *Initialization*. Select initial link costs: usually free flow travel times. Perform all-or-nothing assignment based on t_a = $t_a(0)$, $\forall a$. This yields $\{x_a^1\}$. Set counter n = 1
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$$\|x_a^{n+1} - x_a^n\| \leq \varepsilon$$

Frank-Wolfe Assignment

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MSA – Stochastic User Equilibrium

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Next

This week
- Tutorial
- Problem set released

CIVL6415, Semester 2, 2025

TRAFFIC ANALYSIS AND SIMULATION

MODULE 5

Traffic Assignment

Lecture Week 13

Dr. Mehmet Yildirimoglu
School of Civil Engineering
The University of Queensland
m.yildirimoglu@uq.edu.au

Outline

- Introduction to traffic assignment / equilibrium
- Path choice models
  – Deterministic vs. Probabilistic
- Static traffic assignment
  – User Equilibrium (UE)
  – System Optimum (SO)
  – Methods to reach equilibrium
- Dynamic traffic assignment
  – Dynamic User Equilibrium (DUE),
  – Dynamic System Optimum (DSO)
- ~~Departure time choice~~

FINAL EXAM

Final Exam

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- **All materials**
- **Exam time - *Internal***
  - – Examination Duration: **120** minutes
  - – Reading Time: **10** minutes

Final Exam

- **20** MCQ **– 50pts**
- **4** Short calculation questions – **50pts**
  - **Short calculation questions**
- Closed-book
- Permitted materials
  - One A4 sheet of handwritten or typed notes double sided is permitted
  - Unannotated Bilingual dictionary

Final Exam

- **MCQ**
  - **“Which one of the following statements is true/false?” - 15 questions**
  - **Brief calculation questions – 5 questions**
  - **Understand the concept, do not memorise!**
  - **2.5pts / question**
  - **5 choices for each question**

Final Exam

- Short calculation questions
  - Expect a balanced distribution across the modules
    - Car following,
    - Shockwaves,
    - Traffic assignment,
    - etc.
  - 4 questions in this format
  - Each worth 10-15 pts

Course Teaching Feedback

- What are the benefits of completing the survey?
  - This is your opportunity to provide feedback on your learning experience! Your responses can be used to improve course design and quality. Please provide specific examples where possible.
  - Respectful constructive feedback is always helpful to help improve your learning experience. All responses submitted are strictly confidential.
- How do I complete the survey?
- Link

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Static traffic assignment

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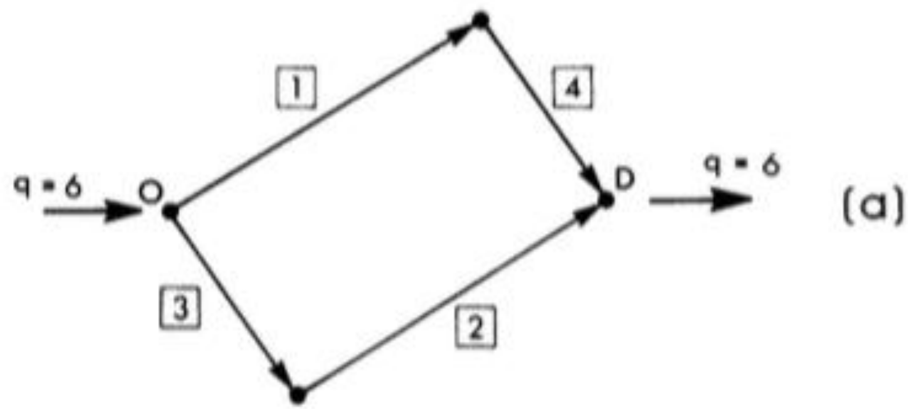

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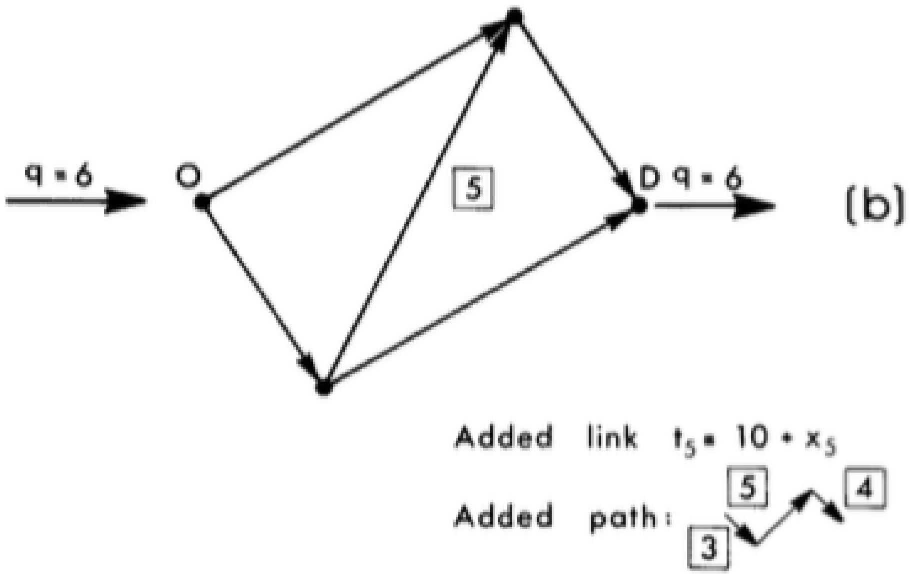

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Dynamic traffic assignment (DTA)

The problems with static assignment

For example...

1. Not all vehicles will experience the same delay
2. What does the flow mean in static assignment?
3. Where does congestion actually lie?
4. What about queue spillback?

$$t = t_f\left(1 + \alpha\left(\frac{v}{c}\right)^{\beta}\right)$$

What this means is that we need a fundamentally different congestion model, based in traffic flow theory rather than arbitrary functions.

Dynamic extension

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Dynamic User Equilibrium (DUE)

- All routes used by travellers leaving the same origin **at the same time** for the same destination have equal and minimal travel time.

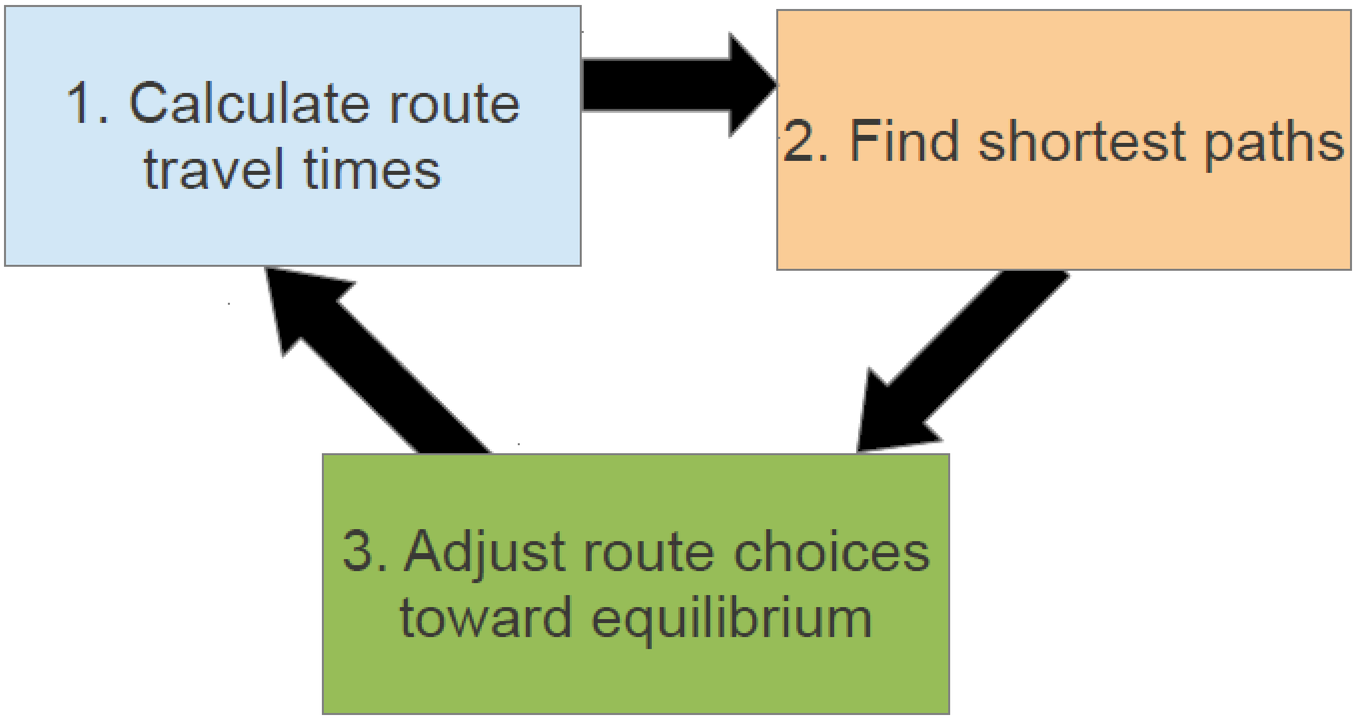
Essential requirements

- A model for how congestion (travel times) varies over time.
- A concept of equilibrium route choice.
- Equilibration based on experienced travel times, not instantaneous travel times.

What most simulation packages including Aimsun does is a one-shot assignment, not DTA!! Although it is presented as DTA..

DTA

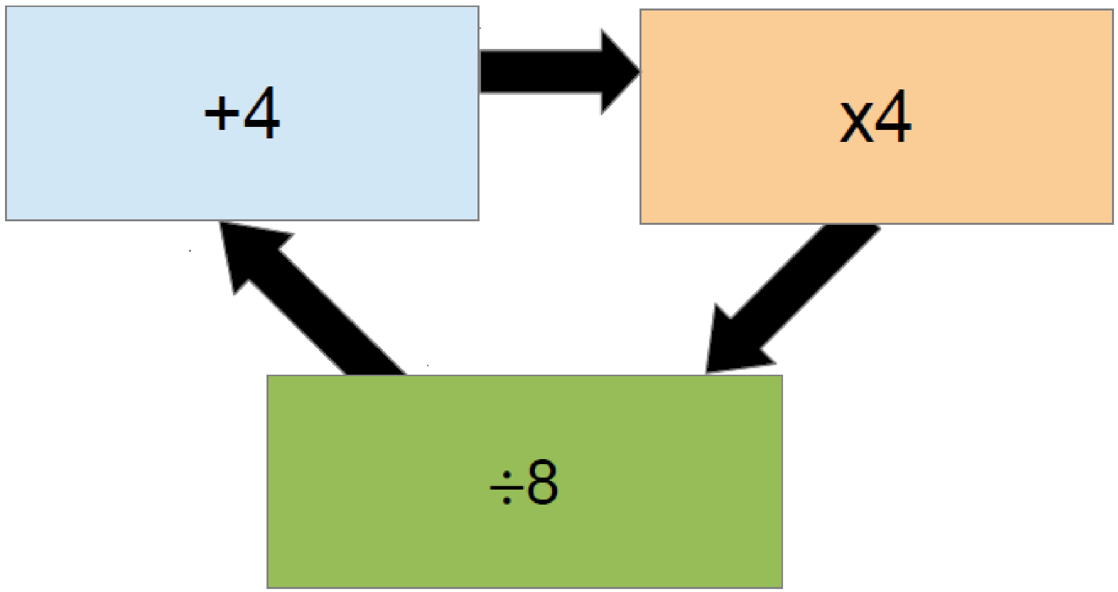
A similar iterative procedure to static traffic assignment



Interestingly, the easiest step in static (updating path travel times) is the hardest one in dynamic.

DTA – Fixed point

- Fixed-point solution can be presented in a mathematical form

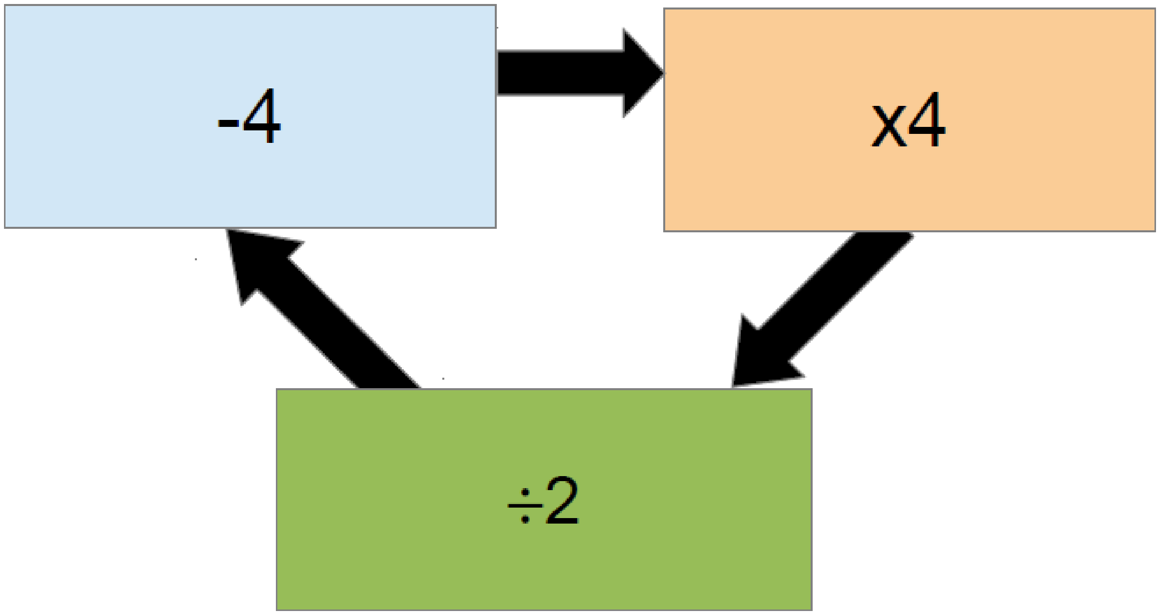


- This problem can be solved by denoting the values in the boxes as x, y, and z.
- We then have $y = x + 4$, $z = 4y$, and $x = z/8$. Substituting them into each other, we have $x = 0.5x + 2$ or $x = 4$.

DTA – Fixed point

- A not-so-clever approach is to pick a starting value, then calculate through the loop iteratively.
- 10 → 7 → 5.5 → 4.75 → 4.375 → … which converges to the correct answer (4).
- Picking a different starting value: 1 → 2.5 → 3.25 → 3.625 → … also converges to the same answer.
- No convergence guarantee particularly in DTA!!

DTA – Fixed point



- Choosing the same starting value, we have 10 → 12 → 16 → 24 → 40 →… which diverges to +Inf.
- The clever approach still works: y = x - 4, z = 4y, and x = z/2, so x = 2x - 8 and x = 8. If we used this as our starting value, the simple approach would work, but for any other value it will diverge.

DTA – Fixed point

- **The moral of the story:** if we are simply iterating between the three steps, we need to be careful to ensure that the solution will converge to a consistent answer.

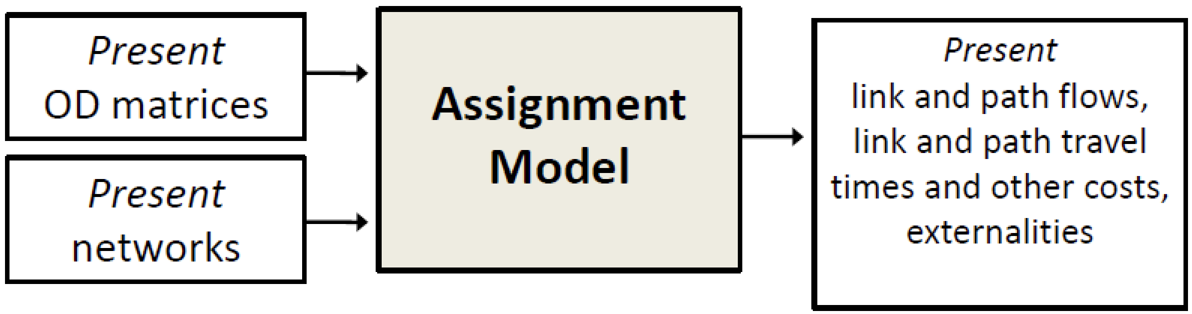
- DTA problems in large-scale networks are complicated enough that **we do not know of a faster or a clever method**.

- We are stuck with iterative methods, which suffer significant convergence issues particularly in congested large-scale networks.

Applications

Applications of Assignment models

- As estimators of the current state of the transport systems



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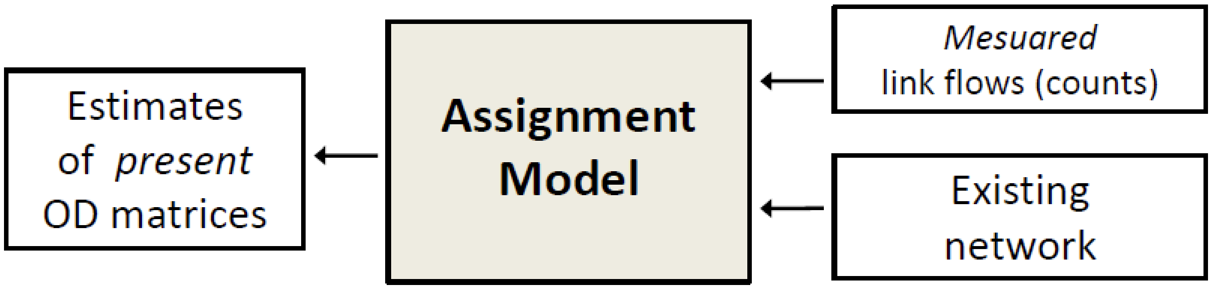

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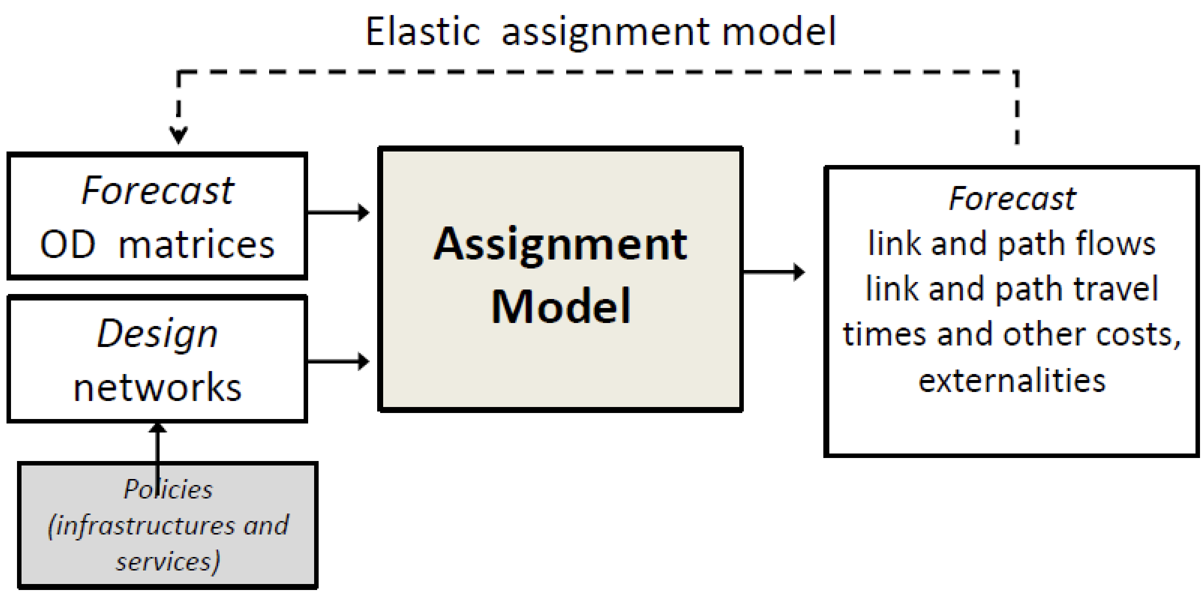

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